Existential Subtype Entailment Is Decidable for Non-Structural Subtyping

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Existential subtype entailment asks whether every solution of a finite system φ of subtype constraints extends to a solution of a second system ψ over additional hidden variables, that is, whether $\varphi \models \exists \bar{z}\, \psi$. Subsumption between constrained type schemes and the correctness of constraint simplification are instances of this question. For non-structural subtyping the problem has been open since 1996, and it has been suggested that it might be undecidable. We prove that it is decidable for one constructor f of every arity and every variance, with bottom and top, over finite, regular, and arbitrary trees, and also with bottom only when f is covariant. In particular, subsumption of constrained types over arrows, bottom and top is decidable, with no restriction on the polarity of the variables. When both systems are satisfiable in the tree class considered and the entailment fails, a counterexample word of length triply exponential in the number of variables after flattening certifies the failure, uniformly in the arity.

We represent each tree by two cones of words, those that meet a leaf of either sign, which turns every system into a Horn system with unary rules, and we characterize the entailment by three conditions on single seeds. Two of them become hole problems for finite readers: one for a single word, and one for two words that share a suffix, in which a word may also obstruct itself through a comparison with its own suffixes. For two words, outside one hard case, finite automata and a pumping argument give singly exponential holes. In the hard case, the mortal holes, we write the two words as one word around a separator, bound its hierarchical depth, and extend the bounded-depth compression from the decision procedure for plain entailment so that it preserves a comparison anchored at the separator. A search over compression summaries decides the problem in 2-EXPSPACE, and in PSPACE for a unary constructor. Shortest counterexamples can be exponentially long, and for a covariant constructor the problem is PSPACE-hard for every fixed arity at least two.

1 Introduction

Type inference with subtyping expresses the requirements of a program as subtype constraints between type expressions with unknowns [Aiken and Wimmers 1993; Eifrig et al. 1995; Fuh and Mishra 1990; Mitchell 1991; Pottier 1998a]. An inferred *constrained type* $\forall \bar{\alpha}.\, C \Rightarrow \tau$ pairs a type τ with a conjunction C of constraints; its instances are the types $\rho(\tau)$ for the solutions ρ of C [Trifonov and Smith 1996]. Inferred constraint sets grow quickly, and practical systems simplify them [Aiken et al. 1999; Fähndrich and Aiken 1996; Flanagan and Felleisen 1999; Pottier 1996, 2001; Trifonov and Smith 1996]. Whether a simplification is correct, and whether one constrained type may be used where another is expected, are questions about the solution sets of constraint systems.

The simplest such question is *entailment*: does every solution of a conjunction φ satisfy an inequality $x \le y$? Constrained types, however, also involve variables that occur on one side only. Let $\varphi(\bar{x})$ and $\psi(\bar{x}, \bar{z})$ be finite conjunctions of subtype constraints, where the *hidden* variables $\bar{z}$ do not occur in φ. *Existential entailment* $\varphi \models \exists \bar{z}\, \psi$ holds if every solution of φ extends to a solution of ψ by some choice of trees for $\bar{z}$ (Definition 2.4); logically, it is the validity of $\forall \bar{x}\, (\varphi \Rightarrow \exists \bar{z}\, \psi)$. Plain entailment is the case without hidden variables, and satisfiability of ψ is the case in which φ is empty and every variable is hidden.

Subsumption and Simplification. Trifonov and Smith [1996] define subtyping between constrained types semantically: $\forall \bar{\alpha}.\, C_1 \Rightarrow \tau_1$ is below $\forall \bar{\beta}.\, C_2 \Rightarrow \tau_2$ if every instance of the right-hand type is a supertype of some instance of the left-hand type. With $\bar{\alpha}$ and $\bar{\beta}$ renamed apart from each other and from the free variables of both types (Proposition 12.1), this is the existential entailment

$$\forall \bar{\alpha}.\, C_1 \Rightarrow \tau_1 \;\le\; \forall \bar{\beta}.\, C_2 \Rightarrow \tau_2 \qquad \Longleftrightarrow \qquad C_2 \models \exists \bar{\alpha}\, (C_1 \wedge \tau_1 \le \tau_2), \tag{1}$$

in which the variables of the right-hand type are universally quantified and the witnesses belong to the smaller, more general type. For example, $\forall\alpha.\,\alpha \to \alpha \le \beta \to \beta$ holds with the witness $\alpha = \beta$, while the converse fails; Proposition 12.1 adds the assumptions on program variables of Trifonov and Smith [1996]. Likewise, replacing a system $C(\bar{x}, \bar{z})$ by $D(\bar{x}, \bar{k})$, where $\bar{x}$ remain observable, is correct when $C \models \exists\bar{k}\, D$ and $D \models \exists\bar{z}\, C$. Removing a redundant constraint needs only plain entailment; eliminating or merging internal variables needs the existential form (Section 12). The arrow is contravariant in its domain, so a variable of a constrained type of a higher-order program typically occurs both co- and contravariantly, as α does in $\alpha \to \alpha$.

Why Plain Entailment Does Not Suffice. Consider trees built from a bottom element $\bot$ and a binary covariant constructor f, and the consequent

$$\psi \;=\; x \le z \;\wedge\; z \le y \;\wedge\; z \le f(z, z).$$

The literal $z \le f(z, z)$ says that all subtrees of z lie below z. Write $x \cdot \eta$ for the subtree of x at the path $\eta \in \{0, 1\}^*$, with $x \cdot \eta = \bot$ when x has no node at η. Then $\exists z\, \psi$ holds exactly when $x \cdot \eta \le y$ for every path η: necessity follows from $x \cdot \eta \le z \cdot \eta \le z \le y$, and the least upper bound of all the subtrees $x \cdot \eta$ is a witness. Thus $\varphi \models \exists z\, \psi$ is an infinite family of plain entailments, and the witness for z depends on the solution of φ. Sound tests replace hidden variables by canonical bounds or search for syntactic derivations [Pottier 2001; Trifonov and Smith 1996]; they are incomplete, and Section 12 recalls a correct simplification of Pottier [1998b] that such a test misses.

Example 1.1 (Shift closure: the running example). Take ψ as above, with x, y visible and z hidden. Let $K(x) \subseteq \{0, 1\}^*$ be the set of positions of the f-nodes of x, so that $x \le y$ iff $K(x) \subseteq K(y)$ (Lemma 2.3), and let $\mathrm{Suf}(L)$ be the set of suffixes of words of L. Since $K(x \cdot \eta) = \eta^{-1}K(x)$, the condition "$x \cdot \eta \le y$ for every η" says that $\exists z\, \psi$ holds iff $\mathrm{Suf}(K(x)) \subseteq K(y)$. For $\varphi_1 = x \le y$ the entailment fails: the solution $x = y = f(f(\bot, f(\bot, \bot)), \bot)$ of φ_1, with $K(x) = \{\varepsilon, 0, 01\}$, has the position 01, whose suffix 1 is not in $K(y)$. For $\varphi_2 = x \le y \wedge x \le f(x, b) \wedge b = \bot$, the entailment holds: $z = x$ is a witness, because $x \le f(x, b) \le f(x, x)$. For $\varphi_3 = x \le y \wedge x \le f(f(\bot, f(\bot, \bot)), \bot)$ the entailment fails with the same countermodel as for φ_1, in which x takes the largest value that φ_3 allows.

Figure 1 draws two solutions of φ_1. For the first, the join of all subtrees of x fits below y and is the least witness for z; for the second it does not, and no witness exists. The choice of z is made after x and y are known, from their values.

An Open Problem. In non-structural subtyping, $\bot \le t \le \top$ relates trees of different shapes. Plain non-structural subtype entailment was posed as an open problem in 1996 [Rehof 1996], and Zhao [2026] decides it for one constructor of any arity and variance. The existential problem has been open since the same year. Trifonov and Smith [1996, Section 1] "leave open the question whether it is decidable" and decide a sound approximation. Fähndrich and Aiken [1996, Section 3.2] write that "no complete algorithm is known". Su [2002, Section 7.2.2] writes that "for existential entailment and subtyping constrained types, virtually nothing is known at this point, except the lower bounds carried over from entailment", and that for non-structural subtyping the problem "may in fact be undecidable", because an alternative undecidability proof for the first-order theory appears to use a single quantifier alternation. Priesnitz [2004, Section 1.4.2] calls subtyping of constrained types a "prominent open problem". The first-order theory is indeed undecidable, already over $\{\bot, f\}$ with f binary and covariant, in all three tree classes [Su et al. 2002], but with six quantifier blocks and a matrix with disjunctions and negations, whereas existential entailment has one alternation, and its antecedent and consequent are conjunctions. Over feature trees, existential entailment is decided

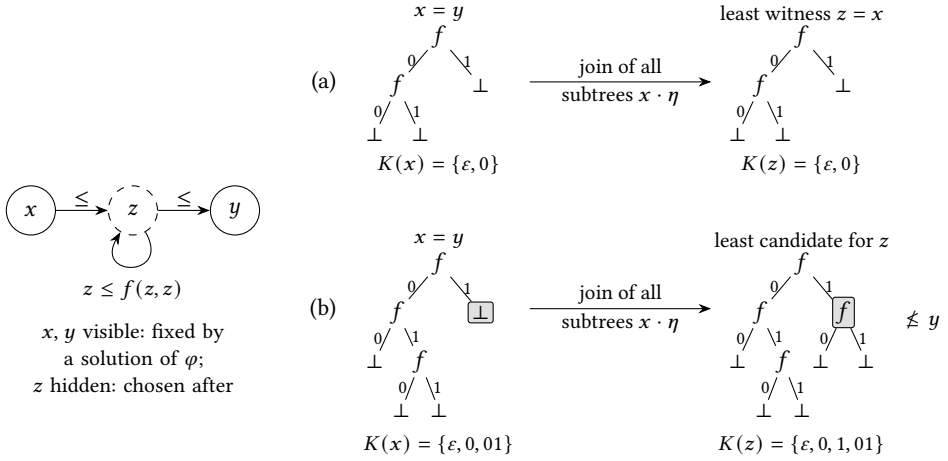


Fig. 1. The shift closure of Example 1.1 over $\{\bot, f\}$. A solution of φ fixes the visible variables x and y; the hidden z is chosen afterwards and depends on it. The literal $z \le f(z,z)$ makes the constructor positions of z closed under suffixes, so z must lie above every subtree of x, and their join is the least candidate. (a) For $x = y = f(f(\bot,\bot),\bot)$ the candidate is x itself, which fits below y. (b) For $x = y = f(f(\bot, f(\bot,\bot)),\bot)$, the countermodel of φ_1, the position 01 of x forces a constructor of z at its suffix 1 (shaded), where y has the leaf $\bot$; no z exists.

by an inclusion of finite automata [Müller et al. 1998, 2001], in a setting where a fresh feature is always available, which a constructor of fixed arity does not offer (Section 15).

Our Result. We fix one constructor f of arity $n \ge 0$ and a *variance* $c : \{0,\dots,n-1\} \to \{0,1\}$, which makes each argument covariant or contravariant; the arrow is $n = 2$ with $c(0) = 1$ and $c(1) = 0$ (Section 2.2). Trees may be infinite unless stated otherwise.

Theorem 1.2 (Main theorem). *For every arity $n \ge 0$ and every variance c, existential entailment $\varphi \models \exists \bar{z}\, \psi$ over $\{\bot, f, \top\}$ is decidable over arbitrary, over regular, and over finite trees. Over $\{\bot, f\}$, the same holds when f is covariant. If φ and ψ are satisfiable in the tree class considered and the entailment fails, then the failure is certified by a* counterexample word *of length at most $E_3(O(s))$: a word over the argument positions of f, or two such words joined by a separator, from which a countermodel is computed (Definition 11.1). Here $E_3(x) = 2^{2^{2^x}}$, s is the number of variables after flattening, every introduced name and every hidden variable included, and the constant does not depend on n.*

Section 11 defines counterexample words (Definition 11.1), states the theorem precisely (Theorem 11.2), shows that every counterexample word yields a countermodel effectively (Lemma 11.3), and assembles the proof. The theorem needs no restriction on the polarity of the variables: a variable may occur both co- and contravariantly, in the antecedent and in the consequent. In particular, it decides the subsumption of Trifonov and Smith [1996] between constrained types over $\bot$, $\to$ and $\top$, and the correctness of simplifications of such types (Section 12). We do not claim the signature $\{\bot, f\}$ with a contravariant argument; Remark 10.18 explains why the reduction to $\{\bot, f, \top\}$ needs covariance.

Bounds and Complexity. Proposition 11.5 and Section 13 refine the theorem. Only two kinds of failure, both described in Section 1.1, have triply exponential bounds: failures of condition (U) and *mortal holes*. The larger bound belongs to the mortal holes, the only case that needs a new compression technique; all other failures have singly exponential witnesses. The decision procedure

does not enumerate words: a nondeterministic search builds a witness level by level, keeping only a summary of each level, and runs in 2-EXPSPACE, hence in 3-EXPTIME (Proposition 13.1 and Theorem 13.2); for plain entailment this saves one exponential in space over the procedure of Zhao [2026] (Remark 13.3). For a unary constructor, $n = 1$, witnesses are singly exponential and the problem is in PSPACE, for both variances and all three tree classes (Theorem 13.4), and for $n = 0$ it is a finite enumeration (Remark 13.5). On the other side, shortest counterexample words can have length $2^{\Omega(s)}$ (Proposition 13.6), and the problem is PSPACE-hard for every fixed arity $n \geq 2$, already with f covariant, by explicit reductions that use the additional arguments of f; bare padding of a binary construction does not suffice (Theorem 13.9 and Example 13.7). The gap between PSPACE and 2-EXPSPACE, the gap between singly and triply exponential witnesses, and hardness for $n = 1$ remain open.

Mechanization. The decidability results are checked in the Lean 4 proof assistant [de Moura and Ullrich 2021] with Mathlib [The mathlib Community 2020]: over $\{\bot, f, \top\}$ for every arity, arity zero included, every variance and all three tree classes, and over $\{\bot, f\}$ with f covariant (Section 14). Each of the six procedures is a computable definition with a correctness theorem without premises, and plain entailment is decided as the special case without hidden variables. The anchored compression of Section 9 is mechanized with an elementary bound of the same form as ours, and all search bounds are explicit functions of the numbers of automaton states. The procedures take flat literals as input. The flattening of arbitrary type expressions (Lemma 2.6), the bounds in terms of the number of variables, the construction of countermodels and the complexity results are not mechanized.

1.1 Overview of the Proof

The proof follows a single route for all arities and variances (Figure 2). It works over $\{\bot, f, \top\}$ and arbitrary trees and transfers the result to the other tree classes and to $\{\bot, f\}$ at the end. We end in the hole problem of the plain-entailment proof [Zhao 2026], whose readers, holes and bounds Section 2 recalls: a *reader* is a deterministic automaton with rejected states and an admission relation on pairs of states; a *comparison* of a word w is a pair of cuts (s, e), $s < e$, such that the suffix after e is a prefix of the suffix after s; and a *hole* is a rejected word without an admitted comparison. That proof bounds a shortest hole in three steps: a short rejected tail, a bound on the *depth*, a measure of nested repetition, and a compression of words of bounded depth. What is new here is that some of our holes must also *contain* a comparison of a particular kind, and that a single word may be dead because of a comparison with itself.

Leaf cones. Over $\{\bot, f\}$ with f covariant, a tree is its set of constructor positions, solution sets are closed under unions and intersections, and every literal becomes a set of implications with one premise (Remark 4.6). With $\top$ and variance, a literal can relate a leaf $\top$ of one tree to a leaf $\bot$ of another, and one set of positions no longer describes a tree. We restore a Horn structure for every arity and variance by describing a tree t through its two *leaf cones*: the words whose path meets a leaf of t of *sign* 0, and those whose path meets a leaf of sign 1, where the sign of a leaf combines its label, $\top$ or $\bot$, with the polarity of its path. The two cones are disjoint and closed under extension, every such pair of sets is the pair of exactly one tree, and the order of variance c adds leaves of sign 0 and removes leaves of sign 1 (Theorem 3.2). An *atom* $(v^{\pm}, w)$ says that w lies in a cone of v. Every literal becomes *pure moves*, implications between atoms that copy, push or pop the first letter of the word as on a stack, together with *facts*, atoms that every solution contains, and *forbidden roots*, atoms that none contains (Lemma 3.5). The only all-negative binary clauses are the exclusions of opposite atoms, which are the same for every system. So least models are closures of seeds, every atom of a closure has a single seed (Lemma 3.8), every pure move has a dual of

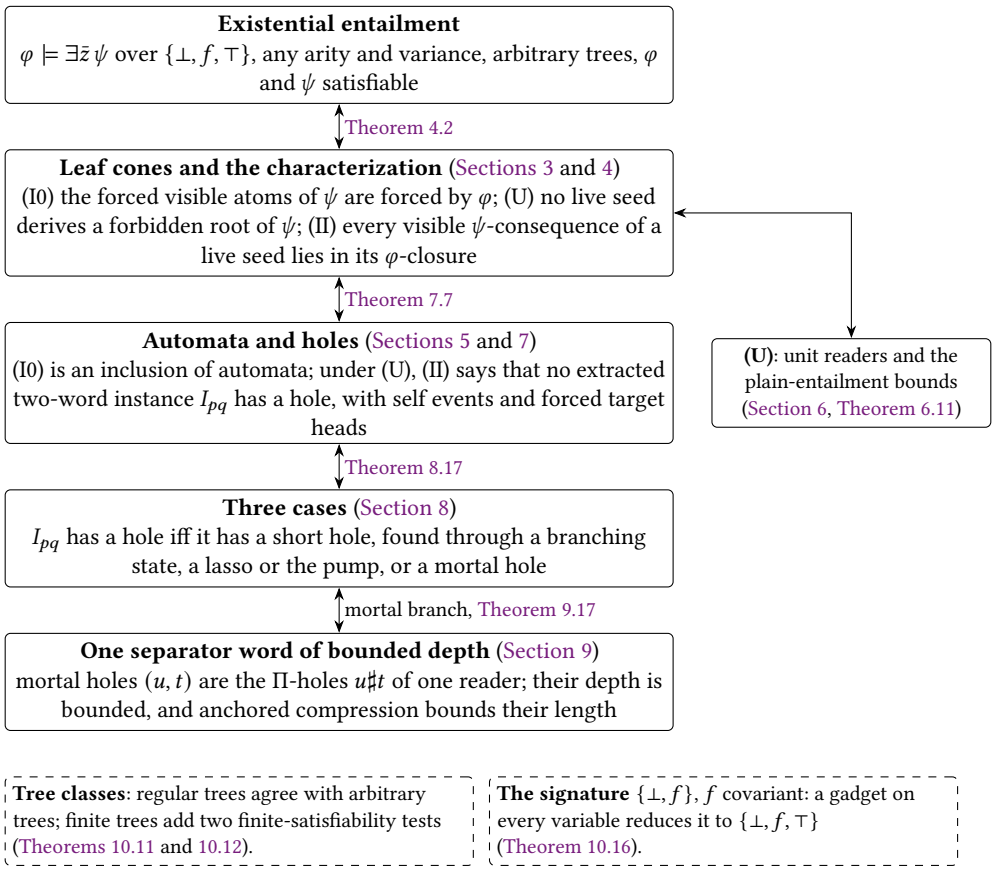


Fig. 2. The route of the proof, for satisfiable φ and ψ over $\{\bot, f, \top\}$ and arbitrary trees. Each double arrow is an equivalence, labeled with the result that proves it, or for condition (U) with its box; the last vertical one concerns only the mortal branch of the case split, whose short holes are found directly. Condition (U) leaves the main route: it is a hole problem of the kind solved for plain entailment, decided by the bounds of Zhao [2026]. Here p, q range over the visible signed variables. The dashed boxes describe the transfers to the other tree classes and to $\{\bot, f\}$. Section 11 assembles the steps.

opposite sign (Lemma 3.10), and two derivations that collide in an exclusion can be merged into one (Lemma 3.12).

Three conditions. Collision return turns every obstruction to extending a solution of φ into an obstruction at a single seed. If both systems are satisfiable, $\varphi \models \exists \bar{z}\, \psi$ holds iff three conditions hold (Theorem 4.2): (I0), every visible atom that ψ forces is forced by φ; (U), ψ derives no forbidden root from a visible atom that some solution of φ contains; and (II), every visible atom that ψ derives from such an atom lies in the least solution of φ that contains it.

Pushdown summaries. The pure moves form a pushdown system whose control states are the signed variables. Two deterministic automata summarize it: after reading a word γ, the *upper automaton* of p holds the signed variables reached from p by popping γ, and after reading β the *lower automaton* of q holds those from which q is reached by pushing β. Every pure derivation splits at its lowest point into a *pop head*, read by the first, and a *push head*, read by the second,

above a common suffix (Lemma 5.3). Variance adds a new reason for an atom to be dead. Besides a reachable forbidden root, or an opposite extension forced by the facts, which make an atom *dead by support*, the closure of a seed (p, w) may contain the opposite of an extension of the seed. For a seed that is not already dead by support, this happens exactly when two cuts of the single word w form a comparison whose automaton states are related: a *self event* (Theorem 5.12). Self events occur only through contravariant directions (Remark 5.14), and the set of live words need not be regular (Example 5.13). Condition (I0) is an inclusion of automata, decidable in polynomial space (Proposition 5.17).

Condition (U) as a hole problem. A failure of (U) is a word w such that (p, w) is live in φ and leads to a forbidden root in ψ. Liveness is the absence of deadness by support, a regular property, and of self events, which are comparisons of w. So the failures are exactly the holes of a single reader, the *unit reader*, whose admitted comparisons are the self events (Lemma 6.5). It has the rejected-path property of Zhao [2026], and the bounds proved there decide (U) (Theorem 6.11). A smaller reader with the same holes, together with a clock that reads only the antecedent and bounds the depth, keeps the length bound for (U) below the one for mortal holes.

Two-word holes with self events. A failure of (II) consists of a live seed (p, t), a pure derivation of ψ from it to a visible atom (q, u), and the fact that (q, u) lies outside the closure of the seed in φ. Each part is read by automata on the two words: the derivation is a *head event*, a pair of cuts at which t and u have equal suffixes; membership in the closure is an *event*, a pair of cuts (s, e) at which the suffix of t after e is a prefix of the suffix of u after s; and liveness of the seed forbids self events of t. So, once (U) holds, (II) fails iff an *extracted instance* I_{pq} of a hole problem for two words has a *hole*: a safe spine u, a rejected target t, a head event, and neither an event nor a self event (Definition 7.1 and Theorem 7.7). The target automaton marks a cut of the seed as rejected only while the seed is still a candidate, neither dead by support in φ nor root-dead in ψ. Under (U) this enriched rejection loses nothing, and it forces every pop head of ψ to end before the first rejected cut, condition (FH) (Lemma 7.6).

Holes that are not mortal. Cut a hole (u, t) at the first rejected cut of its target. By (FH) the rest θ of the target is a common suffix of both words, and what remains is a *pre-hole*. Appending θ to a pre-hole creates no new event and no new self event, and it keeps an old one exactly when θ continues its *overhang*, the part of one suffix that the other does not cover, periodically (Lemma 8.1). So the question is whether one safe continuation breaks every such periodicity obligation. If the spine ends in a *branching* state, which reaches a cycle with a choice of letters, a breaking continuation exists by the theorem of Fine and Wilf [1965] (Lemma 8.6). If it ends in a *thin* state, long safe continuations follow finitely many lassos; a lasso that is not purely periodic breaks everything, and for a purely periodic lasso the *pump* shortens the pre-hole itself, keeping the obligations broken (Theorem 8.14). Short pre-holes are found by a finite graph (Proposition 8.12). In the remaining case the hole is *mortal*: its spine ends in a state without an infinite safe continuation, and its target ends shortly after its first rejected cut (Theorem 8.17). Outside the mortal case, holes are polynomial in the sizes of the automata, hence singly exponential in s.

Mortal holes. In a mortal hole the continuation is short, but the common suffix of the two words can be long, so the pair itself must be shortened, with both words together. We pack them into one word $u\sharp t$ around a separator. Events and self events become comparisons of this word, and a head event becomes an *anchored* comparison, one whose matched factor is followed by the separator (Lemma 9.4). Mortal holes are exactly the Π*-holes* of one reader, its holes with an anchored comparison whose state pair lies in a relation Π (Lemma 9.5), and the short rejected tail bounds their depth by the depth theorem of Zhao [2026] (Lemma 9.6). The bounded-depth compression

Table 1. The running examples.

Example	Sections	What it shows
Shift closure: $\psi = x \le z \wedge z \le y \wedge z \le f(z,z)$, with f binary and covariant		
Example 1.1	Section 1	an existential entailment that is an infinite family of plain entailments
Example 5.5	Section 5	the signed system over $\{\bot, f, \top\}$ and its pure derivations
Example 6.7	Section 6	(U) fails at $x = y = \top$, and holds once y is restricted to trees without $\top$
Example 7.11	Section 7	(II) fails by the hole $(01, 1)$ of an extracted instance, whose spine ends in a branching state
Example 8.4	Section 8	the pre-hole $(0, \varepsilon)$, its single obligation, and, after Lemma 8.6, a breaker
Example 11.4	Section 11	the procedure end to end: the hole word $01\sharp 1$, its two pins, and the finite countermodel of Example 1.1
The arrow: the antecedent $x \le x \to x$		
Examples 3.7 and 3.9	Section 3	a single-seed inconsistency
Example 4.4	Section 4	the entailment $x \le x \to x \models \exists z\,(x \le z \to z)$, whose failure without the antecedent needs collision return
Example 5.13	Section 5	a nonregular set of dead words
Examples 6.8 and 7.12	Sections 6 and 7	why admitted self events and the enrichment of the target automaton are needed

of that paper keeps every negative condition but can delete the letters on which the anchored comparison depends (Example 9.13). A *positive channel* records the anchored witness at each level of the hierarchy (Lemma 9.10). The channel is added to the information that this compression compares before it merges letters or deletes columns, so every shortening keeps some anchored witness. This gives the anchored compression theorem (Theorem 9.14) and a triply exponential bound on mortal holes (Corollary 9.19), the largest of our bounds, which decides the two-word problem under (FH) (Theorem 9.17).

Tree classes and the signature without top. Over regular trees the closure of regular seeds is regular, and a failure has at most two seeds, so regular and arbitrary trees give the same answer (Theorem 10.11). Least models are infinite in general, so finite trees need a different argument: the at most two observations behind a failure are expressed by finite ground bounds, and adding finite ground bounds preserves finite satisfiability, for every variance (Theorem 10.6). This gives the dispatch of Theorem 10.12. For $\{\bot, f\}$ with f covariant, the literals $v \le y_v$ and $y_v \le f(y_v, \dots, y_v)$ force v to be free of $\top$, and placing them on every visible and every hidden variable reduces $\{\bot, f\}$ to $\{\bot, f, \top\}$ in each tree class (Theorem 10.16).

The running examples. Two examples recur through the paper (Table 1). The first is the shift closure of Example 1.1, for a covariant constructor; the second follows the antecedent $x \le x \to x$ for the arrow, in which x occurs at both polarities. For a covariant constructor, and more generally on instances with a joint parity assignment, self events do not occur and every dead atom is dead by support (Remarks 4.5, 5.14 and 7.10); over $\{\bot, f\}$ the three conditions compare unions of solutions, least solutions, and closures of single positions (Remark 4.6).

Contributions.

- *Decidability* (Theorem 1.2). Existential entailment is decidable for one constructor of every arity and every variance over $\{\bot, f, \top\}$, and over $\{\bot, f\}$ with f covariant, in all three tree classes. This settles the decidability question of Trifonov and Smith [1996] for semantic subsumption of constrained types.
- *Signed leaf cones* (Sections 3 and 4). A representation of trees of any variance in which every system is a Horn system with unary rules, and a characterization of existential entailment by three conditions on single seeds.
- *Self events* (Sections 5 to 7). Deadness is deadness by support or a self event, a comparison of one word with itself. Both (U) and (II) become hole problems whose admissions include self events, the second with an enriched rejection that makes target heads forced.
- *Non-mortal holes* (Section 8). Breakers, lassos and a pump with polynomial capacity give singly exponential witnesses outside the mortal case.
- *Anchored compression* (Section 9). Positive channels extend the bounded-depth compression of Zhao [2026] to holes that must contain an anchored comparison.
- *Tree classes and* $\{\bot, f\}$ (Section 10). Regular and finite transfers for every variance, and a reduction of covariant $\{\bot, f\}$ to $\{\bot, f, \top\}$.
- *Complexity* (Section 13). Elementary witness bounds; a 2-EXPSPACE decision by summary search; PSPACE for $n \le 1$; exponential shortest witnesses; and PSPACE-hardness for every fixed arity $n \ge 2$, already with f covariant.
- *Applications* (Section 12). Subsumption of constrained type schemes and correctness of simplification are decidable, including arrow types without any restriction on polarity, and several classes reduce directly to plain entailment.
- *Mechanization* (Section 14). Computable decision procedures for all six cases of Theorem 1.2, checked in Lean 4, including anchored compression with an elementary bound.

Organization. Section 2 fixes notation and recalls the results of Zhao [2026] that we use. Sections 3 to 10 follow the route above, and Section 11 assembles the proof. Sections 12 to 16 discuss applications, complexity, the mechanization, related work, and open problems. Omitted proofs are in the appendices. Sections 3 and 4 give the characterization, and Sections 5 to 7 turn it into hole problems; Sections 8 and 9 are the technical core, and Section 9 assumes Section 2.6. A reader interested in the result and its use may go from Section 4 to Sections 11 and 12.

2 Background

Throughout the paper we fix an arity $n \ge 0$ and a variance c of the constructor f. This section fixes the notation for words, trees, and constraints, defines existential entailment, and states the results that we use from the decidability proof for plain entailment of Zhao [2026]. That paper treats one constructor of every arity and every variance; we cite its results in that generality. Short proofs are repeated; all other cited results are stated exactly and proved there. Appendix A collects the further results of that paper used later: the satisfiability criteria, the cores lemma, the ordinary and symbolic states of derived monitors, the structure of the bounded-depth search, and its constants.

2.1 Words

Put $\Sigma = \{0, \dots, n-1\}$, the alphabet of *directions*; the letter i names the ith argument of f. For $n = 0$ the alphabet is empty and ε is the only word over it. Readers (Section 2.5) also work over other finite alphabets, written Γ, for instance $\Sigma \sqcup \{\sharp\}$.

Words are finite unless they are explicitly called infinite. For a word w, a *cut* is an integer j with $0 \le j \le |w|$; it separates the first j letters from the rest. We write $w[i]$ for the letter at position i, where $0 \le i < |w|$, and $w[:j]$, $w[j{:}h]$, and $w[j:]$ for the prefix, factor, and suffix at the indicated

cuts. The empty word is ε, and $u \preceq w$ means that u is a prefix of w. For a set L of words and a word β, put $\beta L = \{\beta\sigma : \sigma \in L\}$ and $\beta^{-1}L = \{\sigma : \beta\sigma \in L\}$. A positive integer ϖ is a *period* of w if $w[i] = w[i - \varpi]$ for $\varpi \le i < |w|$. For a nonempty word o, we write $\theta \preceq o^\omega$ if $\theta \preceq o^k$ for some $k \ge 0$.

Lemma 2.1 (Periodic prefixes [Zhao 2026, Lemma "Periodic prefixes"]). *For every nonempty word o and every word θ, $\theta \preceq o\theta$ iff $\theta \preceq o^\omega$.*

Proof. If $\theta \preceq o^k$, then $o\theta \preceq o^{k+1}$ and $\theta \preceq o^{k+1}$; of two prefixes of one word the shorter is a prefix of the longer, so $\theta \preceq o\theta$. Conversely, let $\theta \preceq o\theta$. If $|\theta| \le |o|$, then $\theta \preceq o$. Otherwise $\theta = o\theta'$ for a word θ', and $o\theta' \preceq oo\theta'$ gives $\theta' \preceq o\theta'$. By induction on the length, $\theta' \preceq o^k$ for some k, so $\theta \preceq o^{k+1}$. □

2.2 Trees, Variance, and Normalization

Trees. Order the labels by $\bot < f < \top$. A *tree* is a partial map $t : \Sigma^* \rightharpoonup \{\bot, f, \top\}$ whose domain $\mathrm{dom}(t)$ contains ε, is prefix closed, and satisfies, for every $w \in \mathrm{dom}(t)$ and every $i \in \Sigma$,

$$wi \in \mathrm{dom}(t) \iff t(w) = f.$$

The elements of $\mathrm{dom}(t)$ are the *positions* of t. We write $f(t_0, \dots, t_{n-1})$, or $f(t_i)_i$, for the tree with root label f and subtrees t_i at the positions i. The *subtree* of t at a position β is the tree $\beta^{-1}t : w \mapsto t(\beta w)$. The full constructor tree $\Omega = f(\Omega, \dots, \Omega)$ has label f at every word; it is infinite for $n \ge 1$, and $\Omega = f()$ for $n = 0$. At arity zero the only trees are $\bot$, $f()$, and $\top$.

Variance and the order. A *variance* is a map $c : \Sigma \to \{0, 1\}$; argument i is *covariant* if $c(i) = 0$ and *contravariant* if $c(i) = 1$, and f is covariant if $c = 0$. For the arrow, $n = 2$, $f(a_0, a_1) = a_0 \to a_1$, $c(0) = 1$, and $c(1) = 0$. The *polarity* of a word counts its contravariant letters modulo two:

$$\mathrm{par}(w) = \bigoplus_{j < |w|} c(w[j]), \qquad \mathrm{par}(\varepsilon) = 0, \qquad \mathrm{par}(uv) = \mathrm{par}(u) \oplus \mathrm{par}(v). \tag{2}$$

The *non-structural order* of variance c is

$$t \le_c u \iff \forall w \in \mathrm{dom}(t) \cap \mathrm{dom}(u) \quad \begin{cases} t(w) \le u(w), & \mathrm{par}(w) = 0, \\ u(w) \le t(w), & \mathrm{par}(w) = 1. \end{cases} \tag{3}$$

For the covariant variance $c = 0$ we write $\le$ and call the order *covariant*. The relation $\le_c$ is a partial order, and it is the greatest relation R such that $R(t, u)$ implies $t = \bot$, or $u = \top$, or $t = f(a_i)_i$ and $u = f(b_i)_i$ with $R(a_i, b_i)$ for every i with $c(i) = 0$ and $R(b_i, a_i)$ for every i with $c(i) = 1$ [Zhao 2026, Sect. "Non-Structural Subtyping"]. In particular, $\bot \le_c t \le_c \top$ for every tree t, and $f(a_i)_i \le_c f(b_i)_i$ iff each pair of arguments is related in the direction given by its variance. For the arrow this is the usual rule: $a_0 \to a_1 \le_c b_0 \to b_1$ iff $b_0 \le_c a_0$ and $a_1 \le_c b_1$.

Duality and normalization. The *dual* t^∂ of a tree t has the domain of t and the labels of t with $\bot$ and $\top$ exchanged; labels f are unchanged. For a *sign* $s \in \{0, 1\}$, the *normalization* $\nu_s t$ has the domain of t; its label at a position w is that of t with $\bot$ and $\top$ exchanged if $s \oplus \mathrm{par}(w) = 1$, and unchanged otherwise. Thus ν_0 exchanges the extreme labels at the positions of odd polarity and ν_1 at those of even polarity. The polarity is taken at actual positions of t only; a leaf is never followed by further letters. For covariant f, $\nu_0 t = t$ and $\nu_1 t = t^\partial$. We also write $+$ for the sign 0 and $-$ for the sign 1.

Lemma 2.2 (Normalization [Zhao 2026, Lemma "Normalization"]). *For every variance c, all trees $t, u, t_0, \dots, t_{n-1}$, and every sign s:*

(1) $\nu_s f(t_i)_i = f(\nu_{s \oplus c(i)} t_i)_i$, $\nu_1 t = (\nu_0 t)^\partial = \nu_0(t^\partial)$, and $\nu_s \nu_s t = t$;
(2) $t \le_c u$ iff $\nu_0 t \le \nu_0 u$ iff $\nu_1 u \le \nu_1 t$;

(3) $t \le_c u$ *iff* $u^\partial \le_c t^\partial$;
(4) *the subtree of* $v_s t$ *at a position* w *is* $v_{s \oplus \mathrm{par}(w)}(w^{-1}t)$, *and* v_s *and duality preserve domains, finiteness, and regularity.*

Proof. (1) The root of $f(t_i)_i$ is labeled f and is never changed. At a position iw we have $\mathrm{par}(iw) = c(i) \oplus \mathrm{par}(w)$, so the condition $s \oplus \mathrm{par}(iw) = 1$ for v_s at iw is the condition for $v_{s \oplus c(i)}$ at the position w of t_i. The conditions of v_0 and v_1 at a position are complementary, so $v_1 t$ is $v_0 t$ with every extreme label exchanged, and exchanging all extreme labels commutes with exchanging some of them. Applying v_s twice exchanges the same labels twice.

(3) Exchanging $\bot$ and $\top$ reverses the chain $\bot < f < \top$, and t and t^∂ have the same domain, so every comparison in Eq. (3) is reversed.

(2) The trees t and $v_0 t$ have the same domain. At a common position w with $\mathrm{par}(w) = 0$, v_0 changes neither label; at one with $\mathrm{par}(w) = 1$ it applies the order-reversing exchange to both labels, so $u(w) \le t(w)$ iff $(v_0 t)(w) \le (v_0 u)(w)$. Hence Eq. (3) for $\le_c$ is the covariant comparison of $v_0 t$ and $v_0 u$ at every common position. The second equivalence follows from (1) and (3) for the covariant order.

(4) The first assertion follows from (1) by induction on w. So the subtrees of $v_s t$ are among the trees $v_0 t'$ and $v_1 t'$ for subtrees t' of t, which are finitely many if t is regular. Domains are preserved by definition, and duality is handled in the same way. □

Thus v_0 is an order isomorphism from the trees ordered by $\le_c$ onto the trees ordered covariantly, and v_1 is an order anti-isomorphism. This is how variance is removed: a constraint of variance c about t becomes a pair of covariant constraints about $v_0 t$ and $v_1 t$ (Lemma 2.5).

Tree classes. We consider three *tree classes*: the class $\mathcal{T}$ of all trees, possibly infinite; the class $\mathcal{T}_{\mathrm{reg}}$ of *regular* trees, which have finitely many distinct subtrees; and the class $\mathcal{T}_{\mathrm{fin}}$ of *finite* trees, which have finite domain. Then $\mathcal{T}_{\mathrm{fin}} \subseteq \mathcal{T}_{\mathrm{reg}} \subseteq \mathcal{T}$, and Ω is regular but not finite for $n \ge 1$. A tree is regular iff it is the unfolding of a finite graph whose constructor nodes have n ordered successors [Zhao 2026, Sect. "Words and Trees"]. Constructor application and the constants preserve each class. At arity zero the three classes coincide.

Trees over $\{\bot, f\}$. A *tree over* $\{\bot, f\}$ is a tree without the label $\top$. For a tree class $\mathcal{D}$, its trees over $\{\bot, f\}$ form the corresponding class over $\{\bot, f\}$; constructor application and $\bot$ preserve it. We use the signature $\{\bot, f\}$ only with f covariant. Then a tree t over $\{\bot, f\}$ is determined by the set $K(t) = \{w : t(w) = f\}$ of its constructor positions, which is prefix closed, and every prefix-closed set of words is $K(t)$ for exactly one tree t over $\{\bot, f\}$.

Lemma 2.3 (Order over $\{\bot, f\}$). *For covariant trees* t, u *over* $\{\bot, f\}$, $t \le u$ *iff* $K(t) \subseteq K(u)$.

Proof. Suppose $K(t) \subseteq K(u)$ and let w be a common position. If $t(w) = f$, then $u(w) = f$; otherwise $t(w) = \bot$. In both cases $t(w) \le u(w)$. Conversely, suppose $t \le u$ and $w \in K(t)$. Every prefix of w lies in $K(t)$. By induction on the prefixes w' of w, each of them is a position of u, and $f = t(w') \le u(w')$ forces $u(w') = f$, since u has no label $\top$. Hence $w \in K(u)$. □

2.3 Constraints and Existential Entailment

Flat systems. Fix a set of variables. A *flat literal* has one of the forms

$$u \le v, \qquad f(a_i)_i \le u, \qquad u \le f(a_i)_i, \qquad u = \bot, \qquad \top \le u,$$

where $u, v, a_0, \ldots, a_{n-1}$ are variables, not necessarily distinct. We call $u \le v$ a *copy literal*, $f(a_i)_i \le u$ a *lower constructor literal* and $u \le f(a_i)_i$ an *upper constructor literal*, both with *root* u and *children* a_i, and $u = \bot$ and $\top \le u$ *extreme literals*. Over $\{\bot, f\}$ the literal $\top \le u$ is not used. A *system* is a

Table 2. The signed literals of a literal of variance c; here $\bar{c}(i) = 1 \oplus c(i)$.

Literal of χ	Literals of $\chi^{\pm}$	
$u \le v$	$u^+ \le v^+$	$v^- \le u^-$
$f(a_i)_i \le u$	$f(a_i^{c(i)})_i \le u^+$	$u^- \le f(a_i^{\bar{c}(i)})_i$
$u \le f(a_i)_i$	$u^+ \le f(a_i^{c(i)})_i$	$f(a_i^{\bar{c}(i)})_i \le u^-$
$u = \bot$	$u^+ = \bot$	$\top \le u^-$
$\top \le u$	$\top \le u^+$	$u^- = \bot$

finite conjunction of flat literals, also read as a finite set of literals; a system *over* a set V has its variables in V.

Literals are always written with $\le$ and are read with the order of the variance at hand: an assignment ρ of trees to the variables of a system χ is a *solution* of χ for the variance c if $\rho(u) \le_c \rho(v)$ for every $u \le v \in \chi$, $f(\rho(a_i))_i \le_c \rho(u)$ for every $f(a_i)_i \le u \in \chi$, $\rho(u) \le_c f(\rho(a_i))_i$ for every $u \le f(a_i)_i \in \chi$, $\rho(u) = \bot$ for every $u = \bot \in \chi$, and $\rho(u) = \top$ for every $\top \le u \in \chi$. A *covariant system* is read with the covariant order. For a tree class $\mathcal{D}$, $\mathrm{Sol}_{\mathcal{D}}(\chi)$ is the set of solutions with values in $\mathcal{D}$, and χ is *$\mathcal{D}$-satisfiable*, written $\mathrm{Sat}_{\mathcal{D}}(\chi)$, if this set is nonempty. We write Sat, $\mathrm{Sat}_{\mathrm{reg}}$, and $\mathrm{Sat}_{\mathrm{fin}}$ for the three tree classes, and add the subscript c, as in Sat_c, when the variance must be named. Over $\{\bot, f\}$, all values are trees over $\{\bot, f\}$.

Definition 2.4 (Existential entailment). Let X and Z be disjoint finite sets of variables, let φ be a system over X, and let ψ be a system over $X \cup Z$. For a tree class $\mathcal{D}$, the *existential entailment*

$$\varphi \models_{\mathcal{D}} \exists \bar{z}\, \psi$$

holds if every solution $\rho : X \to \mathcal{D}$ of φ extends to a solution $\rho' : X \cup Z \to \mathcal{D}$ of ψ. The variables in X are *visible*, those in Z *hidden*. Over $\{\bot, f\}$, both ρ and ρ' take values in the trees of $\mathcal{D}$ over $\{\bot, f\}$.

We omit the subscript for $\mathcal{D} = \mathcal{T}$, and write $\models_{\mathrm{reg}}$ and $\models_{\mathrm{fin}}$ for the other two classes. As predicates of the pair of systems, $E_{\mathrm{all}}(\varphi, \psi)$, $E_{\mathrm{reg}}(\varphi, \psi)$ and $E_{\mathrm{fin}}(\varphi, \psi)$ denote the existential entailment $\varphi \models_{\mathcal{D}} \exists \bar{z}\, \psi$ for $\mathcal{D} = \mathcal{T}$, $\mathcal{T}_{\mathrm{reg}}$ and $\mathcal{T}_{\mathrm{fin}}$, respectively. The subscript of $\models$ always names a tree class; the signature, $\{\bot, f\}$ or $\{\bot, f, \top\}$, and the variance are stated in words. Without loss of generality X contains every variable of φ and every variable of ψ outside Z; a visible variable that does not occur in φ is unconstrained there. If φ is not $\mathcal{D}$-satisfiable, the entailment holds vacuously; if φ is $\mathcal{D}$-satisfiable and ψ is not, it fails. *Plain entailment* $\varphi \models_{\mathcal{D}} x \le y$ is the special case $\psi = \{x \le y\}$ with $Z = \varnothing$; it is the non-structural subtype entailment problem, which Zhao [2026] decides.

The size parameter. All size bounds of this paper are stated in terms of

$$s = |X| + |Z|,$$

the total number of variables, counted after flattening (see below) with every introduced name included. The bounds on the lengths of witness words do not depend on the arity n.

Signed systems. The *signed variables* of a set V are the pairs $v^s = (v, s)$ with $v \in V$ and a sign s; we write $v^+ = v^0$, $v^- = v^1$, $V^{\pm}$ for their set, and $\overline{v^s} = v^{1-s}$ for the *flip* of a signed variable. The *signed system* $\chi^{\pm}$ of a system χ over V, for the variance c, is the covariant system over $V^{\pm}$ that contains, for every literal of χ, the two literals listed in Table 2. It extends the signed translation of the plain-entailment proof [Zhao 2026, Sect. "The Signed System"] by copy literals.

Lemma 2.5 (Signed correspondence [Zhao 2026, Lemma "Signed correspondence"]). *Let χ be a system over V and c a variance. For an assignment ρ of trees to V, define $\rho^{\pm}$ on $V^{\pm}$ by $\rho^{\pm}(v^s) = \nu_s \rho(v)$. Then ρ is a solution of χ for the variance c iff $\rho^{\pm}$ is a solution of the covariant system $\chi^{\pm}$. The map $\rho \mapsto \rho^{\pm}$ is a bijection from the assignments to V onto the assignments σ to $V^{\pm}$ with $\sigma(v^-) = \sigma(v^+)^{\partial}$ for every v, and both $\rho \mapsto \rho^{\pm}$ and its inverse preserve each tree class.*

Proof. Each literal of χ is equivalent to each of its two signed literals under $\rho^{\pm}$, by Lemma 2.2. For $u \le v$, part (2) gives $\rho u \le_c \rho v$ iff $\nu_0 \rho u \le \nu_0 \rho v$ iff $\nu_1 \rho v \le \nu_1 \rho u$. For $f(a_i)_i \le u$, part (1) gives $\nu_0 f(\rho a_i)_i = f(\rho^{\pm}(a_i^{c(i)}))_i$ and $\nu_1 f(\rho a_i)_i = f(\rho^{\pm}(a_i^{\bar{c}(i)}))_i$, and part (2) applies again; the upper constructor literals are symmetric. For the extreme literals, ν_0 maps the one-node trees $\bot$ and $\top$ to themselves and ν_1 exchanges them. By part (1), $\rho^{\pm}(v^-) = \rho^{\pm}(v^+)^{\partial}$; conversely, if σ has this property, then $\rho(v) = \nu_0 \sigma(v^+)$ defines the unique assignment with $\rho^{\pm} = \sigma$, because ν_0 is an involution. Part (4) gives the preservation of the tree classes. □

A solution σ of $\chi^{\pm}$ need not satisfy $\sigma(v^-) = \sigma(v^+)^{\partial}$, but χ and $\chi^{\pm}$ are satisfiable together, in each tree class (Lemma A.2).

Flattening. Inequalities and equalities between arbitrary finite type expressions reduce to flat literals by naming subexpressions. Let φ and ψ be finite conjunctions of inequalities $e \le e'$ and equalities $e = e'$ between finite expressions built from variables, $\bot$, f of arity n, and, over $\{\bot, f, \top\}$, $\top$; the variables of ψ are visible or hidden as in Definition 2.4. Treat φ and ψ separately. In each, give every distinct nonvariable subexpression e a fresh name v_e, visible for φ and hidden for ψ; a subexpression of both systems receives one name of each kind. Let $v_e = e$ when e is a variable. For $e = f(e_i)_i$ add the *defining literals* $v_e \le f(v_{e_i})_i$ and $f(v_{e_i})_i \le v_e$, and for a constant add $v_e = \bot$ or $\top \le v_e$. Replace $e \le e'$ by $v_e \le v_{e'}$, and $e = e'$ by $v_e \le v_{e'}$ and $v_{e'} \le v_e$. Write φ' and ψ' for the resulting systems, and $\bar{z}'$ for the hidden variables of ψ together with the names introduced in ψ.

Lemma 2.6 (Flattening). *For every variance c, both signatures, and every tree class $\mathcal{D}$, $\varphi \models_{\mathcal{D}} \exists \bar{z}\, \psi$ iff $\varphi' \models_{\mathcal{D}} \exists \bar{z}'\, \psi'$. The flattened systems have size linear in the size of φ and ψ.*

Proof. The order $\le_c$ is antisymmetric, so the two defining literals of v_e say $v_e = f(v_{e_i})_i$, and the defining literal of a constant fixes its value. By induction on subexpressions, every assignment to the variables of a system has exactly one extension to its names that satisfies the defining literals, namely the one that assigns to v_e the value of e, and under it the replaced literals say exactly the original ones. These values lie in $\mathcal{D}$, and are trees over $\{\bot, f\}$ if the assignment is, because constructor application and the constants preserve each class.

Suppose $\varphi \models_{\mathcal{D}} \exists \bar{z}\, \psi$, and let ρ' be a $\mathcal{D}$-solution of φ'. Its restriction to the original visible variables solves φ, since ρ' assigns to each name of φ the value of its subexpression. Extend this restriction to the hidden variables of ψ so that ψ holds, and then to the names of ψ by the values of their subexpressions; the result solves ψ' and agrees with ρ' on the visible variables of φ'. Conversely, suppose $\varphi' \models_{\mathcal{D}} \exists \bar{z}'\, \psi'$, and let ρ be a $\mathcal{D}$-solution of φ. Its extension to the names of φ solves φ', so it extends further to a solution of ψ', which assigns to each name of ψ the value of its subexpression; hence its restriction to the variables of ψ solves ψ. For the size, there is one name and at most two defining literals, each with at most $n + 1$ variables, per distinct subexpression. □

From now on all systems are flat, and s counts the variables *after flattening*. The bounds of this paper would be false in terms of the variables of unflattened expressions. For example, let f be covariant with $n \ge 1$, and put $T_0 = \bot$ and $T_{k+1} = f(T_k, \dots, T_k)$, so that $K(T_k)$ is the set of words of length less than k. Over $\{\bot, f\}$, the query true $\models x \le T_k$ has one variable, and by Lemma 2.3 a solution violates $x \le T_k$ exactly when x has a constructor position of length k; so

every countermodel has a constructor at depth k. Flattening names the $k+1$ distinct subexpressions $T_0, \ldots, T_k$, so that $s = k + 2$, and the bounds in s then apply.

2.4 Results on Constraints from the Plain-Entailment Proof

The flat syntax of Zhao [2026] has no copy literals: its literals are $f(a_i)_i \le u$, $u \le f(a_i)_i$, $u = \bot$, and $u = \top$, the last being our $\top \le u$. At positive arity copy literals are eliminated as follows [Zhao 2026, Proposition "Flattening"].

Lemma 2.7 (Sandwich elimination). *Let $n \ge 1$. Replace a literal $u \le v$ by the two literals $f(a_i)_i \le y$ and $y \le f(b_i)_i$, where y and o are fresh variables, $a_i = b_i = o$ for $i \ge 1$, and $(a_0, b_0) = (u, v)$ if $c(0) = 0$ and $(a_0, b_0) = (v, u)$ if $c(0) = 1$. An assignment satisfies $u \le v$ iff it extends to y and o so as to satisfy the two new literals. If the assignment takes values in a tree class $\mathcal{D}$, over $\{\bot, f, \top\}$ or over $\{\bot, f\}$, the extension can be chosen in the same class.*

Proof. If the two literals hold, transitivity gives $f(a_i)_i \le_c f(b_i)_i$, whose argument 0 gives $u \le_c v$ in both cases of $c(0)$. Conversely, put $o = \bot$ and $y = f(a_i)_i$; then $f(a_i)_i \le_c f(b_i)_i$, because the arguments 0 compare in the direction of their variance and the other arguments are equal. This extension lies in the class of the assignment. □

Applying Lemma 2.7 to every copy literal of a system χ gives a system $\chi^\flat$ in the syntax of Zhao [2026] whose solutions in each tree class are exactly the extensions of the solutions of χ in that class, and whose size is linear in that of χ.

Remark 2.8 (Arity zero). For $n = 0$ the trees are $\bot < f() < \top$, the only word is ε, the order $\le_c$ is this chain for the empty variance, and the three tree classes coincide. Satisfiability of a system over V, and existential entailment $\varphi \models \exists \bar{z}\, \psi$, are then decided by enumerating the $3^{|V|}$, respectively 3^s, assignments.

Theorem 2.9 (Decidability for constraints [Zhao 2026, Theorem "Main theorem", Proposition "Transfer", Theorem "Least shape and satisfiability", Lemma "Finite signed correspondence"]). *For every arity $n \ge 0$ and every variance c, the following hold for systems over $\{\bot, f, \top\}$, copy literals included.*

(1) Satisfiability, regular satisfiability, and finite satisfiability are decidable, and $\mathrm{Sat}_{\mathrm{reg}}(\chi) \iff \mathrm{Sat}(\chi)$.

(2) Plain entailment $\varphi \models_{\mathcal{D}} x \le y$ is decidable for each tree class $\mathcal{D}$, and

$$\varphi \models_{\mathrm{reg}} x \le y \iff \varphi \models x \le y, \qquad \varphi \models_{\mathrm{fin}} x \le y \iff \neg\, \mathrm{Sat}_{\mathrm{fin}}(\varphi) \ \vee\ \varphi \models x \le y.$$

Proof. For $n = 0$ use Remark 2.8. For $n \ge 1$, replace χ and φ by $\chi^\flat$ and $\varphi^\flat$. Their solutions in each tree class are the extensions of those of χ and φ, so satisfiability in each class and plain entailment in each class are unchanged. Zhao [2026] decides satisfiability and finite satisfiability in that syntax and proves that a satisfiable system has a regular solution (Theorem A.1); it decides plain entailment over the three tree classes and proves the two displayed transfers. □

These procedures concern solutions over $\{\bot, f, \top\}$. A system without literals $\top \le u$ can still have solutions that use $\top$, so they do not decide finite satisfiability over $\{\bot, f\}$: for $n \ge 1$ and f covariant, $\{f(x, \ldots, x) \le x\}$ has the finite solution $x = \top$, but its only solution over $\{\bot, f\}$ is $x = \Omega$. The signature $\{\bot, f\}$ is reduced to $\{\bot, f, \top\}$ in Section 10.

2.5 Readers and Holes

Of this subsection and the next, Sections 3 to 5 use only comparisons (Definition 2.10). Readers and the tail, depth and search theorems are used from Section 6 on; lettered monitors, derivatives and the outline of the search at the end of Section 2.6 are used only in Section 9, and the first two also in the summary search of Section 13.

Zhao [2026] reduces plain entailment to a word problem for finite automata, the hole problem below, and proves bounds that hold for every such automaton. We use the same notions and bounds. Throughout, Γ is a finite alphabet.

Definition 2.10 (Comparison). A *comparison* of a word w is a pair of cuts (s, e) with $0 \le s < e \le |w|$ and $w[e{:}] \preceq w[s{:}]$. Its *period* is $e - s$, and it is *terminal* if $e = |w|$.

Equivalently, (s, e) is a comparison iff the suffix $w[s{:}]$ has period $e - s$: reading w from e repeats what it reads from s, until the word ends. Every pair $(s, |w|)$ with $s < |w|$ is a terminal comparison. For example, the comparisons of 0100 are $(0, 3)$, $(2, 3)$, and the four terminal pairs $(s, 4)$; the pair $(0, 2)$ is not one, since 00 is not a prefix of 0100. If (s, e) is a comparison of w and $e \le h \le |w|$, then (s, e) is a comparison of $w[{:}h]$.

Definition 2.11 (Reader). A *reader* $(\mathcal{M}, \mathsf{Adm}, \mathsf{Rej})$ consists of a deterministic finite automaton $\mathcal{M} = (Q, q_0, \delta)$ over Γ, with $\delta : Q \times \Gamma \to Q$ extended to words, an *admission* relation $\mathsf{Adm} \subseteq Q \times Q$, and a set $\mathsf{Rej} \subseteq Q$ of *rejected* states. Its *size* is $N = |Q| \ge 1$. For a word w, we write $q_i = \delta(q_0, w[{:}i])$ for the state at cut i.

Definition 2.12 (Hole). A word w is *rejected* if $q_{|w|} \in \mathsf{Rej}$; then $J(w)$ is the least cut J with $q_J \in \mathsf{Rej}$, the *first rejected cut*, and $r(w) = |w| - J(w)$ is the *rejected tail* of w. A rejected word w is a *hole* of the reader if no comparison (s, e) of w satisfies $\mathsf{Adm}(q_s, q_e)$; a comparison that satisfies it is *admitted*.

A reader thus reads a word once and tests every comparison by the pair of states at its two cuts; a hole is a rejected word that passes all these tests. The names follow Zhao [2026]. Rejection is the regular part of a failure, and an admitted comparison is a non-regular exception that rescues the word; a hole is a word that the regular test condemns and no comparison rescues. In the readers of later sections, a rejected word is a candidate failure, and an admitted comparison shows that the candidate is not one. The empty word has no comparison, so it is a hole iff $q_0 \in \mathsf{Rej}$. The tail bound below needs two properties that the readers for plain entailment have and that our readers will be shown to have.

Definition 2.13 (Rejected-path property [Zhao 2026, Sect. "The Rejected-Path Property"]). A reader has the *rejected-path property* if it satisfies:

Betweenness
if two prefixes of a word end in Rej, every intermediate prefix also ends in Rej;

Admission cone
in a rejected word w, every admitted comparison (s, e) has $e \le J(w)$.

Lemma 2.14 (Rejected extension [Zhao 2026, Lemma "Rejected extension"]). *Let a reader satisfy the admission cone. If v is a hole and w is a rejected word with $v \preceq w$, then w is a hole, and $J(w) = J(v)$.*

Proof. The words v and w have the same states at the cuts of v, and v is rejected, so $J(w) = J(v) \le |v|$. An admitted comparison (s, e) of w has $e \le J(w) \le |v|$ by the admission cone, so it is a comparison of v with the same states, contradicting the assumption that v is a hole. □

Theorem 2.15 (Bounded rejected tail [Zhao 2026, Theorem "Bounded rejected tail"]). *Let* $(\mathcal{M}, \mathrm{Adm}, \mathrm{Rej})$ *be a reader with N states over a finite alphabet that has the rejected-path property. If it has a hole, then it has a hole w with* $r(w) \le 4N - 2$.

The theorem selects a hole; it bounds neither the tail of every hole nor a continuation of a given rejected prefix. Its proof distinguishes whether some reachable rejected state on a cycle of rejected states has two distinct letters that keep the run rejected; the bound holds for every alphabet size.

2.6 The Canonical Hierarchy, Depth, and Derivatives

Definition 2.16 (Canonical hierarchy, depth [Zhao 2026, Definition "Hierarchy and depth"]). Let v be a nonempty word over an alphabet Γ with last letter ℓ, the *marker*. Then $v = X_0\ell X_1\ell\cdots X_{k-1}\ell$ for unique *blocks* $X_i \in (\Gamma \setminus \{\ell\})^*$, and the *desubstitution* of v is the word $[X_0][X_1]\cdots[X_{k-1}]$ of length k over the alphabet $(\Gamma \setminus \{\ell\})^*$ of blocks, each block being one letter; v is its *expansion*. The *canonical hierarchy* of a nonempty word w is $w_0 = w$ and w_{j+1} = the desubstitution of w_j, whose marker ℓ_j is the last letter of w_j. A word is *unary* if all its letters are equal. The *depth* $\mathrm{depth}(w)$ is the least j such that w_j is unary.

A level that is not unary is strictly longer than the next one, so the depth is well defined and $\mathrm{depth}(w) \le |w| - 1$. For example, for $n = 2$, $w = 01001101$ has $w_1 = [0][00][\varepsilon][0]$, then $w_2 = [\varepsilon][[00][\varepsilon]]$, and w_3 is a single letter, so $\mathrm{depth}(w) = 3$. Block alphabets can be infinite even when Γ is finite, and are infinite when $\Gamma \setminus \{\ell\} \neq \emptyset$; every word uses only finitely many of their letters. Over a one-letter alphabet, in particular for $n = 1$, every nonempty word has depth 0.

The hierarchy is compatible with comparisons: a comparison that does not end at the last cut is a comparison of the next level, refined by a common suffix of two blocks (Figure 3).

Lemma 2.17 (Alignment [Zhao 2026, Lemma "Alignment"]). *Let* $v = X_0\ell X_1\ell\cdots X_{k-1}\ell$ *be the expansion of a nonempty word* $v' = [X_0]\cdots[X_{k-1}]$ *of blocks avoiding* ℓ*, and let* $a_i = |X_0\ell\cdots X_{i-1}\ell|$ *be the cut at which* X_i *begins. For cuts* $s < e < |v|$*, the pair* (s, e) *is a comparison of* v *iff there are indices* $i < i' < k$ *and decompositions* $X_i = xy$ *and* $X_{i'} = x'y$ *with* $s = a_i + |x|$, $e = a_{i'} + |x'|$, *and* $(i + 1, i' + 1)$ *a comparison of* v'.

Proof. Every cut $h < |v|$ lies in exactly one interval $[a_i, a_i + |X_i|]$; write $X_i = xy$ with $h = a_i + |x|$, so that $v[h{:}] = y\ell X_{i+1}\ell\cdots X_{k-1}\ell$. Let s and e correspond in this way to $X_i = xy$ and $X_{i'} = x'y'$. Since y and y' avoid ℓ, the first marker of $v[s{:}]$ is at offset $|y|$ and that of $v[e{:}]$ at offset $|y'|$. If $v[e{:}] \preceq v[s{:}]$, then $y = y'$, and canceling $y\ell$ shows that the expansion of $[X_{i'+1}]\cdots[X_{k-1}]$ is a prefix of the expansion of $[X_{i+1}]\cdots[X_{k-1}]$. Expansions determine their block words, since the markers delimit the blocks, so $v'[i' + 1{:}] \preceq v'[i + 1{:}]$. Moreover $i < i'$: if $i = i'$, then $y = y'$ gives $s = e$, and if $i > i'$, then $s \ge a_i > a_{i'} + |X_{i'}| \ge e$. So $(i + 1, i' + 1)$ is a comparison of v'. The converse reverses these steps: $v[e{:}] = y\ell X_{i'+1}\ell\cdots \preceq y\ell X_{i+1}\ell\cdots = v[s{:}]$. □

For example, the comparison $(0, 6)$ of $w = 01001101$, with $w[6{:}] = 01$, starts in the blocks $X_0 = 0$ and $X_3 = 0$ of $w = (0)1(00)1(\varepsilon)1(0)1$, both at distance one from their markers and with the common suffix $y = 0$, and $(1, 4)$ is the terminal comparison of $w_1 = [0][00][\varepsilon][0]$ that ends at its last cut. A comparison of one level thus reappears at the next level as a comparison of blocks, together with a test at the same distance from the ends of two blocks. The derivatives of lettered monitors (Definition 2.23) check comparisons in exactly this form.

Depth measures how deeply such repetitions are nested. It matters because the compression behind Theorem 2.20 works one level of the hierarchy at a time, so its length bound grows with the depth. The depth of a rejected word is bounded in terms of its rejected tail, for every reader. The key step is the following lemma.

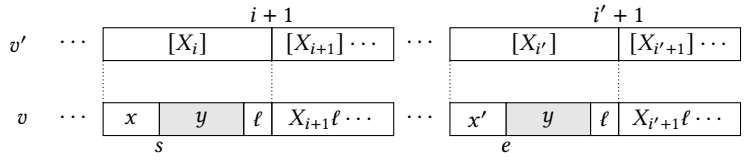


Fig. 3. The alignment lemma. Bottom: a comparison (s, e) of the expansion v with $e < |v|$ starts in two blocks $X_i = xy$ and $X_{i'} = x'y$ that have the common suffix y (shaded) after s and after e, so both cuts lie at the same distance from the markers that close their blocks. Top: what follows these two markers is the expansion of the block word v' after the cuts $i + 1$ and $i' + 1$, so (s, e) is a comparison of v iff, in addition, $(i + 1, i' + 1)$ is a comparison of v'.

Lemma 2.18 (Return gap [Zhao 2026, Lemma "Return gap"]). *Let w be a rejected word of a reader. If (s, e) is a comparison of w and $q_s = q_e$, then $e - s \le r(w)$.*

Proof. Put $\varpi = e - s$. The comparison gives $w[:s]\, w[e:] = w[:|w| - \varpi]$. Since $q_s = q_e$, reading $w[e:]$ from q_s ends in $q_{|w|}$, so the state at the cut $|w| - \varpi$ is $q_{|w|} \in$ Rej. By minimality of $J(w)$, $|w| - \varpi \ge J(w)$, so $\varpi \le r(w)$. □

Theorem 2.19 (Depth from the rejected tail [Zhao 2026, Theorem "Depth from tail"]). *For every reader with N states, every nonempty rejected word w satisfies* $\mathrm{depth}(w) \le 2N(r(w) + 1) - 1$.

Theorem 2.20 (Bounded-depth search [Zhao 2026, Theorem "Bounded-depth search"]). *Let $(\mathcal{M}, \mathrm{Adm}, \mathrm{Rej})$ be a reader with N states over a finite alphabet, and let $d \ge 0$. If it has a nonempty hole of depth at most d, then it has a nonempty hole of depth at most d and length at most* $\mathsf{Len}(N, d)$*, the explicit elementary function of Eq. (15). The same holds for every lettered monitor with N states over any alphabet (defined below, Definition 2.22).*

Theorem 2.19 applies to every nonempty rejected word, whether or not it is a hole, and neither theorem uses the rejected-path property or a property of subtype constraints; the alphabet size does not enter the bounds. Combined with Theorem 2.15, they decide the hole problem for readers with the rejected-path property.

Corollary 2.21 (Finite search for holes [Zhao 2026, Theorem "Finite search for holes", Lemma "Elementary estimate"]). *Let a reader with N states over a finite alphabet have the rejected-path property, and put $d_N = 2N(4N-1)-1$. It has a hole iff it has a hole of length at most* $\mathsf{Len}(N, d_N)$*, and* $\mathsf{Len}(N, d_N) \le 2^{2^{1280(N+1)^5}}$.

Proof. By Theorem 2.15, some hole w has $r(w) \le 4N - 2$. If w is empty, it meets the bound. Otherwise Theorem 2.19 gives $\mathrm{depth}(w) \le d_N$, and Theorem 2.20 gives a hole of length at most $\mathsf{Len}(N, d_N)$. The converse is trivial, and the estimate is Lemma A.13. □

Lettered monitors. The proof of Theorem 2.20 reads each level of the hierarchy of a hole with its own automaton, obtained from the automaton of the level below by a derivative. These automata evaluate their tests at positions inside blocks, so the tests see the letters as well as the states.

Definition 2.22 (Lettered monitor [Zhao 2026, Sect. "Lettered Monitors and Their Derivatives"]). A *lettered monitor* $\mathcal{M}$ over an alphabet Γ, possibly infinite, has a finite state set Q, a start state ζ_0, a transition function $\delta : Q \times \Gamma \to Q$, a *final-target* predicate $T^{\mathrm{fin}} \subseteq Q \times \Gamma$, and two relations Λ and Λ^{fin} on *labels*, the second for terminal comparisons. Labels are a symbol $\triangleright$, the *start label*, and the pairs $(\zeta, a) \in Q \times \Gamma$. Reading a word v gives the states $\zeta_{i+1} = \delta(\zeta_i, v[i])$ and the labels $L_0 = \triangleright$

and $L_i = (\zeta_{i-1}, v[i-1])$ for $1 \le i \le |v|$. A nonempty word v is a *hole* of $\mathcal{M}$ if $T^{\mathrm{fin}}(L_{|v|})$ holds, no terminal comparison $(s, |v|)$ has $\Lambda^{\mathrm{fin}}(L_s, L_{|v|})$, and no other comparison (s, e) has $\Lambda(L_s, L_e)$.

So the label of a cut $i \ge 1$ is the last transition before it: the state at the cut $i-1$ and the letter read from it. A reader $(\mathcal{M}, \mathsf{Adm}, \mathsf{Rej})$ with states Q, start q_0, and transition δ determines the lettered monitor with the same states and transitions, $T^{\mathrm{fin}}(\zeta, a)$ iff $\delta(\zeta, a) \in \mathsf{Rej}$, and $\Lambda = \Lambda^{\mathrm{fin}}$ given by Adm on the *denoted* states, where $\mathrm{den}(\triangleright) = q_0$ and $\mathrm{den}(\zeta, a) = \delta(\zeta, a)$. Since $\mathrm{den}(L_i) = q_i$, the reader and this monitor have the same nonempty holes. The tests take labels, and not only the states they denote, because the monitors of higher levels evaluate them at positions inside blocks, which requires the letters.

Shifted blocks and derivatives. Fix a marker $\ell \in \Gamma$, a block $X \in (\Gamma \setminus \{\ell\})^*$, and $\zeta \in Q \sqcup \{\star\}$, where $\star$ is a fresh symbol. The *shifted block* of X read from $\zeta \in Q$ is the word ℓX read from ζ; read from $\star$, it is X read from ζ_0. It has a label $lab_g(\zeta, X)$ at each position $g = 0, \dots, |X|$: for $\zeta \in Q$, the label of the cut after $\ell X[:g]$, so that position 0 carries the *marker label* (ζ, ℓ); for $\zeta = \star$, the label $\triangleright$ at position 0 and the label of the cut after $X[:g]$ at positions $g \ge 1$. The *exit* $\delta_\ell(\zeta, X) \in Q$ is the state at the end of the shifted block. In a word of blocks read by the derivative below, the *entry state* of a position is the state at the cut before it, from which the shifted block of its letter is read.

Definition 2.23 (Derivative [Zhao 2026, Definition "Derivative"]). The *derivative* $\mathcal{M}_\ell$ of $\mathcal{M}$ along ℓ is the lettered monitor over $(\Gamma \setminus \{\ell\})^*$ with states $Q \sqcup \{\star\}$, start state $\star$, transitions δ_ℓ, and

$$\begin{aligned}
\Lambda_\ell((\zeta, X), (\zeta', X')) &\iff X = xy,\ X' = x'y,\ \Lambda\big(lab_{|x|}(\zeta, X), lab_{|x'|}(\zeta', X')\big) \text{ for some } x, x', y,\\
\Lambda^{\mathrm{fin}}_\ell((\zeta, X), (\zeta', X')) &\iff \Lambda_\ell((\zeta, X), (\zeta', X')) \text{ or } \Lambda^{\mathrm{fin}}\big(lab_g(\zeta, X), (\delta_\ell(\zeta', X'), \ell)\big) \text{ for some } g \le |X|,\\
T^{\mathrm{fin}}_\ell(\zeta, X) &\iff T^{\mathrm{fin}}(\delta_\ell(\zeta, X), \ell) \text{ and } \neg\Lambda^{\mathrm{fin}}\big(lab_g(\zeta, X), (\delta_\ell(\zeta, X), \ell)\big) \text{ for all } g \le |X|;
\end{aligned}$$

every relation with the start label $\triangleright$ of $\mathcal{M}_\ell$ as an argument is false.

In words: Λ_ℓ compares two positions at the same distance from the ends of two blocks, with equal letters after them, as in Lemma 2.17; the second disjunct of $\Lambda^{\mathrm{fin}}_\ell$ compares a position anywhere in the block of its first argument with the final cut, which carries the label of the closing marker; and T^{fin}_ℓ checks the original target and all terminal comparisons that start in the last block. Comparisons of the expansion that start at its cut 0 are represented inside the first shifted block, whose entry is $\star$, so setting the relations at the new start label to false discards none of them.

The label facts. Let $v = X_0\ell X_1\ell\cdots X_{k-1}\ell$ be the expansion of a nonempty word $v' = [X_0]\cdots[X_{k-1}]$ of blocks avoiding ℓ. Let $\zeta'_0 = \star$ and $\zeta'_{i+1} = \delta_\ell(\zeta'_i, X_i)$ be the run of v' in $\mathcal{M}_\ell$, put $a_i = |X_0\ell\cdots X_{i-1}\ell|$, and let $b_i = a_i + |X_i|$ be the position of the marker closing X_i. The intervals $[a_i, b_i]$ partition the cuts $0, \dots, |v| - 1$, and the labels L of v in $\mathcal{M}$ and L' of v' in $\mathcal{M}_\ell$ satisfy the *label facts* [Zhao 2026, proof of Lemma "Holes under desubstitution"]

$$L_{a_i+g} = lab_g(\zeta'_i, X_i) \quad (0 \le g \le |X_i|), \qquad L_{|v|} = (\zeta'_k, \ell), \qquad L'_{i+1} = (\zeta'_i, X_i). \tag{4}$$

In particular ζ'_{i+1} is the state of $\mathcal{M}$ at the cut b_i.

Lemma 2.24 (Holes under desubstitution [Zhao 2026, Lemma "Holes under desubstitution"]). *With v and v' as above, v is a hole of $\mathcal{M}$ iff v' is a hole of $\mathcal{M}_\ell$. Consequently, if $\mathcal{M}_0$ is a lettered monitor with N states and $\mathcal{M}_{j+1} = (\mathcal{M}_j)_{\ell_j}$ along the canonical hierarchy of a nonempty word w, then w is a hole of $\mathcal{M}_0$ iff w_j is a hole of $\mathcal{M}_j$, for every j, and $\mathcal{M}_j$ has $N + j$ states.*

Example 2.25 (A derivative). Let a lettered monitor $\mathcal{M}$ over $\{0, 1\}$ read $w = 01001101$, with the state ζ_h and the label L_h at the cut h. With the marker $\ell = 1$, the next level $w_1 = [0][00][\varepsilon][0]$ has the blocks $X_0 = 0$, $X_1 = 00$, $X_2 = \varepsilon$ and $X_3 = 0$, which begin at the cuts 0, 2, 5 and 6 of w:

i	X_i	cuts a_i–b_i	entry ζ'_i	label L'_{i+1}
0	0	0–1	$\star$	$(\star, 0)$
1	00	2–4	ζ_1	$(\zeta_1, 00)$
2	ε	5	ζ_4	(ζ_4, ε)
3	0	6–7	ζ_5	$(\zeta_5, 0)$

The marker is 1, so $\mathcal{M}_1$ is both the derivative along 1 and the monitor of level 1. By the label facts, the run of w_1 in $\mathcal{M}_1$ is $\star, \zeta_1, \zeta_4, \zeta_5, \zeta_7$, the states before the four markers, and the labels of w_1 are those in the last column; the shifted block of X_3, for instance, is the factor 10 of w between the cuts 5 and 7.

The comparisons of w_1 that start after its cut 0 are the terminal pairs $(1, 4)$, $(2, 4)$ and $(3, 4)$, which $\mathcal{M}_1$ tests with Λ_1^{fin}. For $(1, 4)$, the first disjunct $\Lambda_1(L'_1, L'_4)$ takes the two common suffixes 0 and ε of X_0 and X_3 and reads $\Lambda(L_0, L_6)$ and $\Lambda(L_1, L_7)$: the comparisons $(0, 6)$ and $(1, 7)$ of w (Figure 3). The second disjunct reads $\Lambda^{\mathrm{fin}}(L_0, L_8)$ and $\Lambda^{\mathrm{fin}}(L_1, L_8)$: the terminal comparisons of w that start in X_0. In the same way, $(2, 4)$ stands for the comparisons $(3, 6)$ and $(4, 7)$ and for the terminal comparisons from the cuts 2, 3 and 4, and $(3, 4)$ for $(5, 7)$ and the terminal comparison from the cut 5. Finally, $T_1^{\mathrm{fin}}(L'_4)$ reads the final target $T^{\mathrm{fin}}(L_8)$ and excludes the terminal comparisons from the cuts 6 and 7, which start in the last block. These are all the comparisons of w, so w is a hole of $\mathcal{M}$ iff w_1 is a hole of $\mathcal{M}_1$. The comparisons $(0, 3)$ and $(0, 4)$ of w_1 start at its start label, where every relation of $\mathcal{M}_1$ is false.

How the bounded-depth search works. Later sections extend the proof of Theorem 2.20, which has five steps; Appendix A.4 states them precisely. Let w be a nonempty hole of depth $D \le d$ with hierarchy (w_j, ℓ_j).

(1) *Derivatives.* Let $\mathcal{M}_0$ be the lettered monitor of the reader, and $\mathcal{M}_{j+1} = (\mathcal{M}_j)_{\ell_j}$. By Lemma 2.24, w_j is a hole of $\mathcal{M}_j$ for every j. The $N + j$ states of $\mathcal{M}_j$ are the N *ordinary* states of $\mathcal{M}_0$ and the *symbolic* start states $\star_1, \ldots, \star_j$ added by the derivatives; ordinary states lead to ordinary states (Appendix A.3).

(2) *Dictionary normal form.* At each level $1 \le i < D$, two isolated letters with the same exchange type are merged: one replaces the other, which keeps the comparisons and the state run of the level, and the level stays a hole (Lemma A.7). The result is again a hole of depth D, no longer. Repeating this bounds the dictionary at every level $i \ge 1$ by $\mathrm{G}(N, d)$. The terms are the following.

Dictionary. The *dictionary* of level i is the set of letters of w_i.

Protected and unprotected letters. The *protected* letters of level i are the marker ℓ_i and the letters of the *start prefix*, the first at most i positions of w_i, which are read from symbolic states; every other letter is *unprotected* and is read only from ordinary states.

Isolated letters. An unprotected letter is *isolated* if all its unequal comparison partners are protected: whenever it precedes one cut of a comparison and a different letter precedes the other cut, that letter is protected.

Exchange type. The *exchange type* of an unprotected letter is the finite table of how $\mathcal{M}_i$ treats it when it is read from an ordinary state: its transitions and final targets, and the values of Λ_i and Λ_i^{fin} when it is compared with itself (the *self table*), with a protected letter (the *protected rows*), and with the start label (the *initial row*).

(3) *The top level.* The unary top level w_D is shortened to at most $N + D$ letters by deleting the letters between two nonfinal cuts with equal states.

(4) *Simultaneous compression.* Going down from the top, the at most $\mathrm{G}(N, d)$ distinct blocks in use at each level are written as the tracks of one word of *columns*, aligned at their right ends,

and shortened together by deleting the columns between two cuts with equal *summaries*. The summary of a tuple of blocks records, for every entry state, the exit of each block and the final marker labels to which Λ^{fin} relates its labels, which blocks are equal, and which pairs of blocks the derived admission relates; so it determines the derived monitor on the letters in use, and the shortened word is again a hole. Precisely, the new derived monitors *agree* with the old ones on the letters in use, the *support*: up to the renaming of each letter by its shortened version, they have the same states, transitions, targets and relations (Eq. (13)). The induction that produces the new markers level by level from the top is the *supported compression* of Lemma A.12.

(5) *The bound.* Each shortened block has fewer than $\mathrm{col}(N + d + 1, \mathrm{G}(N, d))$ letters, so each of the at most d expansions multiplies the length by at most this factor. With the top level this gives $\mathsf{Len}(N, d)$.

Shortening each block separately would fail, because comparisons relate different blocks through equal suffixes. The compression keeps a word a hole, that is, it preserves the absence of admitted comparisons, but it may destroy any particular comparison; positive conditions need the extension of Section 9.

3 Signed Leaf Cones

This section represents trees, and the solutions of systems, by sets of propositional atoms. In these coordinates every system over $\{\bot, f, \top\}$, of any arity and any variance, becomes a set of Horn clauses: implications with one premise, unit clauses (facts and forbidden roots), and one fixed family of binary exclusions. The representation has two consequences on which the rest of the paper rests. Least solutions are computed by closing a set of seeds, and every atom of a closure has a single seed (Section 3.3). Every pure move has a dual with both signs flipped, so two derivations that collide can be merged into one (Section 3.4).

We use the notation of Sections 2.2 and 2.3, in particular the directions $\Sigma = \{0, \dots, n-1\}$, the variance c, the polarity $\mathrm{par}(w)$, the order $\leq_c$, the normalizations ν_s, the dual t^∂ and the full constructor tree Ω, and we put $\Sigma^+ = \Sigma^* \setminus \{\varepsilon\}$. In this section and the next, all trees are trees over $\{\bot, f, \top\}$, and assignments range over arbitrary trees; regular and finite trees are treated in Section 10. Proofs omitted here are in Appendix B.

The positions of a tree t labeled $\bot$ or $\top$ are its *leaves*. We use the *chain property*: for every tree t and every word w, at most one prefix of w is a leaf of t, and if none is, then w is a position of t labeled f. Indeed, by induction on its prefixes, a prefix of w is a position as long as no shorter prefix is a leaf, and a position that is not a leaf has all its children.

3.1 Leaf Cones

Definition 3.1 (Signed leaf cones). For a tree T, the *top-leaf cone* A_T is the set of words w such that some prefix of w is a position of T labeled $\top$. For a tree t and a sign $s \in \{0, 1\}$, the *leaf cone of sign* s of t is

$$L_t^s = A_{\nu_s t}.$$

We also write $L_t^+ = L_t^0$ and $L_t^- = L_t^1$, and call (L_t^0, L_t^1) the *leaf pair* of t.

The cones have a direct description. The *sign* of a leaf v of t is $b \oplus \mathrm{par}(v)$, where $b = 0$ if $t(v) = \top$ and $b = 1$ if $t(v) = \bot$. The tree $\nu_s t$ carries the label $\top$ at a leaf v iff v has sign s. By the chain property, therefore,

$$w \in L_t^s \iff \text{the path } w \text{ meets a leaf of } t \text{ of sign } s. \tag{5}$$

Since $\nu_1 t = (\nu_0 t)^\partial$ (Lemma 2.2(1)), the cone L_t^1 is the bottom-leaf cone of $\nu_0 t$: the two cones belong to the same normalized tree. The variance enters only through the sign of each leaf, and the parity that determines this sign accumulates along the path and stops at the leaf. There are two cones per

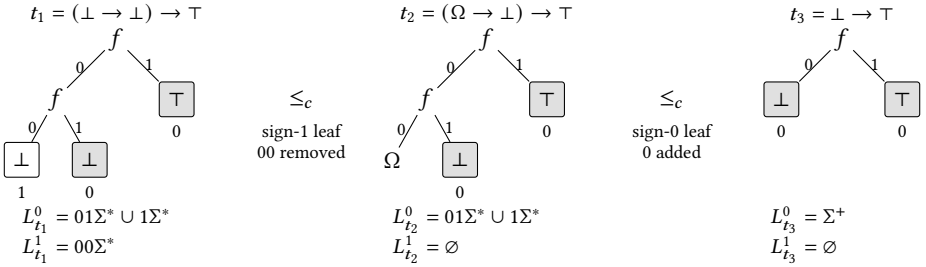


Fig. 4. Signed leaf cones of three arrow types, $c(0) = 1$ and $c(1) = 0$. The number below a leaf is its sign: 0 for $\top$ and 1 for $\bot$, plus the number of letters 0 on its path, modulo two. Leaves of sign 0 are shaded; the leaf $\bot$ at 01 has sign 0, and so does the leaf $\bot$ at 0 in t_3. Going up in the order removes leaves of sign 1 and adds leaves of sign 0 (Theorem 3.2(2)): $t_1 \leq_c t_2$ shrinks L^1, and $t_2 \leq_c t_3$ grows L^0 by replacing the subtree $\Omega \to \bot$ at the contravariant position 0 with the smaller tree $\bot$.

tree, whatever the arity, not one per assignment of signs to the children. In terms of the signed correspondence (Lemma 2.5), $L^s_{\rho(v)}$ is the top-leaf cone of the covariant tree $\rho^{\pm}(v^s)$.

Theorem 3.2 (Leaf representation).

(1) For every tree t, the sets L^0_t and L^1_t are disjoint, and each is extension closed*: $w \in L^s_t$ implies $wi \in L^s_t$ for every letter i. Conversely, every pair (L^0, L^1) of disjoint extension-closed sets of words is the leaf pair of exactly one tree.*

(2) For all trees t and u,

$$t \leq_c u \iff L^0_t \subseteq L^0_u \text{ and } L^1_u \subseteq L^1_t.$$

(3) $L^0_\top = L^1_\bot = \Sigma^*$, $L^1_\top = L^0_\bot = \varnothing$, and $L^s_{t^\partial} = L^{1-s}_t$. If $t = f(t_i)_i$, then neither cone of t contains ε, and for every letter i, word w, and sign s,

$$iw \in L^s_t \iff w \in L^{s \oplus c(i)}_{t_i}.$$

In particular, $\varepsilon \in L^0_t$ iff $t = \top$, and $\varepsilon \in L^1_t$ iff $t = \bot$.

(4) The positions of t labeled f are exactly the words outside $L^0_t \cup L^1_t$. Hence t is finite iff $\Sigma^* \setminus (L^0_t \cup L^1_t)$ is finite, and t is regular iff L^0_t and L^1_t are regular languages.

By (2), the order adds leaves of sign 0 and removes leaves of sign 1 as it goes up: after normalization, larger trees have more top leaves and fewer bottom leaves. The order is therefore not the inclusion of both cones. Inclusion of both cones is an *information order*, whose least element is the tree without leaves, Ω, with the leaf pair $(\varnothing, \varnothing)$. The least solutions of Section 3.3 are least in the information order.

Example 3.3 (Leaf cones of arrows). For the arrow, $n = 2$ and $f(a_0, a_1) = a_0 \to a_1$ with $c(0) = 1$ and $c(1) = 0$, so $\mathrm{par}(w)$ is the parity of the number of letters 0 in w. The leaf pairs of $\top$, $\bot$ and Ω are $(\Sigma^*, \varnothing)$, $(\varnothing, \Sigma^*)$ and $(\varnothing, \varnothing)$. In $\top \to \bot$, the leaf 0 is labeled $\top$ and has polarity 1, and the leaf 1 is labeled $\bot$ and has polarity 0; both have sign 1, so the leaf pair is $(\varnothing, \Sigma^+)$. Likewise $\bot \to \top$ has the leaf pair $(\Sigma^+, \varnothing)$, and $\bot \to \Omega$ has $(0\Sigma^*, \varnothing)$. By Theorem 3.2(3), both cones of every tree $f(a_0, a_1)$ lie in Σ^+, so (2) gives $\top \to \bot \leq_c f(a_0, a_1) \leq_c \bot \to \top$: these are the least and the greatest arrow types.

Figure 4 follows three arrow types up to the greatest one, $\bot \to \top$. The sign of each leaf is read off its label and its path, and each step up either removes a leaf of sign 1 or adds a leaf of sign 0, even where a subtree at a contravariant position becomes smaller.

Table 3. The clause form of a literal, for all words w and all letters $i \in \Sigma$; here $\bar{c}(i) = 1 \oplus c(i)$. Every system also has the extensions $(p, w) \to (p, wi)$ for all $p \in V^{\pm}$, $w \in \Sigma^*$ and $i \in \Sigma$.

Literal	Pure moves	Root unit
$u \le v$	copies: $(u^0, w) \to (v^0, w)$ and $(v^1, w) \to (u^1, w)$	none
$f(a_i)_i \le u$	push_i: $(a_i^{c(i)}, w) \to (u^0, iw)$ pop_i: $(u^1, iw) \to (a_i^{\bar{c}(i)}, w)$	forbidden root (u^1, ε)
$u \le f(a_i)_i$	pop_i: $(u^0, iw) \to (a_i^{c(i)}, w)$ push_i: $(a_i^{\bar{c}(i)}, w) \to (u^1, iw)$	forbidden root (u^0, ε)
$u = \bot$	none	fact (u^1, ε)
$\top \le u$	none	fact (u^0, ε)

3.2 Atoms and Clauses

Let V be a finite set of variables, and recall the signed variables v^s, their set $V^{\pm}$, and the flip $\overline{v^s} = v^{1-s}$ (Section 2.3); we put $\mathrm{sgn}(v^s) = s$. An *atom* is a pair $(p, w) \in C_V = V^{\pm} \times \Sigma^*$, and its *opposite* is $\overline{(p, w)} = (\bar{p}, w)$. An assignment ρ of trees to V determines the set of atoms

$$S_\rho = \{(v^s, w) : w \in L^s_{\rho(v)}\},$$

and an atom *holds* in ρ if it lies in S_ρ. An atom of sign 0 is *positive*, and an atom of sign 1 is *negative*. Opposite atoms are incompatible but not complementary: in Ω neither holds. For $U \subseteq V$ and $S \subseteq C_V$ we put $S|_U = S \cap C_U$. We read the word of an atom as a stack whose top is its first letter.

A set S of atoms is *extension closed* if $(p, w) \in S$ implies $(p, wi) \in S$ for every letter i, and an *exclusion pair* of S is a pair $a, \bar{a}$ of opposite atoms that both lie in S. By Theorem 3.2(1), applied to each variable, $\rho \mapsto S_\rho$ is a bijection from the assignments of trees to V onto the extension-closed subsets of C_V without exclusion pairs.

The literals of a system χ over V determine the *pure moves*, *facts* and *forbidden roots* listed in Table 3; facts and forbidden roots are the unit clauses. A push or a pop of direction i changes the sign by $c(i)$, and a copy keeps it. The *rules* of χ are its pure moves and the *extensions* $(p, w) \to (p, wi)$. We write Fact_χ and Neg_χ for the sets of facts and forbidden roots. Pure moves read and write only the top of the stack: push_i puts the letter i on top, pop_i removes it, and a copy leaves the stack unchanged. They form a pushdown system with control states $V^{\pm}$ and stack alphabet Σ, which Section 5 summarizes by finite automata. Extensions, in contrast, add a letter at the bottom of the stack.

The table is the signed system $\chi^{\pm}$ of Table 2 read in top-leaf cones. Every copy literal $p \le q$ of $\chi^{\pm}$ gives the pure moves $(p, w) \to (q, w)$, every lower constructor literal $f(p_i)_i \le q$ gives $(p_i, w) \to (q, iw)$, and every upper constructor literal $q \le f(p_i)_i$ gives $(q, iw) \to (p_i, w)$. The forbidden roots are the atoms (q, ε) for the roots q of the upper constructor literals of $\chi^{\pm}$, and the facts are the atoms (q, ε) for its literals $\top \le q$.

Definition 3.4 (Models). A set $S \subseteq C_V$ is *consistent* for χ if it contains no forbidden root of χ and no exclusion pair. A *model* of χ is a consistent set of atoms that contains Fact_χ and is closed under all rules of χ. We write $\mathrm{Mod}(\chi)$ for the set of models of χ.

Lemma 3.5 (Clause form). *An assignment ρ satisfies a literal iff S_ρ is closed under the pure moves of the literal, contains its facts, and contains none of its forbidden roots. Consequently, $\rho \mapsto S_\rho$ is a bijection from the solutions of χ onto* $\mathrm{Mod}(\chi)$.

The proof (Appendix B) substitutes the identities of Theorem 3.2(3) into the inclusions of Theorem 3.2(2). Every literal thus contributes implications with one premise and units only; the only all-negative binary clauses are the exclusions $\neg(a \wedge \bar{a})$, which are the same for all systems, and no clause is a positive disjunction. A constructor literal contributes $2n$ schemes of pure moves and one forbidden root, and a copy literal two schemes. For $n = 0$, the literal $f() \le u$ says $u \neq \bot$ and $u \le f()$ says $u \neq \top$.

We write $a \to_\chi b$ if a pure move of χ leads from a to b.

Lemma 3.6 (Pure moves). *Let $(p, u) \to_\chi (q, t)$.*

(1) For every word σ, $(p, u\sigma) \to_\chi (q, t\sigma)$, by the same scheme of Table 3.
(2) $\mathrm{sgn}(q) \oplus \mathrm{par}(t) = \mathrm{sgn}(p) \oplus \mathrm{par}(u)$.

Item (1) holds because each scheme is stated for all words w, and (2) is a direct check of the six schemes (Appendix B). Both extend to sequences of pure moves. By (2), along such a sequence the sign of the current signed variable is determined by the first atom and the parity of the current word; we call (2) the *sign invariant*.

Example 3.7 (The system $x \le x \to x$). Let $\varphi = \{x \le x \to x\}$ for the arrow of Example 3.3, that is, the literal $x \le f(x, x)$. By Table 3, its pure moves are

$$\mathrm{pop}_0 : (x^0, 0w) \to (x^1, w), \quad \mathrm{push}_0 : (x^0, w) \to (x^1, 0w),$$
$$\mathrm{pop}_1 : (x^0, 1w) \to (x^0, w), \quad \mathrm{push}_1 : (x^1, w) \to (x^1, 1w),$$

it has the forbidden root (x^0, ε), and it has no facts. A move in the contravariant direction 0 changes the sign, and a move in the covariant direction 1 keeps it. In each row, the push is the pop read backwards with both signs reversed; Lemma 3.10 shows that this holds for every system. The tree $\top \to \bot$, with leaf pair $(\varnothing, \Sigma^+)$, is a solution: its atoms are the (x^1, w) with $w \neq \varepsilon$, and this set is closed under push_1 and under extensions.

3.3 Least Closures

A *derivation* of χ is a finite sequence of atoms $a_0, \ldots, a_k$, with $k \ge 0$, in which each a_{j+1} follows from a_j by a rule of χ; it is *pure* if all its rules are pure moves. We write $a \Rightarrow_\chi^* b$ if a derivation leads from a to b, and $a \to_\chi^* b$ if a pure derivation does; we drop the subscript when χ is clear. For an atom a and a set A of atoms put

$$\mathrm{Reach}_\chi(a) = \{b : a \Rightarrow_\chi^* b\}, \qquad \mu_\chi = \bigcup_{e \in \mathrm{Fact}_\chi} \mathrm{Reach}_\chi(e), \qquad H_\chi(A) = \mu_\chi \cup \bigcup_{a \in A} \mathrm{Reach}_\chi(a),$$

and $\mathrm{Cl}_\chi(a) = H_\chi(\{a\})$. An atom is *live* in χ if some model of χ contains it, and *dead* otherwise; Live_χ is the set of live atoms. Every rule has a single premise, so every atom b of $H_\chi(A)$ has a *seed*: an atom $x \in \mathrm{Fact}_\chi \cup A$ with $b \in \mathrm{Reach}_\chi(x)$.

Lemma 3.8 (Least closures). *Let χ be a system over V and $A \subseteq C_V$.*

(1) $H_\chi(A)$ is the least set of atoms that contains $\mathrm{Fact}_\chi \cup A$ and is closed under the rules of χ.
(2) Some model of χ contains A iff $H_\chi(A)$ is consistent, and then $H_\chi(A)$ is the least model that contains A. In particular, χ is satisfiable iff μ_χ is consistent, and then μ_χ is the least model of χ; and an atom a is live iff $\mathrm{Cl}_\chi(a)$ is consistent, and then $\mathrm{Cl}_\chi(a)$ is the least model that contains a.
(3) If $H_\chi(A)$ is not consistent, then there are seeds $x, y \in \mathrm{Fact}_\chi \cup A$, possibly equal, such that $\mathrm{Reach}_\chi(x)$ contains a forbidden root, or $b \in \mathrm{Reach}_\chi(x)$ and $\bar{b} \in \mathrm{Reach}_\chi(y)$ for some atom b.
(4) For every atom (p, u) and every word α, $(p, u\alpha) \in \mathrm{Reach}_\chi(p, u)$; hence $\mathrm{Cl}_\chi(p, u\alpha) \subseteq \mathrm{Cl}_\chi(p, u)$, and if (p, u) is live, then so is $(p, u\alpha)$.

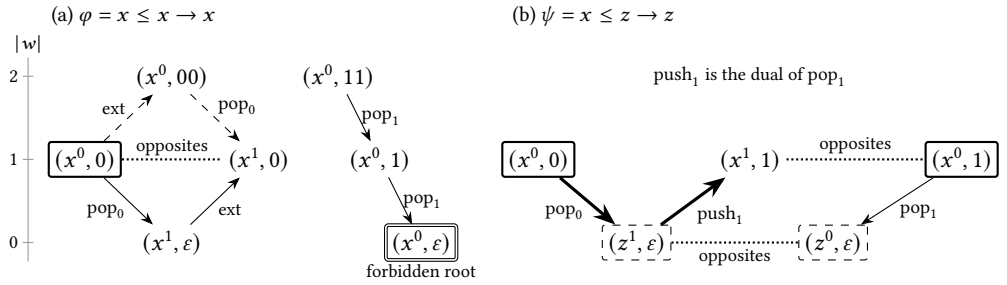


Fig. 5. Pure moves and extensions for the arrow, with each atom drawn at the height $|w|$ of its word: a pop lowers the stack, a push raises it, and an extension adds a letter at the bottom. Seeds are framed, hidden atoms dashed. (a) In $x \le x \to x$ (Examples 3.7 and 3.9), the seed $(x^0, 0)$ derives its own opposite $(x^1, 0)$, by a pop and an extension or, dashed, with the extension first (Lemma 3.11): a single-seed inconsistency without any unit. The atom $(x^0, 11)$ is dead for a different reason: it pops down to the forbidden root (x^0, ε). (b) In $x \le z \to z$ (Example 4.4), the seeds $(x^0, 0)$ and $(x^0, 1)$ derive the opposite atoms (z^1, ε) and (z^0, ε) of the hidden z. Reversing the second branch with flipped signs (Lemma 3.10) continues the first one (thick): the single seed $(x^0, 0)$ derives $(x^1, 1)$, the opposite of the second seed, which is collision return (Lemma 3.12).

(5) Let $V = X \cup Z$ with $X \cap Z = \varnothing$, and let $S \subseteq C_X$ be extension closed and without exclusion pairs. Then $S = T|_X$ for some model T of χ iff $H_\chi(S)$ is consistent and $H_\chi(S)|_X = S$; in this case $H_\chi(S)$ is the least such model.

Item (3) says that every inconsistency has at most two seeds; it bounds neither the words of the seeds nor the lengths of the derivations. Item (5) reduces the extension of a given visible assignment to a single closure; Theorem 4.2 and the transfers of Section 10 use the lemma in this form.

Example 3.9 (Liveness in $x \le x \to x$). We continue Example 3.7. The system φ has no facts, so $\mu_\varphi = \varnothing$ is consistent: φ is satisfiable, and its least solution is $x = \Omega$. Every atom (x^1, u) is live, since the solution $x = \bot$ contains it. The atoms $(x^0, 1^k)$ are dead: k pops in direction 1 lead to the forbidden root (x^0, ε). The atom $(x^0, 0)$ is dead for a different reason. The pop $(x^0, 0) \to (x^1, \varepsilon)$ followed by an extension derives $(x^1, 0)$, the opposite of the seed, so $\text{Cl}_\varphi((x^0, 0))$ contains an exclusion pair. No forbidden root is derivable from $(x^0, 0)$: the only atoms (x^0, w) that it derives are its extensions $(x^0, 0w)$, and no move leads from a negative atom of x to a positive one. So this inconsistency has a single seed and uses no unit. Finally, $(x^0, 01)$ is live: its least model is the solution $((\Omega \to \bot) \to \bot) \to \bot$, whose atoms are the (x^0, w) with $w \in 01\Sigma^*$ and the (x^1, w) with $w \in 1^+\Sigma^* \cup 1^*001\Sigma^*$.

Figure 5(a) draws these derivations with each atom at the height of its word, so that the moves act as on a stack: pops go down, pushes go up, and an extension adds a letter at the bottom without changing the top.

3.4 Duality, Extensions First, and Collision Return

Lemma 3.10 (Duality). *If $a \to_\chi b$, then $\bar{b} \to_\chi \bar{a}$. Consequently, $a \to_\chi^* b$ implies $\bar{b} \to_\chi^* \bar{a}$.*

Proof. By Table 3: the two copies of a literal $u \le v$ are each other's duals, and so are push_i and pop_i of the same constructor literal, for each direction i separately; for instance, the dual of $(a_i^{c(i)}, w) \to (u^0, iw)$ is $(u^1, iw) \to (a_i^{\bar{c}(i)}, w)$. This holds also when variables are repeated among $u, a_0, \ldots, a_{n-1}$, since the pairing is by scheme. For derivations, reverse the sequence and dualize each move. □

Extensions have no duals: the reverse of an extension would delete the bottom letter of a stack, and no rule does that. The next lemma moves all extensions of a derivation to its beginning, so that duality applies to the rest.

Lemma 3.11 (Extensions first). *For all atoms (p,u) and b, $b \in \mathrm{Reach}_\chi(p,u)$ iff $(p,u\alpha) \to_\chi^* b$ for some word α.*

The proof (Appendix B) inducts on the derivation: an extension at the end appends a letter to the bottom of every stack of the pure derivation obtained so far, which Lemma 3.6(1) allows.

Lemma 3.12 (Collision return). *Let a and (q,v) be atoms. If $b \in \mathrm{Reach}_\chi(a)$ and $\bar{b} \in \mathrm{Reach}_\chi(q,v)$ for some atom b, then $(\bar{q},v\delta) \in \mathrm{Reach}_\chi(a)$ for some word δ. If moreover $a = (p,u)$, then $(p,u\alpha) \to_\chi^* (\bar{q},v\delta)$ for some words α and δ.*

Proof. By Lemma 3.11, $(q,v\delta) \to^* \bar{b}$ for some δ, and by Lemma 3.10, $b \to^* (\bar{q},v\delta)$. Appending this pure derivation to a derivation from a to b gives $(\bar{q},v\delta) \in \mathrm{Reach}_\chi(a)$. For the second claim apply Lemma 3.11 to this membership. □

Thus two derivations that end in an exclusion pair, possibly at a hidden variable, merge into one derivation from the first seed to the opposite of an extension of the second seed (Figure 5(b)): the first derivation continues along the second one, read backwards with flipped signs. The extension is necessary: in general the merged derivation does not reach the opposite of the second seed itself (Example 4.4). The first consequence concerns a single seed.

Corollary 3.13 (Deadness). *Let χ be satisfiable and let (p,u) be an atom. Then (p,u) is dead in χ iff at least one of the following holds:*

(a) $\mathrm{Reach}_\chi(p,u)$ *contains a forbidden root;*
(b) $(\bar{p},u\delta) \in \mu_\chi$ *for some word* δ*;*
(c) $(\bar{p},u\delta) \in \mathrm{Reach}_\chi(p,u)$ *for some word* δ*.*

Moreover, (a) holds iff $(p,u\alpha) \to_\chi^* e$ for some word α and some $e \in \mathrm{Neg}_\chi$, and (c) holds iff $(p,u\alpha) \to_\chi^* (\bar{p},u\delta)$ for some words α and δ.

Causes (a) and (b) make the atom dead by support (Definition 5.10). Cause (c), a *self-collision*, may occur without such support: in Example 3.9, $(x^0,0)$ is dead by $(x^0,00) \to (x^1,0)$, which is (c) with $\alpha = 0$ and $\delta = \varepsilon$. By the sign invariant, (c) requires $\mathrm{par}(\alpha) \neq \mathrm{par}(\delta)$, so self-collisions occur only through contravariant directions; in the covariant case every rule keeps the sign, and only (a) and (b) remain (Remark 4.5). The causes may overlap: for unary contravariant f and $\chi = \{x \le f(x)\}$, the atom (x^0,ε) is a forbidden root, and also $(x^0,0) \to (x^1,\varepsilon)$. Section 5 expresses (a)–(c) by finite automata and reduces a self-collision of an atom that is not already dead by support to a *self event* (Definition 5.10): two cuts $i < j$ of the seed word, the suffix after j a prefix of the suffix after i, such that some signed variable that the seed reaches by popping the letters before i has its flip reached by popping the letters before j.

4 The Characterization

This section characterizes existential entailment over arbitrary trees by three conditions on single seeds. Let X and Z be disjoint finite sets of variables, let φ be a system over X, and let ψ be a system over $X \cup Z$ (Definition 2.4); every variable of ψ outside Z belongs to X. An atom (v^s,w) is *visible* if $v \in X$ and *hidden* if $v \in Z$. Restricting an assignment to X restricts its set of atoms to C_X, so by Lemma 3.5

$$\varphi \models \exists \bar{z}\, \psi \quad\Longleftrightarrow\quad \mathrm{Mod}(\varphi) \subseteq \mathcal{F}_\psi, \qquad \text{where } \mathcal{F}_\psi = \{T|_X : T \in \mathrm{Mod}(\psi)\}. \tag{6}$$

Here $\mathrm{Mod}(\varphi)$ consists of sets of visible atoms, and $\mathcal{F}_\psi$ is the *projected family* of ψ.

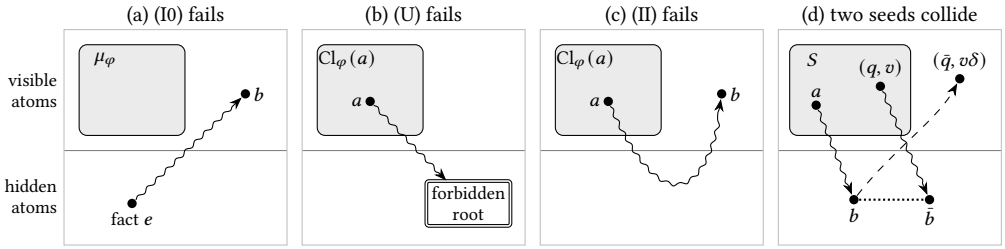


Fig. 6. When both systems are satisfiable, the four ways in which a model S of φ can fail to extend to a model of ψ, in the proof of Theorem 4.2. Upper halves hold visible atoms, lower halves hidden ones; wavy arrows are derivations of ψ, and the shaded box is a model of φ. (a) A fact of ψ derives a visible atom outside the least model μ_φ. (b) A seed a derives a forbidden root; then the least model $\mathrm{Cl}_\varphi(a)$ of φ that contains a cannot be extended. (c) A seed a derives a visible atom outside $\mathrm{Cl}_\varphi(a)$. (d) Two seeds of S derive opposite atoms b and $\bar{b}$, possibly hidden. Collision return (Lemma 3.12) reverses the second derivation by duality (dashed): the single seed a derives the opposite of the extension $(q, v\delta) \in S$ of the second seed, an atom outside $S \supseteq \mathrm{Cl}_\varphi(a)$, so this is case (c). If a is a fact of ψ instead, it is case (a).

4.1 Three Conditions

Definition 4.1 (The three conditions). For φ and ψ as above, consider:

(I0) $\mu_\psi|_X \subseteq \mu_\varphi$;
(U) for every visible atom a that is live in φ, $\mathrm{Reach}_\psi(a)$ contains no forbidden root of ψ;
(II) for every visible atom a that is live in φ, $\mathrm{Reach}_\psi(a)|_X \subseteq \mathrm{Cl}_\varphi(a)$.

The names are mnemonic: (I0) is the inclusion for zero seeds, comparing the consequences of the facts alone; (U) concerns the unit clauses that are forbidden roots; and (II) is the inclusion for a single seed a.

When both systems are satisfiable, μ_ψ and μ_φ are their least models and $\mathrm{Cl}_\varphi(a)$ is the least model of φ that contains a (Lemma 3.8(2)). So (I0) says that every visible atom forced by ψ is forced by φ. Condition (U) says that ψ never derives a forbidden root from an atom that some solution of φ contains. Condition (II) says that every solution of φ that contains an atom a also contains every visible atom that ψ derives from a.

Theorem 4.2 (Characterization). *Let φ be a system over X and ψ a system over $X \cup Z$, for any arity and any variance, and let assignments range over arbitrary trees.*
(1) If φ is unsatisfiable, then $\varphi \models \exists \bar{z}\, \psi$. If φ is satisfiable and ψ is not, then $\varphi \not\models \exists \bar{z}\, \psi$.
(2) If φ and ψ are satisfiable, then $\varphi \models \exists \bar{z}\, \psi$ iff (I0), (U) *and* (II) *hold.*
Equivalently, $E_{\mathrm{all}}(\varphi, \psi) \iff \neg\,\mathrm{Sat}(\varphi) \vee \big(\mathrm{Sat}(\psi) \wedge (\mathrm{I0}) \wedge (\mathrm{U}) \wedge (\mathrm{II})\big)$.

Each condition concerns a single seed. An obstruction to extending a model S of φ may a priori involve two atoms of S whose closures under ψ collide in an exclusion pair, possibly at a hidden variable. The proof below uses collision return (Lemma 3.12) to turn every such collision into a violation of (I0) or of (II) at one seed. This step uses duality, which holds for every variance; no condition on pairs of seeds remains. When both systems are satisfiable, Figure 6 shows the four kinds of obstruction that the proof meets: three of them are the failures of the three conditions, and the fourth, a collision of two seeds, is reduced to the first or the third.

Proof. Item (1) is immediate from Definition 2.4.

For (2), assume that φ and ψ are satisfiable. We first show that the three conditions are necessary. Assume $\varphi \models \exists \bar{z}\, \psi$. The least model μ_φ of φ is $T|_X$ for some model T of ψ by Eq. (6), and $T \supseteq \mu_\psi$

by Lemma 3.8(2); hence $\mu_\psi|_X \subseteq T|_X = \mu_\varphi$, which is (I0). Let a be a visible atom that is live in φ. Its least model $\mathrm{Cl}_\varphi(a)$ is $T|_X$ for a model T of ψ. Since T contains a and is closed under the rules of ψ, it contains $\mathrm{Reach}_\psi(a)$. As T contains no forbidden root, this gives (U), and $\mathrm{Reach}_\psi(a)|_X \subseteq T|_X = \mathrm{Cl}_\varphi(a)$ gives (II).

Conversely, assume (I0), (U) and (II), let S be a model of φ, and put $T = H_\psi(S)$, where S is read as a set of atoms over $X \cup Z$. By Lemma 3.8(5) and Eq. (6), it suffices to show that $T|_X = S$ and that T is consistent. Every atom of S is live in φ, and $\mathrm{Cl}_\varphi(a) \subseteq S$ for every $a \in S$ by Lemma 3.8(1), since S is closed under the rules of φ and contains its facts.

Visible part. Clearly $S \subseteq T|_X$. A visible atom of T lies in $\mu_\psi|_X \subseteq \mu_\varphi \subseteq S$, by (I0), or in $\mathrm{Reach}_\psi(a)|_X$ for some $a \in S$, which lies in $\mathrm{Cl}_\varphi(a) \subseteq S$ by (II).

Consistency. Suppose that T is not consistent, and take seeds $x, y \in \mathrm{Fact}_\psi \cup S$ as in Lemma 3.8(3). If $\mathrm{Reach}_\psi(x)$ contains a forbidden root, then $x \notin \mathrm{Fact}_\psi$, because μ_ψ is consistent; so $x \in S$, contradicting (U). Otherwise $b \in \mathrm{Reach}_\psi(x)$ and $\bar{b} \in \mathrm{Reach}_\psi(y)$ for some atom b; since an exclusion pair is unordered, we may assume that $y \in S$ if one of x, y lies in S.

- If $x, y \in \mathrm{Fact}_\psi$, then $b, \bar{b} \in \mu_\psi$, contradicting the consistency of μ_ψ.
- If $x \in \mathrm{Fact}_\psi$ and $y = (q, v) \in S$, then Lemma 3.12 gives $(\bar{q}, v\delta) \in \mathrm{Reach}_\psi(x) \subseteq \mu_\psi$ for some δ. This atom is visible, so it lies in $\mu_\varphi \subseteq S$ by (I0). But $(q, v\delta) \in S$ because S is extension closed, and S has an exclusion pair.
- If $x \in S$ and $y = (q, v) \in S$, possibly $x = y$, then Lemma 3.12 gives $(\bar{q}, v\delta) \in \mathrm{Reach}_\psi(x)|_X$ for some δ, which lies in $\mathrm{Cl}_\varphi(x) \subseteq S$ by (II). Again $(q, v\delta) \in S$, and S has an exclusion pair.

Each case contradicts an assumption. Hence T is a model of ψ with $T|_X = S$, so $S \in \mathcal{F}_\psi$. □

The proof shows that, when the conditions hold, $H_\psi(S)$ is the least extension of every model S of φ; when a condition fails, the least model μ_φ or $\mathrm{Cl}_\varphi(a)$ that refutes it is a countermodel. The two qualifications of the theorem are needed.

- *The satisfiability dispatch.* Let $X = \{x\}$, φ empty, and $\psi = \{z = \bot,\ \top \leq z\}$ with z hidden. Then μ_ψ consists of atoms of z, and ψ has no rule that involves x other than its extensions, so the three conditions hold; but ψ is unsatisfiable, and the entailment fails.
- *Condition* (U). Let $n \geq 0$, $X = \{y\}$, φ empty, and $\psi = \{y \leq f(z_i)_i\}$ with distinct hidden variables z_i. Then ψ has no facts, and its only pure moves from visible atoms are the pops $(y^0, iw) \to (z_i^{c(i)}, w)$, after which no pure move applies. By Lemma 4.3 below, (I0) and (II) hold. But $y = \top$ has no extension: (U) fails at the live atom (y^0, ε), which is a forbidden root of ψ.

The theorem also depends on duality. For a set of one-premise rules without duals, say $(x^0, w) \to (z^0, w)$ and $(y^0, w) \to (z^1, w)$ with z hidden, the three conditions hold for the empty antecedent, but $x = y = \top$ produces an exclusion pair at z. No system has exactly these rules: by Lemma 3.10, a system with these rules also has the duals $(z^1, w) \to (x^1, w)$ and $(z^0, w) \to (y^1, w)$, which expose the collision as the visible consequences $(x^0, w) \Rightarrow^* (y^1, w)$ and $(y^0, w) \Rightarrow^* (x^1, w)$, and (II) fails.

The conditions only need pure derivations of ψ, from extended seeds.

Lemma 4.3 (Pure forms).

(1) (U) *holds iff there are no visible atom* (p, u) *that is live in* φ*, word* α *and forbidden root* e *of* ψ *with* $(p, u\alpha) \to_\psi^* e$.

(2) (II) *holds iff* $b \in \mathrm{Cl}_\varphi(p, u)$ *for every visible atom* (p, u) *that is live in* φ *and every visible atom* b *with* $(p, u) \to_\psi^* b$.

Proof. If (p, u) is live in φ, so is $(p, u\alpha)$, and $\mathrm{Cl}_\varphi(p, u\alpha) \subseteq \mathrm{Cl}_\varphi(p, u)$ (Lemma 3.8(4)). By Lemma 3.11, $b \in \mathrm{Reach}_\psi(p, u)$ iff $(p, u\alpha) \to_\psi^* b$ for some α. Item (1) follows directly, since $(p, u\alpha)$ is again a live visible atom. For (2), the condition is clearly necessary. Conversely, if $b \in \mathrm{Reach}_\psi(p, u)|_X$

for a live (p, u), then $(p, u\alpha) \to_{\psi}^{*} b$ for some α, and the condition, applied to the live atom $(p, u\alpha)$, gives $b \in \mathrm{Cl}_{\varphi}(p, u\alpha) \subseteq \mathrm{Cl}_{\varphi}(p, u)$. □

Example 4.4 (The consequent $x \le z \to z$). For the arrow of Example 3.3, let $X = \{x\}$, $Z = \{z\}$ and $\psi = \{x \le z \to z\}$, the literal $x \le f(z, z)$. By Table 3, ψ has the forbidden root (x^0, ε), no facts, and the pure moves

$$(x^0, 0w) \to (z^1, w) \to (x^1, 1w), \qquad (x^0, 1w) \to (z^0, w) \to (x^1, 0w).$$

No pure move leaves a negative atom of x. So the pure derivations of ψ of positive length between visible atoms are exactly $(x^0, iw) \to^{*} (x^1, \bar{\imath}w)$, where $\bar{\imath} = 1 - i$: ψ exchanges the first letter and the sign. Since $\mu_{\psi} = \varnothing$, (I0) holds for every antecedent, and ψ is satisfiable, with $x = \bot$.

The empty antecedent. Let φ_0 be empty. Then (U) fails at (x^0, ε), with the countermodel $x = \top$. Condition (II) fails at $(x^0, 0)$: ψ derives $(x^1, 1)$, but $\mathrm{Cl}_{\varphi_0}((x^0, 0))$ consists of the atoms $(x^0, 0w)$, the leaf pair $(0\Sigma^*, \varnothing)$ of the countermodel $x = \bot \to \Omega$ (Example 3.3). Indeed $\bot \to \Omega \le_c z \to z$ would require $z \le_c \bot$ and $\Omega \le_c z$. The tree $\bot \to \top$, with leaf pair $(\Sigma^+, \varnothing)$, fails for a reason that a priori involves two seeds: the closures of $(x^0, 0)$ and $(x^0, 1)$ under ψ contain the exclusion pair (z^1, ε) and (z^0, ε). Collision return reverses the second branch $(x^0, 1) \to (z^0, \varepsilon)$ into $(z^1, \varepsilon) \to (x^1, 1)$, which continues the first branch: the single seed $(x^0, 0)$ derives $(x^1, 1)$, the opposite of the second seed, and (II) fails at $(x^0, 0)$. The extension in Lemma 3.12 is needed in general. The atom (x^0, ε) derives (z^0, ε) through its extension $(x^0, 1)$, which collides with the atom (z^1, ε) derived from $(x^0, 0)$; collision return gives $(x^1, 1)$, the opposite of an extension of (x^0, ε), whereas (x^1, ε) is not derivable from $(x^0, 0)$.

The antecedent $x \le x \to x$. Let $\varphi_1 = \{x \le x \to x\}$ be the system of Example 3.7. It is satisfiable. Condition (U) holds, because (x^0, ε) is a forbidden root of φ_1 as well, hence dead. For (II), by Lemma 4.3 it suffices that $(x^1, \bar{\imath}w) \in \mathrm{Cl}_{\varphi_1}((x^0, iw))$ for every live (x^0, iw), and φ_1 derives it: $(x^0, 0w) \to (x^1, w) \to (x^1, 1w)$ by pop_0 and push_1, and $(x^0, 1w) \to (x^0, w) \to (x^1, 0w)$ by pop_1 and push_0. By Theorem 4.2, $\varphi_1 \models \exists z\, \psi$; indeed $z = x$ is a witness.

4.2 Special Cases

Remark 4.5 (Covariant and joint-parity instances). A *parity assignment* of a system χ is a map $\pi : V \to \{0, 1\}$ with $\pi(u) = \pi(v)$ for every literal $u \le v$ and $\pi(a_i) = \pi(u) \oplus c(i)$ for every i and every literal $f(a_i)_i \le u$ or $u \le f(a_i)_i$ of χ. By Table 3, every rule of χ then preserves the bit $\kappa(v^s, w) = s \oplus \pi(v)$ of an atom; for instance, push_i of $f(a_i)_i \le u$ leads from $\kappa = c(i) \oplus \pi(a_i) = \pi(u)$ to $\kappa = \pi(u)$. Opposite atoms have different bits. Hence no derivation leads from an atom to the opposite of one of its extensions: self-collisions, cause (c) of Corollary 3.13, do not occur, and in a satisfiable system every dead atom is dead by support. Likewise, in Lemma 3.12 the two colliding seeds have different bits, so in the proof of Theorem 4.2 two branches from the same seed never collide.

More is true. Rename every atom (v^s, w) to $(v^{s \oplus \pi(v)}, w)$. By Table 3, this maps the rules, facts and forbidden roots of χ onto those of the following system χ^{π} for the covariant variance $c = 0$: χ^{π} keeps every literal of χ whose root has $\pi = 0$ and reverses every other one, where the root of $u \le v$, $u = \bot$ and $\top \le u$ is u. Reversal turns $u \le v$ into $v \le u$, a lower constructor literal into the upper one with the same variables and conversely, and exchanges $u = \bot$ and $\top \le u$. On trees the renaming is $\rho(v) \mapsto \nu_{\pi(v)}\rho(v)$, read covariantly: the covariant leaf cone of sign $s \oplus \pi(v)$ of $\nu_{\pi(v)}\rho(v)$ is the cone L^s of $\rho(v)$, because $(\nu_r t)^{\partial} = \nu_{1 \oplus r} t$ by Lemma 2.2(1). So ρ solves χ iff this assignment solves χ^{π}, by Lemma 3.5. If $\varphi \wedge \psi$ has a *joint* parity assignment, one renaming serves both systems and keeps visibility, and the instance is equivalent to a covariant instance of the same size, in every tree class, since ν_r is an involution that preserves finiteness and regularity (Lemma 2.2). The covariant

case itself is $\pi = 0$, where every rule keeps the sign and the rules of sign 1 are the duals of the rules of sign 0.

Theorem 4.2 needs no parity assignment, and the system $x \le x \to x$ has none, since it would require $\pi(x) = \pi(x) \oplus c(0)$. Its self-collision $(x^0, 00) \to (x^1, 0)$ (Example 3.9) is exactly what a parity assignment excludes.

Remark 4.6 (One sign: the signature $\{\bot, f\}$). Let f be covariant, and consider trees over $\{\bot, f\}$ and systems without literals $\top \le u$ (Section 2.2). Every leaf of such a tree has sign 1, so $L_t^0 = \varnothing$, and L_t^1 is the complement of the set $K(t)$ of positions labeled f (Theorem 3.2(4)). The trees over $\{\bot, f\}$ thus correspond to the extension-closed sets of negative atoms. All facts of such a system come from literals $u = \bot$ and are negative atoms, and every rule keeps the sign, so the closure of a set of negative atoms consists of negative atoms, and an exclusion pair never arises. The models over $\{\bot, f\}$ are therefore the sets of negative atoms that contain the facts, are closed under the rules, and avoid the forbidden roots (u^1, ε) of the lower constructor literals $f(a_i)_i \le u$. Lemma 3.8 and Theorem 4.2 hold over $\{\bot, f\}$, with satisfiability over $\{\bot, f\}$, by the same proofs, in which the exclusion case is void, once every set of atoms is taken in the universe $C_V^- = \{v^1 : v \in V\} \times \Sigma^*$ of negative atoms: the seed sets, the visible sets, the models, the closures, liveness, and the atoms quantified in the three conditions are all negative. (Without this restriction the statement fails: for the empty system the closure of (x^0, ε) is consistent, but no $\top$-free assignment contains it.)

Read through complements, a negative atom (v^1, w) says $w \notin K(v)$, and each rule on negative atoms is the contrapositive of a clause on constructor positions: $u \le v$ says $K(u) \subseteq K(v)$, $f(a_i)_i \le u$ says $\varepsilon \in K(u)$ and $iK(a_i) \subseteq K(u)$, $u \le f(a_i)_i$ says that $iw \in K(u)$ implies $w \in K(a_i)$, $u = \bot$ says $\varepsilon \notin K(u)$, and extensions say that $K(v)$ is prefix closed. In these terms, (I0) says that every visible position that some solution of φ puts into a variable is put there by some solution of ψ; (U) says that every visible position that every solution of ψ puts into a variable is put there by every solution of φ; and (II), read contrapositively, says that every solution of φ is closed under the visible consequences of the clauses of ψ. Over $\{\bot, f\}$ the three conditions therefore compare, respectively, the unions of all solutions, the least solutions, and the closures of single positions, for φ and for the projection of ψ. In this picture the solution sets are closed under nonempty unions and intersections of constructor positions, no exclusions and no self-collisions occur, and the only obstructions are single seeds. The decision procedure of this paper does not treat $\{\bot, f\}$ separately; it reduces $\{\bot, f\}$ to $\{\bot, f, \top\}$ (Section 10).

A forbidden root has the empty word, so the pure derivations in Lemma 4.3(1) end with an empty stack. A failure of (II) with a live seed a and a consequence b is thus witnessed by a pure derivation of ψ from a live extension of a to b, with b outside the least model of φ that contains that extension. Sections 5 to 7 turn the satisfiability tests, (I0), (U) and this form of (II) into problems about finite automata.

5 Pushdown Summaries

The three conditions of Theorem 4.2 concern infinite sets of atoms: least closures, the sets $\mathrm{Reach}_\chi(a)$, and live atoms. This section describes them with finite automata over the alphabet of directions. The pure moves of a system form a pushdown system whose control states are the signed variables and whose stack is the word of an atom (Section 3.2). They are the moves of the signed system $\chi^\pm$, a covariant system, so the upper and lower bound automata of covariant systems summarize the parts of a pure derivation that pop and push its stack, and every pure derivation splits at its lowest point into one part of each kind (Section 5.1). With them we describe least closures, also of regular sets of seeds (Section 5.2).

Two consequences of the signs have no counterpart for a covariant constructor. First, the lower automata are the upper automata with all signs flipped (Lemma 5.4). Second, an atom can be dead because the closure of its seed contains the opposite of an extension of the seed, the self-collision of Corollary 3.13(c). We show that a self-collision of an atom that is not already dead by support is found by comparing two suffixes of the seed word: it yields a strict *self event*, and deadness is the union of deadness by support, which is regular, and of the self events (Theorem 5.12). Deadness need not be regular (Example 5.13). The section ends with closure membership for a live seed (Lemma 5.15), the heads of pure derivations (Lemma 5.16), and the decision of satisfiability and of condition (I0) (Proposition 5.17). Proofs omitted here are in Appendix C.

Conventions. Throughout, χ is a system over a finite set V of variables, we use the notation of Sections 2 and 3 and write $\bar{P} = \{\bar{p} : p \in P\}$ for $P \subseteq V^{\pm}$, and for $n = 0$, where ε is the only word, every statement below holds trivially.

5.1 Summaries and the Lowest Point

By Section 3.2, the pure moves of χ are the moves of the copy and constructor literals of $\chi^{\pm}$, read as a covariant system over $V^{\pm}$: a literal $p \le q$ gives $(p, w) \to (q, w)$, a literal $f(p_i)_i \le q$ gives $(p_i, w) \to (q, iw)$, and a literal $q \le f(p_i)_i$ gives $(q, iw) \to (p_i, w)$; and the forbidden roots are the atoms (q, ε) for the roots q of the upper constructor literals of $\chi^{\pm}$. The extreme literals of $\chi^{\pm}$ give no moves. The *height* of an atom (p, w) is $|w|$, and a pure derivation is *above* a word σ if every atom on it has the form $(p, \eta\sigma)$. The automata below are the bound automata of a covariant system [Zhao 2026], for an arbitrary arity;[1] we write χ instead of $\chi^{\pm}$ in subscripts.

The closure D_χ summarizes the pure derivations that return to their starting height without going below it. Besides copies, such a derivation from a_i to b_i can push the letter i through a literal $f(a_j)_j \le u$, continue with a derivation of the same kind from u to v, one level higher, and pop i again through a literal $v \le f(b_j)_j$; this is the decomposition rule below, and transitivity strings such pieces together. The upper automaton of p, after reading γ, holds the signed variables reached from $(p, \gamma\sigma)$ by popping γ, and the lower automaton of q, after reading β, those from which $(q, \beta\sigma)$ is reached by pushing β (Lemma 5.2).

Definition 5.1 (Closure and bound automata). The *closure* D_χ is the least reflexive and transitive relation on $V^{\pm}$ that contains every pair (p, q) with $p \le q$ in $\chi^{\pm}$ and satisfies the decomposition rule

$$\frac{f(a_j)_j \le u \in \chi^{\pm} \qquad D_\chi(u, v) \qquad v \le f(b_j)_j \in \chi^{\pm}}{D_\chi(a_i, b_i)} \qquad (i \in \Sigma).$$

Put $\uparrow_\chi p = \{r : D_\chi(p, r)\}$ and $\downarrow_\chi q = \{r : D_\chi(r, q)\}$. For $P, Q \subseteq V^{\pm}$ and $i \in \Sigma$, let

$$\delta_i^{\uparrow}(P) = \{b : \exists r \in P,\ \exists\, (r \le f(b_j)_j) \in \chi^{\pm},\ D_\chi(b_i, b)\},$$
$$\delta_i^{\downarrow}(Q) = \{a : \exists r \in Q,\ \exists\, (f(a_j)_j \le r) \in \chi^{\pm},\ D_\chi(a, a_i)\}.$$

The *upper automaton* of $p \in V^{\pm}$ has state set $2^{V^{\pm}}$, start state $\uparrow_\chi p$, and transitions $\delta_i^{\uparrow}$; its state after a word w is $P_p^\chi(w)$. The *lower automaton* of q has start state $\downarrow_\chi q$ and transitions $\delta_i^{\downarrow}$; its state after w is $Q_q^\chi(w)$. At cuts we write $P_{p,j}^\chi(w) = P_p^\chi(w[:j])$ and $Q_{q,j}^\chi(w) = Q_q^\chi(w[:j])$.

Both automata are deterministic and complete, and they read their word from left to right: $P_p^\chi(wi) = \delta_i^{\uparrow}(P_p^\chi(w))$. They have $4^{|V|}$ states each, and D_χ and every transition are computable in time polynomial in the size of χ. Every set $P_p^\chi(w)$ is upward closed under D_χ and every $Q_q^\chi(w)$

[1] In the notation of Appendix A.1, D_χ is the closure $D^{\mathrm{cov}}_{\chi^{\pm}}$, with the copy pairs included in the saturation.

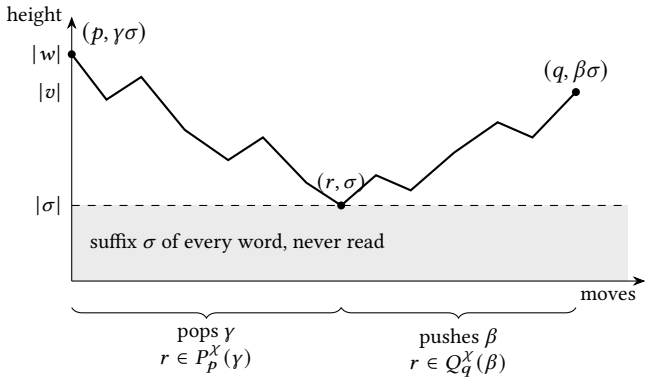


Fig. 7. The lowest point of a pure derivation from (p, w) to (q, v) (Lemma 5.3). No atom on the derivation goes below the least height $|\sigma|$, so the suffix σ is never read: $w = \gamma\sigma$ and $v = \beta\sigma$. The first atom (r, σ) of least height splits the derivation into a part summarized by the upper automaton of p and a part summarized by the lower automaton of q.

is downward closed, the empty set is absorbing in both automata, and $\delta_i^{\uparrow}$ and $\delta_i^{\downarrow}$ distribute over unions.

Lemma 5.2 (Pushdown summaries). *For all signed variables a, b, p, q and words w, β, γ, σ:*

(a) $D_{\chi}(a, b)$ iff $(a, w) \to_{\chi}^{*} (b, w)$ by a pure derivation above w;
(b) $b \in P_p^{\chi}(\gamma)$ iff $(p, \gamma\sigma) \to_{\chi}^{*} (b, \sigma)$ by a pure derivation above σ;
(c) $a \in Q_q^{\chi}(\beta)$ iff $(a, \sigma) \to_{\chi}^{*} (q, \beta\sigma)$ by a pure derivation above σ.

In particular, the existence of these derivations does not depend on w or σ.

Lemma 5.3 (Lowest point). *For signed variables p, q and words w, v, $(p, w) \to_{\chi}^{*} (q, v)$ iff there are cuts $i \le |w|$ and $e \le |v|$ with $w[i{:}] = v[e{:}]$ and $P_{p,i}^{\chi}(w) \cap Q_{q,e}^{\chi}(v) \neq \varnothing$.*

Cut the derivation at its first atom of least height (Figure 7) and apply Lemma 5.2(b,c) to the two parts; the signed variable r at the lowest point may belong to a visible or to a hidden variable, with either sign. In the notation of Figure 7, γ is the *pop head* and β the *push head* of the derivation.

Lemma 5.4 (Duality and sign coherence). *For all signed variables a, b, p, q and words w:*

(1) $D_{\chi}(a, b)$ iff $D_{\chi}(\bar{b}, \bar{a})$; hence $\downarrow_{\chi}\bar{q} = \overline{\uparrow_{\chi} q}$;
(2) $Q_{\bar{q}}^{\chi}(w) = \overline{P_q^{\chi}(w)}$;
(3) $D_{\chi}(a, b)$ implies $\mathrm{sgn}(a) = \mathrm{sgn}(b)$; every $r \in P_p^{\chi}(w)$ has $\mathrm{sgn}(r) = \mathrm{sgn}(p) \oplus \mathrm{par}(w)$; and every $r \in Q_q^{\chi}(w)$ has $\mathrm{sgn}(r) = \mathrm{sgn}(q) \oplus \mathrm{par}(w)$.

Items (1) and (2) reverse pure derivations by duality (Lemma 3.10): the reversed derivation visits the same words, so it stays above the same suffix. By item (2), Lemma 5.3 can be read with upper automata only. Item (3) is *sign coherence*, the sign invariant of Lemma 3.6(2) applied to the derivations of Lemma 5.2.

Example 5.5 (Shift closure: the signed system). Let $n = 2$ with f covariant, and let $\psi = \{x \le z,\ z \le y,\ z \le f(z, z)\}$ be the consequent of Example 1.1, now over $\{\bot, f, \top\}$, with z hidden. The copy and constructor literals of its signed system are

$$x^0 \le z^0, \quad z^0 \le y^0, \quad z^0 \le f(z^0, z^0), \quad y^1 \le z^1, \quad z^1 \le x^1, \quad f(z^1, z^1) \le z^1.$$

It has no facts, and its only forbidden root is (z^0, ε): the root of z is not $\top$. Since f is covariant, (x^0, w) says that ρx has a leaf $\top$ at a prefix of w, and (x^1, w) that it has a leaf $\bot$ at a prefix of w. The decomposition rule never applies, so D_ψ is the reflexive and transitive closure of the copies. Hence $\uparrow_\psi x^0 = \{x^0, z^0, y^0\}$ and $P^\psi_{x^0}(w) = \{z^0, y^0\}$ for $w \neq \varepsilon$; dually, $\downarrow_\psi x^1 = \{x^1, z^1, y^1\}$ and $Q^\psi_{x^1}(w) = \{z^1, y^1\}$ for $w \neq \varepsilon$, as Lemma 5.4(2) predicts. By Lemma 5.3, $(y^1, \sigma) \to^*_\psi (x^1, \beta\sigma)$ for all words β and σ: in every solution of ψ, if y has a leaf $\bot$ at a prefix of σ, then x has a leaf $\bot$ at a prefix of $\beta\sigma$. For trees over $\{\bot, f\}$ this says that every suffix of a constructor position of x is a constructor position of y. Similarly $(x^0, w) \to^*_\psi (z^0, \varepsilon)$ for every w, a pure derivation to the forbidden root: ψ forbids leaves $\top$ in x.

5.2 Least Closures of Regular Seeds

Let

$$\mathsf{Fact}^\uparrow_\chi = \{r : \ r \in P^\chi_e(\gamma) \text{ for some } (e, \varepsilon) \in \mathsf{Fact}_\chi \text{ and some word } \gamma\}$$

be the set of signed variables that a fact reaches at the empty word after popping a word.

Lemma 5.6 (The least closure). *For every signed variable q and word v,*

$$(q, v) \in \mu_\chi \iff Q^\chi_{q,e}(v) \cap \mathsf{Fact}^\uparrow_\chi \neq \varnothing \text{ for some cut } e \leq |v|.$$

The set $\mathsf{Fact}^\uparrow_\chi$ is the least set that contains $\uparrow_\chi p$ for every fact (p, ε) and contains $\delta^\uparrow_i(\{r\})$ for each of its elements r and each $i \in \Sigma$; it is computable in polynomial time.

A fact (p, ε) derives (r, σ) for every σ: extend it to $(p, \gamma\sigma)$ and pop γ. So μ_χ contains every atom that a pure derivation pushing a word reaches from such an (r, σ), and Lemmas 3.11 and 5.3 show that there is nothing else. The same argument applies to any regular set of seeds.

Lemma 5.7 (Closure of regular seeds). *Let S be a set of atoms whose slices $S_p = \{w : (p, w) \in S\}$ are regular for all $p \in V^\pm$. For signed variables p, r, q, put $U_{p,r} = \{\gamma : r \in P^\chi_p(\gamma)\}$, $V_{r,q} = \{\beta : r \in Q^\chi_q(\beta)\}$, and $\mathrm{Res}_{p,r} = \{\sigma : \gamma\sigma \in S_p\Sigma^* \text{ for some } \gamma \in U_{p,r}\}$. Then the slice of $H_\chi(S)$ at q is*

$$H_\chi(S)_q = \bigcup_{r \in \mathsf{Fact}^\uparrow_\chi} V_{r,q}\,\Sigma^* \ \cup \bigcup_{p, r \in V^\pm} V_{r,q}\,\mathrm{Res}_{p,r}.$$

All these languages are regular, and finite automata for them are computable from finite automata for the slices of S. Consequently, whether $H_\chi(S)$ is consistent is decidable.

In particular, μ_χ (the case $S = \varnothing$) and the closures of one or two seeds have regular slices, and their consistency is decidable. The lemma does not say that the set of live atoms is regular, which is false in general (Example 5.13).

5.3 Forbidden Roots and the Pop Bound

A pure derivation pops a letter only at the root of an upper constructor literal of $\chi^\pm$, and every such root is forbidden at the empty word. This gives the *pop bound*, the counterpart of the cone lemma of a covariant system [Zhao 2026], stated with the following notion and with $\mathrm{Neg}^\downarrow_\chi = \{r : D_\chi(r, m) \text{ for some } (m, \varepsilon) \in \mathrm{Neg}_\chi\}$.

Definition 5.8 (Root deadness). An atom (p, w) is *root-dead* in χ, written $\mathsf{DeadU}_\chi(p, w)$, if $\mathrm{Reach}_\chi(p, w)$ contains a forbidden root of χ.

Lemma 5.9 (Roots and the pop bound). *For every signed variable p and all words w, v:*

(1) if $P^{\chi}_{p}(wi) \neq \emptyset$ for some letter i, then some $r \in P^{\chi}_{p}(w)$ is the root of an upper constructor literal of $\chi^{\pm}$, $(p, w) \to^{*}_{\chi} (r, \varepsilon)$, and $(r, \varepsilon) \in \mathsf{Neg}_{\chi}$; in particular $\mathsf{DeadU}_{\chi}(p, w)$;
(2) (pop bound) *if $\mathsf{DeadU}_{\chi}(p, w[:k])$ fails, then $P^{\chi}_{p,j}(w) = \emptyset$ for every cut j with $k < j \leq |w|$;*
(3) $\mathsf{DeadU}_{\chi}(p, w)$ iff $P^{\chi}_{p}(wz) \cap \mathsf{Neg}^{\downarrow}_{\chi} \neq \emptyset$ for some word z;
(4) if $\mathsf{DeadU}_{\chi}(p, wv)$, then $\mathsf{DeadU}_{\chi}(p, w)$.

Proof. (1) If $b \in \delta^{\uparrow}_{i}(P^{\chi}_{p}(w))$, some $r \in P^{\chi}_{p}(w)$ is the root of an upper constructor literal $r \leq f(b_j)_j$ of $\chi^{\pm}$; then $(r, \varepsilon) \in \mathsf{Neg}_{\chi}$ by Section 3.2, and $(p, w) \to^{*}_{\chi} (r, \varepsilon)$ by Lemma 5.2(b) with $\sigma = \varepsilon$. (2) If $P^{\chi}_{p,j}(w) \neq \emptyset$ with $j > k$, then $P^{\chi}_{p,k+1}(w) \neq \emptyset$, because the empty set is absorbing, and (1) applied to $w[:k]$ and the letter $w[k]$ gives $\mathsf{DeadU}_{\chi}(p, w[:k])$. (3) By Lemma 3.11, $(m, \varepsilon) \in \mathsf{Reach}_{\chi}(p, w)$ iff $(p, w\alpha) \to^{*}_{\chi} (m, \varepsilon)$ for some α; by Lemma 5.3 with the empty word $v = \varepsilon$, whose only cut is 0, this holds iff $P^{\chi}_{p}(w\alpha) \cap \downarrow_{\chi} m \neq \emptyset$. (4) By Lemma 3.8(4), $(p, wv) \in \mathsf{Reach}_{\chi}(p, w)$, so $\mathsf{Reach}_{\chi}(p, wv) \subseteq \mathsf{Reach}_{\chi}(p, w)$. □

5.4 Deadness

Corollary 3.13 lists three causes of deadness for a satisfiable system: (a) a forbidden root in $\mathsf{Reach}_{\chi}(p, w)$, which is $\mathsf{DeadU}_{\chi}(p, w)$; (b) the opposite of an extension of the seed in μ_{χ}; and (c) a self-collision, the opposite of an extension of the seed in $\mathsf{Reach}_{\chi}(p, w)$. The causes may overlap. Causes (a) and (b) together are deadness by support, which automata read. A self-collision of an atom that is not dead by support is read by a strict comparison of the seed word.

Definition 5.10 (Deadness by support, self events). An atom (p, w) is *dead by support* in χ, written $\mathsf{DeadT}_{\chi}(p, w)$, if $\mathsf{DeadU}_{\chi}(p, w)$ or $(\bar{p}, w\delta) \in \mu_{\chi}$ for some word δ. A *self event* of (p, w) in χ is a comparison (i, j) of w (Definition 2.10), that is, a pair of cuts $0 \leq i < j \leq |w|$ with $w[j:] \preceq w[i:]$, such that

$$P^{\chi}_{p,i}(w) \cap \overline{P^{\chi}_{p,j}(w)} \neq \emptyset.$$

Self events are strict: $i < j$. We write $\mathsf{SelfEv}_{\chi}(p, w, i, j)$ if (i, j) is a self event of (p, w).

Self events have a counterpart in the plain-entailment proof, whose readers already compare a root with its own signed copy [Zhao 2026]. Here both cuts belong to the word of one seed: the state of the upper automaton of p at the cut i and the flipped state at the cut j share a signed variable, which serves as the lowest point of a pure derivation from an extension of (p, w) to the opposite of an extension of (p, w) (Theorem 5.12).

Both kinds of deadness by support are read by the upper automaton of p together with one bit. Put

$$\mathsf{DeadU}^{\varepsilon}_{\chi} = \{r \in V^{\pm} : \mathsf{DeadU}_{\chi}(r, \varepsilon)\}, \qquad \mathsf{DeadT}^{\varepsilon}_{\chi} = \{r \in V^{\pm} : \mathsf{DeadT}_{\chi}(r, \varepsilon)\}.$$

Lemma 5.11 (Deadness from states). *For every signed variable p and all words w, v:*
(1) $\mathsf{DeadU}_{\chi}(p, w)$ iff $P^{\chi}_{p}(w) \cap \mathsf{DeadU}^{\varepsilon}_{\chi} \neq \emptyset$;
(2) $(\bar{p}, w) \in \mu_{\chi}$ iff $\overline{P^{\chi}_{p,e}(w)} \cap \mathsf{Fact}^{\uparrow}_{\chi} \neq \emptyset$ for some cut $e \leq |w|$;
(3) $\mathsf{DeadT}_{\chi}(p, w)$ iff $(\bar{p}, w) \in \mu_{\chi}$ or $P^{\chi}_{p}(w) \cap \mathsf{DeadT}^{\varepsilon}_{\chi} \neq \emptyset$;
(4) if $\mathsf{DeadT}_{\chi}(p, wv)$, then $\mathsf{DeadT}_{\chi}(p, w)$;
(5) the sets $\mathsf{DeadU}^{\varepsilon}_{\chi}$ and $\mathsf{DeadT}^{\varepsilon}_{\chi}$ are computable in polynomial time.

So the words w with $\mathsf{DeadU}_{\chi}(p, w)$, and those with $\mathsf{DeadT}_{\chi}(p, w)$, form regular languages that are closed under prefixes: the first is read by the upper automaton of p, and the second by that automaton together with a bit that records whether some state P read so far has $\bar{P} \cap \mathsf{Fact}^{\uparrow}_{\chi} \neq \emptyset$. Self events, in contrast, compare two cuts of one word. They are what remains of deadness.

Theorem 5.12 (Deadness). *Let χ be satisfiable. An atom (p, w) is dead iff* $\mathsf{DeadT}_\chi(p, w)$ *or (p, w) has a self event in χ. More precisely, every self event of (p, w) yields a self-collision of (p, w), and if (p, w) has a self-collision but* $\mathsf{DeadT}_\chi(p, w)$ *fails, then (p, w) has a self event.*

By Corollary 3.13, it remains to treat a self-collision, a pure derivation from an extension $(p, w\alpha)$ to an opposite atom $(\bar{p}, w\delta)$, of an atom (p, w) that is not dead by support. Split it at its lowest point. If its pop head or its push head is longer than w, then the upper automaton of p is nonempty after some proper extension of w (for the push head, by Lemma 5.4(2)), and the pop bound (Lemma 5.9) gives $\mathsf{DeadU}_\chi(p, w)$. Otherwise both heads end at cuts i, j of w, the suffixes $w[i{:}]$ and $w[j{:}]$ are comparable in the prefix order, and by Lemma 5.4(2) the lowest point lies in $P^\chi_{p,i}(w) \cap \overline{P^\chi_{p,j}(w)}$. Sign coherence excludes $i = j$, and the condition is symmetric in i and j, so the shorter suffix is a prefix of the longer one and (i, j) or (j, i) is a strict self event. Conversely, every self event yields a self-collision. The proof is in Appendix C.

Example 5.13 (A nonregular dead language). Let $\varphi = \{x \le x \to x\}$ for the arrow, as in Examples 3.7 and 3.9. The copy and constructor literals of $\varphi^\pm$ are $x^0 \le f(x^1, x^0)$ and $f(x^0, x^1) \le x^1$, there are no facts, so $\mu_\varphi = \varnothing$, and the only forbidden root is (x^0, ε). The closure D_φ is the identity, and

$$P^\varphi_{x^0}(1^k) = \{x^0\}, \qquad P^\varphi_{x^0}(1^k 0) = \{x^1\}, \qquad P^\varphi_{x^0}(w) = \varnothing \text{ if } 1^k 0 \text{ is a proper prefix of } w.$$

Hence $\mathsf{DeadT}^\varepsilon_\varphi = \{x^0\}$, and $\mathsf{DeadT}_\varphi(x^0, w)$ holds iff $w \in 1^*$. The guard of a self event (i, j) requires $P^\varphi_{x^0,i}(w) = \{x^0\}$ and $P^\varphi_{x^0,j}(w) = \{x^1\}$, so $w = 1^k 0v$, $i \le k$, and $j = k + 1$; the comparison requires $v \preceq 1^m 0v$ with $m = k - i$, that is, $v \preceq (1^m 0)^\omega$ (Lemma 2.1). By Theorem 5.12, the dead words of x^0 are

$$1^* \ \cup \ \{1^k 0v : \ v \preceq (1^m 0)^\omega \text{ for some } m \le k\}.$$

This language is not regular: its intersection with 1^*01^*0 is $\{1^k 01^m 0 : m \le k\}$. For example, $(x^0, 0)$ is dead by the self event $(0, 1)$ of the word 0, although $\mathsf{DeadT}_\varphi(x^0, 0)$ fails. This is the self event that the proof of Theorem 5.12 extracts from the self-collision $(x^0, 00) \to (x^1, 0)$ of Example 3.9: its lowest point $(x^1, 0)$ lies at the cut 1 of 00 and the cut 0 of 0. The atom $(x^0, 01)$ is live (Example 3.9).

Remark 5.14 (No self events without mixed parity). If χ has a parity assignment π (Remark 4.5), then every pure move preserves $\mathrm{sgn}(r) \oplus \pi(r)$, where $\pi(x^\pm) = \pi(x)$, so every $r \in P^\chi_p(w)$ has $\mathrm{sgn}(r) \oplus \pi(r) = \mathrm{sgn}(p) \oplus \pi(p)$. Hence no set $P^\chi_{p,i}(w)$ meets a flipped set $\overline{P^\chi_{p,j}(w)}$, χ has no self events, deadness in a satisfiable χ is DeadT_χ, and the live words of each signed variable form a regular language. This is the automaton form of Remark 4.5; it includes covariant systems, with $\pi = 0$.

5.5 Closure Membership

Lemma 5.15 (Closure membership). *Let (p, w) be live in χ, and let (q, v) be an atom. Then $(q, v) \in \mathrm{Cl}_\chi(p, w)$ iff $(q, v) \in \mu_\chi$ or there are cuts $i \le |w|$ and $e \le |v|$ with*

$$w[i{:}] \preceq v[e{:}] \qquad \text{and} \qquad P^\chi_{p,i}(w) \cap Q^\chi_{q,e}(v) \neq \varnothing.$$

Proof. By Lemmas 3.11 and 5.3, $(q, v) \in \mathrm{Reach}_\chi(p, w)$ iff there are a word α and cuts $i \le |w\alpha|$, $e \le |v|$ with $(w\alpha)[i{:}] = v[e{:}]$ and $P^\chi_p((w\alpha)[{:}i]) \cap Q^\chi_{q,e}(v) \neq \varnothing$. If $i > |w|$, then $P^\chi_p(w\alpha') \neq \varnothing$ for the nonempty prefix α' of α of length $i - |w|$, and Lemma 5.9(2) with $k = |w|$ gives $\mathsf{DeadU}_\chi(p, w)$, so $\mathrm{Cl}_\chi(p, w)$ contains a forbidden root, contradicting liveness by Lemma 3.8(2). Hence $i \le |w|$, and the equation reads $w[i{:}]\ \alpha = v[e{:}]$, which gives $w[i{:}] \preceq v[e{:}]$. Conversely, if $w[i{:}]\ \alpha = v[e{:}]$ and the intersection is nonempty, Lemma 5.3 gives $(p, w\alpha) \to^*_\chi (q, v)$, and $(p, w\alpha) \in \mathrm{Reach}_\chi(p, w)$. □

The two cuts belong to different words, so there is no restriction $i < e$. The prefix relation points from the seed to the consequence: the seed word may be continued to the word of the consequence, because extensions are rules.

5.6 Heads of Derivations

The pop head of a pure derivation passes only through root-dead atoms. Immediately before a pop, the signed variable of the current atom is the root of an upper constructor literal, whose empty-word atom is forbidden.

Lemma 5.16 (Head prefixes). *If $P_p^\chi(\gamma) \neq \emptyset$, then $\mathsf{DeadU}_\chi(p, \gamma[:i])$ for every $i < |\gamma|$.*

Proof. Since the empty set is absorbing, $P_p^\chi(\gamma[:i+1]) \neq \emptyset$, and Lemma 5.9(1) applies to $\gamma[:i]$ and the letter $\gamma[i]$. □

The lemma uses neither satisfiability nor any condition relating two systems. Applied to ψ, it says that every proper prefix γ' of the pop head γ of a pure derivation of ψ from a visible atom $(p, \gamma\sigma)$ has $\mathsf{DeadU}_\psi(p, \gamma')$; under condition (U), the atom (p, γ') is then dead in φ (Definitions 4.1 and 5.8). Section 7 uses this to end every pop head of ψ at or before the first rejected cut of the seed (Lemmas 7.4 and 7.6).

5.7 Satisfiability and Condition (I0)

Proposition 5.17 (Satisfiability and (I0)).

(1) A system χ is satisfiable iff $\downarrow_\chi m \cap \mathsf{Fact}_\chi^\uparrow = \emptyset$ for every forbidden root (m, ε) and there are no variable x and word w with $(x^+, w), (x^-, w) \in \mu_\chi$.

(2) Let φ be a system over X and ψ a system over $X \cup Z$. Condition (I0), $\mu_\psi|_X \subseteq \mu_\varphi$, fails iff some $q \in X^\pm$ and some word v have $(q, v) \in \mu_\psi$ and $(q, v) \notin \mu_\varphi$.

Satisfiability and condition (I0) are decidable in polynomial space.

Proof. (1) By Lemma 3.8(2), χ is satisfiable iff μ_χ is consistent, that is, contains no forbidden root and no exclusion pair. By Lemma 5.6 with $v = \varepsilon$, $(m, \varepsilon) \in \mu_\chi$ iff $\downarrow_\chi m \cap \mathsf{Fact}_\chi^\uparrow \neq \emptyset$. (2) is the definition of (I0). For the complexity, Lemma 5.6 reads membership in μ_χ at q by the lower automaton of q together with a bit that records whether some state read so far met $\mathsf{Fact}_\chi^\uparrow$. The second condition of (1) is therefore reachability, for some $x \in V$, of a state with both bits set in the product of the lower automata of x^+ and x^-, each with its bit; and the condition of (2) is reachability, for some $q \in X^\pm$, of a state whose ψ-bit is set and whose φ-bit is not, in the product of the lower automata of q in ψ and in φ. A state of these products has a description of polynomial size, its successors are computable in polynomial time, and reachability is decided by guessing a path state by state, in nondeterministic polynomial space, hence in polynomial space. The set $\mathsf{Fact}_\chi^\uparrow$ is computable in polynomial time by Lemma 5.6. □

A shortest witness of the second condition of (1), or of a failure of (I0), is shorter than the number of states of the product automaton. Condition (U) needs more than reachability, because liveness involves self events; it is the subject of Section 6.

6 Condition (U)

From now on, φ is a system over the visible variables X and ψ a system over $X \cup Z$, as in Section 4. Condition (U) of Definition 4.1 asks that $\mathsf{Reach}_\psi(a)$ contain no forbidden root of ψ for every visible atom a that is live in φ. In the terms of Definition 5.8, this reads as follows.

Lemma 6.1 (Condition (U)). *Condition* (U) *holds iff no* $p \in X^{\pm}$ *and no word* w *satisfy* $\mathsf{DeadU}_{\psi}(p, w)$ *while* (p, w) *is live in* φ.

The words w with $\mathsf{DeadU}_{\psi}(p, w)$ form a regular language (Lemma 5.11), but the words w with (p, w) live in φ need not (Example 5.13), so (U) is not an inclusion of regular languages. When φ is satisfiable, however, Theorem 5.12 identifies liveness with the absence of DeadT_{φ}, which is regular, and of self events, which are comparisons of a single word whose two cuts are tested by a pair of automaton states. A failure of (U) is therefore a hole of a reader in the sense of Zhao [2026] (Definitions 2.11 and 2.12), and the reader has the rejected-path property (Definition 2.13). The bounded rejected tail, depth, and bounded-depth search theorems of that paper then decide (U).

6.1 The Source Automaton

One automaton reads the word of a visible seed for both systems. The two-word instances of Section 7 use it again, on the seed word.

Definition 6.2 (Source automaton). Let $p \in X^{\pm}$. The *source automaton* $\mathcal{B}_p$ is the deterministic automaton over Σ whose states are the triples $\zeta = (P, b, P')$ with $P \subseteq X^{\pm}$, $b \in \{0, 1\}$, and $P' \subseteq (X \cup Z)^{\pm}$, with start state $\big(\uparrow_{\varphi} p,\ [\overline{\uparrow_{\varphi} p} \cap \mathsf{Fact}_{\varphi}^{\uparrow} \neq \varnothing],\ \uparrow_{\psi} p\big)$ and transitions

$$\delta\big((P, b, P'), i\big) = \big(\delta_i^{\uparrow}(P),\ b \vee [\overline{\delta_i^{\uparrow}(P)} \cap \mathsf{Fact}_{\varphi}^{\uparrow} \neq \varnothing],\ \delta_i^{\uparrow}(P')\big),$$

where the first component uses the transitions of the upper automata of φ and the third those of ψ, and $[\,\cdot\,]$ is 1 for a true condition and 0 otherwise. For a state $\zeta = (P, b, P')$ put

$$\mathrm{dt}(\zeta) \iff b = 1 \text{ or } P \cap \mathsf{DeadT}_{\varphi}^{\varepsilon} \neq \varnothing, \qquad \mathrm{du}(\zeta) \iff P' \cap \mathsf{DeadU}_{\psi}^{\varepsilon} \neq \varnothing,$$

and let the *self guard* G_{s} consist of the pairs of states $((P_1, b_1, P_1'), (P_2, b_2, P_2'))$ with $P_1 \cap \overline{P_2} \neq \varnothing$.

Lemma 6.3 (Runs of the source automaton). *Let* $p \in X^{\pm}$, *let* w *be a word, and let* ζ_j *be the state of* $\mathcal{B}_p$ *after* $w[:j]$. *Then:*

(1) $\zeta_j = \big(P_{p,j}^{\varphi}(w),\ [(\bar{p}, w[:j]) \in \mu_{\varphi}],\ P_{p,j}^{\psi}(w)\big)$;
(2) $\mathrm{dt}(\zeta_j)$ *iff* $\mathsf{DeadT}_{\varphi}(p, w[:j])$, *and* $\mathrm{du}(\zeta_j)$ *iff* $\mathsf{DeadU}_{\psi}(p, w[:j])$; *if* $\mathrm{dt}(\zeta_j)$ *fails, it fails at every later cut, and if* $\mathrm{du}(\zeta_j)$ *holds, it holds at every earlier cut;*
(3) for $i < j \leq |w|$, *the pair* (i, j) *is a self event of* (p, w) *in* φ *iff it is a comparison of* w *and* $(\zeta_i, \zeta_j) \in G_{\mathrm{s}}$.

The automaton $\mathcal{B}_p$ *has* $N_{\mathrm{U}} = 2^{4|X|+2|Z|+1}$ *states.*

Item (1) holds by induction on j, using Lemma 5.11(2) for the bit; item (2) is Lemma 5.11(1,3) together with Lemma 5.9(4) and Lemma 5.11(4); and item (3) is Definition 5.10. The sets $\mathsf{Fact}_{\varphi}^{\uparrow}$, $\mathsf{DeadT}_{\varphi}^{\varepsilon}$, and $\mathsf{DeadU}_{\psi}^{\varepsilon}$ are computable in polynomial time (Lemmas 5.6 and 5.11), so $\mathcal{B}_p$, dt, du, and G_{s} are computable.

6.2 The Unit Reader

Definition 6.4 (Unit reader). For $p \in X^{\pm}$, the *unit reader* is the reader $\mathcal{R}_{\mathrm{U}}(p) = (\mathcal{B}_p, G_{\mathrm{s}}, \mathrm{Rej}_{\mathrm{U}})$ with the source automaton, the self guard as its admission relation, and the rejected states

$$\mathrm{Rej}_{\mathrm{U}} = \{\zeta :\ \mathrm{dt}(\zeta) \text{ fails and } \mathrm{du}(\zeta) \text{ holds}\}.$$

Lemma 6.5 (Holes of the unit reader). *Let* φ *be satisfiable and* $p \in X^{\pm}$. *A word* w *is a hole of* $\mathcal{R}_{\mathrm{U}}(p)$ *iff* (p, w) *is live in* φ *and* $\mathsf{DeadU}_{\psi}(p, w)$. *Consequently,* (U) *holds iff none of the* $2|X|$ *unit readers has a hole.*

Proof. By Lemma 6.3, w is rejected iff $\mathsf{DeadT}_\varphi(p,w)$ fails and $\mathsf{DeadU}_\psi(p,w)$ holds, and a comparison of w is admitted iff it is a self event of (p,w) in φ. So w is a hole iff $\mathsf{DeadU}_\psi(p,w)$ holds and (p,w) is neither dead by support nor has a self event in φ, that is, by Theorem 5.12, iff $\mathsf{DeadU}_\psi(p,w)$ holds and (p,w) is live in φ. The second claim follows by Lemma 6.1. □

Lemma 6.6 (Rejected paths). *Every unit reader $\mathcal{R}_\mathrm{U}(p)$ has the rejected-path property: it satisfies betweenness and the admission cone. No satisfiability assumption is needed.*

Proof. *Betweenness.* Let $w[{:}a]$ and $w[{:}b]$ be rejected, with $a < k < b$. Since dt fails at a, it fails at k, and since du holds at b, it holds at k (Lemma 6.3(2)). So $w[{:}k]$ is rejected.

Admission cone. Let w be rejected, with first rejected cut J, and let (i,j) be an admitted comparison of w. The guard $P^\varphi_{p,i}(w) \cap \overline{P^\varphi_{p,j}(w)} \neq \varnothing$ requires $P^\varphi_{p,j}(w) \neq \varnothing$. At the cut J, dt fails, so $\mathsf{DeadT}_\varphi(p, w[{:}J])$ fails, and in particular $\mathsf{DeadU}_\varphi(p, w[{:}J])$ fails. The pop bound (Lemma 5.9(2)) gives $P^\varphi_{p,j'}(w) = \varnothing$ for every $j' > J$. Hence $j \le J$. □

The proof gives more than the admission cone: no admission, comparison or not, has its second cut after J.

Example 6.7 (Shift closure: condition (U)). Let $\psi = \{x \le z,\ z \le y,\ z \le f(z,z)\}$ as in Example 5.5, with f binary and covariant, and let $\varphi_1 = \{x \le y\}$. Over $\{\bot, f, \top\}$, condition (U) fails. Indeed, $\mathsf{DeadU}^\varepsilon_\psi = \{x^0, z^0\}$ and $z^0 \in P^\psi_{x^0}(w)$ for every w, so $\mathsf{DeadU}_\psi(x^0, w)$ holds for every word w. On the other side, φ_1 has neither facts nor forbidden roots, so $\mathsf{DeadT}_{\varphi_1}$ never holds, and it has no upper constructor literal, so $P^{\varphi_1}_{x^0}(w) = \varnothing$ for $w \neq \varepsilon$ and no comparison is admitted. Every word is a hole of $\mathcal{R}_\mathrm{U}(x^0)$, the empty word included. Its countermodel is the least model $\mathrm{Cl}_{\varphi_1}(x^0, \varepsilon)$, the solution $x = y = \top$: no z satisfies $\top \le z \le f(z,z)$.

Over $\{\bot, f\}$, the variables cannot take the value $\top$. Within $\{\bot, f, \top\}$, this can be expressed by a variable g with $y \le g$ and $g \le f(g,g)$, which allow exactly the trees y without leaves $\top$; Section 10 turns this observation into a reduction. That reduction puts such a gadget on every variable, but for this example one on y suffices: since $x \le y$, it also keeps x free of $\top$, and $z \le f(z,z)$ already keeps z free of $\top$. So let $\varphi_1' = \{x \le y,\ y \le g,\ g \le f(g,g)\}$, with g visible. Its only forbidden root is (g^0, ε), and $x^0 \to y^0 \to g^0$ by copies, after which g^0 pops every word. So $\mathsf{DeadT}_{\varphi_1'}(x^0, w)$ holds for every w, and $\mathcal{R}_\mathrm{U}(x^0)$ rejects no word. For the other signed variables $p \in \{x^1, y^0, y^1, g^0, g^1\}$, the set $P^\psi_p(w)$ never meets $\mathsf{DeadU}^\varepsilon_\psi$, so $\mathcal{R}_\mathrm{U}(p)$ rejects no word either. Thus (U) holds for φ_1'; Example 7.11 shows that the entailment fails by condition (II).

Example 6.8 (Admitted self events are needed). Let $\varphi = \{x \le x \to x\}$ as in Example 5.13, and let $\psi = \{x \le a \to x,\ c \to y \le a\}$ with a, c hidden. The antecedent is that of Example 4.4, but the consequent is new. For $x \le z \to z$, (U) holds because its only root-dead atom, (x^0, ε), is a forbidden root of φ as well; this consequent is chosen so that (U) holds only because of admitted self events. The signed system of ψ has the upper constructor literals $x^0 \le f(a^1, x^0)$ and $a^1 \le f(c^0, y^1)$ and the lower ones $f(a^0, x^1) \le x^1$ and $f(c^1, y^0) \le a^0$; its closure is the identity, and $\mathsf{DeadU}^\varepsilon_\psi = \{x^0, a^1\}$. Reading from x^0,

$$P^\psi_{x^0}(1^k) = \{x^0\}, \quad P^\psi_{x^0}(1^k 0) = \{a^1\}, \quad P^\psi_{x^0}(1^k 00) = \{c^0\}, \quad P^\psi_{x^0}(1^k 01) = \{y^1\},$$

and the empty set after longer words. So $\mathsf{DeadU}_\psi(x^0, w)$ holds iff $w \in 1^* \cup 1^*0$, and $\mathcal{R}_\mathrm{U}(x^0)$ rejects exactly the words 1^k0, which are not dead by support in φ. Each of them has the admitted comparison $(0, k+1)$: $P^\varphi_{x^0}(\varepsilon) = \{x^0\}$ meets $\overline{P^\varphi_{x^0}(1^k0)} = \{x^0\}$. So $\mathcal{R}_\mathrm{U}(x^0)$ has no hole. The other unit readers reject no word, because x^1, y^0, and y^1 have no root-dead atoms in ψ. Hence (U) holds.

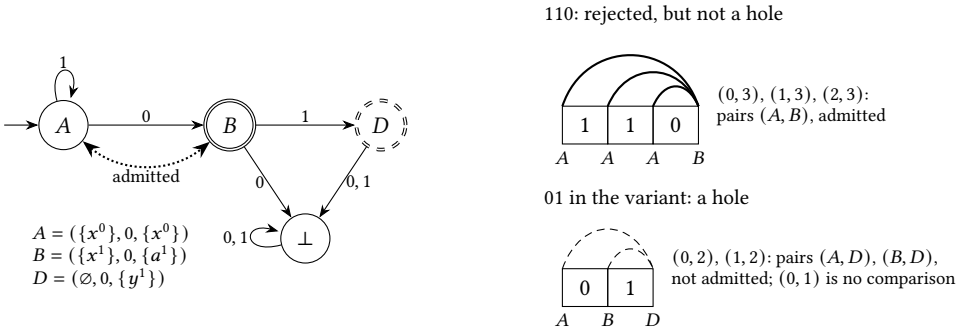


Fig. 8. The unit reader $\mathcal{R}_{\mathrm{U}}(x^0)$ of Example 6.8, after merging the states that cannot occur on a rejected word into a sink $\bot$ (Lemma 6.9). Rejected states are doubly circled; the only admitted pairs are (A, B) and (B, A), the self guard of the antecedent $x \le x \to x$. Every rejected word 1^k0, such as 110, ends in B and has the terminal comparisons $(i, k+1)$ with the admitted pair (A, B): it is rescued, because $(x^0, 1^k0)$ is dead in φ by a self event. In the example itself the state D, dashed, is not rejected and would be merged into $\bot$. In a variant in which y is hidden and ψ also contains $d \to e \le y$, with d, e hidden, the atom (y^1, ε) is a forbidden root, D is rejected, and the word 01 is a hole: none of its comparisons is admitted, so $(x^0, 01)$ is live in φ and root-dead in ψ, and (U) fails.

Without the admitted self events, the word 0 would be a hole: $(x^0, 0)$ is dead in φ only through its self event (Example 5.13).

Figure 8 draws this reader. The admission relation acts as a rescue: a rejected word, a candidate failure of (U), is discarded as soon as two of its cuts form a comparison with an admitted pair of states, because the seed is then dead in φ. A small change of the consequent, also shown there, leaves a rejected word that no comparison rescues, a hole.

6.3 Deciding Condition (U)

Zhao [2026] decides the hole problem for every reader with the rejected-path property (Corollary 2.21): a reader with N states over a finite alphabet that has this property has a hole iff it has a hole of length at most $\mathsf{Len}(N, d_N)$, where $d_N = 2N(4N-1) - 1$. The proof selects a hole with rejected tail at most $4N - 2$ (Theorem 2.15), bounds its depth (Theorem 2.19), and shortens it (Theorem 2.20). Applied to the unit reader, with its N_{U} states, this already decides (U). Two observations give a much smaller bound. First, most states of the source automaton can never occur on a rejected word, so they can be merged into one sink. Second, the depth bound needs only the first rejected cut, and on the words that the unit reader rejects this cut is determined by the antecedent alone, which is read by a much smaller automaton.

Lemma 6.9 (The reduced unit reader). *Let $p \in X^{\pm}$, $v = |X|$ and $s = |X| + |Z|$. Let $\widehat{\mathcal{R}}_p$ arise from the reachable part of $\mathcal{R}_{\mathrm{U}}(p)$ by replacing every state $\zeta = (P, b, P')$ with $b = 1$ or with $\mathrm{du}(\zeta)$ false by one new absorbing state $\bot \notin \mathrm{Rej}$: every transition into a replaced state is redirected to $\bot$, and no pair that involves $\bot$ is admitted. Then:*

(1) a word is rejected by $\widehat{\mathcal{R}}_p$ iff it is rejected by $\mathcal{R}_{\mathrm{U}}(p)$, and then both readers have the same state at every cut, so they have the same holes;

(2) $\widehat{\mathcal{R}}_p$ has the rejected-path property;

(3) $\widehat{\mathcal{R}}_p$ has at most $1 + 2^{v+1}(2^s - 1) < 2^{v+s+1}$ states.

No satisfiability assumption is needed.

The replaced states are closed under transitions, because the bit b never returns to 0 and du, once false, stays false (Lemma 6.3(2)); so a rejected word never meets them. The count uses sign coherence: on a kept state, P' is nonempty and fixes the common sign of all elements of P and P'. The proof is in Appendix C.

Lemma 6.10 (Antecedent rejection clock). *Let $p \in X^{\pm}$ and $v = |X|$, and let $\mathcal{A}_p$ be the reader whose automaton is the reachable part of the upper automaton of p in φ, whose admission relation is empty, and whose rejected states are the sets P with $P \cap \mathsf{DeadT}^{\varepsilon}_{\varphi} = \emptyset$. It has at most $2^{v+1} - 1$ states. If w is rejected by $\mathcal{R}_{\mathrm{U}}(p)$, then w is rejected by $\mathcal{A}_p$ with the same first rejected cut, and if moreover $w \neq \varepsilon$, then $\mathrm{depth}(w) \leq 2(2^{v+1} - 1)(r(w) + 1) - 1$.*

On a word rejected by $\mathcal{R}_{\mathrm{U}}(p)$, the bit and du are constant, so only the first component of the source automaton decides rejection; the depth bound is then Theorem 2.19 for $\mathcal{A}_p$, which uses neither admissions nor holes. The proof is in Appendix C.

Theorem 6.11 (Condition (U) is decidable). *Let φ be satisfiable. If $X = \emptyset$, then* (U) *holds. Otherwise let $v = |X| \geq 1$ and $s = |X| + |Z|$, and put*

$$A_{\mathrm{U}} = 2^{v+1}, \qquad K_{\mathrm{U}} = 2^{v+s+1}, \qquad D_{\mathrm{U}} = 2A_{\mathrm{U}}(4K_{\mathrm{U}} - 1) - 1.$$

Condition (U) *fails iff there are $p \in X^{\pm}$ and a word w with $|w| \leq \mathsf{Len}(K_{\mathrm{U}}, D_{\mathrm{U}})$ such that (p, w) is live in φ and $\mathsf{DeadU}_{\psi}(p, w)$. Moreover $\mathsf{Len}(K_{\mathrm{U}}, D_{\mathrm{U}}) \leq E_3(5v + 3s + 16) \leq E_3(8s + 16)$, where $E_3(x) = 2^{2^{2^x}}$. Hence* (U) *is decidable. The bound depends neither on the arity nor on the variance.*

Proof. If X is empty, there is no visible atom, and (U) holds vacuously. Let $X \neq \emptyset$. If (U) fails, some $\mathcal{R}_{\mathrm{U}}(p)$ has a hole (Lemma 6.5), hence so does $\widehat{\mathcal{R}}_p$, which has $m \leq K_{\mathrm{U}}$ states and the rejected-path property (Lemma 6.9). By Theorem 2.15 it has a hole w_0 with $r(w_0) \leq 4m - 2$; if $w_0 = \varepsilon$, we are done. Otherwise w_0 is rejected by $\mathcal{R}_{\mathrm{U}}(p)$, and Lemma 6.10 gives $\mathrm{depth}(w_0) \leq 2(2^{v+1} - 1)(4m - 1) - 1 \leq D_{\mathrm{U}}$. Theorem 2.20 for $\widehat{\mathcal{R}}_p$ with $d = D_{\mathrm{U}}$ gives a hole of length at most $\mathsf{Len}(m, D_{\mathrm{U}}) \leq \mathsf{Len}(K_{\mathrm{U}}, D_{\mathrm{U}})$, since Len is nondecreasing in each argument (Appendix A.5). This is a hole of $\mathcal{R}_{\mathrm{U}}(p)$ by Lemma 6.9(1), hence a word as stated by Lemma 6.5. The converse is Lemma 6.1.

For the estimate, write $A = A_{\mathrm{U}}$, $K = K_{\mathrm{U}}$ and $D = D_{\mathrm{U}}$. Then $D + 1 \leq 8AK$ and $K + D \leq 9AK$, so the quantity Ξ of Lemma A.13 satisfies $\Xi(K, D) \leq 3K^2 + 288A^2K^3 + 3K + 24AK + 4 \leq 322A^2K^3$, and $5\Xi(K, D) + 3 \leq 1613A^2K^3 < 2^{11}A^2K^3 = 2^{5v+3s+16}$. By Lemma A.13, $\mathsf{Len}(K, D) \leq 2^{2^{5\Xi(K,D)+3}} \leq E_3(5v + 3s + 16)$, and $v \leq s$ gives the second inequality.

For the decision, enumerate the $2|X|$ signed variables and the finitely many words up to this length, and test each pair by Lemma 6.3: the states of $\mathcal{B}_p$ along w are computable, and so are rejection and the admission of each of the finitely many comparisons of w. □

Remark 6.12 (Singly exponential cases). In two cases the witnesses of Theorem 6.11 are much shorter. Let φ be satisfiable and $p \in X^{\pm}$.

- *No self events.* If φ has a parity assignment (Remark 4.5), in particular if f is covariant, then no pair of reachable states of $\mathcal{B}_p$ is admitted (Remark 5.14). The holes of $\mathcal{R}_{\mathrm{U}}(p)$ are then exactly its rejected words, and a shortest one visits no state of $\widehat{\mathcal{R}}_p$ twice, so it is shorter than 2^{v+s+1}.
- *Arity one.* If $n = 1$, every pair of cuts $i < j$ of a word is a comparison. If w is a hole of $\mathcal{R}_{\mathrm{U}}(p)$, then so is its prefix $w_J = w[:J(w)]$, because the comparisons of w_J are comparisons of w with the same states. By Lemma 6.10, w_J is rejected by $\mathcal{A}_p$ with the first rejected cut $J(w)$, and no two of its cuts $i < j \leq J(w)$ have the same state of $\mathcal{A}_p$: the unary word would then repeat the run of $\mathcal{A}_p$ with period $j - i$, and the cut $J(w) - (j - i)$ would already be rejected. Hence $J(w) < 2^{v+1} - 1$, and (U), if it fails, fails at a word of length at most $2^{v+1} - 2$.

A different route reduces (U) to plain entailment: the words w with $\mathrm{DeadU}_\psi(p, w)$ form a regular language W_p, and (U) says that in every solution of φ the leaf cone of p avoids W_p, which is a comparison of the normalized value of the variable of p with a regular ground tree determined by W_p. This alternative proves decidability by naming a finite graph for the regular ground tree; the direct unit-reader construction gives an explicit witness bound without introducing those additional variables. Section 7 turns to condition (II).

7 Two-Word Holes with Self Events

Condition (II) fails when a visible atom that is live in φ has a visible ψ-consequence outside its closure in φ; by Lemma 4.3(2), the consequence can be taken to be derived by a pure derivation. By the lowest point (Lemma 5.3), the seed and the consequence then share a suffix, which follows a pop head in the seed and a push head in the consequence. Liveness of the seed is read by the source automaton of Section 6 together with the comparisons of the seed word that are self events (Theorem 5.12); membership in the closure is read by the lower automata of φ together with comparisons between the two words (Lemma 5.15); and the derivation is read by the bound automata of ψ. This section abstracts these tests into a hole problem for pairs of words read by four automata (Section 7.1), extracts its instances from φ and ψ (Section 7.2), and proves that, once satisfiability and (U) are known, (II) holds iff no extracted instance has a hole (Section 7.3). The extracted instances read the seed with a rejection test that already excludes the root-dead words of ψ. With this *enrichment* they have forced target heads by construction, the property on which Section 8 relies. Proofs omitted here are in Appendix C.

7.1 The Hole Problem

Definition 7.1 (Two-word hole problem with self events). An *instance* of the *two-word hole problem with self events* is a tuple

$$I = (A, \mathsf{Bad}, B, \mathsf{Rej}, G, G_{\mathrm{s}}, A', B', H)$$

that consists of complete deterministic automata A, B, A', B' over Σ, with state sets S_A, S_B, $S_{A'}$, $S_{B'}$ and a start state 0 each, together with sets $\mathsf{Bad} \subseteq S_A$, $\mathsf{Rej} \subseteq S_B$, $G \subseteq S_A \times S_B$, $G_{\mathrm{s}} \subseteq S_B \times S_B$, and $H \subseteq S_{A'} \times S_{B'}$. We call A the *spine automaton*, B the *target automaton*, A' and B' the *head automata*, G the *guard*, G_{s} the *self guard*, and H the *head relation*, and we put $N_A = |S_A|$, and similarly N_B, $N_{A'}$, $N_{B'}$. For words u and t, let $A_s(u)$ be the state of A after $u[{:}s]$ and $B_e(t)$ the state of B after $t[{:}e]$, define $A'_s(u)$ and $B'_e(t)$ in the same way, and write $M(w)$ for the state of an automaton M after the whole word w. If some cut e of t has $B_e(t) \in \mathsf{Rej}$, the least one is the *first rejected cut* $J(t)$. A pair of words (u, t) is a *hole* of I iff

(a) $A_s(u) \notin \mathsf{Bad}$ for every cut $s \le |u|$;
(b) $J(t)$ is defined;
(c) there are no cuts $s \le |u|$ and $e \le J(t)$ with $t[e{:}] \preceq u[s{:}]$ and $(A_s(u), B_e(t)) \in G$;
(d) there are cuts $s \le |u|$ and $e \le |t|$ with $u[s{:}] = t[e{:}]$ and $(A'_s(u), B'_e(t)) \in H$;
(e) there are no cuts $i < j \le J(t)$ with $t[j{:}] \preceq t[i{:}]$ and $(B_i(t), B_j(t)) \in G_{\mathrm{s}}$.

The *hole problem* asks whether a given instance has a hole.

We call u the *spine* and t the *target*. Unlike rejection in Definition 2.12, condition (b) requires a rejected prefix of t, not necessarily a rejected final state; for extracted instances these conditions coincide, as shown in Section 7.2. A state of A outside Bad is *safe*; a word is *safe from* a state r if A visits only safe states while reading it from r, including r and the last state; and the spine u is *safe* if it is safe from 0, which is condition (a). The other conditions are stated in terms of the following pairs of cuts.

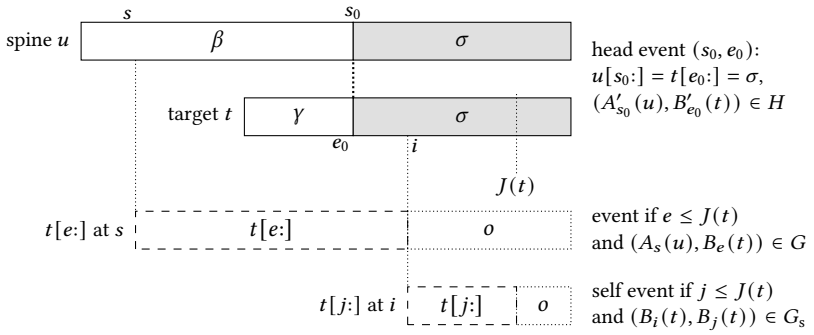


Fig. 9. The pairs of cuts that decide whether (u, t) is a hole (Definitions 7.1 and 7.2). Top: the spine and the target, aligned at their ends; a head event (s_0, e_0) splits them into the heads β and γ and the common suffix σ (shaded). Middle: a cross comparison (s, e) places $t[e:]$ at the cut s of u, where it must be a prefix of $u[s:]$, leaving the overhang o. Bottom: a self comparison (i, j) places $t[j:]$ at the cut i of t in the same way. A hole needs a head event, a safe spine and a first rejected cut $J(t)$ of the target, and it must have no event and no self event: every cross comparison (s, e) with $e \le J(t)$ must carry a state pair outside the guard G, and every self comparison (i, j) with $j \le J(t)$ a pair outside the self guard G_s.

Definition 7.2 (Comparisons and events). Let u and t be words. A *cross comparison* of (u, t) is a pair of cuts (s, e) with $0 \le s \le |u|$, $0 \le e \le |t|$, and $t[e:] \preceq u[s:]$. Its *offset* is $\Delta(s, e) = (|u|-s)-(|t|-e) \ge 0$, its *overhang* $o(s, e)$ is the suffix of u of length $\Delta(s, e)$, so that $u[s:] = t[e:]\, o(s, e)$, and it is *aligned* if $\Delta(s, e) = 0$. A *self comparison* of t is a comparison (i, j) of t (Definition 2.10): $i < j \le |t|$ and $t[j:] \preceq t[i:]$; its overhang is the word o of length $j - i$ with $t[i:] = t[j:]\, o$. For an instance I and a target t with $J(t)$ defined:

- an *event* of (u, t) is a cross comparison (s, e) with $e \le J(t)$ and $(A_s(u), B_e(t)) \in G$;
- a *self event* of t is a self comparison (i, j) of t with $j \le J(t)$ and $(B_i(t), B_j(t)) \in G_s$;
- a *head event* of (u, t) is an aligned cross comparison (s, e) with $(A'_s(u), B'_e(t)) \in H$.

Thus a hole is a pair with a safe spine, a target with $J(t)$ defined, no event, no self event, and a head event. A self comparison (i, j) of t is the cross comparison (i, j) of the pair (t, t), with offset $j - i > 0$; in particular, it is never aligned. Events relate two words; self events relate two cuts of the target. An instance with $G_s = \varnothing$ has no self events, and its holes are those of the suffix-sharing two-word problem of a covariant constructor; Remark 7.10 explains when the extracted instances have this form.

Head form. A head event (s, e) splits the two words as

$$u = \beta\sigma, \qquad t = \gamma\sigma, \qquad \beta = u[:s], \quad \gamma = t[:e], \quad \sigma = u[s:] = t[e:],$$

with $(A'(\beta), B'(\gamma)) \in H$. We call β the *spine head*, γ the *target head*, and σ the *common suffix*. In the extracted instances of Section 7.2, the target head is the pop head of Lemma 5.3 and the spine head its push head. Figure 9 draws the head form together with the two kinds of pairs that a hole must avoid. A cross comparison and a self comparison look alike: in both, a suffix of the target is laid at a cut of a word, the spine or the target itself, and agrees with that word letter by letter until the suffix ends.

Definition 7.3 (Forced target heads). An instance I has *forced target heads*, condition *(FH)*, if for every $(r, r') \in H$ with r reachable in A' and every word γ with $B'(\gamma) = r'$, no proper prefix of γ is rejected: $B_i(\gamma) \notin \mathrm{Rej}$ for all $i < |\gamma|$.

Table 4. The extracted instance I_{pq} for a seed (p,t) and a derived atom (q,u). The state of B at a cut e of t is $(P^{\varphi}_{p,e}(t), [(\bar{p}, t[{:}e]) \in \mu_\varphi], P^{\psi}_{p,e}(t))$, by Lemma 6.3.

Part	Reads	Meaning
A, Bad	$Q^{\varphi}_{q,s}(u)$ on the spine u	safe spine: $(q,u) \notin \mu_\varphi$ (Lemma 5.6)
B, Rej	the source automaton on the target t	rejected: $\neg\mathsf{DeadT}_\varphi(p,t[{:}e])$ and $\neg\mathsf{DeadU}_\psi(p,t[{:}e])$
G	$Q^{\varphi}_{q,s}(u) \cap P^{\varphi}_{p,e}(t)$	event: $(q,u) \in \mathrm{Reach}_\varphi(p,t)$ (Lemma 5.15)
G_{s}	$P^{\varphi}_{p,i}(t) \cap \overline{P^{\varphi}_{p,j}(t)}$	self event: (p,t) is dead (Theorem 5.12)
A', B', H	$Q^{\psi}_{q,s}(u) \cap P^{\psi}_{p,e}(t)$	head event: $(p,t) \to^*_\psi (q,u)$ (Lemma 5.3)

Lemma 7.4 (Heads end before the first rejected cut). *Let I satisfy (FH), and let (s,e) be a head event of (u,t), where $J(t)$ is defined. Then $e \le J(t)$: the target head ends at or before the first rejected cut, and $t[J(t){:}]$ is a suffix of the common suffix.*

Proof. The state $A'_s(u)$ is reachable, and $B'(t[{:}e]) = B'_e(t)$ is paired with it by H. By (FH), $B_i(t) = B_i(t[{:}e]) \notin \mathrm{Rej}$ for every $i < e$, so the first rejected cut is at least e. □

Section 8 uses Lemma 7.4 to split a hole at its first rejected cut. In the extracted instances, (FH) holds by construction (Lemma 7.6).

7.2 The Extracted Instances

The instance for a pair of visible signed variables $p, q \in X^{\pm}$ tests a pure derivation of ψ from a seed (p,t) to an atom (q,u). The names follow the readers of Zhao [2026]: the *target* is the word whose rejection is tested, here the seed word t, read by the source automaton $\mathcal{B}_p$ of Section 6; the *spine* is the word u of the derived atom. The derivation therefore runs from the target to the spine.

Definition 7.5 (Extracted instance). For $p, q \in X^{\pm}$, the *extracted instance* $I_{pq} = I_{pq}(\varphi,\psi)$ consists of:

- A, the lower automaton of q in φ, with $\mathrm{Bad} = \{Q : Q \cap \mathrm{Fact}^{\uparrow}_{\varphi} \neq \varnothing\}$;
- $B = \mathcal{B}_p$, the source automaton of p (Definition 6.2), with

$$\mathrm{Rej} = \{\zeta : \ \mathrm{dt}(\zeta) \text{ and } \mathrm{du}(\zeta) \text{ both fail}\};$$

- $G = \{(Q,(P,b,P')) : Q \cap P \neq \varnothing\}$, and G_{s} the self guard of Definition 6.2;
- A', the lower automaton of q in ψ, and B', the upper automaton of p in ψ, with $H = \{(Q',P') : Q' \cap P' \neq \varnothing\}$.

The start states $\downarrow_\varphi q$, the start state of $\mathcal{B}_p$, $\downarrow_\psi q$, and $\uparrow_\psi p$ are renamed 0.

Table 4 summarizes the instance. The run of B' on t is the third component of the run of B, so the reachable pairs of states of B and B' are at most N_B in number. The automata have $N_A = 2^{2|X|}$, $N_B = 2^{4|X|+2|Z|+1}$, and $N_{A'} = N_{B'} = 2^{2(|X|+|Z|)}$ states, and the $4|X|^2$ instances are computable from φ and ψ in time polynomial in $2^{|X|+|Z|}$ and the size of φ and ψ.

Rejection is enriched: a cut of the seed is rejected only if it is neither dead by support in φ nor root-dead in ψ. Both properties are inherited by prefixes (Lemma 5.9(4), Lemma 5.11(4)), so a rejected cut stays rejected on every extension: $J(t)$ is defined iff t itself is rejected, that is, iff neither $\mathsf{DeadT}_\varphi(p,t)$ nor $\mathsf{DeadU}_\psi(p,t)$ holds.

Lemma 7.6 (Forced target heads). *Every extracted instance I_{pq} satisfies (FH). No hypothesis on φ or ψ is needed.*

Proof. Let $(Q', P') \in H$ and $B'(\gamma) = P'$. Then $P_p^{\psi}(\gamma) = P' \neq \varnothing$, and by Lemma 5.16, $\mathrm{DeadU}_{\psi}(p, \gamma[:i])$ for every $i < |\gamma|$. By Lemma 6.3(2), du holds at every such cut, so $B_i(\gamma) \notin \mathrm{Rej}$. □

Without the enrichment, that is, with Rej defined by the failure of dt alone, the target head can extend beyond the first rejected cut (Example 7.12).

7.3 Holes and Condition (II)

Theorem 7.7 (Holes and condition (II)). *Let φ be satisfiable, let* (U) *hold, let $p, q \in X^{\pm}$, and let u, t be words. Then (u, t) is a hole of I_{pq} iff (p, t) is live in φ, $(p, t) \to_{\psi}^{*} (q, u)$, and $(q, u) \notin \mathrm{Cl}_{\varphi}(p, t)$. Consequently:*

(1) condition (II) *holds iff no extracted instance I_{pq} has a hole;*
(2) for every hole (u, t) of I_{pq}, the least model $\mathrm{Cl}_{\varphi}(p, t)$ of φ that contains (p, t) is not the restriction to X of any model of ψ, so its solution has no extension to a solution of ψ.

The conditions of Definition 7.1 match the three requirements one by one. A head event is the lowest point of the pure derivation of ψ. A safe spine and the absence of events say that (q, u) lies neither in μ_{φ} nor in $\mathrm{Reach}_{\varphi}(p, t)$ (Lemma 5.15). A rejected target and the absence of self events say that (p, t) is neither dead by support nor has a self event, that is, that it is live (Theorem 5.12). Here (U) is used: a live seed must also be rejected, which requires that it not be root-dead in ψ. The bounds $e \leq J(t)$ and $j \leq J(t)$ in Definition 7.1 lose nothing: the guards require a nonempty set $P_{p,e}^{\varphi}(t)$, and the pop bound (Lemma 5.9(2)) empties these sets after $J(t)$. The proof is in Appendix C.

Corollary 7.8 (Entailment and holes). *Let φ and ψ be satisfiable. Then $\varphi \models \exists \bar{z}\, \psi$ iff condition* (I0) *holds, condition* (U) *holds, and no extracted instance I_{pq} with $p, q \in X^{\pm}$ has a hole.*

Proof. If (U) fails, the entailment fails by Theorem 4.2. Otherwise, by Theorem 4.2, the entailment holds iff (I0) and (II) hold, and by Theorem 7.7(1), (II) holds iff no I_{pq} has a hole. □

Conditions (I0) and (U) are decidable (Proposition 5.17 and Theorem 6.11), and every extracted instance satisfies (FH) (Lemma 7.6). It remains to decide the hole problem for instances with forced target heads, which Sections 8 and 9 do.

Remark 7.9 (Condition (U) cannot be dropped). Without (U), a failure of (II) need not be a hole. Let φ be empty over $X = \{x, y\}$, and let $\psi = \{x \leq y,\ x \leq f(a, b)\}$ with a, b hidden and f binary and covariant. The seed (x^0, ε) is live in φ, the copy $(x^0, \varepsilon) \to (y^0, \varepsilon)$ is a pure move of ψ, and $(y^0, \varepsilon) \notin \mathrm{Cl}_{\varphi}(x^0, \varepsilon)$, so (II) fails at this seed. But x^0 is the root of the upper constructor literal $x^0 \leq f(a^0, b^0)$ of $\psi^{\pm}$, so $\mathrm{DeadU}_{\psi}(x^0, \varepsilon)$ holds and the target ε of $I_{x^0 y^0}$ is not rejected: $(\varepsilon, \varepsilon)$ is not a hole. Here (U) fails, with the countermodel $x = \top$. The enrichment trades (U) for (FH): with the rejection test given by the failure of dt alone, the holes still characterize the failures of (II) when φ is satisfiable, without assuming (U), but (FH) can fail (Example 7.12).

Remark 7.10 (Covariant and joint-parity instances). If φ has a parity assignment (Remark 4.5), in particular if f is covariant, then no pair of reachable states of the target automaton lies in G_{s} (Remark 5.14), and every extracted instance is an instance of the suffix-sharing two-word problem without self events.

Example 7.11 (Shift closure: a hole). Continue Example 6.7 with $\psi = \{x \leq z,\ z \leq y,\ z \leq f(z, z)\}$ and $\varphi_1' = \{x \leq y,\ y \leq g,\ g \leq f(g, g)\}$, both satisfiable. Condition (U) holds, and (I0) holds because ψ has no facts. Consider the seed variable y^1 and the derived variable x^1. The copy and constructor literals of $(\varphi_1')^{\pm}$ are $x^0 \leq y^0$, $y^0 \leq g^0$, $g^0 \leq f(g^0, g^0)$, $y^1 \leq x^1$, $g^1 \leq y^1$, and $f(g^1, g^1) \leq g^1$. In $I_{y^1 x^1}$:

- A starts at $\downarrow_{\varphi_1'} x^1 = \{x^1, y^1, g^1\}$ and moves to $\{g^1\}$ on every letter; φ_1' has no facts, so every spine is safe; the state $\{g^1\}$ lies on a loop at which both letters are safe, so it is *branching* (Definition 8.5), the case in which Section 8 completes every pre-hole directly (Lemma 8.6);
- $P^{\varphi_1'}_{y^1}$ is $\{y^1, x^1\}$ at the empty word and empty afterwards, $\mu_{\varphi_1'} = \varnothing$, and $\mathsf{DeadT}^{\varepsilon}_{\varphi_1'} = \{x^0, y^0, g^0\}$; with Example 6.7, no cut of any target is dead by support or root-dead, so $J(t) = 0$ for every target t;
- hence the only possible event is $(0,0)$, a cross comparison iff $t \preceq u$, and there are no self events, since $P^{\varphi_1'}_{y^1,j}(t) = \varnothing$ for $j \geq 1$;
- $Q^{\psi}_{x^1}(\beta)$ always meets $\uparrow_{\psi} y^1 = \{y^1, z^1, x^1\}$, and $P^{\psi}_{y^1}(\gamma) = \varnothing$ for $\gamma \neq \varepsilon$, so the head events are the pairs $(s,0)$ with $u[s{:}] = t$.

So the holes of $I_{y^1 x^1}$ are the pairs (u,t) in which t is a suffix but not a prefix of u, for example $(01,1)$, with head event $(1,0)$. By Theorem 7.7, ψ derives $(x^1, 01)$ from the live seed $(y^1, 1)$, and $(x^1, 01) \notin \mathrm{Cl}_{\varphi_1'}(y^1, 1)$. In words: if y has a leaf $\bot$ at a prefix of 1, then every z between x and y with $z \leq f(z,z)$ has a leaf $\bot$ at a prefix of 1 and hence at a prefix of 01, and so does x. A finite countermodel is $x = y = f(f(\bot, f(\bot,\bot)),\bot)$, whose constructor positions are ε, 0, and 01, with g the complete binary tree whose constructor positions are the words of length at most 2. The least model $\mathrm{Cl}_{\varphi_1'}(y^1, 1)$ of Theorem 7.7(2) is the solution $x = y = f(\Omega, \bot)$ and $g = \Omega$. Since $J(t) = 0$, the target head is empty, and (FH) holds trivially for this instance.

Example 7.12 (A target head beyond the cuts dead by support). Continue Example 6.8 with $\varphi = \{x \leq x \to x\}$ and $\psi = \{x \leq a \to x,\ c \to y \leq a\}$, both satisfiable: $x = \bot$ solves φ, and $x = y = \bot$, $a = c = \top$ solve ψ. Conditions (I0) and (U) hold. Consider the seed variable x^0, the derived variable y^1, and the pair $(u,t) = (\varepsilon, 01)$ in $I_{x^0 y^1}$. Along the target $t = 01$, the upper automata give

$$P^{\varphi}_{x^0} :\ \{x^0\},\ \{x^1\},\ \varnothing, \qquad P^{\psi}_{x^0} :\ \{x^0\},\ \{a^1\},\ \{y^1\}$$

at the cuts 0, 1, 2. With $\mathsf{DeadT}^{\varepsilon}_{\varphi} = \{x^0\}$ and $\mathsf{DeadU}^{\varepsilon}_{\psi} = \{x^0, a^1\}$, cut 0 is dead by support, cut 1 is not dead by support but root-dead in ψ, and cut 2 is rejected: $J(01) = 2$. The head event is $(0,2)$, since $Q^{\psi}_{y^1}(\varepsilon) = \{y^1\} = P^{\psi}_{x^0}(01)$: the derivation pops 0 from x^0 to a^1 and then 1 from a^1 to y^1. The only cross comparison (s,e) with $e \leq 2$ is $(0,2)$, whose guard $\{y^1\} \cap P^{\varphi}_{x^0}(01)$ is empty; the self comparisons of 01 ending at 2 have an empty self guard, and $(0,1)$ is not a self comparison. The spine is safe because φ has no facts. So $(\varepsilon, 01)$ is a hole: ψ derives (y^1, ε), that is $y = \bot$, from the live seed $(x^0, 01)$. A countermodel is $x = (\bot \to \bot) \to \bot$ and $y = \top$: from $x \leq a \to x$ and $c \to y \leq a$, any extension would have $c \to \top \leq_c a \leq_c \bot \to \bot$, hence $\top \leq_c \bot$.

The target head is $\gamma = 01$. The rejection test given by the failure of dt alone would reject already at cut 1, inside the head: $(x^0, 0)$ is not dead by support, although it is dead by the self event $(0,1)$ of the word 0 (Example 5.13). So the unenriched instance does not satisfy (FH). The enrichment rejects only at cut 2, because $(x^0, 0)$ is root-dead in ψ.

8 Non-Mortal Holes

Let I be an instance of the two-word hole problem with self events (Definition 7.1) that satisfies condition (FH) (Definition 7.3). This section shows that if I has a hole, then it has a hole whose length is polynomial in the sizes of its four automata, or it has a *mortal hole*: a hole (u,t) whose spine ends in a state without infinite safe continuations and whose target ends fewer than N_A letters after its first rejected cut (Theorem 8.17). Mortal holes are treated in Section 9.

The argument starts from a simple observation. Cutting a hole (u,t) at the first rejected cut $J(t)$ of its target splits off the *continuation* $\theta = t[J(t){:}]$. Condition (FH) places every target head before

$J(t)$, so θ is a common suffix of both words, and what remains is a shorter pair (u_0, t_0) with $u = u_0\theta$ and $t = t_0\theta$, a *pre-hole* (Lemma 8.3 and Figure 10). Appending the same word to both words creates no new events and no new self events, and it keeps an existing one exactly when it continues the overhang of that event periodically (Lemma 8.1). The question is therefore whether a single continuation that keeps the spine automaton out of Bad can break every pending obligation of a pre-hole: its events, which are never aligned, and the self events of its target. The answer depends on the state $q = A(u_0)$ at the end of the spine of the pre-hole.

- If a cycle that offers a choice of two letters is reachable from q, a breaking continuation of length at most $\max(|u_0|, |t_0|) + 3N_A$ always exists (Lemma 8.6).
- If every reachable cycle is forced, then the long continuations follow finitely many ultimately periodic words, the *lassos* of q (Lemma 8.8). A lasso that is not purely periodic breaks every obligation (Lemma 8.13). A purely periodic lasso may let obligations survive, and then the pre-hole itself has to be shortened. The *pump* (Theorem 8.14) deletes factors of the two words while keeping the endpoint q and the absence of surviving obligations, down to a length polynomial in the sizes of the automata.
- In the remaining cases the continuation reaches a state without infinite safe continuations within fewer than N_A letters, and the hole is mortal.

Conventions. We use the instance, safety, comparison, event, and forced-head conventions of Definitions 7.1 to 7.3. For an automaton M, let $M(r, w)$ denote its state after reading w from r, and let $M_s(w) = M(0, w[:s])$ be its state at the cut s of w. We abbreviate the state counts as

$$a = N_A, \qquad b = N_B, \qquad a' = N_{A'}, \qquad b' = N_{B'};$$

all four are at least 1.

An *infinite word* $\xi' = \xi'[0]\xi'[1]\cdots$ over Σ has prefixes $\xi'[:k]$ and suffixes $\xi'[k:]$ as for finite words, and $v\xi'$ is the infinite word that starts with the finite word v. A number $\varpi \geq 1$ is a *period* of ξ' if $\xi'[i + \varpi] = \xi'[i]$ for all $i \geq 0$; ξ' is *purely periodic* if it has a period, and its *least period* is then the least of its periods. For a nonempty finite word o, $o^\omega = ooo\cdots$ is the infinite word with period $|o|$ that begins with o; a finite word is a prefix of o^ω exactly when it is a prefix of some o^k, in agreement with the notation $\theta \preceq o^\omega$ of Section 2.1. Holes are written (u, t), pre-holes (u_0, t_0), continuations θ, and infinite words ξ.

8.1 The Continuation Identity

For a cross comparison (s, e) of (u, t) and a word θ, we say that θ *keeps* (s, e) if $t[e:]\ \theta \preceq u[s:]\ \theta$, and that θ *breaks* it otherwise; for a self comparison (i, j) of t, θ keeps it if $t[j:]\ \theta \preceq t[i:]\ \theta$. Non-aligned cross comparisons can be unboundedly many: if $u = \ell^k$ and $t = \ell^m$ for a letter ℓ, every pair of cuts with a nonnegative offset is one. We therefore do not count them but control them through common continuations.

Lemma 8.1 (Continuation identity). *Let u, t and θ be words such that $J(t)$ is defined.*

(a) $J(t\theta) = J(t)$, and each of the four automata has the same state at every cut $s \leq |u|$ of $u\theta$ as at the cut s of u, and at every cut $e \leq |t|$ of $t\theta$ as at the cut e of t.

(b) Let $s \leq |u\theta|$ and $e \leq J(t)$. Then (s, e) is a cross comparison of $(u\theta, t\theta)$ if and only if $s \leq |u|$, (s, e) is a cross comparison of (u, t), and θ keeps it. Likewise, for $i < j \leq J(t)$, the pair (i, j) is a self comparison of $t\theta$ if and only if it is a self comparison of t and θ keeps it.

(c) A word θ keeps a cross or self comparison with overhang o if and only if $\theta \preceq o\theta$. If $o \neq \varepsilon$, this holds if and only if $\theta \preceq o^\omega$. Every word keeps every aligned cross comparison.

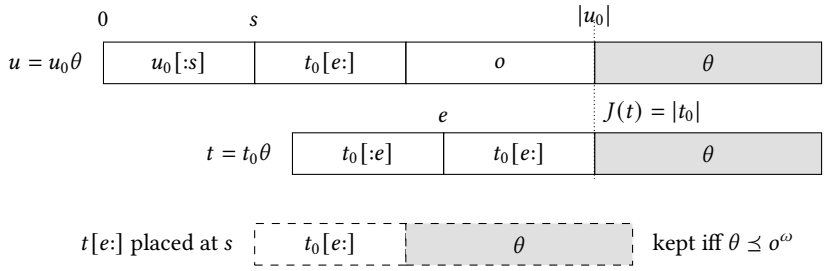


Fig. 10. A hole $(u, t) = (u_0\theta, t_0\theta)$ cut at the first rejected cut $J(t) = |t_0|$ of its target (Lemma 8.3); the continuation θ is shaded. An event (s, e) of the pre-hole (u_0, t_0) places $t_0[e:]$ at the cut s of u_0, followed by the overhang $o = o(s, e)$. After θ is appended, (s, e) is still a cross comparison if and only if the shifted copy of θ (bottom row) agrees with $o\theta$, that is, if and only if $\theta \preceq o^\omega$ (Lemma 8.1). A self event (i, j) of t_0 behaves in the same way, with t_0 in the top row. A hole requires θ to break every event and every self event.

In words: a common continuation creates no new comparison among the cuts up to $J(t)$, and it keeps an existing one exactly when it continues the overhang periodically. The proof is in Appendix D.2. We use the lemma through the following *continuation facts*, which hold for all words u and t with $J(t)$ defined and every word θ:

- (K1) the events of $(u\theta, t\theta)$ are exactly the events of (u, t) that θ keeps, and the self events of $t\theta$ are exactly the self events of t that θ keeps;
- (K2) every word keeps every aligned cross comparison, so aligned events can never be removed;
- (K3) if θ breaks a cross or self comparison, so does every word with prefix θ;
- (K4) every head event of (u, t) is a head event of $(u\theta, t\theta)$;
- (K5) if u is safe, then $u\theta$ is safe if and only if θ is safe from $A(u)$.

A non-aligned event or a self event with overhang o is thus a periodicity obligation: a continuation removes it exactly when it departs from o^ω.

8.2 Pre-Holes and the Decomposition at the First Rejected Cut

Definition 8.2 (Pre-hole). A *pre-hole* of I is a pair (u_0, t_0) such that u_0 is safe, $J(t_0) = |t_0|$, (u_0, t_0) has a head event, and (u_0, t_0) has no aligned event. Its *obligations* are its events and the self events of t_0. A word θ *completes* (u_0, t_0) if θ is safe from $A(u_0)$ and breaks every obligation of (u_0, t_0).

No obligation of a pre-hole is aligned, so every obligation has a nonempty overhang. The overhang of an event has length at most $|u_0|$ and that of a self event at most $|t_0|$, so every obligation has an overhang of length at most

$$W = \max(|u_0|, |t_0|).$$

A pre-hole without obligations is a hole, completed by the empty word.

Lemma 8.3 (Decomposition at the first rejected cut). *If (u_0, t_0) is a pre-hole and θ completes it, then $(u_0\theta, t_0\theta)$ is a hole. Conversely, let I satisfy* (FH) *and let (u, t) be a hole. Then $\theta = t[J(t):]$ is a suffix of u, and if $u = u_0\theta$ and $t = t_0\theta$, then (u_0, t_0) is a pre-hole and θ completes it.*

In one direction, appending a completing word keeps the spine safe, keeps the head event, and leaves no event and no self event, by (K5), (K4) and (K1). In the other, (FH) places the target head of a head event before $J(t)$ (Lemma 7.4). Hence θ is no longer than the common suffix and can be split off both words, and (K1) and (K2) show that θ breaks every obligation of (u_0, t_0) and that (u_0, t_0) has no aligned event. The proof is in Appendix D.2. Condition (FH) cannot be dropped (Remark 8.20).

Example 8.4 (Shift closure: decomposition). In the instance $I_{y^1x^1}$ of Example 7.11, every target t has $J(t) = 0$. So the hole $(01, 1)$ splits as $\theta = t[0{:}] = 1$ and $(u_0, t_0) = (0, \varepsilon)$. The pre-hole $(0, \varepsilon)$ has the head event $(1, 0)$, and this aligned pair is not an event, because $A_1(0) = \{g^1\}$ misses $P^{\varphi_1'}_{y^1}(\varepsilon) = \{y^1, x^1\}$. Its only obligation is the event $(0, 0)$, with overhang 0; the target ε has no self comparison. By Lemma 8.1(c), a continuation keeps this obligation iff it is a prefix of 0^ω. So the continuation 0 keeps it, and indeed $(00, 0)$ has the event $(0, 0)$, while every continuation that contains the letter 1 breaks it; the shortest one, $\theta = 1$, gives back the hole $(01, 1)$.

8.3 Kinds of State

Let $\mathcal{G}_A$ be the *safe graph* of A: its vertices are the safe states of A, with an edge from r to $A(r, \ell)$, labeled by the letter ℓ, whenever $A(r, \ell)$ is safe. The labels of the paths from r are exactly the words safe from r. A safe state is *cyclic* if it lies on a cycle of $\mathcal{G}_A$, and a letter ℓ is *safe at* a safe state r if $A(r, \ell)$ is safe.

Definition 8.5 (Kinds of state). A safe state q of A is

- *mortal* if no infinite word is safe from q, equivalently, if no cyclic state is reachable from q in $\mathcal{G}_A$;
- *branching* if some cyclic state r reachable from q in $\mathcal{G}_A$ has two distinct safe letters $\ell \neq \ell'$;
- *thin* if it is neither mortal nor branching.

From a mortal state every safe run dies within N_A letters. From a branching state some reachable cycle offers a choice of letters, so a safe continuation can depart from any given period. From a thin state infinite safe runs exist, but once they reach a cycle they are forced, letter by letter. The two descriptions of mortality agree because an infinite path in the finite graph $\mathcal{G}_A$ repeats a vertex, and a reachable cyclic state can be visited forever. A branching state reaches a cyclic state, so it is not mortal; the three kinds are therefore pairwise disjoint, and every safe state has exactly one of them. Each kind is decided by reachability in $\mathcal{G}_A$.

Branching asks only for two distinct safe letters at a reachable cyclic state; the states they lead to may coincide, and the letter that leaves the cycle need not admit any further safe continuation, since the breaker stops right after it. Requiring all letters, or two specific letters, to be safe would misclassify states over larger alphabets. For arity $n \leq 1$ no state has two distinct safe letters, so none is branching (Remark 10.20); for $n = 0$ there are no letters at all, and every safe state is mortal.

8.4 The Breaker

Lemma 8.6 (Breaker). *Let q be a branching state and $W \geq 0$. There is a word θ, safe from q, with $|\theta| \leq W + 3N_A$, that is not a prefix of o^ω for any word o with $1 \leq |o| \leq W$. In particular, every pre-hole (u_0, t_0) with $A(u_0)$ branching is completed by a word of length at most $\max(|u_0|, |t_0|) + 3N_A$.*

The idea is to steer the spine into a cycle that offers a choice of letters, to run around it for longer than any overhang, and then to leave it. The word $\theta = \alpha\lambda^k\ell$ follows a shortest path α to a cyclic state r with two distinct safe letters, runs k times around a shortest cycle λ through r, with $k|\lambda| \geq W + |\lambda|$, and leaves the cycle by a safe letter $\ell \neq \lambda[0]$; one of the two safe letters at r differs from $\lambda[0]$. If θ were a prefix of o^ω, then over the stretch λ^k, of length at least $|o| + |\lambda|$, it would have both periods $|o|$ and $|\lambda|$, hence the period $\gcd(|o|, |\lambda|)$ by the theorem of Fine and Wilf [Fine and Wilf 1965]. So after λ^k it would have to continue with $\lambda[0]$, not with ℓ. This uses only equality of letters and holds over every alphabet. Every obligation of a pre-hole has a nonempty overhang of length at most $W = \max(|u_0|, |t_0|)$, so θ breaks it by Lemma 8.1(c). The proof is in Appendix D.3.

In Example 8.4, the spine state $A(0) = \{g^1\}$ lies on a loop at which both letters are safe, so it is branching, and the breaker also completes the pre-hole $(u_0, t_0) = (0, \varepsilon)$. Here $W = 1$, and the

construction may take the cyclic state $r = \{g^1\}$ itself, $\alpha = \varepsilon$, the cycle $\lambda = 0$ and $k = 2$; this gives $\theta = \alpha\lambda^k\ell = 001$ with the breaker letter $\ell = 1$ and the hole $(0001, 001)$, while the shortest completion $\theta = 1$ gives back $(01, 1)$.

The bound uses $\max(|u_0|, |t_0|)$ because the overhang of a self event of t_0 can be as long as t_0, which may be longer than u_0. The maximum cannot be replaced by $|u_0|$, as the following example shows.

Example 8.7 (Self events need the longer word). Let $\Sigma = \{0, 1\}$ and $t_0 = 0001011100$. Let A and A' have a single state 0, with Bad $= \varnothing$, so that $N_A = 1$ and the state 0 of A is branching, and let $G = \varnothing$. Let B and B' both be the automaton whose states are the prefixes of t_0, with start state ε, and one sink; it moves from v to $v\ell$ while $v\ell$ is a prefix of t_0, and to the sink otherwise. Let Rej $= \{t_0\}$, let H consist of the pair of the state of A' and the state t_0 of B', and let G_s consist of the pairs $(t_0[:i], t_0)$ of states of B with $0 \le i \le 7$. Condition (FH) holds, since t_0 is the only word that leads B' to the state t_0, and its proper prefixes are not rejected. The pair (ε, t_0) is a pre-hole, and its obligations are the self events $(i, 10)$ with $0 \le i \le 7$, whose overhangs are the suffixes $t_0[i:]$, of length at least 3. The eight factors $t_0[i{:}i+3]$, $0 \le i \le 7$, are the eight words of length 3 over Σ. So every word θ of length at most 3 is a prefix of some $t_0[i:]$ with $i \le 7$ and keeps the self event $(i, 10)$. Hence no word of length at most $|u_0| + 3N_A = 3$ completes (ε, t_0), whereas Lemma 8.6 provides a completion of length at most $|t_0| + 3N_A = 13$.

8.5 Thin States and Lassos

A *lasso* of a safe state q is an infinite word $\alpha\lambda^\omega$ with $\lambda \neq \varepsilon$, $|\alpha| + |\lambda| \le N_A$, $\alpha\lambda$ safe from q, and $A(q, \alpha\lambda) = A(q, \alpha)$. Every lasso of q is safe from q, and there are finitely many.

Lemma 8.8 (Thin states). *Let q be a safe state of A.*

(a) If q is mortal, every word safe from q has length less than N_A and leads to a mortal state.
(b) If q is thin, every cyclic state reachable from q has exactly one safe letter, and the edge along it leads to a cyclic state.
(c) If q is thin, every infinite word safe from q is a lasso of q, and q has at least one lasso.
(d) If q is thin, θ is safe from q, and $A(q, \theta)$ is mortal, then $|\theta| < N_A$.

Thus a thin region consists of finitely many lassos together with acyclic paths into mortal states. Each part follows a safe run until it repeats a state; the proof is in Appendix D.3.

Definition 8.9 (Survival). A cross or self comparison *survives* an infinite word ξ if every finite prefix of ξ keeps it. An infinite word ξ is *survivor-free* for a pre-hole (u_0, t_0) if no obligation of (u_0, t_0) survives ξ.

Lemma 8.10 (Survival). *Let u and t be words and ξ an infinite word.*

(a) A cross or self comparison with nonempty overhang o survives ξ if and only if $\xi = o^\omega$. Every aligned cross comparison survives ξ.
(b) Let $s \le |u|$ and $e \le |t|$ with $|t| - e \le |u| - s$. Then (s, e) is a cross comparison of (u, t) that survives ξ if and only if $u[s:]\,\xi = t[e:]\,\xi$. Likewise, for $i < j \le |t|$, the pair (i, j) is a self comparison of t that survives ξ if and only if $t[i:]\,\xi = t[j:]\,\xi$.
(c) Let $\xi = \alpha\lambda^\omega$ with $\lambda \neq \varepsilon$. If a comparison with nonempty overhang o does not survive ξ, then the prefix of ξ of length $|\alpha| + |\lambda| + |o| - 1$ breaks it.

Part (a) is Lemma 8.1(c) applied to every prefix of ξ. Part (b) turns survival into an equality of infinite words, which is the form used by the pump; it holds because two finite words are prefixes of the same infinite word exactly when the shorter is a prefix of the longer. Part (c) compares two ultimately periodic words by the theorem of Fine and Wilf. The proofs are in Appendix D.4.

Corollary 8.11 (Lasso completion). *Let (u_0, t_0) be a pre-hole, and let ξ be a lasso of $A(u_0)$ that is survivor-free for (u_0, t_0). Then the prefix of ξ of length $\max(|u_0|, |t_0|) + N_A - 1$ completes (u_0, t_0).*

Proof. The prefix is safe from $A(u_0)$, since ξ is. Write $\xi = \alpha\lambda^\omega$ with $|\alpha|+|\lambda| \le N_A$. Each obligation has a nonempty overhang o with $|o| \le \max(|u_0|, |t_0|)$ and does not survive ξ. By Lemma 8.10(c) the prefix of ξ of length $|\alpha| + |\lambda| + |o| - 1 \le \max(|u_0|, |t_0|) + N_A - 1$ breaks it, and by (K3) so does the longer prefix. □

8.6 Short Pre-Holes

Pre-holes form a recognizable set of pairs. A pre-hole asks for a safe spine, a target rejected exactly at its end, and conditions on the aligned pairs of cuts only, and the aligned pairs are the pairs at equal distance from the ends within the longest common suffix. A finite graph that reads both words forward, aligned at their right ends, can therefore track the four automata and two bits: whether some aligned pair seen so far carries H, and whether some carries G. Both bits are reset whenever the two words read different letters. Its paths spell exactly the pre-holes, and a shortest path visits no node twice.

Proposition 8.12 (Short pre-holes). *Let E be a set of safe states of A. If some pre-hole (u_0, t_0) has $A(u_0) \in E$, then some pre-hole (u_1, t_1) has $A(u_1) \in E$ and $|u_1|, |t_1| < 12aba'b'$. In particular, it is decidable whether a pre-hole with $A(u_0) \in E$ exists.*

The graph has the nodes $(r_A, r_B, r_{A'}, r_{B'}, h, g, \phi)$, with a state of each automaton, two bits h and g, and a phase $\phi \in \{0, 1, 2\}$ that records whether only the spine, only the target, or both words are being read; hence $12aba'b'$ nodes. The construction and its correctness are in Appendix D.5. The proposition does not preserve survivor-freeness for a given lasso; it is used where every pre-hole with the given endpoint is good enough, namely for branching endpoints and for lassos that are not purely periodic.

8.7 Aperiodic Lassos

Lemma 8.13 (Aperiodic lassos). *If an infinite word ξ is not purely periodic, then no cross or self comparison with a nonempty overhang survives ξ. In particular, ξ is survivor-free for every pre-hole.*

Proof. By Lemma 8.10(a), a comparison with nonempty overhang o survives ξ only if $\xi = o^\omega$, which is purely periodic. Every obligation of a pre-hole has a nonempty overhang. □

8.8 The Pump

The remaining non-mortal case is a pre-hole whose spine ends in a thin state q and a purely periodic lasso of q that is survivor-free for it. Here survivor-freeness depends on the pre-hole, and it can fail for short pre-holes and hold for long ones, so Proposition 8.12 does not apply. The pump shortens the pre-hole directly, keeping its endpoint and its survivor-freeness. Put

$$\mathrm{P}(a, b, a', b') = 2aba'b' + a^2a' + ab + aa' + bb'.$$

Theorem 8.14 (Pump). *Let ξ be a purely periodic infinite word with least period ϖ, and let (u, t) be a pre-hole for which ξ is survivor-free. Then there is a pre-hole (u', t') for which ξ is survivor-free, with $A(u') = A(u)$ and*

$$|u'|, |t'| < 2aba'b' + aa'\varpi + b\varpi + aa' + bb'.$$

In particular, if $\varpi \le N_A$, then $|u'|, |t'| < \mathrm{P}(a, b, a', b')$.

The theorem uses no property of the state $A(u)$, and a lasso enters only through the bound on its period: if a lasso $\xi = \alpha\lambda^\omega$ is purely periodic, then its least period is at most $|\lambda| \le N_A$ (Lemma D.1(b)),

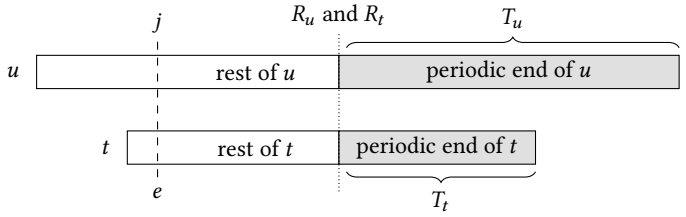


Fig. 11. The rest and the periodic end of the spine and of the target for a purely periodic ξ with least period ϖ, drawn with the two boundaries R_u and R_t on the dotted line. Two cuts in the periodic ends have equal ξ-suffixes if and only if their end distances agree modulo ϖ. A rest cut j of u has an equal ξ-suffix with a cut e of t only if e is a rest cut at the same distance before the boundary, $R_u - j = R_t - e$ (dashed), and a rest cut never matches a cut of a periodic end (Lemma D.7).

so the second bound applies. The proof rests on three observations and a count; the details are in Appendix D.6.

Survival as an equality of infinite words. By Lemma 8.10(b), a pre-hole (u, t) is survivor-free for ξ if and only if no pair of cuts (s, e) with $|t| - e \le |u| - s$ and $u[s{:}]\,\xi = t[e{:}]\,\xi$ carries G, and no pair of cuts $i < j$ of t with $t[i{:}]\,\xi = t[j{:}]\,\xi$ carries G_s; the aligned pairs are among the former. Call such pairs *ξ-matched*. The pump only has to make sure that a deletion creates no new ξ-matched pair.

Rests and periodic ends. Call a cut s of a word w *periodic* if $w[s{:}]\,\xi$ has period ϖ. The periodic cuts form an interval $R_w \le s \le |w|$; the factor $w[{:}R_w]$ is the *rest* of w and $w[R_w{:}]$, of length $T_w = |w| - R_w$, its *periodic end*. The ξ-matched pairs have a rigid structure (Figure 11). Two periodic cuts have equal ξ-suffixes exactly when their *end distances*, the distances to the ends of their words, agree modulo ϖ. A periodic cut never matches a rest cut. A rest cut j of u matches at most one cut of t, its *partner*, which is then a rest cut at the same distance before the boundary of the rest: $R_u - j = R_t - e$. Finally, the infinite word $\xi' = u[j{:}]\,\xi$ at a rest cut is *non-self-similar*: $z\xi' \ne \xi'$ for every nonempty finite word z. Indeed, $z\xi' = \xi'$ would give ξ' the period $|z|$, and then also the period ϖ of its suffix ξ, which would make j a periodic cut. Moreover, the periodic end of a target t with $J(t) = |t|$ is short, $T_t < N_B\,\varpi$ (Lemma D.8): two of its cuts with the same state of B and end distances that agree modulo ϖ are followed by the same letters up to the end of t, so the earlier one would lead B to the rejected state $B(t)$ before the end.

Deletions. Delete a factor w from $u = xwy$ and a factor w' from $t = x'w'y'$, at least one of them nonempty, such that each automaton is in the same state at both ends of each deleted factor. Then the states at the remaining cuts are unchanged, so safety, rejection and the endpoint $A(u)$ are kept. The shortened pair is again a pre-hole that is survivor-free for ξ, provided some head event of (u, t) survives the deletion and every ξ-matched pair of the shortened words comes from a ξ-matched pair of the original words (Lemma D.10). Two mechanisms provide this lifting. A factor of the periodic end of u whose length is a multiple of ϖ is *absorbed*: $w\,(y\xi) = y\xi$, so inserting it back changes no infinite suffix. *Rigidity*, the non-self-similarity of the suffix after a rest cut, handles two specific configurations: the synchronous deletion of the factor between two partnered rest cuts of u together with the factor between their partners, and the deletion of a factor that ends at an unpartnered rest cut (Lemmas D.11 and D.12). A factor that merely ends at a rest cut can create new matches: for $\xi = 0^\omega$, $u = 1010$ and $t = 110$, deleting the factor $u[1{:}2] = 0$, which ends at the rest cut 2, turns u into t and creates the matched pair $(0, 0)$. Four deletions, listed in Table 5, are found by pigeonhole on the states at the ends of the factors, the end distance modulo ϖ, and one bit recording whether a head event lies beyond the cut. Write $N_1, \ldots, N_4$ for their signature counts and

Table 5. The four deletions of the pump (Lemma D.12). Each applies to a pre-hole (u, t) for which ξ is survivor-free when the condition in the second column holds; it deletes the factors in the third column, which are found by pigeonhole on the signature in the fourth column, of size N_i.

Deletion	Condition	Deleted factors	Signature	Count
1, spine end	$T_u \geq T_t + N_1$	a factor of the periodic end of u at end distance at least T_t	states of A, A'; end distance mod ϖ	$N_1 = aa'\varpi$
2, synchronous rest	$\lvert C_u\rvert > N_2$	the factor between two partnered rest cuts of u, and the one between their partners	four states, head bit	$N_2 = 2aba'b'$
3, spine rest	$\lvert U_u\rvert > N_3$	the factor between two unpartnered rest cuts of u	states of A, A'	$N_3 = aa'$
4, target rest	$\lvert U_t\rvert > N_4$	the factor between two unpartnered rest cuts of t	states of B, B'	$N_4 = bb'$

C_u, U_u and U_t for the sets of partnered rest cuts of u, unpartnered rest cuts of u, and unpartnered rest cuts of t.

If no deletion applies, then the rests are short, since partners inject the partnered rest cuts of t into those of u, so that $R_u \leq N_2 + N_3$ and $R_t \leq N_2 + N_4$; the periodic end of t is shorter than $b\varpi$; and $T_u < T_t + N_1$. Hence both words are shorter than $N_1 + N_2 + N_3 + N_4 + b\varpi = 2aba'b' + aa'\varpi + b\varpi + aa' + bb'$ (Lemma D.13). Induction on $|u| + |t|$ proves the theorem. No deletion inside the periodic end of t is needed, and in particular no factors of both periodic ends are deleted together.

Remark 8.15 (An automaton instead of the pump). One could instead recognize the survivor-free pre-holes of a periodic lasso by a two-tape automaton that records a set of spine states for every residue class of end distances modulo the period. Such an automaton has a number of states exponential in N_A^2, whereas the pump needs no automaton and gives a polynomial bound.

8.9 The Three Cases

Definition 8.16 (Mortal hole). A *mortal hole* of I is a hole (u, t) such that $A(u)$ is mortal and $|t| - J(t) < N_A$.

Put

$$\mathsf{L}_{\mathrm{nm}}(a, b, a', b') = 4 \max\big(12aba'b',\ \mathsf{P}(a, b, a', b')\big) + 6a.$$

Since $a, b, a', b' \geq 1$, this is at most $54a^2ba'b'$, a polynomial in the four state counts (Appendix D.7).

Theorem 8.17 (Three cases). *Let I satisfy* (FH). *Then I has a hole if and only if at least one of the following holds:*

(1) some pre-hole (u_0, t_0) has $A(u_0)$ branching;
(2) some pre-hole (u_0, t_0) has $A(u_0)$ thin, and some infinite word safe from $A(u_0)$ is survivor-free for (u_0, t_0);
(3) I has a mortal hole.

More precisely, let $(u, t) = (u_0\theta, t_0\theta)$ be a hole decomposed as in Lemma 8.3. If $A(u_0)$ is branching, (u_0, t_0) satisfies (1). *If $A(u_0)$ is thin and $A(u)$ is not mortal, then (u_0, t_0) satisfies* (2) *with an infinite word that begins with θ. Otherwise (u, t) is a mortal hole. Moreover, if* (1) *or* (2) *holds, then I has a hole (u, t) with $|u| + |t| < \mathsf{L}_{\mathrm{nm}}(a, b, a', b')$; this part does not need* (FH).

A pre-hole as in (1) is shortened by Proposition 8.12, keeping its endpoint, and completed by Lemma 8.6. In case (2) the infinite word is a lasso ξ of the thin state $q = A(u_0)$ (Lemma 8.8(c)). If ξ is not purely periodic, every pre-hole with endpoint q is survivor-free for ξ (Lemma 8.13), so Proposition 8.12 shortens the pre-hole; if ξ is purely periodic, its least period is at most N_A and the pump shortens it. In both cases Corollary 8.11 completes the short pre-hole. Conversely, Lemma 8.3 decomposes a hole as $(u_0\theta, t_0\theta)$. If $A(u_0)$ is branching, case (1) holds. Otherwise, if $A(u_0)$ is thin and $A(u)$ has an infinite safe continuation ξ', then $\theta\xi'$ is safe from $A(u_0)$ and survivor-free, because its prefix θ already breaks every obligation. In the remaining cases $A(u)$ is mortal, and Lemma 8.8(a) and (d) give $|t| - J(t) = |\theta| < N_A$. The proofs are in Appendix D.7.

Corollary 8.18 (Short holes outside the mortal case). *Let I satisfy* (FH). *Then I has a hole if and only if it has a hole (u, t) with $|u| + |t| < \mathsf{L}_{\mathrm{nm}}(a, b, a', b')$ or it has a mortal hole.*

Proof. A hole of either kind is a hole. Conversely, if I has a hole, then by Theorem 8.17 one of (1)–(3) holds; in cases (1) and (2) the theorem provides a hole with $|u| + |t| < \mathsf{L}_{\mathrm{nm}}(a, b, a', b')$. □

The bound L_{nm} depends on neither the alphabet nor the self guard. The alphabet enters only the number of candidate pairs, and the self guard is absorbed by the use of $\max(|u_0|, |t_0|)$ in Lemma 8.6 and Corollary 8.11 and by the pump, whose signatures already contain the states of B.

Remark 8.19 (When mortal holes cannot occur). If no state reachable from 0 in $\mathcal{G}_A$ is mortal, for example if $\mathsf{Bad} = \varnothing$ and $|\Sigma| \geq 1$, then case (3) cannot occur, because $A(u)$ is reachable from 0 in $\mathcal{G}_A$ for every safe word u. Then Corollary 8.18 alone decides whether I has a hole.

Remark 8.20 (Condition (FH) is needed). Let $\Sigma = \{\ell\}$, let A, A' and B have the single state 0, with $\mathsf{Bad} = \varnothing$, $\mathsf{Rej} = \{0\}$ and $G = G_{\mathrm{s}} = \varnothing$, let B' have the states 0 and 1, with $B'(0, \ell) = B'(1, \ell) = 1$, and let $H = \{(0, 1)\}$. The pair (ε, ℓ) is a hole, with the head event $(0, 1)$, but I has no pre-hole: every target has $J(t) = 0$, so a pre-hole has $t_0 = \varepsilon$, whose only aligned pair $(|u_0|, 0)$ carries the state 0 of B'. Since the only state of A is thin, none of the cases (1)–(3) holds. Condition (FH) fails: the head $\gamma = \ell$ has $B'(\gamma) = 1$, and its proper prefix ε is rejected.

9 Mortal Holes

Let $I = (A, \mathsf{Bad}, B, \mathsf{Rej}, G, G_{\mathrm{s}}, A', B', H)$ be an instance of the two-word hole problem with self events (Definition 7.1), with the notation of Definitions 7.1 and 7.2. A *mortal hole* of I is a hole (u, t) whose spine ends in a mortal state and whose target ends fewer than N_A letters after its first rejected cut (Definition 8.16). By Theorem 8.17, an instance with forced target heads has a hole iff it has a short hole or a mortal hole. This section bounds the length of a shortest mortal hole and thereby decides the hole problem under (FH) (Theorem 9.17).

Where mortal holes come from. In an extracted instance I_{pq}, mortality depends on φ alone. The spine automaton is the lower automaton of q in φ, and its bad states are those that meet $\mathsf{Fact}^{\uparrow}_{\varphi}$, so a spine u is safe iff $(q, u) \notin \mu_\varphi$ (Table 4): the derived atom is not one of the atoms that every solution of φ contains. Facts come only from extreme literals (Table 3). Hence, if φ has no extreme literal, then μ_φ, $\mathsf{Fact}^{\uparrow}_{\varphi}$ and Bad are empty, and for $n \geq 1$ no state is mortal, so I_{pq} has no mortal hole (Remark 8.19). Mortal holes arise only when φ forces atoms. Since every safe run from a mortal state dies within N_A letters (Lemma 8.8), a safe spine u ends in a mortal state iff $(q, uv) \in \mu_\varphi$ for every word v of length N_A: the derived atom is not forced, but all its long extensions are. A mortal hole therefore admits no arbitrarily long safe continuation. The argument of Section 8, which replaces the continuation of a hole by a new one taken from the infinite safe continuations of the spine, does not apply, and the pair has to be shortened as it stands.

Example 9.1 (A mortal hole). Let $n = 2$ with f covariant, let $X = \{x, y, a, b, c\}$ and $Z = \emptyset$, and

$$\varphi = \{x \le f(x, b),\ b \le f(a, a),\ a = \bot,\ f(y, c) \le y\}, \qquad \psi = \{x \le f(y, y)\}.$$

Both systems are satisfiable. Condition (I0) holds, since ψ has no facts, and so does (U): the only atom that is root-dead in ψ is its forbidden root (x^0, ε), which is also a forbidden root of φ and hence dead in φ. Consider $I_{y^1 x^1}$.

- The only fact of φ is (a^1, ε), and $\mathrm{Fact}_\varphi^{\uparrow} = \{a^1\}$. The spine automaton starts at $\downarrow_\varphi x^1 = \{x^1\}$, loops there on 0, moves to $\{b^1\}$ on 1, and from $\{b^1\}$ to the bad state $\{a^1\}$ on either letter; so $\{b^1\}$ is mortal. In terms of trees, $x \le f(x, b)$, $b \le f(a, a)$ and $a = \bot$ force a leaf $\bot$ of x at a prefix of every word $0^k 1 i$.
- The φ-component of the target automaton is $\{y^1\}$ along 0^*, $\{c^1\}$ after the first letter 1, and empty afterwards. The atoms $(y^1, 0^k)$ are dead by support, because (y^1, ε) is a forbidden root of φ and $(y^1, 0^k)$ pops to it, while no atom (y^1, w) is root-dead in ψ. So $J(t)$ is defined iff t contains the letter 1, and it is then the cut after the first 1.
- The guard and the self guard never apply, because the sets that they intersect are disjoint.
- The pop head is empty: ψ derives $(x^1, 0\sigma)$ from (y^1, σ) by a single push, and the head events are the pairs $(1, 0)$ with $u[1{:}] = t$.

Hence the holes of $I_{y^1 x^1}$ are exactly the pairs $(0^{j+1}1, 0^j 1)$ with $j \ge 0$, all of them mortal, with $J(t) = |t|$; the spine and the common suffix $0^j 1$ are unbounded. These pairs are also the only pre-holes, and their spines end in the mortal state, so the instance has holes, but only in case (3) of Theorem 8.17. In the corresponding failure of (II), ψ derives $(x^1, 0^{j+1}1)$ from the live seed $(y^1, 0^j 1)$, but the least model of φ that contains the seed does not contain $(x^1, 0^{j+1}1)$. For $j = 0$, the solution $a = c = \bot$, $b = f(\bot, \bot)$, $x = f(f(\bot, b), b)$, $y = f(\top, \bot)$ of φ is not a solution of ψ: $x \le f(y, y)$ would require $f(\bot, b) \le f(\top, \bot)$, hence $b \le \bot$.

In a mortal hole the target ends shortly after its first rejected cut, but the spine and the common suffix of a head event can be arbitrarily long, as in the example. They cannot be shortened independently: events relate cuts of the spine to cuts of the target, self events relate two cuts of the target, and a head event demands that two suffixes be *equal*. We proceed in four steps.

(1) We pack the pair into the *separator word* $u\sharp t$ over $\Sigma \sqcup \{\sharp\}$. Events and self events become comparisons of this word, and a head event becomes an *anchored* comparison, one whose matched factor is followed by the separator. Mortal holes are then exactly the Π-*holes* of one reader, its holes with an anchored comparison whose state pair lies in a relation Π. Mortality and the short rejected tail enter through the reader itself, and self events enter through its admission relation (Sections 9.1 and 9.2).
(2) The short rejected tail bounds the depth of the separator word, by the depth theorem of Zhao [2026] (Section 9.3).
(3) We attach to each level of the hierarchy a *positive channel*, which records the anchored witness in the form in which it appears at that level, and show that channels have derivatives exactly as lettered monitors do (Section 9.4). The bounded-depth search of that paper does not look at the channel and can lose the witness (Example 9.13); once its exchange types, its shortening of the top level and its column summaries record the channel, it gives the anchored compression theorem (Section 9.5).
(4) Combining the three steps bounds the length of a shortest mortal hole by a doubly exponential function of the automaton state counts, hence a triply exponential function of s for extracted instances, which decides the hole problem (Section 9.6).

Condition (FH) is used only in the last theorem (Remark 9.20). The proofs are in Appendix E.

9.1 Anchored Comparisons

Fix a letter $\sharp$, the *separator*. Comparisons in this section are comparisons of one word, in the end the separator word $u\sharp t$ of Section 9.2, which is a different word from the pair (u, t); we therefore write them (x, y) (here x and y are cuts, not variables), whereas the cross comparisons (s, e) of Definition 7.2 relate cuts of the two words of the pair. For a comparison (x, y) of a word w, the factor $w[x{:}x + |w| - y] = w[y{:}]$ is its *matched factor*, and the letter $w[x + |w| - y]$ that follows the matched factor inside $w[x{:}]$ is its *continuation letter*; it exists because $x < y$. For a terminal comparison $(x, |w|)$ the matched factor is empty and the continuation letter is $w[x]$.

Definition 9.2 (Anchored comparison, Π-hole). A comparison of a word w is *anchored* if its continuation letter is $\sharp$. Let $(\mathcal{M}, \mathsf{Adm}, \mathsf{Rej})$ be a reader with states Q over a finite alphabet $\Gamma \ni \sharp$ (Definition 2.11), and let $\Pi \subseteq Q \times Q$. A Π-*hole* of the reader is a nonempty hole w (Definition 2.12) that has an anchored comparison (x, y) with $(q_x, q_y) \in \Pi$.

If w contains $\sharp$ exactly once, at the position m, then a pair of cuts (x, y) is an anchored comparison iff

$$x \le m, \qquad y = x + |w| - m, \qquad w[y{:}] = w[x{:}m].$$

Indeed, the continuation letter of a comparison (x, y) is $\sharp$ iff $x + |w| - y = m$, and then the matched factor $w[y{:}]$ has length $m - x$, so the prefix relation $w[y{:}] \preceq w[x{:}]$ says $w[y{:}] = w[x{:}m]$; conversely these three conditions make (x, y) a comparison, since $x < y$ and $w[x{:}m]$ is a prefix of $w[x{:}]$. The terminal pair $(m, |w|)$ is always anchored, with empty matched factor. Every other anchored comparison matches a nonempty factor that ends immediately before the separator with a suffix of the whole word.

The equation $y = x + |w| - m$ is what matters. A comparison (x, y) with $y > m$ only says that the suffix after y is a *prefix* of the suffix after x; the equation forces the two matched factors to have the same length, and the prefix relation becomes an equality. In the separator word this is exactly the difference between an event, a prefix relation $t[e{:}] \preceq u[x{:}]$, and a head event, an equality $t[e{:}] = u[x{:}]$ (Lemma 9.4). A compression that preserves only "some comparison with a state pair in Π" is therefore not enough.

Example 9.3 (An unanchored positive comparison). Over $\Gamma = \{0, 1, \sharp\}$ take the reader whose states $0, \dots, 4$ form the prefix chain of $01\sharp0$, that is, $\delta(0, 0) = 1$, $\delta(1, 1) = 2$, $\delta(2, \sharp) = 3$ and $\delta(3, 0) = 4$, with all other transitions into a sink 5. Let $\mathsf{Rej} = \{4\}$, $\mathsf{Adm} = \varnothing$ and $\Pi = \{(0, 3)\}$. The only rejected word is $w = 01\sharp0$, so it is the only hole. It has the comparison $(0, 3)$, since $w[3{:}] = 0 \preceq w$, with state pair $(0, 3) \in \Pi$, but the continuation letter of $(0, 3)$ is $w[1] = 1$. The only anchored comparison of w is the terminal pair $(2, 4)$, whose state pair $(2, 4)$ is not in Π. Hence the reader has no Π-hole, although its only hole has a comparison labeled by Π.

9.2 The Separator Word

For words $u, t \in \Sigma^*$ put $w = u\sharp t$, a word over $\Gamma = \Sigma \sqcup \{\sharp\}$, and $m = |u|$, so that $|w| = |u| + |t| + 1$. The cuts $x \le m$ of w are the cuts of u, and the cut $m + 1 + e$ of w is the cut e of t.

Lemma 9.4 (Comparisons of the separator word). *Let $u, t \in \Sigma^*$, $w = u\sharp t$ and $m = |u|$.*

(i) No pair (x, y) with $y \le m$ is a comparison of w.

(ii) For $x \le m$ and $e \le |t|$, the pair $(x, m + 1 + e)$ is a comparison of w iff (x, e) is a cross comparison of (u, t).

(iii) For $i < j \le |t|$, the pair $(m + 1 + i, m + 1 + j)$ is a comparison of w iff (i, j) is a comparison of t.

(iv) For $x \le m$ and $e \le |t|$, the pair $(x, m + 1 + e)$ is an anchored comparison of w iff (x, e) is an aligned cross comparison of (u, t), that is, iff $u[x{:}] = t[e{:}]$.

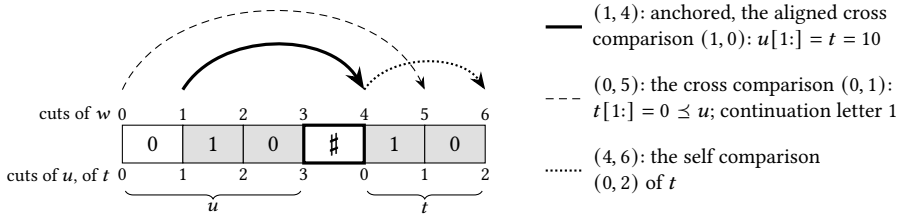


Fig. 12. The separator word $w = u\sharp t$ for $u = 010$ and $t = 10$, the word of Example 9.12; each arrow points from the first cut of a comparison to its second. The cut $x \le 3$ of w is the cut x of u, and the cut $4 + e$ is the cut e of t. Every comparison of w joins a cut of u to a cut of t, as the cross comparison $(0,5)$ does, or two cuts of t, as $(4,6)$ does (Lemma 9.4). The comparison $(1,4)$ is anchored: its matched factor $w[1{:}3] = 10$ (shaded) ends right before the separator and equals the suffix $w[4{:}]$. This is how a head event, an equality of two suffixes, appears in w; a cross comparison that is not aligned, such as $(0,5)$, has a continuation letter inside u. The anchored comparisons of w are $(1,4)$, $(2,5)$ and $(3,6)$.

So every comparison of w either joins a cut of the spine to a cut of the target, and is then a cross comparison, or joins two cuts of the target (Figure 12). The separator keeps the two words apart: every matched factor is a suffix of t, so none runs across the separator, and a matched factor that starts in u stops either inside u or right before the separator; the comparisons of the second kind are the anchored ones. The proof is in Appendix E.1. All conditions of a mortal hole, mortality included, are therefore read by one reader, which we now construct.

The reader $\mathcal{R}_I^{\Pi}$. Mortality of a safe state is decided by reachability in the safe graph of A (Definition 8.5). The reader $\mathcal{R}_I^{\Pi}$ over $\Gamma = \Sigma \sqcup \{\sharp\}$ reads $u\sharp t$ in three phases.

- On the spine, its states $\mathrm{pre}(q, q')$, with $q \in S_A \setminus \mathsf{Bad}$ and $q' \in S_{A'}$, follow A and A'.
- On the target up to its first rejected cut, its states $\mathrm{act}(r, r')$, with $r \in S_B$ and $r' \in S_{B'}$, follow B and B'.
- After the first rejected cut, its states $\mathrm{done}(r', k)$, with $r' \in S_{B'}$ and $1 \le k \le N_A$, follow B' and count the letters read, capped at N_A.

A state sink absorbs every failure. The automata A, B, A', B' start in their state 0 (Definition 7.1); the start state of $\mathcal{R}_I^{\Pi}$ is $\mathrm{pre}(0,0)$, or sink if $0 \in \mathsf{Bad}$. For a letter $i \in \Sigma$ the transitions are

$$\mathrm{pre}(q,q') \xrightarrow{i} \mathrm{pre}(A(q,i), A'(q',i)) \text{ if } A(q,i) \notin \mathsf{Bad}, \text{ else sink},$$

$$\mathrm{pre}(q,q') \xrightarrow{\sharp} \mathrm{act}(0,0) \text{ if } q \text{ is mortal}, \text{ else sink},$$

$$\mathrm{act}(r,r') \xrightarrow{i} \mathrm{done}(B'(r',i), 1) \text{ if } r \in \mathsf{Rej}, \text{ else } \mathrm{act}(B(r,i), B'(r',i)),$$

$$\mathrm{done}(r',k) \xrightarrow{i} \mathrm{done}(B'(r',i), \min(k+1, N_A)),$$

and every other transition, in particular every transition on $\sharp$ after the separator, leads to sink. The rejected states are $\mathrm{act}(r,r')$ with $r \in \mathsf{Rej}$ and $\mathrm{done}(r',k)$ with $k \le N_A - 1$. The admission relation consists of the pairs

$$\big(\mathrm{pre}(q,q'), \mathrm{act}(r,r')\big) \text{ with } (q,r) \in G, \qquad \big(\mathrm{act}(r_1,r_1'), \mathrm{act}(r_2,r_2')\big) \text{ with } (r_1,r_2) \in G_{\mathrm{s}},$$

and no others. The relation Π relates $\mathrm{pre}(q,q')$ to $\mathrm{act}(r,r')$ and to $\mathrm{done}(r',k)$ iff $(q',r') \in H$, and relates no other pairs.

Thus $\mathcal{R}_I^{\Pi}$ reads the separator only from a mortal state, rejects a word only if fewer than N_A letters follow the first rejected cut of the target, and lets Π compare the head automata on the two sides of

the separator. The target automaton is kept up to and including the first rejected cut, because events and self events may end there; after it, events and self events no longer exist (Definition 7.2), so only B' and the counter remain. The admissions from the self guard G_s are the only difference from a reader for instances without self events; the compression below allows an arbitrary admission relation.

For a reader, only the states reachable from its start state matter: the restriction to them is again a reader, with the same run on every word and hence the same holes and Π-holes. Let r_I and z_I be the numbers of states of the product automata $A \times A'$ and $B \times B'$ reachable from their start states, so that $r_I \le N_A N_{A'}$ and $z_I \le N_B N_{B'}$, and put

$$\bar{N}_I = 1 + r_I + z_I + N_A N_{B'}, \qquad d_I = 2\bar{N}_I N_A - 1.$$

Lemma 9.5 (The Π-reader). *The reader $\mathcal{R}_I^{\Pi}$ and the relation Π are computable from I, and:*

(i) at most $\bar{N}_I$ states of $\mathcal{R}_I^{\Pi}$ are reachable from its start state;
(ii) a word w over $\Sigma \sqcup \{\sharp\}$ is a Π-hole of $\mathcal{R}_I^{\Pi}$ iff $w = u\sharp t$ with $u, t \in \Sigma^*$ and (u, t) a mortal hole of I;
(iii) for such w the first rejected cut is $|u| + 1 + J(t)$, so the rejected tail of w is $|t| - J(t)$.

On $u\sharp t$, the state at a cut $x \le |u|$ carries $A_x(u)$ and $A'_x(u)$ while u is safe, and the state at the cut $|u| + 1 + e$ carries $B_e(t)$ up to $J(t)$ and $B'_e(t)$ throughout. By Lemma 9.4, the admitted comparisons are then exactly the events and the self events, and Π on the anchored comparisons is exactly the head condition. The proof is in Appendix E.1.

9.3 The Depth of the Separator Word

The length bound of the bounded-depth search grows with the depth of the word (Theorem 2.20). For a mortal hole, the depth of the separator word is bounded at once.

Lemma 9.6 (Depth of mortal holes). *Every mortal hole (u, t) of I satisfies* $\mathrm{depth}(u\sharp t) \le d_I$.

Proof. By Lemma 9.5, $u\sharp t$ is a nonempty rejected word of the reachable part of $\mathcal{R}_I^{\Pi}$, a reader with at most $\bar{N}_I$ states, and its rejected tail is $|t| - J(t) \le N_A - 1$. Theorem 2.19 gives $\mathrm{depth}(u\sharp t) \le 2\bar{N}_I((N_A - 1) + 1) - 1 = d_I$. □

The bound is coarse, since it involves the spine and head automata through $\bar{N}_I$, but it enters the final estimate only through a polynomial in $\bar{N}_I$ and N_A, and changes only the constant in the exponent (Lemma 9.18).

9.4 Positive Channels

The bounded-depth search sees each level of the hierarchy through a lettered monitor, the derivative of the monitor one level below (Definitions 2.22 and 2.23); the derivatives carry the negative conditions, because a comparison of one level reappears at the next as a comparison of blocks or as a test in the final target. We carry the positive condition in the same way. At level 0 it is an anchored comparison in Π, a comparison together with a test of its continuation letter; one level up, the continuation letter is the first letter of the expansion of a block, or a letter inside the last block, and the comparison may have become a test of the final label. Five components cover all the forms that arise.

Definition 9.7 (Positive channel). Let $\mathcal{M}$ be a lettered monitor over an alphabet Γ with state set Q. A *positive channel* $\Theta = (F, R, S, S', \Phi)$ on $\mathcal{M}$ consists of a predicate $F \subseteq Q \times \Gamma$, three relations R, S and S' on labels, and a set $\Phi \subseteq \Gamma$ of letters. For a nonempty word v with labels $L_0, \dots, L_{|v|}$, the *positive condition* $\mathrm{Pos}_\Theta(v)$ holds iff one of the following clauses holds:

(F) $F(L_{|v|})$;

(R) some comparison (x, y) of v with $y < |v|$ has $R(L_x, L_y)$ and $v[x + |v| - y] \in \Phi$;
(S) some cut $x < |v|$ has $S(L_x, L_{|v|})$;
(S′) some cut $x < |v|$ has $S'(L_x, L_{|v|})$ and $v[x] \in \Phi$.
A *positive hole* of $(\mathcal{M}, \Theta)$ is a hole v of $\mathcal{M}$ with $\mathrm{Pos}_\Theta(v)$.

Clauses (R) and (S′) together say that some comparison has its labels in a relation, R if it is not terminal and S' if it is, and its continuation letter in Φ; the continuation letter of the terminal comparison $(x, |v|)$ is $v[x]$. Clauses (F) and (S) are tests without a continuation letter. They are empty at level 0, but derivatives move witnesses into them, as the derivatives of lettered monitors move admissions into the terminal relation and the final target.

For a reader $(\mathcal{M}, \mathrm{Adm}, \mathrm{Rej})$ over $\Gamma \ni \sharp$ with start q_0 and $\Pi \subseteq Q \times Q$, the *base channel* on the lettered monitor of the reader (Definition 2.22) is

$$\Theta_\Pi = (\varnothing, \widehat{\Pi}, \varnothing, \widehat{\Pi}, \{\sharp\}), \qquad \widehat{\Pi}(L, L') \iff (\operatorname{den} L, \operatorname{den} L') \in \Pi.$$

Lemma 9.8 (The base channel). *A nonempty word w satisfies $\mathrm{Pos}_{\Theta_\Pi}(w)$ iff it has an anchored comparison (x, y) with $(q_x, q_y) \in \Pi$. Hence the Π-holes of the reader are exactly the positive holes of its lettered monitor with the base channel.*

Proof. Since $\operatorname{den} L_i = q_i$, clause (R) is an anchored comparison (x, y) with $y < |w|$ and $(q_x, q_y) \in \Pi$, and clause (S′) is the same for a terminal comparison $(x, |w|)$, whose continuation letter is $w[x]$; clauses (F) and (S) are empty. The reader and its lettered monitor have the same nonempty holes. □

The derivative of a channel uses the shifted blocks of Definition 2.23. The letter that follows the position g of the shifted block of X is $(X\ell)[g]$: the letter $X[g]$ for $g < |X|$, and the marker that opens the next shifted block for $g = |X|$.

Definition 9.9 (Derivative of a channel). Let $\Theta = (F, R, S, S', \Phi)$ be a positive channel on $\mathcal{M}$ and $\ell \in \Gamma$. The *derivative* $\Theta_\ell = (F_\ell, R_\ell, S_\ell, S'_\ell, \Phi_\ell)$ is the positive channel on $\mathcal{M}_\ell$ given by

$$\begin{aligned}
\Phi_\ell &= \{X : (X\ell)[0] \in \Phi\},\\
R_\ell((\zeta, X), (\zeta', X')) &\iff X = xy,\ X' = x'y,\ R\big(lab_{|x|}(\zeta, X), lab_{|x'|}(\zeta', X')\big) \text{ for some } x, x', y,\\
S'_\ell &= R_\ell,\\
S_\ell((\zeta, X), (\zeta', X')) &\iff pt(\zeta, X; \delta_\ell(\zeta', X')),\\
F_\ell(\zeta, X) &\iff F(\delta_\ell(\zeta, X), \ell) \text{ or } pt(\zeta, X; \delta_\ell(\zeta, X)),
\end{aligned}$$

where the *positive terminal test* is

$$\begin{aligned}
pt(\zeta, X; \zeta'') \iff &\text{for some } g \le |X| :\ S\big(lab_g(\zeta, X), (\zeta'', \ell)\big),\\
&\text{or } S'\big(lab_g(\zeta, X), (\zeta'', \ell)\big) \text{ and } (X\ell)[g] \in \Phi,
\end{aligned}$$

and every relation with the start label $\triangleright$ of $\mathcal{M}_\ell$ as an argument is false.

The relation R_ℓ is built from R exactly as the derived admission Λ_ℓ is built from Λ, and the positive terminal test is the analog of the terminal test in $\Lambda^{\mathrm{fin}}_\ell$ and T^{fin}_ℓ: it relates a position of a block, possibly the closing marker, to the final label, which is (ζ'', ℓ) when the final marker is read from the state ζ''. The predicate F concerns only the final label, so it enters F_ℓ but not S_ℓ. The set Φ_ℓ tests the first letter of the expansion of a block, because the continuation letter of a comparison of the expansion is the first letter of the expansion of the continuation letter of the corresponding comparison of blocks (Lemma E.1). This is how the anchoring equation travels up the hierarchy: "followed by $\sharp$" at level 0 becomes "followed by a block whose expansion starts with $\sharp$" at level 1, and so on.

Lemma 9.10 (Positive desubstitution). *Let Θ be a positive channel on $\mathcal{M}$, let $\ell \in \Gamma$, and let v be the expansion of a nonempty word v' of blocks avoiding ℓ. Then* $\mathrm{Pos}_{\Theta}(v) \iff \mathrm{Pos}_{\Theta_\ell}(v')$. *Consequently v is a positive hole of $(\mathcal{M}, \Theta)$ iff v' is a positive hole of $(\mathcal{M}_\ell, \Theta_\ell)$.*

The second claim follows from the first and Lemma 2.24. The proof of the first, in Appendix E.2, translates each clause by the label facts Eq. (4) and the alignment lemma (Lemma 2.17): a nonterminal comparison of v becomes a comparison of blocks, read by R_ℓ or, if it ends at the last cut, by S'_ℓ; a terminal comparison of v that starts in the block X_i becomes a test of S_ℓ or, if X_i is the last block, of F_ℓ.

Corollary 9.11 (Channel towers). *Let $\mathcal{M}_0$ be a lettered monitor with N states, Θ_0 a positive channel on it, and w a nonempty word with canonical hierarchy (w_j, ℓ_j). Put $\mathcal{M}_{j+1} = (\mathcal{M}_j)_{\ell_j}$ and $\Theta_{j+1} = (\Theta_j)_{\ell_j}$. Then w is a positive hole of $(\mathcal{M}_0, \Theta_0)$ iff w_j is a positive hole of $(\mathcal{M}_j, \Theta_j)$, for every j.*

Proof. Each w_{j+1} is a nonempty word of blocks avoiding ℓ_j, with expansion w_j. Apply Lemma 9.10 at each level. □

Neither the lemma nor the corollary assumes that ♯ occurs once, or at all, and the reader may have an arbitrary admission relation.

Example 9.12 (A witness travels up the tower). Take $\Gamma = \{0, 1, \sharp\}$, the five states $0, \dots, 4$ with start 0, the transitions

state	0	1	2	3	4
on 0	3	0	0	3	0
on 1	1	1	4	0	2
on ♯	4	3	2	4	0

$\mathrm{Adm} = \varnothing$, $\mathrm{Rej} = \{0, 1\}$ and $\Pi = \{(3, 4), (4, 2)\}$. The word $w = 010\sharp10$ has the run $0, 3, 0, 3, 4, 2, 0$ and ends in a rejected state, so it is a hole. Its anchored comparisons are $(1, 4)$, $(2, 5)$ and $(3, 6)$, with state pairs $(3, 4)$, $(0, 2)$ and $(3, 0)$. Only the first, which matches the factor 10 before the separator with the final 10, is in Π, so w is a Π-hole; for the base channel, its witness $(1, 4)$ is clause (R). The hierarchy of w is $w_1 = abc$ with $a = [\varepsilon]$, $b = [1]$ and $c = [\sharp1]$, and $w_2 = [ab]$, a single letter. At level 1, with the marker 0, the cut 1 lies in the block $X_1 = 1$ and the cut 4 in the last block $X_2 = \sharp1$, both with the common suffix 1; so the comparison of blocks is the terminal pair $(2, 3)$ of w_1, and the witness becomes clause (S′) at the cut 2, with continuation letter c, which lies in Φ_1 because its expansion starts with ♯. At level 2, with the marker c, the cut 2 of w_1 is the end of the only block ab of w_2; so the witness becomes the second disjunct of the final target F_2, through the positive terminal test at the position $|ab|$, whose letter is the marker $c \in \Phi_1$. Clause (R) at level 0 has become clause (S′) at level 1 and clause (F) at level 2.

Figure 13 draws the three levels of this example. Each level sees the witness only through its own letters: once the matched factor 10 has become the single letter b and the separator the first letter of c, the comparison turns into a test at the boundary between b and c, and one level higher into a test of the final label.

9.5 Anchored Compression

What the compression for plain entailment misses. Steps 2 and 4 of the bounded-depth search (Section 2.6, and precisely Appendix A.4) replace the word by another one: step 2 merges isolated letters with the same exchange type, and step 4 deletes the columns between two cuts of a column word with equal summaries. Each of them only has to keep the word a hole. A Π-hole must also *contain* an anchored comparison in Π. The exchange types and summaries record nothing about Π, so a column deletion, and similarly a merge, can destroy every such comparison.

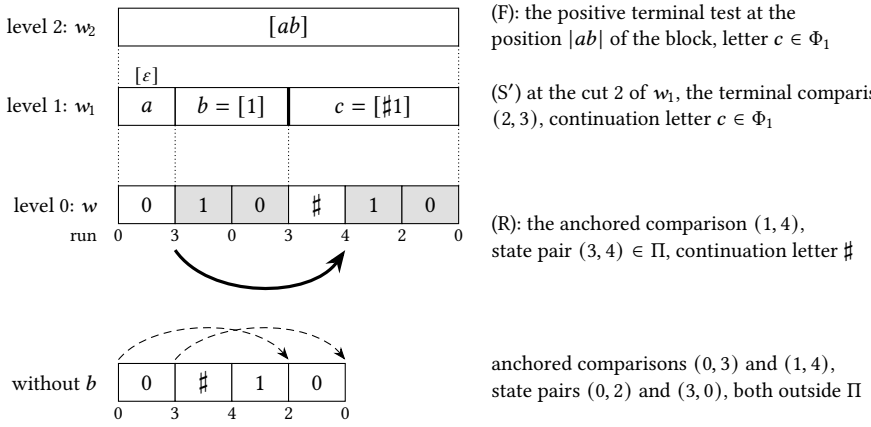


Fig. 13. A positive witness travels up the hierarchy (Examples 9.12 and 9.13). Bottom up: the hole $w = 010\sharp10$ of the reader of Example 9.12 with its run, the word $w_1 = abc$ of blocks for the marker 0, each letter drawn over its block and the marker that closes it, and the one-letter word $w_2 = [ab]$ for the marker c. The anchored witness $(1, 4)$ in Π, whose matched factor 10 is shaded, is clause (R) of the base channel at level 0, clause (S′) of the derived channel at level 1, and clause (F) at level 2, where it is a test of the final label (Definition 9.7 and Lemma 9.10). The compression for plain entailment sees no difference between the prefixes a and ab of the block and deletes the column of b, which carries the matched factor 10; the result $0\sharp10$ (last row) is still a hole, but no longer a Π-hole. Recording the channel in the summaries keeps b.

Example 9.13 (The compression for plain entailment loses the anchored witness). Continue Example 9.12. Step 4 shortens the block ab over the level-1 monitor with the marker c. In this monitor c leads every state to 2, and the exits of the shifted blocks of the prefixes a and ab are 0 from every entry state; since Adm is empty, all terminal and admission bits are false. So the summaries after a and after ab coincide, and step 4 deletes the column holding b. Expanding $[a]$ gives $0\sharp10$ (Figure 13, last row), with run $0, 3, 4, 2, 0$: a hole, whose anchored comparisons $(0, 3)$ and $(1, 4)$ have the state pairs $(0, 2)$ and $(3, 0)$, both outside Π. The deleted letter b is exactly the letter whose expansion is the matched factor 10 before the separator.

The bound of the next theorem is an explicit function $\mathsf{Len}^\sharp(N, d)$. Like $\mathsf{Len}(N, d)$ of Zhao [2026], it is built from constants $\mathsf{C}^\sharp$, $\mathsf{A}^\sharp$, $\mathsf{G}^\sharp$ and $\mathsf{col}^\sharp$, which bound the anchored exchange types, the protected and isolated letters, the dictionaries and the positive column summaries. They are defined in Eq. (18), increase in each argument, and do not depend on the alphabet; the theorem states their closed form.

Theorem 9.14 (Anchored compression). *Let $(\mathcal{M}, \mathsf{Adm}, \mathsf{Rej})$ be a reader with $N \geq 1$ states over a finite alphabet $\Gamma \ni \sharp$, let $\Pi \subseteq Q \times Q$, and let $d \geq 0$. If the reader has a Π-hole of depth at most d, then it has a Π-hole of depth at most d and length at most $\mathsf{Len}^\sharp(N, d)$. More generally, for every lettered monitor $\mathcal{M}_0$ with N states over any alphabet and every positive channel Θ_0 on it, if $(\mathcal{M}_0, \Theta_0)$ has a positive hole of depth at most d, then it has one of depth at most d and length at most $\mathsf{Len}^\sharp(N, d)$. Moreover, with $\Xi^\sharp(N, d) = 6N^2 + 10N(N + d)(d + 1) + 7N + 3d + 4$,*

$$\mathsf{G}^\sharp(N, d) \leq 2^{\Xi^\sharp(N,d)} \qquad \text{and} \qquad \mathsf{Len}^\sharp(N, d) \leq 2^{2^{5\Xi^\sharp(N,d)}}.$$

The theorem is the analog of Theorem 2.20 for Π-holes. It preserves the *existence* of an anchored comparison in Π, not its coordinates or its state pair; to keep a particular state pair, apply it to the

singleton relation of that pair. It needs no uniqueness of the separator. The first claim is the second one for the lettered monitor of the reader and the base channel, by Lemma 9.8.

Why recording the channel suffices. The positive condition is a property of the labels at the cuts of a witness and of one letter, the continuation letter. The shortenings of Zhao [2026] (steps 2–4 in Section 2.6) preserve such data once they record the channel. The merges preserve the state run, the comparison positions, and the relevant channel tests; the top shortening preserves the labels of the cuts of the witness, which it never deletes, and the letter, which is constant on a unary word; and the summaries determine the derived channel, so that agreement preserves labels and letters up to renaming. This is why recording the channel suffices, with no further condition on the separator.

Proof outline. The complete proof is in Appendix E.8. The proof of Theorem 2.20 merges letters with equal exchange types and deletes columns between equal summaries. We add the five components of the channel to both, so that each shortening keeps some anchored witness, though not necessarily the original one; every argument about the negative conditions is reused unchanged. Let w be a positive hole of depth $D \le d$, with hierarchy (w_j, ℓ_j); we follow the five steps of that proof (Section 2.6).

(1) *Towers.* By Corollary 9.11, w_j is a positive hole of $(\mathcal{M}_j, \Theta_j)$ for every j; the clause of the witness may change from level to level (Example 9.12).
(2) *Merging letters.* In the terms of step 2 in Section 2.6, a letter of w_i is unprotected if it is neither the marker nor a letter of the start prefix, and isolated if, moreover, all its unequal comparison partners are protected; its exchange type is the table of how $\mathcal{M}_i$ treats it when it is read from an ordinary state. The *anchored exchange type* of an unprotected letter a at a level $1 \le i < D$ adds the same table for the channel: the row of F_i at a, the self table, the protected rows and the initial row of each of R_i, S_i and S'_i, and the bit $[a \in \Phi_i]$ (Definition E.2). There are at most $\mathrm{C}^\sharp(N, d)$ anchored types (Lemma E.3). Merging two isolated letters with the same anchored type keeps the comparisons and the run of the level, so it keeps every value of Λ_i, Λ_i^{fin}, R_i, S_i and S'_i at the comparisons of the level, exactly as in the argument of Zhao [2026] for Λ_i and Λ_i^{fin}; it keeps the final label, because the marker is protected; and it keeps the position of every continuation letter and changes that letter at most from a to b, which have the same bit. So the merged level is again positive, and expanding it gives a positive hole of $(\mathcal{M}_0, \Theta_0)$ of the same depth, no longer, with a smaller dictionary (Lemma E.4). Repeated merges bound the dictionary, the set of letters, of every level $i \ge 1$ by $\mathrm{G}^\sharp(N, d)$ (Lemma E.5). A merge is not injective and gives no agreement; after each merge the hierarchy and the towers are recomputed.
(3) *The top level.* The top level w_D is unary. Deleting a loop between two cuts with equal states keeps a hole (Lemma A.9), and it keeps the positive witness if no cut of the witness is deleted. Protecting the at most three cuts of one witness leaves at most three stretches of pairwise distinct states, so $|w_D| \le 3(N + D)$ after shortening (Lemma E.6).
(4) *Deleting columns.* The *positive summary* of a tuple of blocks extends the summary of Appendix A.4 by fields that record the positive terminal test, the last pending S'-test (a test at the end of a block, whose letter is the next one to be read), whether a block is empty, the bit Φ of its first letter, and the lift R_ℓ between pairs of blocks. These fields are updated column by column and determine the derived channel on the blocks, in particular S_ℓ without an F-term and F_ℓ with one (Lemma E.8). Deleting the columns between two cuts with equal positive summaries therefore keeps both the derived monitor and the derived channel (Lemma E.10). In Example 9.13, the pending S'-test is false after a and true after ab, where it records the witness

$(1, 4)$; so the positive summaries after a and after ab differ, and the column holding b is kept (Example E.9).

(5) *Agreement and the bound.* Agreement of monitors on the letters in use (step 4 in Section 2.6) is extended to the five components of the channel, which must also take equal values at corresponding letters and labels. Extended agreement preserves positive holes and is preserved under derivation (Lemma E.7). The supported compression of Zhao [2026], which shortens the letters in use level by level from the top, then runs unchanged with the positive summaries (Lemma E.11). Each of the at most d expansions costs a factor $\mathrm{col}^\sharp(N + d + 1, \mathrm{G}^\sharp(N, d))$, and the top level costs $3(N + d + 1)$, which gives $\mathsf{Len}^\sharp(N, d)$. The new top word is a positive hole of the new top monitor, and Lemmas 2.24 and 9.10 reconstruct a positive hole of $(\mathcal{M}_0, \Theta_0)$ of depth at most D.

The closed-form estimate is Lemma E.12. □

9.6 Short Mortal Holes and Decidability

Put

$$\mathsf{Len}_I^\sharp = \mathsf{Len}^\sharp(\bar{N}_I, d_I),$$

with $\bar{N}_I$ and d_I as in Section 9.2, and recall the bound L_{nm} of Theorem 8.17, evaluated at the state counts $(N_A, N_B, N_{A'}, N_{B'})$.

Theorem 9.15 (Short mortal holes). *If I has a mortal hole, then it has a mortal hole (u, t) with $|u| + 1 + |t| \le \mathsf{Len}_I^\sharp$. No assumption on I is needed; in particular (FH) is not used.*

Proof. Let (u, t) be a mortal hole. By Lemma 9.5, $w = u\sharp t$ is a Π-hole of $\mathcal{R}_I^\Pi$, and of its reachable part, a reader with at most $\bar{N}_I$ states over the finite alphabet $\Sigma \sqcup \{\sharp\}$. By Lemma 9.6, $\mathrm{depth}(w) \le d_I$. By Theorem 9.14 and monotonicity of $\mathsf{Len}^\sharp$, the reachable part has a Π-hole $\tilde{w}$ of length at most $\mathsf{Len}_I^\sharp$. By Lemma 9.5(ii), $\tilde{w} = \tilde{u}\sharp\tilde{t}$ with $(\tilde{u}, \tilde{t})$ a mortal hole, and $|\tilde{u}| + 1 + |\tilde{t}| = |\tilde{w}|$. □

Corollary 9.16 (Mortal holes are decidable). *It is decidable whether an instance of the two-word hole problem with self events has a mortal hole.*

Theorem 9.17 (Short holes). *Let I satisfy (FH). Then I has a hole iff it has a hole (u, t) with*

$$|u| + |t| < \max\big(\mathsf{L}_{\mathrm{nm}}(N_A, N_B, N_{A'}, N_{B'}),\ \mathsf{Len}_I^\sharp\big).$$

In particular, it is decidable whether an instance with forced target heads has a hole.

Proof. By Corollary 8.18, I has a hole iff it has a hole with $|u| + |t| < \mathsf{L}_{\mathrm{nm}}(N_A, N_B, N_{A'}, N_{B'})$ or a mortal hole. In the second case Theorem 9.15 gives a mortal hole, which is a hole, with $|u| + |t| < \mathsf{Len}_I^\sharp$. The bound is computable from I, and the alphabet Σ is finite, so there are finitely many candidate pairs; for each, the conditions of Definition 7.1 are checked over finitely many cuts. □

The proof of Corollary 9.16 is the same enumeration with the bound of Theorem 9.15 and the additional tests that $A(u)$ is mortal and $|t| - J(t) < N_A$ (Appendix E.9). Both bounds are independent of the alphabet, and hence of the arity; the alphabet enters only the number of candidate pairs. For $n = 0$ the only candidate is $(\varepsilon, \varepsilon)$.

The size of the bound. Write $E_1(x) = 2^x$ and $E_{k+1}(x) = 2^{E_k(x)}$.

Lemma 9.18 (Mortal bound). *Let $\alpha \ge 3$ and $\beta \ge 0$ be integers with $\bar{N}_I \le 2^\alpha$ and $N_A \le 2^\beta$. Then $\mathsf{Len}_I^\sharp \le E_3(3\alpha + 2\beta + 9)$.*

Corollary 9.19 (Mortal bound for extracted instances). *Let φ be a flat system over X and ψ a flat system over $X \cup Z$, with $v = |X|$, $h = |Z|$ and $s = v + h$. Every extracted instance I_{pq} (Definition 7.5) satisfies*

$$\mathsf{Len}^{\sharp}_{I_{pq}} \le E_3(10v + 6s + 18) \le E_3(16s + 18),$$

for every arity n and every variance. So if I_{pq} has a mortal hole, it has one whose separator word has length at most $E_3(16s + 18)$.

Proof. By Section 7.2, $N_A = 2^{2v}$, $N_B = 2^{4v+2h+1} = 2^{2v+2s+1}$ and $N_{A'} = N_{B'} = 2^{2s}$, and the reachable pairs of states of B and B' number at most N_B. Hence $r_I \le N_A N_{A'} = 2^{2v+2s}$, $z_I \le 2^{2v+2s+1}$ and $N_A N_{B'} = 2^{2v+2s}$, so $\bar{N}_I \le 1 + 2^{2v+2s+2} \le 2^{2v+2s+3}$. Lemma 9.18 with $\alpha = 2v + 2s + 3$ and $\beta = 2v$ gives $E_3(3(2v + 2s + 3) + 4v + 9) = E_3(10v + 6s + 18)$, and $v \le s$. The second claim is Theorem 9.15. None of these sizes depends on n. □

Remark 9.20 (Where (FH) is used). Condition (FH) is used only in Theorem 9.17, through Theorem 8.17. Every other result of this section, in particular Theorem 9.15 and Corollary 9.16, holds for every instance.

10 Tree Classes

The preceding sections decide existential entailment over arbitrary trees. This section transfers the decision to regular and to finite trees, for every arity $n \ge 0$ and every variance c, and it reduces the signature $\{\bot, f\}$ with f covariant to $\{\bot, f, \top\}$. Regular trees are handled by the least closures of Section 3, whose slices are regular for regular seeds by Lemma 5.7; a failed extension is caused by at most two seeds (Lemma 10.3), so $E_{\mathrm{reg}} = E_{\mathrm{all}}$ (Theorem 10.11). Finite trees need a different argument. Least closures describe trees with as few leaves as possible, and these are infinite in general: the empty system has the empty least model, which assigns Ω to every variable. Instead we express observations by finite ground bounds (Lemma 10.4) and show that adding finite ground bounds preserves finite satisfiability, for every variance (Theorem 10.6). This gives the dispatch Eq. (9) of Theorem 10.12. Proofs not given here are in Appendix F.

Throughout, trees are over $\{\bot, f, \top\}$ and ordered by $\le_c$ unless stated otherwise, φ and ψ are as in Definition 2.4, we use the notation of Sections 2 and 3 together with $f(t^n) = f(t, \dots, t)$ and $v_{\pm}$ for v_0 and v_1, and a model T *decodes* to the solution ρ with $S_\rho = T$ (Lemma 3.5).

10.1 Extensions and Two Seeds

An assignment ρ of X is *extensible* if it extends to a solution of ψ over arbitrary trees.

Lemma 10.1 (Extension criterion). *An assignment ρ of X is extensible iff $H_\psi(S_\rho)$ is consistent and $H_\psi(S_\rho)|_X = S_\rho$. In that case $H_\psi(S_\rho)$ decodes to an extension of ρ.*

This restates Lemma 3.8(5) for assignments.

Proof. By Lemma 3.8(5) and Lemma 3.5, applied to the set S_ρ, which is extension closed and has no exclusion pair, ρ is extensible exactly when $H_\psi(S_\rho)$ is consistent and has visible part S_ρ; then that closure decodes to an extension. □

Definition 10.2 (Observations). A *visible observation* is a visible atom $a = (x^{\pm}, w)$ together with one of the words *asserted* or *denied*. An asserted observation *holds* in an assignment ρ of X if $a \in S_\rho$, a denied one if $a \notin S_\rho$. The variable x and the word w are the variable and the word of the observation.

Lemma 10.3 (Two seeds). *Let ρ be an assignment of X that is not extensible. There are a set $A \subseteq S_\rho$ of visible atoms and a set C of visible atoms outside S_ρ, with $|A| + |C| \le 2$, such that no assignment ρ' of X with $A \subseteq S_{\rho'}$ and $C \cap S_{\rho'} = \emptyset$ is extensible. In particular, at most two visible observations hold in ρ and hold together in no extensible assignment.*

Proof. Every atom b of $H_\psi(S_\rho)$ has a seed in $\mathrm{Fact}_\psi \cup S_\rho$, so $b \in H_\psi(A_b)$ for a set $A_b \subseteq S_\rho$ with at most one element. By Lemma 10.1, $H_\psi(S_\rho)$ is not consistent or has a visible atom outside S_ρ. In the first case, Lemma 3.8(3) gives seeds $x, y \in \mathrm{Fact}_\psi \cup S_\rho$ such that $\mathrm{Reach}_\psi(x)$ contains a forbidden root, or $\mathrm{Reach}_\psi(x)$ and $\mathrm{Reach}_\psi(y)$ contain an exclusion pair; let A consist of those of x, y that lie in S_ρ, and $C = \emptyset$. In the second case, let $b \in H_\psi(S_\rho)|_X$ with $b \notin S_\rho$, and put $A = A_b$ and $C = \{b\}$. In both cases $|A| + |C| \le 2$. Let ρ' be an assignment of X with $A \subseteq S_{\rho'}$ and $C \cap S_{\rho'} = \emptyset$. Since H_ψ is monotone, $H_\psi(S_{\rho'}) \supseteq H_\psi(A)$, so $H_\psi(S_{\rho'})$ contains the same forbidden root or the same two opposite atoms, or the visible atom $b \notin S_{\rho'}$. By Lemma 10.1, ρ' is not extensible. The observations asserting the atoms of A and denying those of C prove the last claim. □

The lemma uses no satisfiability hypothesis. If ψ is unsatisfiable, then μ_ψ is inconsistent by Lemma 3.8(2), and $A = C = \emptyset$ will do. The two seeds of A may belong to the same variable.

10.2 Observation Pins

For a word w and trees b, g, let $\mathrm{emb}_b(w, g)$ be the tree with a pin path along w that carries g at its end and b at every child off the pin path: $\mathrm{emb}_b(\varepsilon, g) = g$, and $\mathrm{emb}_b(iw, g) = f(t_j)_j$ with $t_i = \mathrm{emb}_b(w, g)$ and $t_j = b$ for $j \neq i$.

Lemma 10.4 (Observation pins). *For every covariant tree t and every word w,*

$$\begin{aligned} w \in L^0_t &\iff \mathrm{emb}_\bot(w, \top) \le t, \quad w \notin L^0_t \iff t \le \mathrm{emb}_\top(w, f(\top^n)), \\ w \in L^1_t &\iff t \le \mathrm{emb}_\top(w, \bot), \quad w \notin L^1_t \iff \mathrm{emb}_\bot(w, f(\bot^n)) \le t, \end{aligned} \tag{7}$$

and each of these four trees has at most $1 + n(|w| + 1)$ positions. Consequently, for every variance c, every visible observation with variable x and word w is equivalent, for all assignments ρ of X, to a single inequality $g \le_c \rho(x)$ or $\rho(x) \le_c g$, where g is a finite ground tree with at most $1 + n(|w| + 1)$ positions.

For $n \ge 1$ the bound is attained by the two trees with a constructor at the end of the pin path; for $n = 0$ the only word is ε, and the four trees are $\top$, $f()$, $\bot$ and $f()$.

10.3 Finite-Bound Pinning

Intersection of leaf cones. Over arbitrary and regular trees, extensions can be intersected; over finite trees they cannot (item (3) below), which is why the transfer to finite trees pins observations by finite ground bounds instead. For trees t and u, let $t \sqcap u$ be the tree with the leaf pair $(L^0_t \cap L^0_u,\ L^1_t \cap L^1_u)$, which exists by Theorem 3.2(1); its signed leaf cones contain exactly the observations common to t and u. For assignments, $\sqcap$ acts variablewise.

Proposition 10.5 (Closure under intersection).

(1) *The intersection of a nonempty family of models of a system χ is a model of χ, and the intersection of a nonempty subfamily of the projected family $\mathcal{F}_\psi$ (Eq. (6)) belongs to $\mathcal{F}_\psi$. In particular, the solutions of χ and the assignments of X that extend to solutions of ψ are closed under $\sqcap$.*
(2) *The operation $\sqcap$ is commutative, associative, idempotent and monotone for $\le_c$, and $f(t_i)_i \sqcap f(u_i)_i = f(t_i \sqcap u_i)_i$. Moreover, $t \sqcap u$ is the median of t, u and Ω in the lattice of trees ordered by $\le_c$.*
(3) *If t and u are regular, then so is $t \sqcap u$. For $n \ge 1$, $\bot \sqcap \top = \Omega$ is infinite, so the finite trees are not closed under $\sqcap$.*

The proof is in Appendix F. By (1), the model $H_\psi(S)$ of Lemma 3.8(5) is the intersection of all models T of ψ with $T|_X = S$, the least extension of S. By (1) and commutativity, a relation that is the projection of the solutions of a system over arbitrary trees, or over regular trees, and contains two pairs (a, b) and (b, a) also contains the diagonal pair $(a \sqcap b, a \sqcap b)$. So no such relation is nonempty, symmetric and irreflexive; in particular, over arbitrary or regular trees no system with hidden variables defines the disequality of two trees. The decision procedure does not use Proposition 10.5.

Ground bounds. A *ground bound* on a variable x is an inequality $g \leq_c x$, a *lower* bound, or $x \leq_c g$, an *upper* bound, where g is a finite ground tree. The following theorem extends [Zhao 2026, Theorem "Finite countermodel transfer"] from the pin paths of a single word to arbitrary finite bounds.

Theorem 10.6 (Finite-bound pinning). *Let χ be a system over a set V of variables, with any variance, that has a finite solution, and let B be a finite set of ground bounds on variables of V. If $\chi \cup B$ has a solution, then it has a finite solution.*

The proof reduces the theorem to covariant systems by the signed system $\chi^\pm$ of Table 2, and then symmetrizes a covariant solution by a median. A ground bound b is translated into a pair $b^\pm$ of covariant ground bounds on signed variables,

$$(g \leq_c x)^\pm = \{v_+ g \leq x^+,\ x^- \leq v_- g\}, \qquad (x \leq_c g)^\pm = \{x^+ \leq v_+ g,\ v_- g \leq x^-\}, \tag{8}$$

and $B^\pm$ is the union of these pairs. For an assignment ρ of V, $\rho^\pm$ is the assignment $v^\pm \mapsto v_\pm \rho(v)$ of Lemma 2.5. The *sign involution* ι maps an assignment σ of $V^\pm$ to the assignment

$$(\iota\sigma)(v^+) = \sigma(v^-)^\partial, \qquad (\iota\sigma)(v^-) = \sigma(v^+)^\partial.$$

The literals of Table 2 that contain $\bot$ or $\top$ belong to $\chi^\pm$; the proof needs them (Remark 10.10).

Lemma 10.7 (Signed bounds). *Let χ be a system over V and B a finite set of ground bounds on V.*

(1) An assignment ρ of V solves $\chi \cup B$ iff $\rho^\pm$ solves $\chi^\pm \cup B^\pm$.
(2) ι is an involution that maps the solutions of $\chi^\pm \cup B^\pm$ to solutions, and $\iota\rho^\pm = \rho^\pm$ for every assignment ρ of V.
(3) If $\iota\sigma = \sigma$, then $\sigma = \rho^\pm$ for the assignment $\rho(v) = v_+\sigma(v^+)$; and $\rho \mapsto \rho^\pm$ and its inverse preserve finiteness.

For covariant trees t_1, t_2, t_3, the *median* $\mathrm{med}(t_1, t_2, t_3)$ is the covariant tree whose cones consist of the words that lie in the corresponding cones of at least two of the three trees: for both signs,

$$L^\pm_{\mathrm{med}(t_1,t_2,t_3)} = \{w : w \in L^\pm_{t_j} \text{ for at least two } j\}.$$

It exists by Theorem 3.2(1): the sets on the right are extension closed, and they are disjoint, since a word in two of the cones $L^0_{t_j}$ and in two of the cones $L^1_{t_j}$ lies in both cones of one t_j. The median of assignments is taken variablewise. It is the lattice median $(t_1 \wedge t_2) \vee (t_1 \wedge t_3) \vee (t_2 \wedge t_3)$ of the covariant order, but we use only the following properties.

Lemma 10.8 (Median).

(1) The median is symmetric and monotone in each argument, and $\mathrm{med}(t, t, u) = t$.
(2) $\mathrm{med}(f(a^1_i)_i, f(a^2_i)_i, f(a^3_i)_i) = f(\mathrm{med}(a^1_i, a^2_i, a^3_i))_i$ and $\mathrm{med}(t_1, t_2, t_3)^\partial = \mathrm{med}(t_1^\partial, t_2^\partial, t_3^\partial)$. In particular ι commutes with medians of assignments of signed variables.
(3) The median of three finite trees is finite.
(4) If three assignments solve a covariant system together with covariant ground bounds, so does their median.

The covariant case of Theorem 10.6 is proved directly from the finite-satisfiability criterion of Zhao [2026] (Theorem A.1), in Appendix F.3. By that criterion, a system has a finite solution iff it has neither a label clash nor a cycle clash. The proof encodes each ground bound by fresh variables, one for each position of the bound; since a path through fresh variables strictly increases the length of the position they stand for, every cycle clash of the encoded system lies among the original variables, where the finite solution of χ excludes it, and the solution of $\chi \cup B$ excludes label clashes.

Lemma 10.9 (Covariant finite-bound pinning). *Theorem 10.6 holds when f is covariant.*

Proof of Theorem 10.6. If $n = 0$, every tree is finite. Let $n \geq 1$, let δ be a finite solution of χ, and let ρ be a solution of $\chi \cup B$. By Lemma 10.7(1,3), $\delta^{\pm}$ is a finite solution of $\chi^{\pm}$, and $\rho^{\pm}$ is a solution of $\chi^{\pm} \cup B^{\pm}$. By Lemma 10.9, applied to the covariant system $\chi^{\pm}$ and the bounds $B^{\pm}$, the system $\chi^{\pm} \cup B^{\pm}$ has a finite solution α. By Lemma 10.7(2), $\iota\alpha$ is a solution of $\chi^{\pm} \cup B^{\pm}$, and it is finite because duality keeps the domain. Consider the median

$$\operatorname{med}(\alpha, \iota\alpha, \delta^{\pm}).$$

All three arguments solve $\chi^{\pm}$, so $\operatorname{med}(\alpha, \iota\alpha, \delta^{\pm})$ solves $\chi^{\pm}$ by Lemma 10.8(4), and it is finite by Lemma 10.8(3). The assignment $\delta^{\pm}$ need not satisfy the bounds; still $\operatorname{med}(\alpha, \iota\alpha, \delta^{\pm})$ satisfies $B^{\pm}$: for a lower bound $G \leq p$ in $B^{\pm}$, both $\alpha(p)$ and $(\iota\alpha)(p)$ lie above G, so $G = \operatorname{med}(G, G, \delta^{\pm}(p)) \leq \operatorname{med}(\alpha, \iota\alpha, \delta^{\pm})(p)$ by Lemma 10.8(1), and upper bounds are symmetric. By Lemma 10.8(2) and Lemma 10.7(2),

$$\iota \operatorname{med}(\alpha, \iota\alpha, \delta^{\pm}) = \operatorname{med}(\iota\alpha, \iota\iota\alpha, \iota\delta^{\pm}) = \operatorname{med}(\iota\alpha, \alpha, \delta^{\pm}) = \operatorname{med}(\alpha, \iota\alpha, \delta^{\pm}).$$

By Lemma 10.7(3), $\operatorname{med}(\alpha, \iota\alpha, \delta^{\pm}) = \tau^{\pm}$ for a finite assignment τ of V, which solves $\chi \cup B$ by Lemma 10.7(1). □

Remark 10.10 (The extreme literals are needed). The signed literals of $u = \bot$ and $\top \leq u$ in Table 2 cannot be dropped from $\chi^{\pm}$, although they contribute no rules on leaf cones, only facts (Table 3). Let $n \geq 1$, $\chi = \{\top \leq x\}$, $B = \varnothing$, and $\delta(x) = \top$. Without its extreme literals the signed system is empty, and $\alpha(x^{+}) = \bot$, $\alpha(x^{-}) = \top$ is a finite solution with $\iota\alpha = \alpha$. Then $\operatorname{med}(\alpha, \iota\alpha, \delta^{\pm}) = \alpha$, which decodes to $x = \bot$ and violates χ. The third argument of the median must also be a finite solution of the unpinned system: Ω in every coordinate is fixed by ι and solves every signed system without extreme literals, but the median with it is infinite in general.

10.4 Regular and Finite Trees

Theorem 10.11 (Regular trees). *For every arity $n \geq 0$, every variance c, and all systems φ over X and ψ over $X \cup Z$:*

(1) a system is satisfiable iff it has a regular solution;
(2) every regular extensible assignment of X has a regular extension;
(3) if some solution of φ is not extensible, then some regular solution of φ is not extensible.

Consequently $E_{\mathrm{reg}}(\varphi, \psi) \iff E_{\mathrm{all}}(\varphi, \psi)$.

Proof. (1) This restates Theorem 2.9(1) and Lemma A.2.

(2) Let ρ be regular and extensible. By Lemma 10.1, $H_{\psi}(S_{\rho})$ decodes to an extension of ρ. The cones of ρ are regular by Theorem 3.2(4), hence so are those of $H_{\psi}(S_{\rho})$ by Lemma 5.7, and the extension is regular by Theorem 3.2(4).

(3) Let ρ be a solution of φ that is not extensible, and take A and C from Lemma 10.3. The model S_{ρ} of φ contains A, so by Lemma 3.8(2) the set $H_{\varphi}(A)$ is the least model of φ that contains A, and $H_{\varphi}(A) \subseteq S_{\rho}$. It decodes to a solution ρ_0 of φ, which is regular by Lemma 5.7 and Theorem 3.2(4),

since A is finite. Its atom set contains A and, being contained in S_ρ, is disjoint from C. By Lemma 10.3, ρ_0 is not extensible.

For the consequence, if E_{all} holds, every regular solution of φ is extensible and has a regular extension by (2); if E_{all} fails, (3) gives a regular solution of φ without any extension. □

Theorem 10.12 (Finite trees). *For every arity $n \geq 0$, every variance c, and all systems φ over X and ψ over $X \cup Z$,*

$$E_{\text{fin}}(\varphi, \psi) \iff \neg\, \text{Sat}_{\text{fin}}(\varphi) \;\vee\; \big(\text{Sat}_{\text{fin}}(\psi) \wedge E_{\text{all}}(\varphi, \psi)\big). \tag{9}$$

More precisely:

(1) if φ has a finite solution and some solution of φ is not extensible, then some finite solution of φ is not extensible;

(2) if ψ has a finite solution, then every finite extensible assignment of X has a finite extension.

Proof. (1) Let ρ be a solution of φ that is not extensible. By Lemma 10.3, at most two visible observations hold in ρ and hold together in no extensible assignment. By Lemma 10.4, each of them is equivalent to a ground bound on a visible variable; let B be the set of these bounds. Then ρ solves $\varphi \cup B$, and φ has a finite solution, so Theorem 10.6 gives a finite solution ρ_1 of $\varphi \cup B$. The observations hold in ρ_1, so ρ_1 is not extensible.

(2) Let ρ be a finite assignment of X with an extension ρ' that solves ψ. Let B consist of the bounds $\rho(x) \leq_c x$ and $x \leq_c \rho(x)$ for $x \in X$. Then ρ' solves $\psi \cup B$, and ψ has a finite solution, so Theorem 10.6 gives a finite solution τ of $\psi \cup B$. By antisymmetry of $\leq_c$, τ agrees with ρ on X, so it is a finite extension of ρ.

For Eq. (9): if φ has no finite solution, both sides hold. If φ has a finite solution and ψ has none, that solution has no finite extension, and both sides fail. If both have finite solutions, (1) and (2) give $E_{\text{fin}}(\varphi, \psi) \iff E_{\text{all}}(\varphi, \psi)$. □

The bounds in the proof of (1) concern at most two visible variables, and each has at most $1 + n(|w| + 1)$ positions, where w is the word of its observation. The proofs of Theorems 10.11 and 10.12 use no satisfiability hypothesis and no condition on the variance.

Corollary 10.13 (Decisions for regular and finite trees). *For fixed n and c, if existential entailment over $\{\bot, f, \top\}$ is decidable over arbitrary trees, then it is decidable over regular trees and over finite trees: $E_{\text{reg}} = E_{\text{all}}$, and E_{fin} is decided by Eq. (9), whose two finite-satisfiability tests are decidable by Theorem 2.9(1).*

Remark 10.14 (Both finite-satisfiability tests are needed). Let $n \geq 1$ and let c be any variance. For $X = \varnothing$, the empty antecedent and $\psi = \{z \leq f(z^n),\ f(z^n) \leq z\}$, the entailment holds over arbitrary and regular trees, with the unique witness $z = \Omega$, but fails over finite trees, since ψ has no finite solution. For $\varphi = \{x \leq f(x^n),\ f(x^n) \leq x\}$ and $\psi = \{x = \bot\}$, the entailment fails over arbitrary and regular trees, at $x = \Omega$, and holds vacuously over finite trees. Least models are not finite countermodels in general, and by Proposition 10.5(3) the finite trees are not closed under intersection of leaf cones; this is why Theorem 10.12 pins observations instead.

10.5 Trees without Top

In this subsection f is covariant, so $\leq_c$ is the covariant order $\leq$. A tree is $\top$-*free* if no position is labeled $\top$; the trees over $\{\bot, f\}$ are the $\top$-free trees (Section 2.2), and a $\top$-free tree is finite or regular in the same sense over $\{\bot, f\}$ and over $\{\bot, f, \top\}$. We write $\models_{\mathcal{D}}^{\{\bot, f\}}$ and $\models_{\mathcal{D}}^{\{\bot, f, \top\}}$ for existential entailment over the two signatures, with f covariant; over $\{\bot, f\}$ the systems have no literal $\top \leq u$ and all values are $\top$-free (Definition 2.4). We use the trees $T_0 = \bot$ and $T_{k+1} = f(T_k^n)$ of

Section 2.3, now for every $n \geq 0$: the positions of T_k are the words of length at most k, labeled f below length k and $\bot$ at length k, and for $n = 0$ and $k \geq 1$ the tree T_k is $f()$. For a finite tree t, $h(t)$ is the maximal length of a position of t. The *gadget* on a variable v, with a fresh variable y_v, is the pair of literals

$$\mathsf{TF}(v, y_v) = \{v \leq y_v,\ y_v \leq f(y_v^n)\}.$$

Lemma 10.15 (Top-freeness gadget). *Let f be covariant.*

(1) If $y \leq f(y^n)$, then y is $\top$-free, and so is every tree $t \leq y$.

(2) For every $\top$-free tree t, the regular tree $y = \Omega$ satisfies $t \leq y \leq f(y^n)$. If t is finite, then so does the finite tree $y = T_{h(t)+1}$.

Proof. (1) For covariant trees L_t^0 is the set of words with a prefix labeled $\top$ in t, since $\nu_+ t = t$; so t is $\top$-free iff $L_t^0 = \varnothing$. By Theorem 3.2(2,3), $y \leq f(y^n)$ gives

$$L_y^0 \subseteq L_{f(y^n)}^0 = \bigcup_{i \in \Sigma} i\, L_y^0 = \Sigma L_y^0.$$

If L_y^0 were nonempty, a shortest word in it would have the form iv with $v \in L_y^0$, contradicting minimality. So $L_y^0 = \varnothing$; for $n = 0$ this holds because $\Sigma L_y^0 = \varnothing$. If $t \leq y$, then $L_t^0 \subseteq L_y^0 = \varnothing$ by Theorem 3.2(2). The argument concerns finite words only and applies to infinite trees y.

(2) Every position of Ω is labeled f, so $t(w) \leq \Omega(w)$ at every common position w since t is $\top$-free; and $\Omega = f(\Omega^n)$ has a single subtree. If t is finite, then $K(t)$ consists of words of length less than $h(t) + 1$, so $K(t) \subseteq K(T_{h(t)+1})$ and $t \leq T_{h(t)+1}$ by Lemma 2.3. By the same lemma, $T_{h(t)+1} \leq T_{h(t)+2} = f(T_{h(t)+1}^n)$. □

Theorem 10.16 (Reduction of $\{\bot, f\}$ to $\{\bot, f, \top\}$). *Let f be covariant, and let φ over X and ψ over $X \cup Z$ be systems over $\{\bot, f\}$. Take fresh variables y_v for all $v \in X \cup Z$, put $Y_X = \{y_x : x \in X\}$ and $Y_Z = \{y_z : z \in Z\}$, and let*

$$\varphi^\natural = \varphi \cup \bigcup_{x \in X} \mathsf{TF}(x, y_x), \qquad \psi^\natural = \psi \cup \bigcup_{z \in Z} \mathsf{TF}(z, y_z),$$

with visible variables $X \cup Y_X$ and hidden variables $Z \cup Y_Z$. For every tree class $\mathcal{D}$,

$$\varphi \models_{\mathcal{D}}^{\{\bot, f\}} \exists \bar{z}\, \psi \iff \varphi^\natural \models_{\mathcal{D}}^{\{\bot, f, \top\}} \exists \bar{z}\, \bar{y}\, \psi^\natural,$$

where $\bar{z}\, \bar{y}$ lists $Z \cup Y_Z$.

The gadget is placed on every visible variable $x \in X$, including those that do not occur in φ, and on every hidden variable $z \in Z$; the visible variables of ψ are covered by the gadgets of $\varphi^\natural$.

Proof. For a $\top$-free tree t in $\mathcal{D}$ let $y(t) = T_{h(t)+1}$ if $\mathcal{D} = \mathcal{T}_{\mathrm{fin}}$ and $y(t) = \Omega$ otherwise. By Lemma 10.15(2), $y(t)$ is a $\top$-free tree in $\mathcal{D}$ with $t \leq y(t) \leq f(y(t)^n)$. A $\top$-free assignment satisfies a literal over $\{\bot, f\}$ in the sense of $\{\bot, f\}$ iff it satisfies it in the sense of $\{\bot, f, \top\}$, since both signatures use the same trees, constructor and order on $\top$-free trees (Section 2.2).

Assume the left side, and let ρ be a solution of $\varphi^\natural$ with values in $\mathcal{D}$. By Lemma 10.15(1), the values $\rho(y_x)$ and $\rho(x)$, $x \in X$, are $\top$-free. So $\rho|_X$ is a solution of φ over $\{\bot, f\}$ with values in $\mathcal{D}$, and it has an extension σ to Z, with $\top$-free values in $\mathcal{D}$, that solves ψ. Put $\sigma(y_z) = y(\sigma(z))$ for $z \in Z$. Then ρ together with σ solves $\psi^\natural$, which does not constrain the variables y_x.

Assume the right side, and let ρ be a solution of φ over $\{\bot, f\}$ with values in $\mathcal{D}$. Extend it by $\rho(y_x) = y(\rho(x))$ for $x \in X$; this gives a solution of $\varphi^\natural$ over $\{\bot, f, \top\}$ with values in $\mathcal{D}$. It has an extension σ to $Z \cup Y_Z$, with values in $\mathcal{D}$, that solves $\psi^\natural$. By Lemma 10.15(1), every $\sigma(y_z)$ and every $\sigma(z)$ is $\top$-free, so $\sigma|_Z$ extends ρ to a solution of ψ over $\{\bot, f\}$ with values in $\mathcal{D}$. □

Corollary 10.17 (The signature $\{\bot, f\}$). *Let f be covariant, of any arity $n \geq 0$. In each tree class, existential entailment over $\{\bot, f\}$ reduces to existential entailment over $\{\bot, f, \top\}$ in the same tree class, at the same arity and with f covariant. The reduction doubles the numbers of visible and of hidden variables, so the size parameter s becomes $2s$, and it adds two literals per variable. Consequently, existential entailment over $\{\bot, f\}$ with f covariant is decidable over arbitrary, regular and finite trees whenever the covariant problem over $\{\bot, f, \top\}$ is, and $E_{\mathrm{reg}} = E_{\mathrm{all}}$ holds over $\{\bot, f\}$ as well.*

The last claim follows from Theorems 10.11 and 10.16. The finite witness $T_{h(t)+1}$ can be large, but it is used only in the proof; the reduction introduces variables and literals, not ground trees. Both requirements on the gadget are necessary.

Remark 10.18 (Every visible variable, and covariance). The gadget cannot be omitted on a visible variable that does not occur in φ: with $X = \{x\}$, the entailment true $\models_{\mathcal{D}}^{\{\bot,f\}} \exists \bar{z}\ x \leq f(z_i)_i$ holds in every tree class (take the z_i to be the children of x, or $\bot$ if $x = \bot$), whereas without a gadget on x its translation fails at $x = \top$. The gadget also needs covariance. If $c(j) = 1$ for some j, let $y = f(t_i)_i$ with $t_i = \top$ if $c(i) = 1$ and $t_i = \bot$ if $c(i) = 0$. Then $y \leq_c f(y^n)$, since the covariant arguments require $\bot \leq_c y$ and the contravariant ones $y \leq_c \top$, but y has a label $\top$; for the arrow, $y = \top \to \bot$ satisfies $y \leq_c y \to y$. We do not treat $\{\bot, f\}$ with a contravariant argument. That problem is not obtained from $\{\bot, f, \top\}$ by omitting the literals $\top \leq u$: for every variance, the entailment above holds over $\{\bot, f\}$ but fails over $\{\bot, f, \top\}$ at $x = \top$.

10.6 Arities Zero and One

Remark 10.19 (Arity zero). Let $n = 0$. By Remark 2.8, all three tree classes coincide at arity zero, so $E_{\mathrm{fin}} = E_{\mathrm{reg}} = E_{\mathrm{all}}$, and existential entailment is decidable by finite enumeration; each of the 3^s assignments is checked in time linear in $|\varphi| + |\psi|$. The route of Sections 3 and 4 also applies, with no nonempty words, but the enumeration is simpler. Copy literals cannot be eliminated at arity zero (Lemma 2.7 needs $n \geq 1$), which is why the proof of Theorem 10.6 treats $n = 0$ separately.

Remark 10.20 (Arity one). Let $n = 1$. Every tree is $f^k(\bot)$ or $f^k(\top)$ for some $k \geq 0$, or Ω, so every tree is regular and Theorem 10.11 is immediate. Finite and arbitrary trees still differ: the system $\{x \leq f(x),\ f(x) \leq x\}$ has the unique solution Ω. The variance $c(0) = 1$ is allowed, and then mixed polarity already occurs: for $\{x \leq f(x)\}$ there is no parity assignment π as in Remark 4.5, since the condition $\pi(a) = \pi(u) \oplus c(i)$ for a constructor literal with root u and child a at argument i would require $\pi(x) = \pi(x) \oplus 1$. The spine automaton of Section 8 runs over $\Sigma = \{0\}$, so no state has two distinct safe direction letters. Hence its branching case does not occur, and every infinite direction word is 0^ω. The general argument applies without change.

11 The Main Theorem

This section states the main theorem and proves it by assembling the results of Sections 2 to 10. Apart from bookkeeping, only two things are proved here: how a counterexample word yields a countermodel (Lemma 11.3), and two short estimates that turn the length bounds of the earlier sections into closed forms (Proposition 11.5).

Throughout, the arity $n \geq 0$ and the variance c are fixed. For a flat system φ over X and a flat system ψ over $X \cup Z$ we write

$$v = |X|, \qquad h = |Z|, \qquad s = v + h,$$

so that s is the size parameter of Section 2.3, counted after flattening. As in Section 9.6, $E_1(x) = 2^x$ and $E_{k+1}(x) = 2^{E_k(x)}$.

11.1 Statement

When the entailment fails although both systems are satisfiable, the failure is certified by a single word, which comes from one of the three conditions of Theorem 4.2.

Definition 11.1 (Counterexample words). Let φ over X and ψ over $X \cup Z$ be flat systems over $\{\bot, f, \top\}$, both satisfiable. A *counterexample word* of (φ, ψ) is a word of one of the following three kinds, together with the signed variables named in it.

(C1) An (I0)-*word*: a word w with $(q, w) \in \mu_\psi$ and $(q, w) \notin \mu_\varphi$ for some $q \in X^{\pm}$.
(C2) A (U)-*word*: a word w such that (p, w) is live in φ and $\mathrm{DeadU}_\psi(p, w)$ holds (Definition 5.8), for some $p \in X^{\pm}$.
(C3) A *hole word*, used only when (U) holds: the separator word $u\sharp t$ of a hole (u, t) of an extracted instance I_{pq} with $p, q \in X^{\pm}$ (Definition 7.5).

Its *length* is the length of the word, so a hole word has length $|u| + |t| + 1$.

By Proposition 5.17(2), an (I0)-word exists iff (I0) fails. By Lemma 6.1, a (U)-word exists iff (U) fails. If (U) holds, then by Theorem 7.7(1) a hole word exists iff (II) fails.

Theorem 11.2 (Main theorem). *Let $n \geq 0$ be an arity and c a variance of f.*

(1) Existential entailment $\varphi \models_{\mathcal{D}} \exists \bar{z}\, \psi$ over $\{\bot, f, \top\}$ is decidable for each of the tree classes $\mathcal{D} = \mathcal{T}$, $\mathcal{T}_{\mathrm{reg}}$ and $\mathcal{T}_{\mathrm{fin}}$.
(2) If f is covariant, existential entailment over $\{\bot, f\}$ is decidable for each of the three tree classes.
(3) Let φ over X and ψ over $X \cup Z$ be flat systems over $\{\bot, f, \top\}$, with $v = |X|$ and $s = |X| + |Z|$, and let $\mathcal{D}$ be a tree class. If φ and ψ are $\mathcal{D}$-satisfiable and $\varphi \not\models_{\mathcal{D}} \exists \bar{z}\, \psi$, then (φ, ψ) has a counterexample word of length at most $E_3(10v + 6s + 18) \leq E_3(16s + 18)$. Every counterexample word determines, effectively, a solution of φ with values in $\mathcal{D}$ that has no extension with values in $\mathcal{D}$ to a solution of ψ.
(4) Let f be covariant, let φ and ψ be flat systems over $\{\bot, f\}$ with $v = |X|$ and $s = |X| + |Z|$, and let $\mathcal{D}$ be a tree class. If φ and ψ are $\mathcal{D}$-satisfiable over $\{\bot, f\}$ and the entailment fails over $\{\bot, f\}$ in $\mathcal{D}$, then the pair $(\varphi^\natural, \psi^\natural)$ of Theorem 10.16 has a counterexample word of length at most $E_3(20v + 12s + 18) \leq E_3(32s + 18)$, and every counterexample word of this pair determines, effectively, a solution of φ over $\{\bot, f\}$ in $\mathcal{D}$ without an extension over $\{\bot, f\}$ in $\mathcal{D}$ to a solution of ψ.

The bounds in (3) and (4) depend neither on n nor on c.

Unflattened inputs are first flattened (Lemma 2.6), and s then counts the variables after flattening, as the bounds require (Section 2.3). The satisfiability hypotheses in (3) and (4) are needed only to make the certificate a word: if φ is not $\mathcal{D}$-satisfiable the entailment holds, and if φ is $\mathcal{D}$-satisfiable and ψ is not, every $\mathcal{D}$-solution of φ is a countermodel. The bounds concern certificates, not countermodels or witnesses: a countermodel is in general an infinite regular tree assignment, described by finite automata (Lemma 11.3), and when the entailment holds the witnesses for the hidden variables may have to be infinite (Remark 10.14). Part (2) is not claimed for $\{\bot, f\}$ with a contravariant argument; the reduction of Theorem 10.16 needs covariance (Remark 10.18).

11.2 The Decision Procedure

The input consists of the arity n, the variance c, the signature ($\{\bot, f, \top\}$, or $\{\bot, f\}$ with f covariant), a tree class $\mathcal{D}$, and two finite conjunctions φ and ψ of inequalities and equalities between type expressions, with visible variables X and hidden variables Z. The output is whether $\varphi \models_{\mathcal{D}} \exists \bar{z}\, \psi$ holds and, if it fails, a certificate: a counterexample word when both systems are $\mathcal{D}$-satisfiable,

and otherwise a solution of φ in $\mathcal{D}$, which is then a countermodel. Each step cites the result that justifies it; Table 6 lists the sections involved.

Step 1. *Flatten.* Replace φ and ψ by the flat systems of Lemma 2.6. The answer is unchanged in every tree class and for both signatures. From now on s counts the variables after flattening.

Step 2. *Arity zero.* If $n = 0$, the trees are $\bot < f() < \top$, all finite, and the three classes coincide (Remarks 2.8 and 10.19); over $\{\bot, f\}$ only $\bot$ and $f()$ remain. Enumerate the at most 3^s assignments of $X \cup Z$, check φ and ψ for each, and answer by Definition 2.4. Stop.

Step 3. *The signature $\{\bot, f\}$.* If the signature is $\{\bot, f\}$, replace (φ, ψ) by $(\varphi^\natural, \psi^\natural)$, with visible variables $X \cup Y_X$ and hidden variables $Z \cup Y_Z$, and continue over $\{\bot, f, \top\}$. The answer is unchanged in every tree class (Theorem 10.16); the size parameter doubles (Corollary 10.17).

Step 4. *Satisfiability in $\mathcal{D}$.* Decide whether φ and ψ are $\mathcal{D}$-satisfiable: for $\mathcal{D} = \mathcal{T}$ by Proposition 5.17(1), for $\mathcal{D} = \mathcal{T}_{\mathrm{reg}}$ by the same test, since $\mathrm{Sat}_{\mathrm{reg}} = \mathrm{Sat}$ (Theorem 2.9(1)), and for $\mathcal{D} = \mathcal{T}_{\mathrm{fin}}$ by Theorem 2.9(1), with the criteria of Theorem A.1 and Lemma A.2. If φ is not $\mathcal{D}$-satisfiable, answer *holds*. If φ is $\mathcal{D}$-satisfiable and ψ is not, answer *fails*, with a solution of φ in $\mathcal{D}$, which no assignment of Z extends: for $\mathcal{D} = \mathcal{T}$ or $\mathcal{T}_{\mathrm{reg}}$ the least solution, which is regular and decoded from μ_φ with finite automata for its leaf cones (the proof of Lemma 11.3(1) for $A = \varnothing$ uses only that φ is satisfiable), and for $\mathcal{D} = \mathcal{T}_{\mathrm{fin}}$ the first solution of φ in an enumeration of the finite assignments of X, which terminates because one exists. Otherwise both systems are satisfiable over arbitrary trees; continue.

Step 5. *Condition* (I0). For each $q \in X^{\pm}$, decide whether some word w has $(q, w) \in \mu_\psi$ and $(q, w) \notin \mu_\varphi$ (Proposition 5.17(2)). If so, answer *fails*, with the (I0)-word w.

Step 6. *Condition* (U). For each $p \in X^{\pm}$, test the words w of length at most $\mathsf{Len}(K_{\mathrm{U}}, D_{\mathrm{U}})$, with the constants $K_{\mathrm{U}} = 2^{2|X|+|Z|+1}$ and $D_{\mathrm{U}} = 2^{|X|+2}(4K_{\mathrm{U}} - 1) - 1$ of Theorem 6.11, for (p, w) live in φ and $\mathrm{Dead}\mathsf{U}_\psi(p, w)$, using the source automaton $\mathcal{B}_p$ and its self guard (Theorem 6.11 and Lemmas 6.3 and 6.5). If one passes, answer *fails*, with the (U)-word w.

Step 7. *Condition* (II). Now (I0) and (U) hold. For each $p, q \in X^{\pm}$, compute the extracted instance I_{pq} (Definition 7.5) and

$$K_{pq} = \max\bigl(\mathsf{L}_{\mathrm{nm}}(N_A, N_B, N_{A'}, N_{B'}),\ \mathsf{Len}^{\sharp}_{I_{pq}}\bigr),$$

with the state counts of I_{pq}, and test the pairs (u, t) with $|u| + |t| < K_{pq}$ for conditions (a)–(e) of Definition 7.1. This search finds a hole iff I_{pq} has one: by Lemma 7.6, I_{pq} has forced target heads, so Corollary 8.18 leaves only holes shorter than L_{nm} and mortal holes, and Theorem 9.15 shortens the latter (Theorem 9.17; numerically, Corollary 9.19). If some I_{pq} has a hole (u, t), answer *fails*, with the hole word $u\sharp t$; otherwise answer *holds* (Corollary 7.8).

Step 8. *Tree classes.* Return the answer. Steps 5 to 7 decide entailment over arbitrary trees, and their answer is the answer for $\mathcal{D}$: $E_{\mathrm{reg}} = E_{\mathrm{all}}$ (Theorem 10.11), and $E_{\mathrm{fin}} = E_{\mathrm{all}}$ when both systems have finite solutions (Theorem 10.12). A countermodel in $\mathcal{D}$ is obtained from a counterexample word by Lemma 11.3. Over $\{\bot, f\}$, a countermodel of $(\varphi^\natural, \psi^\natural)$, including one returned in Step 4, is restricted to the original visible variables (Lemma 11.3(4)).

Condition (U) is decided in Step 6 before the holes of the extracted instances are used in Step 7. This order matters: Theorem 7.7 assumes (U), and without it a failure of (II) need not be a hole (Remark 7.9). The bounded search of Step 7 itself needs only forced target heads, condition (FH), which every extracted instance satisfies by construction (Lemma 7.6); without (FH) the search would be incomplete (Remark 8.20). Equivalently, after Step 4 one may test all candidate words of the three kinds of Definition 11.1 up to the bound of Theorem 11.2(3): a candidate hole of some

Table 6. The steps of the decision procedure, the results that justify them, and the sections where these are proved. The characterization (Theorem 4.2), built on the leaf cones of Section 3, underlies Steps 4 to 7.

Step	Task	Results used	Sections
Step 1	flatten	Lemma 2.6	Section 2
Step 2	arity zero	Remarks 2.8 and 10.19	Sections 2 and 10
Step 3	$\{\bot, f\}$	Lemma 10.15, Theorem 10.16, and Corollary 10.17	Section 10
Step 4	satisfiability	Proposition 5.17(1), Theorems A.1 and 2.9 and Lemma A.2, Theorem 4.2(1)	Sections 2, 4 and 5, Appendix A
Step 5	(I0)	Lemma 5.6, Proposition 5.17(2)	Section 5
Step 6	(U)	Lemmas 6.5, 6.9 and 6.10 and Theorem 6.11; Theorems 2.15 and 2.20	Sections 2 and 6
Step 7	(II)	Theorem 7.7, Corollary 7.8, and Lemma 7.6; Corollary 8.18; Theorems 9.15 and 9.17 and Corollary 9.19	Sections 7 to 9
Step 8	tree classes	Theorems 10.11 and 10.12 and Corollary 10.13, Lemma 11.3	Sections 10 and 11

I_{pq} found while (U) fails still gives the correct answer *fails*, and if no candidate passes, all three conditions hold by Proposition 11.5.

11.3 Countermodels

A counterexample word determines a countermodel through least closures: the least model of φ that contains the seed of the word is a solution of φ that no solution of ψ extends. For finite trees, the word gives at most two observations, which are pinned by finite ground bounds.

Lemma 11.3 (Countermodels). *Let φ over X and ψ over $X \cup Z$ be flat systems over $\{\bot, f, \top\}$, both satisfiable, and let a counterexample word of (φ, ψ) be given. Define a set A of seeds and a set O of visible observations (Definition 10.2) by*

- *for an* (I0)*-word w with q: $A = \varnothing$, and O denies (q, w);*
- *for a* (U)*-word w with p: $A = \{(p, w)\}$, and O asserts (p, w);*
- *for a hole word $u \sharp t$ of I_{pq}: $A = \{(p, t)\}$, and O asserts (p, t) and denies (q, u).*

(1) $H_\varphi(A)$ is a model of φ, namely μ_φ, $\mathrm{Cl}_\varphi(p, w)$ or $\mathrm{Cl}_\varphi(p, t)$. It decodes to a regular solution ρ_A of φ in which the observations of O hold, and finite automata for the leaf cones of ρ_A are computable from the certificate.

(2) No assignment of X in which the observations of O hold extends to a solution of ψ. In particular, ρ_A is a countermodel over arbitrary and over regular trees.

(3) Suppose that φ has a finite solution. Let B be the set of finite ground bounds equivalent to the observations of O by Lemma 10.4; each has at most $1 + n(L + 1)$ positions, where L is the length of the counterexample word. Then $\varphi \cup B$ has a finite solution, and every finite solution of $\varphi \cup B$ is a finite solution of φ without any extension to a solution of ψ. Such a solution is found by enumerating the finite assignments of X.

(4) Suppose that $(\varphi, \psi) = (\varphi_0^\natural, \psi_0^\natural)$ for systems φ_0 over X_0 and ψ_0 over $X_0 \cup Z_0$ over $\{\bot, f\}$, with f covariant, as in Theorem 10.16. If ρ is a solution of φ with values in a tree class $\mathcal{D}$ that has no extension with values in $\mathcal{D}$ to a solution of ψ, then $\rho|_{X_0}$ is a solution of φ_0 over $\{\bot, f\}$ in $\mathcal{D}$ without an extension over $\{\bot, f\}$ in $\mathcal{D}$ to a solution of ψ_0.

Proof. (1) Since φ is satisfiable, μ_φ is its least model (Lemma 3.8(2)). A (U)-word has (p, w) live in φ by definition, and a hole (u, t) of I_{pq} has (p, t) live in φ by Theorem 7.7, which applies because (U) holds; so $\mathrm{Cl}_\varphi(p, w)$, respectively $\mathrm{Cl}_\varphi(p, t)$, is the least model of φ that contains the seed (Lemma 3.8(2)). The observations hold in $H_\varphi(A)$: $(q, w) \notin \mu_\varphi$ by definition, the seed lies in its own closure, and $(q, u) \notin \mathrm{Cl}_\varphi(p, t)$ by Theorem 7.7. By Lemma 3.5, $H_\varphi(A)$ is S_{ρ_A} for a unique solution ρ_A of φ. The set A is finite, so its slices are regular, and Lemma 5.7 gives computable finite automata for the slices of $H_\varphi(A)$, which are the leaf cones of ρ_A. By Theorem 3.2(4), ρ_A is regular.

(2) Let ρ' be an assignment of X in which the observations hold, and suppose that it extends to a solution of ψ. By Lemma 3.5, $S_{\rho'} = T|_X$ for a model T of ψ. Since T contains Fact_ψ and is closed under the rules of ψ, it contains μ_ψ, and it contains $\mathrm{Reach}_\psi(a)$ for each of its atoms a (Lemma 3.8(1)). For an (I0)-word, $(q, w) \in \mu_\psi \subseteq T$ is visible, so $(q, w) \in S_{\rho'}$, against the denial. For a (U)-word, $(p, w) \in T$, so $T \supseteq \mathrm{Reach}_\psi(p, w)$ contains a forbidden root of ψ, against the consistency of T. For a hole word, $(p, t) \in T$ and $(p, t) \to^*_\psi (q, u)$ by Theorem 7.7, so $(q, u) \in T|_X = S_{\rho'}$, against the denial. Applied to ρ_A, which is regular by (1), this proves the second claim; Theorem 7.7(2) is the case of a hole word.

(3) By Lemma 10.4, an observation holds in an assignment iff the corresponding ground bound does, and the bound has at most $1 + n(|w| + 1)$ positions for the word w of the observation, which is at most L long. So ρ_A solves $\varphi \cup B$, and Theorem 10.6 gives a finite solution. Every solution of $\varphi \cup B$ satisfies the observations, so it has no extension at all by (2), in particular no finite one. Whether a finite assignment solves $\varphi \cup B$ is decidable, since it compares finite trees, so the enumeration terminates.

(4) The values $\rho(x)$, $x \in X_0$, lie below the values $\rho(y_x)$, which satisfy $y_x \le f(y_x^n)$, so they are $\top$-free by Lemma 10.15(1), and $\rho|_{X_0}$ solves φ_0 over $\{\bot, f\}$. Suppose that σ, with values in $\mathcal{D}$, extends $\rho|_{X_0}$ to a solution of ψ_0 over $\{\bot, f\}$. Put $\sigma(y_z) = T_{h(\sigma(z))+1}$ for $z \in Z_0$ if $\mathcal{D} = \mathcal{T}_{\mathrm{fin}}$, and $\sigma(y_z) = \Omega$ otherwise. By Lemma 10.15(2), ρ together with σ solves ψ, which does not constrain the variables y_x; this contradicts the choice of ρ. □

Example 11.4 (Shift closure: the procedure end to end). Run the procedure on Example 1.1 with $\varphi_1 = \{x \le y\}$ and $\psi = \{x \le z,\ z \le y,\ z \le f(z, z)\}$, over $\{\bot, f\}$ with f binary and covariant, in the class of finite trees. Both systems are flat. Step 3 adds $\mathsf{TF}(x, y_x)$ and $\mathsf{TF}(y, y_y)$ to φ_1 and $\mathsf{TF}(z, y_z)$ to ψ; the assignment $\bot$ to every variable solves both new systems, so Step 4 continues.

- Step 5: $\psi^\natural$ has no facts, so $\mu_{\psi^\natural} = \varnothing$ and (I0) holds.
- Step 6: every atom (v^0, w) with $v \in \{x, y, y_x, y_y\}$ is dead by support in $\varphi_1^\natural$, since for $v \in \{x, y\}$ the copy $v^0 \to y_v^0$, and then pops of y_v^0, lead to the forbidden root (y_v^0, ε). Since f is covariant, pure moves preserve signs (Lemma 3.6(2)), and the forbidden roots (z^0, ε) and (y_z^0, ε) of $\psi^\natural$ have sign 0, so no atom of sign 1 is root-dead in $\psi^\natural$. Hence no $\mathcal{R}_{\mathrm{U}}(p)$ rejects a word, and (U) holds, as in Example 6.7.
- Step 7: the computation of Example 7.11 carries over, with y_y in the role of g. The spine automaton of x^1 moves from $\{x^1, y^1, y_x^1, y_y^1\}$ to $\{y_x^1, y_y^1\}$ on every letter and is safe; $J(t) = 0$ for every target t; the only possible event is $(0, 0)$, present iff t is a prefix of u; and the head events are the pairs $(s, 0)$ with $u[s{:}] = t$. So $(01, 1)$ is a hole of $I_{y^1 x^1}$, and the search may answer *fails* with the hole word $01\sharp 1$, of length 4.

By Lemma 11.3, the word gives $A = \{(y^1, 1)\}$, and $\mathcal{O}$ asserts $(y^1, 1)$ and denies $(x^1, 01)$. The least model $\mathrm{Cl}_{\varphi_1^\natural}(y^1, 1)$ is the regular countermodel $x = y = f(\Omega, \bot)$, $y_x = y_y = \Omega$: the closure of the seed consists of the atoms $(y^1, 1\sigma)$ and $(x^1, 1\sigma)$. For finite trees, the pins of Lemma 10.4 for the two

observations are

$$y \le \mathrm{emb}_\top(1,\bot) = f(\top,\bot), \qquad f(f(\bot,f(\bot,\bot)),\bot) = \mathrm{emb}_\bot(01, f(\bot,\bot)) \le x.$$

Every finite solution of $\varphi_1^\natural$ with these two bounds is a countermodel, for instance $x = y = f(f(\bot,f(\bot,\bot)),\bot)$ with $y_x = y_y = T_3$, and its restriction to $\{x,y\}$ (Lemma 11.3(4)) is the countermodel of Example 1.1. On trees without $\top$, the two pins say $1 \notin K(y)$ and $01 \in K(x)$, which is exactly the failure of $\mathrm{Suf}(K(x)) \subseteq K(y)$ found there.

11.4 Counterexample Bounds

The automata of the extracted instances have

$$N_A = 2^{2v}, \qquad N_B = 2^{4v+2h+1} = 2^{2v+2s+1}, \qquad N_{A'} = N_{B'} = 2^{2s}$$

states (Section 7.2). For condition (U), the reduced unit readers have fewer than $K_\mathrm{U} = 2^{v+s+1}$ states, and the antecedent clock that bounds their depth has fewer than $A_\mathrm{U} = 2^{v+1}$ (Lemmas 6.9 and 6.10).

Proposition 11.5 (Counterexample bounds). *Let φ over X and ψ over $X \cup Z$ be flat systems over $\{\bot, f, \top\}$, both satisfiable, with $v = |X|$ and $s = |X| + |Z|$.*

(1) If (I0) *fails, there is an* (I0)*-word of length less than $2^{2v+2s+2}$.*
(2) If (U) *fails, there is a* (U)*-word of length at most $\mathsf{Len}(K_\mathrm{U}, D_\mathrm{U}) \le E_3(5v+3s+16)$, where $D_\mathrm{U} = 2A_\mathrm{U}(4K_\mathrm{U}-1)-1$.*
(3) If (I0) *and* (U) *hold and* (II) *fails, there are $p,q \in X^\pm$ and a hole (u,t) of I_{pq} such that either $|u|+|t| < \mathsf{L}_\mathrm{nm}(N_A, N_B, N_{A'}, N_{B'})$, a bound singly exponential in s, or (u,t) is a mortal hole and $|u\sharp t| \le \mathsf{Len}^\sharp_{I_{pq}} \le E_3(10v+6s+18)$.*

If $v \ge 1$, each of these bounds is at most $E_3(10v+6s+18) \le E_3(16s+18)$, and the mortal bound is the largest. If $v = 0$, there is no counterexample word.

Proof. (1) By Lemma 5.6, $(q,w) \in \mu_\chi$ iff some cut $e \le |w|$ has $Q^\chi_{q,e}(w) \cap \mathsf{Fact}^\uparrow_\chi \ne \varnothing$. The lower automaton of q in χ, extended by a bit that records whether some state read so far meets $\mathsf{Fact}^\uparrow_\chi$, is a deterministic automaton that reads this property; for $\chi = \psi$ it has 2^{2s+1} states, and for $\chi = \varphi$ it has 2^{2v+1}. An (I0)-word for q is a word that leads their product to a state whose ψ-bit is set and whose φ-bit is not. A shortest such word visits no state of the product twice, so it is shorter than $2^{2v+2s+2}$.

(2) This is Theorem 6.11. A reduced unit reader with a hole has one with a rejected tail of at most $4K_\mathrm{U} - 2$ (Theorem 2.15), the clock turns this tail bound into the depth bound D_U (Lemma 6.10), and the bounded-depth search at depth D_U gives the length (Theorem 2.20); Lemma A.13 gives the closed form.

(3) By Theorem 7.7(1), which applies because φ is satisfiable and (U) holds, some I_{pq} has a hole. It satisfies (FH) by Lemma 7.6, so by Corollary 8.18 it has a hole with $|u|+|t| < \mathsf{L}_\mathrm{nm}(N_A, N_B, N_{A'}, N_{B'})$ or a mortal hole. The function L_nm is a polynomial in the four state counts (Section 8.9), each of which is at most $2^{2v+2s+1}$, so the first bound is singly exponential in s. In the second case, Theorem 9.15 gives a mortal hole whose separator word has length at most $\mathsf{Len}^\sharp_{I_{pq}}$, and Corollary 9.19 bounds this by $E_3(10v+6s+18)$.

For the comparison, let $v \ge 1$. The bound of (1) is below 2^{4s+2}, and the non-mortal bound is $\mathsf{L}_\mathrm{nm} \le 54N_A^2N_BN_{A'}N_{B'} < 2^{6v+6s+7}$ (Section 8.9); both are far below the triply exponential bounds. Of the two triply exponential bounds, $E_3(5v+3s+16)$ is below the mortal bound $E_3(10v+6s+18)$, which is therefore the largest, and $E_3(10v+6s+18) \le E_3(16s+18)$ since $v \le s$. If $v = 0$, then $X^\pm$ is empty, and Definition 11.1 admits no word. □

Table 7. Length bounds for shortest counterexample words (Definition 11.1), with $v = |X|$ and $s = |X| + |Z|$, and their growth in s: singly or triply exponential. Over $\{\bot, f\}$ the reduction of Theorem 10.16 replaces v and s by $2v$ and $2s$.

Failing condition	Word	Length bound	Growth	Source
(I0)	w	$< 2^{2v+2s+2}$	single	Proposition 11.5(1)
(U)	w	$\le E_3(5v + 3s + 16)$	triple	Theorem 6.11
(II), not mortal	$u\sharp t$	$\le \mathsf{L}_{\mathrm{nm}}(N_A, N_B, N_{A'}, N_{B'})$	single	Corollary 8.18
(II), mortal	$u\sharp t$	$\le E_3(10v + 6s + 18)$	triple	Theorem 9.15 and Corollary 9.19

Table 7 summarizes the cases. Both triply exponential terms come from the bounded-depth search of Zhao [2026] and its anchored extension, and in both cases the depth is bounded through a short rejected tail (Theorem 2.19). A mortal hole has a rejected tail shorter than $N_A = 2^{2v}$, which involves only visible variables; its reader has $\bar{N}_I \le 2^{2v+2s+3}$ states, and its depth is at most $d_I = 2\bar{N}_I N_A - 1 < 2^{4v+2s+4}$ (Lemma 9.6). A reduced unit reader has fewer than $K_{\mathrm{U}} = 2^{v+s+1}$ states. The tail bound of Theorem 2.15 allows it a tail of up to $4K_{\mathrm{U}} - 2$, but the clock of Lemma 6.10, which reads only the antecedent, bounds the depth by $D_{\mathrm{U}} < 2^{2v+s+5}$. The mortal reader is larger and deeper, which is why the mortal bound is the largest.

11.5 Proof of the Main Theorem

Proof of Theorem 11.2. *Parts (1) and (2).* Each step of the procedure of Section 11.2 terminates and is correct by the result it cites (Table 6); three points need a remark. First, if the procedure passes Step 4, both systems are satisfiable in $\mathcal{D}$, hence over arbitrary trees, so Steps 5 to 7 decide entailment over arbitrary trees by Theorem 4.2(2), and Step 8 transfers the answer to $\mathcal{D}$. Second, Step 6 runs before Step 7, so (U) holds when Step 7 starts, and Theorems 7.7 and 9.17 apply to every I_{pq}, which has forced target heads by Lemma 7.6. Third, Steps 2, 6, and 7 are finite searches: there are finitely many assignments at arity zero, the alphabet Σ is finite, the bounds are computable, and each candidate is tested over finitely many cuts. This proves (1), and (2) follows by Step 3, which preserves the answer by Theorem 10.16.

Part (3). Let φ and ψ be $\mathcal{D}$-satisfiable with $\varphi \not\models_{\mathcal{D}} \exists\bar{z}\,\psi$. Then both are satisfiable over arbitrary trees, and $\varphi \not\models \exists\bar{z}\,\psi$: for $\mathcal{D} = \mathcal{T}_{\mathrm{reg}}$ by Theorem 10.11, and for $\mathcal{D} = \mathcal{T}_{\mathrm{fin}}$ by Eq. (9), since both systems have finite solutions. By Theorem 4.2(2), (I0), (U) or (II) fails. In the first two cases Proposition 11.5(1,2) gives an (I0)-word or a (U)-word; otherwise (I0) and (U) hold and Proposition 11.5(3) gives a hole word. A counterexample word exists only if $v \ge 1$, and then its length is at most $E_3(10v + 6s + 18) \le E_3(16s + 18)$ by Proposition 11.5. This argument uses no hypothesis on n; for $n = 0$ the only word is ε, and every counterexample word has length at most 1. By Lemma 11.3(1,2), every counterexample word yields a regular countermodel, which serves for $\mathcal{T}$ and $\mathcal{T}_{\mathrm{reg}}$, and by Lemma 11.3(3) a finite one when φ has a finite solution, which is the case for $\mathcal{D} = \mathcal{T}_{\mathrm{fin}}$. All of them are computed from the word. None of the bounds involves n or c.

Part (4). Let f be covariant, and let φ and ψ over $\{\bot, f\}$ be $\mathcal{D}$-satisfiable over $\{\bot, f\}$, with failing entailment over $\{\bot, f\}$ in $\mathcal{D}$. For a $\top$-free tree t in $\mathcal{D}$, let $y(t)$ be the tree of Lemma 10.15(2) in $\mathcal{D}$ with $t \le y(t) \le f(y(t)^n)$. Extending a $\{\bot, f\}$-solution of φ by $y_x = y(\rho(x))$, and a $\{\bot, f\}$-solution σ of ψ by $y_z = y(\sigma(z))$, shows that $\varphi^\natural$ and $\psi^\natural$ are $\mathcal{D}$-satisfiable over $\{\bot, f, \top\}$, and by Theorem 10.16 the entailment of $(\varphi^\natural, \psi^\natural)$ fails in $\mathcal{D}$. This pair has $2|X|$ visible and $2|Z|$ hidden variables (Corollary 10.17), so (3), applied with $2v$ and $2s$, gives a counterexample word of length at most $E_3(10 \cdot 2v + 6 \cdot 2s + 18) = E_3(20v + 12s + 18) \le E_3(32s + 18)$. It yields a countermodel for $(\varphi^\natural, \psi^\natural)$ in $\mathcal{D}$, which restricts to a countermodel over $\{\bot, f\}$ in $\mathcal{D}$ by Lemma 11.3(4). □

Remark 11.6 (Complexity). The procedure of Section 11.2 is written to make its correctness evident: Steps 6 and 7 enumerate all words up to a triply exponential length. Section 12 applies the theorem to constrained types and their simplification, and needs only its statement; Section 13 then gives the complexity of the problem, sharper procedures and lower bounds.

12 Constrained Types and Simplification

This section states the subsumption of constrained type schemes as an existential entailment, separates the simplifications that need existential entailment from those that need only plain entailment, lists classes of consequents for which existential entailment reduces directly to plain entailments, and ends with examples, all of which involve variables that occur both co- and contravariantly. Proofs omitted here are in Appendix G.

The signature is $\{\bot, f, \top\}$ with an arity n and a variance c, or $\{\bot, f\}$ with f covariant, and $\mathcal{D}$ is a tree class. For arrow types, $n = 2$, $c(0) = 1$, $c(1) = 0$, and we write $a \to b$ for $f(a, b)$. Constraints are inequalities between finite terms; by Lemma 2.6 they are flattened, with the names of the subterms of the antecedent visible and those of the consequent hidden. Plain entailments are taken in the same signature and tree class.

12.1 Subsumption

A *type scheme* $S = \forall \bar{a}.\, C \Rightarrow (\Gamma \Rightarrow \tau)$ consists of a finite set $\bar{a}$ of bound type variables, a finite conjunction C of inequalities between terms, a *context* Γ, that is, a finite map from program variables to terms, and a term τ. Its other type variables are *free*. For a valuation η of the free variables in $\mathcal{D}$, the *instances* of S under η are the pairs $(\rho \circ \Gamma, \rho(\tau))$, where ρ extends η to $\bar{a}$ with values in $\mathcal{D}$, satisfies C, and $\rho \circ \Gamma$ maps each v in $\operatorname{Dom}\Gamma$ to $\rho(\Gamma(v))$. Following Trifonov and Smith [1996, Section 4.2], an instance (Δ, T) is below an instance (Δ', T') if $T \leq_c T'$, $\operatorname{Dom}\Delta \subseteq \operatorname{Dom}\Delta'$, and $\Delta'(v) \leq_c \Delta(v)$ for every $v \in \operatorname{Dom}\Delta$; contexts are compared contravariantly. Then $S_1 \leq S_2$ if, under every valuation of the free variables, every instance of S_2 is above some instance of S_1.

Proposition 12.1 (Subsumption formula). *Let $S_i = \forall \bar{a}_i.\, C_i \Rightarrow (\Gamma_i \Rightarrow \tau_i)$ for $i = 1, 2$, where the bound variables are renamed so that $\bar{a}_1$ and $\bar{a}_2$ are disjoint from each other and from the free variables of both schemes. Let the visible variables be $\bar{a}_2$ and the free variables, and let $\bar{a}_1$ be hidden.*

(1) If $\operatorname{Dom}\Gamma_1 \subseteq \operatorname{Dom}\Gamma_2$, then

$$S_1 \leq S_2 \iff C_2 \models_{\mathcal{D}} \exists \bar{a}_1\, (C_1 \wedge R_{12}), \qquad R_{12} = (\tau_1 \leq \tau_2) \wedge \bigwedge_{v \in \operatorname{Dom}\Gamma_1} \Gamma_2(v) \leq \Gamma_1(v).$$

(2) If $\operatorname{Dom}\Gamma_1 \not\subseteq \operatorname{Dom}\Gamma_2$, then $S_1 \leq S_2$ iff C_2 has no solution in $\mathcal{D}$.

Proof. A valuation of the visible variables that satisfies C_2 is the same as a valuation η of the free variables together with an instance of S_2 under η. Since $\bar{a}_1$ is disjoint from the visible variables, an extension of this valuation to $\bar{a}_1$ that satisfies C_1 is the same as an instance of S_1 under η, and R_{12} together with the domain condition says that this instance is below the given instance of S_2. This proves (1). If the domain condition fails, no instance of S_1 is below an instance of S_2, so $S_1 \leq S_2$ holds iff S_2 has no instance. □

Both renamings are needed. For $S_1 = b$, with b free, and $S_2 = \forall b.\, b$, with empty contexts, the formula read without renaming $\bar{a}_2$ would be the valid entailment true $\models b \leq b$; but at $b = \top$ the instance $\bot$ of S_2 lies above no instance of S_1, so $S_1 \not\leq S_2$.

This is Eq. (1) with contexts. The formula has no restriction on the occurrences of variables: in a scheme for a higher-order program, a type variable typically occurs in the domain and in the codomain of arrows, and then at both polarities.

Corollary 12.2 (Deciding subsumption). *For $\{\bot, f, \top\}$ with any arity and variance, in particular for arrow types over $\bot$, $\to$ and $\top$, and for $\{\bot, f\}$ with f covariant, subsumption of type schemes is decidable in each tree class, in 2-EXPSPACE. Over regular trees, the setting of Trifonov and Smith [1996], it agrees with subsumption over arbitrary trees.*

Proof. By Proposition 12.1 and Lemma 2.6, subsumption is an existential entailment between flat systems of linear size, or a satisfiability test. The first is decided by Theorem 11.2, within the bound of Theorem 13.2, and the second in polynomial time (proof of Theorem 13.2). Regular and arbitrary trees agree for both questions (Theorem 10.11). □

12.2 Simplification

A system $C(\bar{x}, \bar{h})$ may be replaced by $C'(\bar{x}, \bar{k})$, where $\bar{x}$ remain observable, if $\exists \bar{h}\, C \iff \exists \bar{k}\, C'$. The forward implication is the existential entailment $C \models \exists \bar{k}\, C'$ with $\bar{h}$ visible, and the reverse one exchanges the systems; equivalence of type schemes is subsumption in both directions (Proposition 12.1) and need not preserve the solutions on every type variable. These are the correctness conditions of constraint simplification and minimization [Eifrig et al. 1995; Pottier 1996, 2001; Trifonov and Smith 1996], and by Theorem 11.2 they are decidable, for arrow types without any restriction on polarity.

Some simplifications need only plain entailment. Deleting a literal from a system preserves its full solution set iff the remaining literals entail it, since the other inclusion holds trivially; replacing a term under an entailed equality, or checking a conjunction without hidden variables, likewise reduces to one plain entailment per literal. For the equivalence of projections above, plain entailment of the deleted literal is a sufficient test but not a necessary one: deleting $h = \bot$, for a hidden h that occurs nowhere else, preserves the projection, although $\mathrm{true} \not\models h = \bot$. Existential entailment is needed when a transformation changes the hidden variables: eliminating or merging internal variables, or replacing a scheme by one with different bound variables (Example 12.5).

12.3 Direct Reductions to Plain Entailment

For the following classes of consequents, existential entailment reduces to satisfiability tests and finitely many plain entailments, for every variance and in each tree class, without the machinery of Sections 5 to 9. Visible terms may be arbitrary, including terms in which a variable occurs at both polarities. Plain entailment is decided over $\{\bot, f, \top\}$ by Zhao [2026] (Theorem 2.9) and over $\{\bot, f\}$ by Theorem 11.2. Appendix G states the precise conditions, with the caveats for finite trees, and gives the proofs.

- *Flat bound witnesses* (Proposition G.2). The hidden variables occur only in inequalities among themselves and with terms over X. The entailment holds iff φ entails the other literals and $s \leq_c t$ for every lower bound s and upper bound t joined by a chain of hidden variables; joins of lower bounds are witnesses.
- *Unipolar witnesses* (Proposition G.3). A hidden variable all of whose occurrences are negative can be replaced by $\bot$, and one all of whose occurrences are positive by $\top$; over $\{\bot, f\}$ the greatest tree Ω replaces $\top$, except over finite trees.
- *Functional witnesses* (Proposition G.4). The consequent defines every hidden variable by an equation, and every cycle of definitions passes through a constructor. The hidden variables are then determined by the visible ones, and the entailment is a plain entailment with the hidden variables universally quantified; over finite trees a cycle of definitions instead makes it equivalent to the absence of a finite solution of φ.

- *Mixed sandwich* (Proposition G.5). For the arrow, the entailment of $\exists z\,[\,x \le (z \to z) \wedge (z \to z) \le y\,]$, whose hidden variable occurs at both polarities, reduces by a case split on the shapes of x and y to two satisfiability tests and six plain entailments.

The running example (Example 1.1) lies outside all these classes. Its hidden variable z occurs in $z \le f(z,z)$, so it is bounded by a term that is not over X; it occurs negatively in $z \le y$ and positively in $x \le z$, so it is not unipolar; ψ defines z by no equation; and the mixed sandwich concerns the arrow. The procedure of Sections 5 to 9 handles it (Examples 5.5, 6.7 and 7.11).

12.4 Examples

Example 12.3 (Two identity schemes). Let $S_1 = \forall\alpha.\,\alpha \to \alpha$ and $S_2 = \forall\beta\gamma.\,(\beta \le \gamma) \Rightarrow \beta \to \gamma$, with empty contexts. By Proposition 12.1, $S_1 \le S_2$ iff $\beta \le \gamma \models \exists\alpha\,(\alpha \to \alpha \le \beta \to \gamma)$. Both sides of the inequality are arrows, so it says $\beta \le_c \alpha$ and $\alpha \le_c \gamma$, and $\alpha = \beta$ is a witness. Conversely, $S_2 \le S_1$ iff $\models \exists\beta\gamma\,(\beta \le \gamma \wedge \beta \to \gamma \le \alpha \to \alpha)$, with α visible, that is, iff some β, γ satisfy $\alpha \le \beta \le \gamma \le \alpha$, and $\beta = \gamma = \alpha$ is a witness. So the two schemes are equivalent. The hidden variable α of the first entailment occurs in the domain and in the codomain of the same arrow, so it has both polarities, and after flattening the literal $f(\alpha,\alpha) \le u$ admits no parity assignment (Remark 4.5), which would need $\pi(\alpha) = \pi(u) \oplus 1$ and $\pi(\alpha) = \pi(u)$.

Trifonov and Smith [1996] decide subsumption by a sound approximation through canonical bounds, and simplifiers prove entailments by syntactic reasoning about the bounds of variables; such tests are incomplete [Pottier 2001]. The following example of Pottier [1998b, Section 8.5] shows a correct simplification that reasoning through lower bounds misses.

Example 12.4 (A missed redundant constraint). For arrow types put

$$g = \bot \to \top, \qquad k = g \to \bot, \qquad C(a) = (a \to \bot \le a).$$

Then $C \models k \le a$ in each tree class. Indeed $a \ne \bot$, since $a \to \bot \le_c \bot$ is false. If $a = \top$, then $k \le_c a$. Otherwise a is an arrow $a_0 \to a_1$, so $a \le_c g$, because g is the greatest arrow (Example 3.3); by contravariance $k = g \to \bot \le_c a \to \bot \le_c a$. Thus adding or deleting the literal $k \le a$ is an exact simplification, and equivalently $C \models \exists z\,(k \le z \wedge z \le a)$, with the witness $z = k$ (Proposition G.2). The variable a has the single lower bound $a \to \bot$ and the upper bound $\top$ in C. Proving $k \le a$ through this lower bound requires $k \le a \to \bot$, that is, $a \le g$, and this is not entailed: $a = \top$ satisfies C. A test that proves an inequality $k \le a$ only through a lower bound of a therefore fails on this true simplification.

The antecedent mentions a at both polarities, and it admits no parity assignment. The example also shows why the signed system cannot be read on its own. The signed literals of C (Table 2) are $f(a^-,\bot) \le a^+$ and $a^- \le f(a^+,\top)$, read covariantly. Put $F = f(\bot,\bot)$. The covariant values $a^+ = F$ and $a^- = \bot$ satisfy them, but they violate $a^- = (a^+)^\partial$, the condition of Lemma 2.5. The signed form of $k \le a$ requires $\nu_0 k \le a^+$ (Lemma 2.2(2)), and $\nu_0 k = f(\nu_1 g,\bot) = f(F,\bot)$ by Lemma 2.2(1). Since $f(F,\bot) \not\le F$, a test that drops the coupling between a^+ and a^- reports a countermodel of a true entailment. The leaf cones of Section 3 keep this coupling: the two cones of a are the cones of one tree, and the exclusions and duality of Section 3.4 enforce it.

The last example concerns a merge of type variables, which changes the bound variables of a scheme and so needs existential entailment.

Example 12.5 (Merging the variables of self-application). The term $\lambda x.\,x\,x$ has the inferred scheme

$$S = \forall\alpha\beta.\,(\alpha \le \alpha \to \beta) \Rightarrow \alpha \to \beta,$$

in which α is both a function and its own argument. A simplifier might merge β into α and propose $S' = \forall\gamma.\,(\gamma \le \gamma \to \gamma) \Rightarrow \gamma \to \gamma$. By Proposition 12.1, $S \le S'$ iff

$$\gamma \le \gamma \to \gamma \;\models\; \exists\alpha\beta\,(\alpha \le \alpha \to \beta \;\wedge\; \alpha \to \beta \le \gamma \to \gamma),$$

which holds with the witness $\alpha = \beta = \gamma$. But $S' \le S$ iff

$$\alpha \le \alpha \to \beta \;\models\; \exists\gamma\,(\gamma \le \gamma \to \gamma \;\wedge\; \gamma \to \gamma \le \alpha \to \beta),$$

and the second literal says $\alpha \le_c \gamma \le_c \beta$. The values $\alpha = \top \to \bot$ and $\beta = \bot$ satisfy the antecedent: the domain $\top$ of α lies above α, and its codomain $\bot$ below β. No γ lies between the arrow α and $\bot$, so the entailment fails, and the instance $(\top \to \bot) \to \bot$ of S, the type of $\lambda x.\,x\,x$ applied to arguments that accept everything and return $\bot$, lies above no instance of S'. The merge loses generality. In both entailments every hidden variable has both occurrence signs in the consequent, and a hidden variable is bounded by an arrow in which it occurs itself, $\alpha \le \alpha \to \beta$ and $\gamma \le \gamma \to \gamma$; so none of the classes of Section 12.3 applies, and Theorem 11.2 decides both.

The semantics matters as well: Appendix G.1 gives a covariant entailment over $\{\bot, f, \top\}$ that holds for trees but fails in an extensible constructor algebra, so a test that is sound for such algebras cannot certify the corresponding simplification of tree constraints.

13 Complexity

This section bounds the complexity of the decision and proves lower bounds. Throughout, $s = |X| + |Z|$ is the number of variables after flattening (Section 2.3), $v = |X|$, and $E_1(x) = 2^x$, $E_{k+1}(x) = 2^{E_k(x)}$. The upper bounds of Section 13.1 hold for every arity n and every variance c, with constants that do not depend on n; the arity enters only the size of the alphabet and of the input. The lower bounds of Section 13.2 are proved for a covariant constructor of arity at least two.

13.1 Upper Bounds

Witness lengths. When both systems are satisfiable in the tree class considered and the entailment fails, a counterexample word (Definition 11.1) of length at most $E_3(10v + 6s + 18) \le E_3(16s + 18)$ exists over $\{\bot, f, \top\}$, and of length at most $E_3(20v + 12s + 18) \le E_3(32s + 18)$ over $\{\bot, f\}$ with f covariant (Theorem 11.2); every counterexample word yields a countermodel effectively (Lemma 11.3). Proposition 11.5 and Table 7 bound the kinds separately. Words for (I0) and holes that are not mortal are singly exponential in s; for the latter the pump keeps the purely periodic case polynomial in the sizes of the automata (Theorem 8.14), for every instance of the two-word problem with self events that satisfies (FH). The words for (U) and the mortal holes need the triply exponential estimates of compression. All these bounds are existential: they bound a shortest counterexample, not every one.

Searching without storing the word. Enumerating all words up to the bound of Theorem 11.2 decides the problem in triply exponential space. The words need not be stored. The compression of Theorem 9.14 only ever uses a dictionary of boundedly many blocks per level, and a level is described, for the purpose of the levels above it, by the derived monitor and channel on these blocks, which the column summaries determine (Lemma E.8). A nondeterministic search can therefore build the hierarchy from the bottom, one level at a time, keeping only the current derived tables.

Proposition 13.1 (Summary search). *Let $\mathcal{M}_0$ be a lettered monitor with $N \ge 1$ states over a finite alphabet Γ, let Θ_0 be a positive channel on it, both given by their tables, and let $d \ge 0$. Whether $(\mathcal{M}_0, \Theta_0)$ has a positive hole of depth at most d is decidable in nondeterministic space polynomial in $N + d + 1$, $\mathrm{G}^\sharp(N, d)$ and $|\Gamma|$, hence in deterministic space $2^{O((N+d+1)^3)} \cdot \mathrm{poly}(|\Gamma|)$.*

Proof. Put $g = \mathrm{G}^{\sharp}(N, d)$. We use the summaries, column words and agreement of Appendix A.4 and their extensions by channels in Appendix E. The search keeps a level $j \le d$ and a *current pair* $(\mathcal{W}, \Theta)$: a lettered monitor with the N states of $\mathcal{M}_0$ and j symbolic states, and a positive channel on it, over a current alphabet. Initially $j = 0$ and $(\mathcal{W}, \Theta) = (\mathcal{M}_0, \Theta_0)$, over Γ; for $j \ge 1$ the current alphabet is $\{1, \dots, h\}$ with $h \le g$. The search repeats one of two moves.

- *Accept* if some letter z and some $m \le 3(N + j)$ make z^m a positive hole of $(\mathcal{W}, \Theta)$. This is checked by computing the labels of z^m and testing every pair of cuts.
- *Derive*, if $j < d$: choose a marker ℓ and a number $1 \le h \le g$, and guess a valid column word with h tracks over the current alphabet without ℓ, one column at a time, while computing the positive summary of the tracks read so far by the automaton of Lemma E.8, starting from the summary of h empty blocks; a binary counter stops the guess before $\mathrm{col}^{\sharp}(N + j, h)$ columns. Require that the final equality fields declare the h blocks pairwise distinct. Replace $(\mathcal{W}, \Theta)$ by the restriction of $(\mathcal{W}_\ell, \Theta_\ell)$ to the h blocks, named $1, \dots, h$, which the final summary determines at every state (Appendix A.4, step 4, and Lemma E.8(2)), and increase j.

Space. A current pair over $h \le \max(g, |\Gamma|)$ letters with $M \le N + d + 1$ states is given by its transitions, final targets, the relations Λ, Λ^{fin}, R, S, S' on labels, the predicate F and the set Φ, in $O(M^2h^2 + Mh \log M)$ bits. A positive summary has at most $\mathrm{col}^{\sharp}(M, h)$ values, hence $O(M^2h^2 + Mh \log M)$ bits, and so does the counter; a column has h entries. Only the current pair, one summary, one column, the counter and j are stored. Since $g \le 2^{\Xi^{\sharp}(N,d)}$ by Theorem 9.14 and $\Xi^{\sharp}(N, d) \le 30(N + d + 1)^3$, the nondeterministic space is polynomial in $N + d + 1$, g and $|\Gamma|$, and Savitch's theorem squares it, which keeps the stated form.

Soundness. We show by induction on the number of moves that there are markers $\widetilde{\ell_0}, \dots, \widetilde{\ell_{j-1}}$ and an injective map ρ_j from the current alphabet into the letters of level j, with a valid image family for these markers (Appendix A.4), such that $(\mathcal{W}, \Theta) \equiv_{E,\rho_j} (\mathcal{M}_j^{\widetilde{\ell}}, \Theta_j^{\widetilde{\ell}})$, where E is the current alphabet. Initially ρ_0 is the identity. In a derive move, the guessed column word spells blocks $Y_1, \dots, Y_h$ over E that avoid ℓ; by Lemma E.8(1) the computed summary is theirs, so the new pair is the restriction of $(\mathcal{W}_\ell, \Theta_\ell)$ to $\{Y_1, \dots, Y_h\}$. By Lemma E.7, $(\mathcal{W}_\ell, \Theta_\ell) \equiv_{\{Y_a\},\rho_j^*} ((\mathcal{M}_j^{\widetilde{\ell}})_{\rho_j(\ell)}, (\Theta_j^{\widetilde{\ell}})_{\rho_j(\ell)})$, and the images avoid $\rho_j(\ell)$. Put $\widetilde{\ell_j} = \rho_j(\ell)$ and $\rho_{j+1}(a) = \rho_j^*(Y_a)$, which is injective because the Y_a are distinct and ρ_j is injective. If the search accepts with z^m, then $\rho_j(z)^m$ is a positive hole of $(\mathcal{M}_j^{\widetilde{\ell}}, \Theta_j^{\widetilde{\ell}})$, since agreement preserves positive holes (Appendix E.5). Expanding it along the markers gives, by Lemmas 2.24 and 9.10, a positive hole w of $(\mathcal{M}_0, \Theta_0)$. Each expansion ends with its marker and its blocks avoid it, so the canonical hierarchy of w recovers the expanded levels and reaches the unary word $\rho_j(z)^m$ at level j, and $\mathrm{depth}(w) \le j \le d$.

Completeness. Suppose a positive hole of depth at most d exists. By Lemma E.5 there is one, w, of depth $D \le d$ whose dictionaries satisfy $g_i \le g$ for $1 \le i \le D$; let (w_j, ℓ_j) be its hierarchy, $\mathcal{A}_j$ its dictionaries and $(\mathcal{M}_j, \Theta_j)$ its towers. We follow a run in which, for $1 \le j \le D$, the current pair is the restriction of $(\mathcal{M}_j, \Theta_j)$ to $\mathcal{A}_j$, up to the names of the letters. At level $j < D$, choose $\ell = \ell_j$ and $h = g_{j+1}$. The letters of $\mathcal{A}_{j+1}$ are h distinct blocks over $\mathcal{A}_j$ that avoid ℓ_j, and their positive summary depends only on the tables at the letters of $\mathcal{A}_j$, so it is also their summary for the current pair. By the deletion argument in the proof of Lemma E.10, some valid column word with fewer than $\mathrm{col}^{\sharp}(N + j, h)$ columns has the same final summary; the search guesses it, and the new pair is the restriction of $(\mathcal{M}_{j+1}, \Theta_{j+1})$ to $\mathcal{A}_{j+1}$, because the summary determines it. At level D, $w_D = z^m$ is a positive hole of $(\mathcal{M}_D, \Theta_D)$ by Corollary 9.11, hence of the current pair, and Lemma E.6 shortens it to $m \le 3(N + D)$. The search accepts. □

The *trivial channel* $(Q \times \Gamma, \varnothing, \varnothing, \varnothing, \varnothing)$ makes every nonempty word positive, so its positive holes are the holes of the monitor, and Proposition 13.1 also decides whether a monitor has a nonempty hole of depth at most d.

Theorem 13.2 (Complexity of the decision). *Existential entailment over $\{\bot, f, \top\}$, for every arity and variance, and over $\{\bot, f\}$ with f covariant, is decidable in each of the three tree classes in deterministic space $E_2(O(s)) \cdot \mathrm{poly}(m)$, where m is the size of the input, with the arity, the variance and the systems given explicitly. It is therefore in 2-EXPSPACE and in 3-EXPTIME.*

Proof. By Lemma 2.6, flattening is linear, so let the systems be flat. Over $\{\bot, f\}$, apply the reduction of Theorem 10.16, which is polynomial and doubles s (Corollary 10.17). For $n = 0$ enumerate the 3^s assignments (Remark 10.19), in polynomial space. Let $n \geq 1$.

Satisfiability. For each class $\mathcal{D}$, $\mathrm{Sat}_{\mathcal{D}}$ of a system χ is decided in polynomial time: by Lemma A.2, it is satisfiability of the signed system of the system $\chi^\flat$ of Lemma 2.7, and by Theorem A.1(1) this is the absence of a label clash, and over finite trees also of a cycle clash, which are closure and graph tests of polynomial size; regular satisfiability is satisfiability. If φ is not $\mathcal{D}$-satisfiable, the entailment holds; if φ is and ψ is not, it fails. Otherwise both are satisfiable, and Theorems 10.11 and 10.12 reduce the problem to arbitrary trees.

The three conditions. By Corollary 7.8, the entailment holds iff (I0) and (U) hold and no extracted instance has a hole. Condition (I0) is decided in polynomial space (Proposition 5.17). For (U), compute the tables of the reduced unit readers $\widehat{\mathcal{R}}_p$ of Lemma 6.9, with fewer than $K_{\mathrm{U}} = 2^{v+s+1}$ states each, in space $2^{O(s)}\mathrm{poly}(m)$ (Lemma 6.3). By Lemmas 6.5 and 6.9, (U) fails iff some $\widehat{\mathcal{R}}_p$ has a hole. The empty word is tested directly. As in the proof of Theorem 6.11, by Theorem 2.15 and Lemma 6.10, a reduced unit reader with a hole has the empty hole or a nonempty hole of depth at most $D_{\mathrm{U}} = 2^{v+2}(4K_{\mathrm{U}}-1)-1$, and its nonempty holes are those of its lettered monitor (Definition 2.22). So Proposition 13.1 with the trivial channel decides (U) in space $2^{O((K_{\mathrm{U}}+D_{\mathrm{U}}+1)^3)}\mathrm{poly}(n) = E_2(O(s))\,\mathrm{poly}(n)$.

For (II), fix $p, q \in X^{\pm}$. The instance I_{pq} satisfies (FH), so by Corollary 8.18 it has a hole iff it has a hole with $|u| + |t| < \mathsf{L}_{\mathrm{nm}}(N_A, N_B, N_{A'}, N_{B'})$, a bound singly exponential in s (Proposition 11.5), or a mortal hole. The first alternative is decided by enumerating such pairs and checking the conditions of Definition 7.1 cut by cut, in space $2^{O(s)}\mathrm{poly}(m)$. For the second, by Lemmas 9.5 and 9.6, I_{pq} has a mortal hole iff the reachable part of $\mathcal{R}_I^{\Pi}$ has a Π-hole of depth at most d_I; this part has $\bar{N}_I \leq 2^{2v+2s+3}$ states (proof of Corollary 9.19), so $d_I < 2^{4v+2s+4}$, and its tables, including the mortality of the states of A, are computed by reachability in space $2^{O(s)}\mathrm{poly}(m)$. Its Π-holes are the positive holes of its lettered monitor with the base channel (Lemma 9.8), and Proposition 13.1 decides their existence in space $E_2(O(s))\,\mathrm{poly}(n)$.

Total. All tests together use deterministic space $E_2(O(s))\,\mathrm{poly}(m)$, and $s \leq m$. A deterministic machine in this space has at most $2^{E_2(O(s))\mathrm{poly}(m)}$ configurations, which gives the time bound. □

Remark 13.3 (Plain entailment). Plain entailment $\varphi \models_{\mathcal{D}} x \leq y$ is the case $\psi = \{x \leq y\}$, $Z = \varnothing$. Theorem 13.2 therefore also places plain entailment, for every arity and variance and in each tree class, in 2-EXPSPACE. The earlier procedure for plain entailment places non-entailment in N3EXPTIME and entailment in coN3EXPTIME, hence in 3EXPSPACE [Zhao 2026, Corollary "Complexity upper bounds"]; the summary search improves the deterministic space upper bound to 2-EXPSPACE, one exponential lower.

Arity one. For $n = 1$ there is a single letter, words are determined by their lengths, and a suffix comparison is a comparison of lengths. Here a direct shortening gives singly exponential witnesses for every instance, mortal or not.

Theorem 13.4 (Arity one). *Let $n = 1$, with either variance.*

(1) *Let I be an instance of the two-word problem with self events that satisfies (FH), and let $r_I \le N_A N_{A'}$ be the number of states of $A \times A'$ reachable from the start. If I has a hole, it has a hole (u, t) with $|t| \le N_B - 1$ and $|u| + 1 + |t| \le r_I + 2N_B - 2$.*

(2) *Every reader with N states over a one-letter alphabet that has a hole has a hole of length at most $N - 1$.*

(3) *Let φ and ψ be flat systems over $\{\bot, f, \top\}$ with s variables. If they are $\mathcal{D}$-satisfiable and $\varphi \not\models_{\mathcal{D}} \exists \bar{z}\, \psi$, a counterexample word of length less than 2^{4s+3} exists. Over $\{\bot, f\}$ with f covariant, if φ and ψ are $\mathcal{D}$-satisfiable over $\{\bot, f\}$ and the entailment fails, the pair $(\varphi^\natural, \psi^\natural)$ of Theorem 10.16, which has $2s$ variables, has a counterexample word of length less than 2^{8s+3}. Existential entailment at arity one is in PSPACE, for both variances and in each tree class, over $\{\bot, f, \top\}$ and, for covariant f, over $\{\bot, f\}$.*

Proof. Write the letter as 0. For words 0^a and 0^b and cuts $x \le a$, $y \le b$, the pair (x, y) is a cross comparison of $(0^a, 0^b)$ iff $b - y \le a - x$, and it is aligned iff $b - y = a - x$; every pair of cuts $i < j$ of a word is a comparison.

(1) Let (u, t) be a hole, $J = J(t)$, and let (x, e) be a head event; by (FH), $e \le J$ (Lemma 7.4). Since $|u| - x = |t| - e \ge |t| - J$, we may delete $|t| - J$ letters from the end of both words: let u' be the prefix of u of length $L = |u| - (|t| - J)$ and t' the prefix of t of length J. The cuts of u' and t' are cuts of u and t with the same states, and $J(t') = J$. For $x \le L$ and $y \le J$, the pair (x, y) is a cross comparison of (u', t') iff $J - y \le L - x$, iff $|t| - y \le |u| - x$, iff it is one of (u, t); so (u', t') has no event, t' has no self event, and the head event (x, e) is a head event of (u', t'), with $x \le L$. The spine u' is safe. So (u', t') is a hole whose target is first rejected at its end.

The states $B_0(t'), \ldots, B_J(t')$ are pairwise distinct: if $B_i = B_j$ with $i < j \le J$, the run of the single letter repeats with period $j - i$ from i on, so $B_{J-(j-i)}(t') = B_J(t') \in \mathrm{Rej}$ with $J - (j - i) < J$, contradicting the choice of J. Hence $|t'| = J \le N_B - 1$.

If $L < J$, then $|u'| + 1 + |t'| \le 2J \le 2N_B - 2$. Otherwise let m be the least cut of u' at which $A \times A'$ is in the state it has at the cut $L - J$. The states of $A \times A'$ at the cuts $0, \ldots, m$ are pairwise distinct, so $m \le r_I - 1$: if two of them coincided, the run would be periodic from the first of them, and the state at the cut $L - J$ would occur before m, contrary to the choice of m. Let $u'' = 0^{m+J}$. Its states at the cuts $x \le m$ are those of u', and at the cut $m + y$, $y \le J$, those of u' at $L - J + y$. The pair (u'', t') is a hole. Its spine is safe, since all its states occur on u'. The head event (x, e) has $x = L - J + e$ and becomes the aligned pair $(m + e, e)$, with the same states. A cross comparison (x, y) of (u'', t') with $x \ge m$ corresponds to the cross comparison $(L - J + x - m, y)$ of (u', t'), with the same states, and one with $x < m$ has the same states as the pair (x, y) of (u', t'), which is a cross comparison because $L - x > L - m \ge J \ge J - y$; neither is an event. The target and its self events are unchanged. Finally $|u''| + 1 + |t'| \le r_I - 1 + 2J + 1 \le r_I + 2N_B - 2$.

(2) Let w be a hole with first rejected cut J. Its prefix of length J is rejected, and its comparisons are pairs of cuts $i < j \le J$, which are comparisons of w with the same states; so it is a hole. Its states at the cuts $0, \ldots, J$ are pairwise distinct, as in (1), so $J \le N - 1$.

(3) By Proposition 11.5(1), a failure of (I0) has a word of length less than $2^{2v+2s+2}$. A failure of (U) has a (U)-word of length at most $2^{v+1} - 2$ (Remark 6.12). A hole of an extracted instance has one with $|u| + 1 + |t| \le r_I + 2N_B - 2$ by (1), where $r_I \le 2^{2v+2s}$ and $N_B = 2^{2v+2s+1}$. Each bound is less than 2^{4s+3}. Over $\{\bot, f\}$, the pair of Theorem 10.16 is $\mathcal{D}$-satisfiable over $\{\bot, f, \top\}$ and its entailment fails (proof of Theorem 11.2), and it has $2s$ variables, which gives 2^{8s+3}. For membership in PSPACE, follow the proof of Theorem 13.2, which handles satisfiability, the tree classes, $\{\bot, f\}$ and (I0) in polynomial space, and replace the searches for (U) and (II) by an enumeration of the candidates up to these bounds. A unary word is stored by its length, in $O(s)$ bits. The state of an automaton at a cut is a set of signed variables, recomputed from the start in polynomial space, and each condition

of Definitions 2.12 and 7.1 quantifies over cuts, which binary counters enumerate; comparisons are tests of lengths. So all tests use polynomial space. For $\{\bot, f\}$ the gadget doubles s, which keeps the bounds singly exponential. □

Remark 13.5 (Arity zero). For $n = 0$ every tree is $\bot$, $f()$ or $\top$ (Remark 2.8), and the entailment says that every assignment of X that satisfies φ has an assignment of Z that satisfies ψ (Remark 10.19). Assignments have polynomial size and are checked in linear time, so the problem is in Π_2^p, hence in PSPACE, in each tree class, which coincide.

13.2 Lower Bounds

Long witnesses. Shortest counterexamples can be exponentially long, already for plain entailment over $\{\bot, f\}$ with a binary covariant constructor and antecedents without literals $u = \bot$. The construction uses a small automaton for the set of words that are not extensions of one exponentially long word, the record of a binary counter.

Proposition 13.6 (Exponential witnesses). *For every $k \geq 1$ there are a word $W_k \in \{0,1\}^*$ of length $L_k = 2(k+1)2^k$ and a satisfiable flat system φ_k over $\{\bot, f\}$, with f binary and covariant, without hidden variables and without literals $u = \bot$, over $9k + 30$ variables, such that $\varphi_k \models x \leq y$ fails over arbitrary and over regular trees, and:*

(1) every word of $K(\rho x) \setminus K(\rho y)$, for every solution ρ of φ_k, has the prefix W_k, and some regular solution has $W_k \in K(\rho x) \setminus K(\rho y)$;

(2) for the instance of Theorem 10.16 over $\{\bot, f, \top\}$, conditions (I0) *and* (U) *hold, and every hole (u, t) of an extracted instance has $|t| \geq L_k$; so every counterexample word of that instance (Definition 11.1) is a hole word longer than L_k.*

For every arity $n \geq 3$, statements (1) and (2) hold for a system over $\{\bot, f\}$ with f covariant of arity n and $18k + 61$ variables, which may contain literals $u = \bot$. Hence shortest counterexample words can have length $2^{\Omega(s)}$, for every fixed arity $n \geq 2$.

Proof. *A counter.* Over the alphabet $\Gamma = \{0, 1, \#, \$\}$ let $H = C_0 \# C_1 \# \cdots \# C_{2^k-1}\$$, where C_j is the binary expansion of j with k bits, least significant bit first; $|H| = (k+1)2^k$. Consider the nondeterministic automaton $\mathcal{N}$ over Γ, all of whose states are final, with the following states and transitions; a transition that is not listed does not exist.

- E loops on every letter.
- $\mathsf{F}_0, \ldots, \mathsf{F}_k$ check the format: F_i goes on a bit to F_{i+1} if $i < k$ and to E if $i = k$; on # to F_0 if $i = k$ and to E otherwise; on \$ to E if $i < k$.
- I checks that C_0 is zero: it goes on 0 to I and on 1 to E.
- $\mathsf{C}(c)$, $c \in \{0,1\}$, tracks the carry of an increment and spawns checkers: on a bit b it goes to both $\mathsf{C}(c \wedge b)$ and $\mathsf{W}(b \oplus c, k)$, and on # to $\mathsf{C}(1)$.
- $\mathsf{W}(b, j)$, $0 \leq j \leq k$, waits for the corresponding bit of the next block: for $j > 0$ it goes on every letter other than \$ to $\mathsf{W}(b, j-1)$; $\mathsf{W}(b, 0)$ goes on the bit $1 - b$ to E.
- $\mathsf{O}(o)$, $o \in \{0,1\}$, records whether the current block is all ones so far: on a bit b it goes to $\mathsf{O}(o \wedge b)$; on # to E if $o = 1$ and to $\mathsf{O}(1)$ if $o = 0$; $\mathsf{O}(0)$ goes on \$ to E.

The initial states are F_0, I, $\mathsf{C}(1)$ and $\mathsf{O}(1)$, and there are $3k + 9$ states. We claim that the words with a run of $\mathcal{N}$ are exactly the words in $\Gamma^* \setminus H\Gamma^*$.

No branch reaches E on a prefix of H: the format is right; C_0 is zero; a checker spawned at the bit of offset p in block C_i reads the k letters between it and the bit of offset p in C_{i+1}, which is $b \oplus c$ for the bit b and the incoming carry c, because $C_{i+1} = C_i + 1$; a # follows only blocks that are not all ones; and \$ follows the all-ones block C_{2^k-1}. After H no state remains: F_k, $\mathsf{C}(\cdot)$, $\mathsf{O}(1)$ and the

checkers have no transition on \$, and I has none on #. So no word in $H\Gamma^*$ has a run. Every proper prefix of H has a run through the format states. Let w first differ from H at position $i < |H|$. If $H[i]$ is a bit of C_j and $w[i] \in \{\#, \$\}$, the format branch, at some F_p with $p < k$, reaches E. If $w[i]$ is the other bit, then I reaches E if $j = 0$, and otherwise the checker spawned at the same offset of C_{j-1} is at $\mathsf{W}(H[i], 0)$ and reaches E. If $H[i] = \#$, a bit leads F_k to E, and \$ leads $\mathsf{O}(0)$ to E, since C_j is not all ones. If $H[i] = \$$, a bit leads F_k to E and # leads $\mathsf{O}(1)$ to E. Since E loops, w has a run, which proves the claim.

Binary words. Code the tokens by $0 \mapsto 00$, $1 \mapsto 01$, $\# \mapsto 10$, $\$ \mapsto 11$, and let W_k be the code of H, of length L_k. Let $\mathcal{L}$ have the states q, $(q, 0)$ and $(q, 1)$ for every state q of $\mathcal{N}$, all final, the initial states of $\mathcal{N}$, a transition on the bit b from q to (q, b), and a transition on the bit b' from (q, b) to every successor of q on the token coded by bb'. It has $9k + 27$ states. By induction on the number of complete codes, the base states reachable on the code of a token word are the states reachable on that word in $\mathcal{N}$, and after one further bit the reachable set is nonempty iff that set is. So a binary word has a run of $\mathcal{L}$ iff its longest prefix of even length is the code of a token word with a run of $\mathcal{N}$, that is, iff it does not begin with W_k. The language of $\mathcal{L}$ is therefore $\{0,1\}^* \setminus W_k\{0,1\}^*$.

The system. Let φ_k have the variables x, y, d and Y_q for the states q of $\mathcal{L}$, and the literals

$$f(d,d) \le Y_q, \qquad f(Y_r,d) \le Y_q \ \ (q \xrightarrow{0} r), \qquad f(d,Y_r) \le Y_q \ \ (q \xrightarrow{1} r), \qquad Y_q \le y \ \ (q \text{ initial}).$$

In every solution, $K(Y_q)$ contains every word that has a run of $\mathcal{L}$ from q, by induction on its length, using Lemma 2.3; hence $K(y) \supseteq \{0,1\}^* \setminus W_k\{0,1\}^*$, and every word of $K(x) \setminus K(y)$ begins with W_k. Conversely, let $K(Y_q)$ be the set of words with a run from q, $d = \bot$, $K(y)$ the language of $\mathcal{L}$, and $K(x)$ the set of prefixes of W_k. These sets are regular and prefix closed, since all states are final, and they satisfy every literal, so this is a regular solution with $W_k \in K(x) \setminus K(y)$. This proves (1); the antecedent is satisfiable and the entailment fails over arbitrary and regular trees (Theorem 10.11).

(2) The consequent is $\psi = \{x \le y\}$ and $Z = \varnothing$, so $\psi^\natural = \psi$ has no facts and no forbidden roots: (I0) and (U) hold. Its pure moves are the copies $(x^0, w) \to (y^0, w)$ and $(y^1, w) \to (x^1, w)$. Let (u, t) be a hole of an extracted instance I_{pq} of $\varphi^\natural$ and ψ. By Theorem 7.7, $(p, t) \to^* (q, u)$ by pure moves of ψ, and the least solution ρ of $\varphi^\natural$ that contains (p, t) omits (q, u); so $u = t$ and (p, q) is (x^0, y^0) or (y^1, x^1). By Lemma 10.15(1), ρx and ρy are $\top$-free, so their positive cones are empty, and $p = y^1$, $q = x^1$. For $\top$-free covariant trees the negative cone is the complement of the constructor positions (Remark 4.6), so $t \in K(\rho x) \setminus K(\rho y)$. The restriction of ρ to X solves φ_k over $\{\bot, f\}$, and (1) gives $|t| \ge L_k$.

For $n \ge 3$, apply Proposition 13.8 below: the solutions of the translated antecedent are, on the original variables, the images of the solutions of φ_k, with the same sets of constructor positions, and the translation has $2(9k + 30) + 1$ variables; it contains the literal $o = \bot$. Statement (2) carries over by the same argument. Finally $s \le 18k + 61$ and $\log_2 L_k \ge k + 1$, so $L_k \ge 2^{(s-61)/18}$. □

The antecedents have no finite solutions, since $K(y)$ is infinite, so the proposition gives no lower bound over finite trees.

Larger arities. A lower bound for a binary constructor does not transfer to a constructor of larger arity by *bare padding*, that is, by filling the new arguments of every constructor with $\bot$. The translated antecedent constrains values that are not translations of binary trees, and such values may refute the translated entailment.

Example 13.7 (Bare padding fails). Over $\{\bot, f\}$ with f_2 binary and covariant, $\text{true} \models \exists a\, b \ \ x \le f_2(a, b)$ holds in every tree class: take the children of x, or $a = b = \bot$ if $x = \bot$. With f_3 ternary and the new argument filled by $\bot$, the entailment $\text{true} \models \exists a\, b \ \ x \le f_3(a, b, \bot)$ fails at the finite tree $x = f_3(\bot, \bot, f_3(\bot, \bot, \bot))$. In the same way, padding the antecedent of Proposition 13.6 leaves

x unconstrained, and a value of x with a constructor at the one-letter word 2, which lies outside $K(y)$, is a countermodel with a witness of length one.

The repair restricts every variable to the image of the binary trees, by a variant of the gadget of Section 10.5.

Proposition 13.8 (Embedding binary trees). *Let $n \geq 2$, let f be covariant of arity n, and let f_2 be binary and covariant. For a tree t over $\{\bot, f_2\}$ let $E(t)$ be the tree over $\{\bot, f\}$ with $K(Et) = K(t)$. For a system χ over $\{\bot, f_2\}$, let χ^E replace every literal $f_2(a_0, a_1) \leq u$ by $f(a_0, a_1, o, \ldots, o) \leq u$ and every literal $u \leq f_2(a_0, a_1)$ by $u \leq f(a_0, a_1, o, \ldots, o)$, for a fresh variable o. For φ over X and ψ over $X \cup Z$, put*

$$\varphi^{(n)} = \varphi^E \cup \{o = \bot\} \cup \bigcup_{x \in X} B(x, m_x), \qquad \psi^{(n)} = \psi^E \cup \bigcup_{z \in Z} B(z, m_z),$$

where $B(v, m) = \{v \leq m,\ m \leq f(m, m, o, \ldots, o)\}$, with the visible variables $X \cup \{o\} \cup \{m_x\}_{x \in X}$ and the hidden variables $Z \cup \{m_z\}_{z \in Z}$. Then:

(1) E is a bijection from the trees over $\{\bot, f_2\}$ onto the trees over $\{\bot, f\}$ whose constructor positions lie in $\{0,1\}^*$; it preserves and reflects the order, maps each tree class onto its part of the image, and $E(f_2(a, b)) = f(Ea, Eb, \bot, \ldots, \bot)$;

(2) for every tree class $\mathcal{D}$, $\varphi \models_{\mathcal{D}} \exists \bar{z}\, \psi$ over $\{\bot, f_2\}$ iff $\varphi^{(n)} \models_{\mathcal{D}} \exists \bar{z}\, \bar{m}\, \psi^{(n)}$ over $\{\bot, f\}$, and the restrictions to X of the solutions of $\varphi^{(n)}$ are exactly the images under E of the solutions of φ.

The translation has $2s + 1$ variables and size linear in the input for fixed n.

Proof. (1) Prefix-closed subsets of $\{0,1\}^*$ are exactly the sets of constructor positions of binary trees and of the trees in the image, and the order on both sides is inclusion of these sets (Lemma 2.3). A tree over $\{\bot, f\}$ is finite iff this set is finite, and regular iff it is a regular language (Theorem 3.2(4)), on both sides.

(2) Let m be a tree over $\{\bot, f\}$ with $m \leq f(m, m, \bot, \ldots, \bot)$. Then $K(m) \subseteq \{\varepsilon\} \cup 0K(m) \cup 1K(m)$, and by induction on length every word of $K(m)$ lies in $\{0,1\}^*$. Conversely, the regular tree Θ with $K(\Theta) = \{0,1\}^*$ and the finite trees Θ_h with $K(\Theta_h) = \{0,1\}^{<h}$ satisfy this literal, and every tree in the image of E lies below Θ, and below Θ_h if it is finite with all positions shorter than h. So, with $o = \bot$, a value of v admits a value of m with $B(v, m)$ in the class $\mathcal{D}$ iff it lies in the image of E. On this image, a literal of χ^E with $o = \bot$ holds for $E \circ \sigma$ iff the corresponding literal of χ holds for σ, by (1). Both claims of (2) now follow as in the proof of Theorem 10.16, with E in place of the identity on $\top$-free trees. □

Hardness. Henglein and Rehof [1998, Theorems 1 and 2] proved PSPACE-hardness of plain entailment over $\{\bot, f, \top\}$ for a binary covariant constructor. We prove PSPACE-hardness for every fixed arity $n \geq 2$, in each tree class, by explicit reductions from the universality and the inclusion problems of nondeterministic automata over $\{0,1\}$, which are PSPACE-complete [Stockmeyer and Meyer 1973]. For a word $w \in \{0,1\}^*$ let $\mathrm{code}(w) = e(w)1$, where e replaces each letter 0 by 00 and each 1 by 01. The code is prefix free: if $\mathrm{code}(w') \preceq \mathrm{code}(w)$ and $|w'| < |w|$, then position $2|w'|$ carries the final 1 of $\mathrm{code}(w')$ but the letter 0 in $\mathrm{code}(w)$. From an automaton for L one obtains in linear time an automaton for $\mathrm{code}(L)$: replace each transition by two, the first labeled 0, and add a transition labeled 1 from every final state to one new final state.

Theorem 13.9 (PSPACE-hardness). *Let $n \geq 2$ be fixed and f covariant. In each of the three tree classes:*

(1) existential entailment over $\{\bot, f\}$ is PSPACE-hard;

(2) plain entailment over $\{\bot, f, \top\}$ is PSPACE-hard;

(3) *consequently existential entailment over* $\{\bot, f, \top\}$ *is PSPACE-hard, and so is the problem with the variance as part of the input, already on covariant instances.*

We claim no hardness for a fixed variance with a contravariant argument, and none for $n = 1$, where the problem is in PSPACE by Theorem 13.4. The reductions use the extra arguments of f either for an unconstrained filler, whose positions every condition ignores, or for $\bot$, under literals that already confine the relevant values to the letters 0 and 1; neither is bare padding.

Proof. Let $\Sigma = \{0, \dots, n-1\}$. For an automaton $\mathcal{N}$ over $\{0,1\}$ with states Q, initial state q_0 and final states F, write L_q for the words accepted from q.

(1) Let $\mathcal{F}(\mathcal{N})$ have the variables x, d and X_q for $q \in Q$, and the literals

$$x \le X_{q_0}, \quad X_q \le f(X_{q'}, d, \dots, d)\ (q \xrightarrow{0} q'), \quad X_q \le f(d, X_{q'}, d, \dots, d)\ (q \xrightarrow{1} q'), \quad X_q = \bot\ (q \in F).$$

We claim: a tree t over $\{\bot, f\}$ in a class $\mathcal{D}$ is the value of x in some $\mathcal{D}$-solution iff no word of L_{q_0} is a constructor position of t. If $\pi \in K(t)$ and $q_0 \xrightarrow{\pi} q$, then by induction on π the subtree of t at π lies below X_q: an upper literal at a constructor node passes to the child in the direction of the transition. For $\pi \in L_{q_0}$ with $q \in F$ this contradicts $X_q = \bot$. Conversely, let h bound the lengths of the positions of t if t is finite, and put $h = \infty$ otherwise, with $\Sigma^{\le\infty} = \Sigma^*$. Let $K(d) = \Sigma^{\le h}$ and let $K(X_q)$ be the words of $\Sigma^{\le h}$ with no prefix in L_q. These sets are prefix closed, finite if $h < \infty$ and regular otherwise. If $0w \in K(X_q)$ and $q \xrightarrow{0} q'$, then no prefix of w lies in $L_{q'}$, so $w \in K(X_{q'})$, and every $iw \in K(X_q)$ with $i \neq 0$ has $w \in K(d)$; so the upper literals hold by Lemma 2.3, and symmetrically for the letter 1. For $q \in F$, $\varepsilon \in L_q$, so $X_q = \bot$. Finally $K(t) \subseteq K(X_{q_0})$, since $K(t)$ is prefix closed and avoids L_{q_0}.

Given automata $\mathcal{N}_1, \mathcal{N}_2$, let $\mathcal{N}_i'$ accept $\mathrm{code}(L(\mathcal{N}_i))$. Let $\varphi = \mathcal{F}(\mathcal{N}_2')$, with all its variables visible, and let $\psi = \mathcal{F}(\mathcal{N}_1')$ with fresh variables except x, all of them hidden. Both systems are satisfiable, by the assignment $\bot$. By the claim, $\varphi \models_{\mathcal{D}} \exists \bar{z}\, \psi$ iff every $\top$-free $t \in \mathcal{D}$ whose constructor positions avoid $\mathrm{code}(L(\mathcal{N}_2))$ also avoids $\mathrm{code}(L(\mathcal{N}_1))$. If $L(\mathcal{N}_1) \subseteq L(\mathcal{N}_2)$, this holds. If $w \in L(\mathcal{N}_1) \setminus L(\mathcal{N}_2)$, let t be the finite tree whose constructor positions are the prefixes of $\mathrm{code}(w)$; a prefix of $\mathrm{code}(w)$ in $\mathrm{code}(L(\mathcal{N}_2))$ would equal $\mathrm{code}(w)$ by prefix freeness, so t is a countermodel. Inclusion of automata is PSPACE-hard, and the reduction is polynomial.

(2) Let $\mathcal{N}$ accept $\mathrm{code}(L(\mathcal{N}_0))$ for a given automaton $\mathcal{N}_0$, and let $C = (00 \mid 01)^*1$ be the set of all codes and P the set of proper prefixes of codes, the words $e(w)$ and $e(w)0$. Let χ have the variables x, d, b, r, r' and X_q for $q \in Q$, and the literals

$$x \le X_{q_0}, \quad X_q \le f(X_{q'}, d, b, \dots, b)\ (q \xrightarrow{0} q'), \quad X_q \le f(d, X_{q'}, b, \dots, b)\ (q \xrightarrow{1} q'), \quad X_q = \bot\ (q \in F),$$
$$b = \bot, \qquad f(r', b, \dots, b) \le r, \qquad f(r, r, b, \dots, b) \le r'.$$

We show that $\chi \models_{\mathcal{D}} x \le r$ iff $L(\mathcal{N}_0) = \{0,1\}^*$.

Suppose $\mathcal{N}_0$ is universal, so that $\mathcal{N}$ accepts every code, and let ρ be a solution. *The values of* r. Call r the state of the words $e(w)$ and r' that of the words $e(w)0$. By induction along P, if $\pi \in P$ is a position of ρr, then its subtree lies above the value of its state: from r, the literal $f(r', b, \dots, b) \le r$ gives a node f or $\top$ whose child 0, if it exists, lies above r', and from r' the other literal gives children 0 and 1 above r. In particular ρr is not $\bot$ at any position in P. *The values of* x. Every $\pi \in P$ is a proper prefix of an accepted code, so an accepting run reaches, on π, a state q with an outgoing transition; as in (1), if π is a position of ρx, its subtree lies below X_q, which lies below a constructor tree, so ρx is not $\top$ at π, and the children of that subtree in the directions $i \ge 2$ are $\bot$. Every code c is accepted, so if c is a position of ρx, its subtree lies below $X_q = \bot$ for a final q. The children of a word of P are words of P, codes, and words πi with $\pi \in P$ and $i \ge 2$, and ρx is $\bot$ at the last two

kinds; so every position of ρx is of one of these three kinds. Hence at every common position either ρx is $\bot$, or the position lies in P, where ρx has a label in $\{\bot, f\}$ and ρr one in $\{f, \top\}$. So $\rho x \le \rho r$.

Suppose $w \notin L(\mathcal{N}_0)$, and let $c = \mathrm{code}(w)$, so that no prefix of c is accepted by $\mathcal{N}$, by prefix freeness. Let x be the finite tree whose constructor positions are the prefixes of c, c included. For $q \in Q$ let X_q be the join, the union of constructor positions, of the subtrees of x at the prefixes π of c with $q_0 \xrightarrow{\pi} q$; each is the path tree along the rest of c. Then $x \le X_{q_0}$, $X_q = \bot$ for $q \in F$, and the upper literals hold with $b = \bot$ and d the finite tree with $K(d) = \{0,1\}^{\le |c|}$, as in (1). Let $r_0 = r'_0 = \top$, $r_{h+1} = f(r'_h, \bot, \dots, \bot)$ and $r'_{h+1} = f(r_h, r_h, \bot, \dots, \bot)$. By induction $r_{h+1} \le r_h$ and $r'_{h+1} \le r'_h$, so (r_h, r'_h) satisfies the two lower literals, and for $h > |c|$ the tree r_h has the label $\bot$ at c, where x has f. This finite solution refutes $x \le r$.

Universality is PSPACE-hard and the reduction is polynomial; since PSPACE is closed under complement, this proves (2). Item (3) follows from (1) by Theorem 10.16, or from (2), since plain entailment is the case $Z = \varnothing$ and the covariant instances have the variance $c = 0$. □

The reduction of (2) reproves, for $n = 2$, the theorem of Henglein and Rehof [1998], in all three tree classes, and extends it to every fixed arity at least two. Together with Theorems 13.2 and 13.4, the problem is PSPACE-hard and in 2-EXPSPACE for every fixed $n \ge 2$ and covariant f, and in PSPACE for $n \le 1$; whether it is PSPACE-hard for $n = 1$ is open.

14 Mechanization

The artifact, a development in Lean 4 [de Moura and Ullrich 2021] with Mathlib [The mathlib Community 2020], contains six decision procedures, one for each case of Theorem 11.2(1,2): existential entailment over $\{\bot, f, \top\}$ for every arity $n \ge 0$ and every variance, and over $\{\bot, f\}$ for every arity with f covariant, each over arbitrary, regular, and finite trees. Arity zero is included. The procedures are computable definitions, not classical choices of decidability instances, and each has a correctness theorem without premises: none assumes satisfiability or positive arity. They take flat systems (Section 2.3) whose visible and hidden variables are finite ordinals. Classical reasoning enters the proofs of the bounds and of correctness, not the searches. Lean and Mathlib are both pinned to `v4.35.0`, and the correctness theorems depend only on the axioms `propext`, `Classical.choice`, and `Quot.sound`.

Plain entailment is the instance with no hidden variables and the consequent $x \le y$ (Definition 2.4). The artifact defines decision procedures for plain entailment in the three tree classes with the same names and types as those of the artifact of Zhao [2026], and obtains them from the existential procedures through `nsse_iff_ensse` and its regular and finite versions; it uses no other procedure for plain entailment. The same reduction decides covariant plain entailment over $\{\bot, f\}$. The two artifacts share, as a library, the definitions, the results on satisfiability, and the hole search (Section 2.5), but not the reduction of plain entailment to holes, so the decidability of plain entailment is proved a second time, through the characterization of Section 4.

Every search bound is an explicit function of the state counts of the automata involved. Anchored compression (Theorem 9.14) is mechanized with an elementary bound: `Anchored.exists_short_posHole_elementary` shortens a positive hole of a lettered monitor with a positive channel to length at most `Anchored.anchoredLengthBound`, and `Anchored.anchoredLengthBound_le` proves the estimate $2^{2^{5\Xi^\sharp(N,d)+6}}$, against $2^{2^{5\Xi^\sharp(N,d)}}$ in Theorem 9.14, with the same exponent $\Xi^\sharp$. The dictionary bound is $\mathrm{G}^\sharp$ itself; the column count is larger than $\mathrm{col}^\sharp$ because the formal positive summary stores a second complete column summary where the paper adds single fields (Appendix H.4). The mortal bound evaluates this function at the separator reader with all its states, not only the reachable ones, and the non-mortal bound has the form of L_{nm} with a larger pump capacity. Their maximum,

`Core.Words.Self.selfHoleBound`, bounds a shortest hole of every instance with forced target heads, as in Theorem 9.17.

Three principal differences in the formal route are the following. For $\{\bot, f\}$ with f covariant the formalization does not use the reduction of Theorem 10.16: a direct covariant core represents a tree by its set of constructor positions (Lemma 2.3), proves three conditions in which preservation of liveness takes the place of (U) and no self events occur, reduces the third to two-word holes without self guard, transfers to regular and finite trees through regular Horn hulls and finite least solutions, and identifies sets of constructor positions with $\top$-free trees. For regular trees, (2) and (3) of Theorem 10.11 are proved like Theorem 10.12: the at most two observations of Lemma 10.3 become ground bounds; these bounds, and in (2) the regular visible values, are expressed by literals over fresh variables whose values are fixed by the graph equations of regular presentations; and the extended system has a regular solution by (1). Lemma 5.7 is not used. Condition (U) is decided by the hole search of Zhao [2026] on the unit reader with all its N_{U} states, without the reductions of Lemmas 6.9 and 6.10.

Not mechanized are the flattening of type expressions (Lemma 2.6); the bounds in terms of the number s of variables, that is, Theorem 11.2(3,4), Proposition 11.5, Lemma 9.18 and Corollary 9.19, and the closed form in Theorem 6.11; the effective countermodels of Lemma 11.3, since the procedures return only the answer; the results of Section 12; and the complexity results of Section 13, together with the polynomial-space and polynomial-time claims of Proposition 5.17 and Lemmas 5.6 and 5.11, whose tests the artifact performs by finite searches.

The trusted statement consists of the definitions of trees, the variance order, finite terms and regular graphs with their interpretations as trees, flat literals and their satisfaction, and the six entailment relations. Trees, the order, and the two representations are those of Zhao [2026], whose characterization module relates the order to the greatest fixed point of the constructor rule and shows that the representations denote the regular and the finite trees. The kernel checks the proof terms; running the compiled procedures also relies on the compiler. Appendix H gives the public statement, the module layout, the correspondence with the results of this paper, and the differences in detail. The artifact is included in the supplementary material.

15 Related Work

Table 8 summarizes selected results on entailment questions that involve existential quantifiers or projections.

Constrained Types and Their Simplification. Constrained types with recursive constraints were used for type inference for objects by Eifrig et al. [1995]. Trifonov and Smith [1996] defined subtyping between constrained types semantically, over regular trees with arrows, $\bot$, and $\top$; Eq. (1) is their definition, for types without assumptions on program variables, written as an existential entailment. Their decision procedure is a sound approximation based on canonical constraint maps and primitive derivations, and Corollary 12.2 decides their relation exactly. Pottier [1996] described an incomplete entailment algorithm, together with the removal of unreachable constraints and simplifying substitutions. Pottier [2001] developed polarity-based garbage collection, canonization, and minimization, and proved completeness in the equality setting; completeness is lost for subtyping. Pottier [1998b, Section 8.5] gives a correct simplification that canonical bounds miss; Example 12.4 recalls it. Aiken et al. [1999] eliminate quantified variables and obtain optimal representations for unconstrained polymorphic types with subtyping; their extension to constrained types is sound but not complete. Fähndrich and Aiken [1996] use polarity-based minimization, removal of intermediate variables, and cycle collapse in an implemented analysis, and they define the comparison of constrained types by a universal-existential condition. For constrained types, all

Table 8. Selected results related to existential subtype entailment. "Open alphabet" means that a feature not occurring in the formula is always available.

Problem	Trees	Result	Reference
Structural subtyping, first-order theory (hence existential entailment)	finite	decidable	Kuncak and Rinard [2003]
Non-structural subtyping over $\{\bot, f\}$, f binary and covariant, first-order theory	finite, regular, arbitrary	undecidable, with six quantifier blocks	Su et al. [2002]
Feature-tree ordering constraints, existential entailment, open alphabet	finite, arbitrary	PSPACE-complete	Müller et al. [1998, 2001]
Restricted entailment (projection inclusion) in a set-based selector domain	complete infinite	exponential time; PSPACE-hard	Flanagan and Felleisen [1999]
Subtyping of constrained types with $\to$, $\bot$, $\top$	regular	posed as open; sound decidable approximation	Trifonov and Smith [1996]
Subsumption of polar type schemes	extensible algebra	decidable	Dolan [2016]
Plain NSSE, one constructor of any arity and variance	finite, regular, arbitrary	decidable	Zhao [2026]
Existential entailment, one constructor of any arity and variance over $\{\bot, f, \top\}$, and over $\{\bot, f\}$ with f covariant	finite, regular, arbitrary	decidable; in 2-EXPSPACE; PSPACE-hard for each fixed arity ≥ 2, already with f covariant	this paper

these works use sound transformations or approximations; the classes of Section 12.3 are cheap complete cases, and Theorem 11.2 covers the rest.

Flanagan and Felleisen [1999] decide a closely related problem. Their restricted entailment $S_1 \models_E S_2$ is the inclusion $\mathrm{Sol}(S_1)|_E \subseteq \mathrm{Sol}(S_2)|_E$ of projections onto a set E of external variables, which after renaming is the existential entailment $S_1 \models \exists \bar{z}\, S_2$ with $\bar{z}$ the variables of S_2 outside E. They decide it completely, in exponential time, and show PSPACE-hardness. Their domain, however, is the set-based domain $D = \mathcal{P}(\mathrm{Const}) \times D \times D$ of complete infinite trees with total selectors and a contravariant domain component, not the ground trees of non-structural subtyping, so their decision procedure does not apply to our setting.

Feature Trees and Set Constraints. For ordering constraints over feature trees, the full first-order theory is undecidable, while existential entailment is PSPACE-complete over finite and over possibly infinite trees [Müller et al. 1998, 2001]. Their procedure saturates the constraints, builds polynomial-size automata that recognize consequences expressed by paths, hides the existential variables while keeping their states, and decides $\varphi \models \exists \bar{z}\, \psi$ by an inclusion of the resulting languages. Our three conditions and the pure derivations through hidden variables play the role of their consequence languages and hidden states. Their setting differs from ours in one essential respect: the feature alphabet is open, so a feature that does not occur in the formula is always available. Their PSPACE-hardness proof, for instance, builds countermodels with such a fresh feature [Müller et al. 2001, Lemma 5.5], and Müller et al. [2000] use fresh features to construct countermodels for ordering constraints. A ranked signature has no fresh direction: a constructor node has exactly its prescribed children, and $\bot$ and $\top$ have none.

The alphabet matters already for feature trees, as the following example over arbitrary trees shows. Over the features $\{0, 1\}$, the constraints $a(x) \wedge x[0]x \wedge x[1]x \wedge x \leq y$ make x the complete infinite tree labeled a everywhere, so every y above x equals x, and the constraints entail $y \leq x$. Once a third feature is available, y may extend x along it, and the entailment fails. The open-alphabet and the fixed-alphabet problems thus have different answers on this instance, and a decision procedure for the first does not decide the second. The same phenomenon appears in our setting when the arity grows: an entailment for a binary constructor need not survive bare padding to a larger arity (Example 13.7). In set-constraint languages, entailment of atomic set constraints is PSPACE-complete over infinite signatures [Niehren et al. 1999]; these constraints have no existential conclusions, and sets of trees are a different domain from trees under the subtype order.

The First-Order Theory. The undecidability proof of Su et al. [2002], recalled in Section 1, reduces the Post correspondence problem to a sentence whose matrix uses implications, disjunctions, and negations; for example, it expresses that the subtypes of a tree form a chain. Our problem is the validity of $\forall \bar{x}\, (\varphi \Rightarrow \exists \bar{z}\, \psi)$ with conjunctive φ and ψ, a proper restriction of the $\forall^*\exists^*$ fragment. The same paper decides the first-order theory over signatures with only unary and nullary constructors. Su [2002] and the extended version of the paper [Su et al. 2006] reduce entailment to emptiness of constrained automata on tuples of trees, which is undecidable in general. The remark of Su [2002, Section 7.2.2] quoted in Section 1 concerns an adaptation of that undecidability proof; no formula with a single alternation is displayed, and undecidability of a fragment with one alternation but an unrestricted matrix would still not settle the case of conjunctions. Kuncak and Rinard [2003] showed that the first-order theory of structural subtyping of non-recursive types is decidable, by a term-power construction. This covers existential entailment for structural subtyping, which compares only trees of the same shape, over finite trees.

The Automata Route to Plain Entailment. Niehren and Priesnitz [1999] proved PSPACE-completeness of plain non-structural entailment for the fragment without literals $x = \bot$ and $x = \top$, by a reduction to universality of finite automata. Niehren and Priesnitz [2001, 2003] introduced cap automata, decided the unary case, and identified the role of word equations in the general problem. Henglein and Rehof [1998] proved PSPACE-hardness of non-structural entailment for a binary covariant constructor, which Theorem 13.9 extends to every fixed arity at least two, and the theses of Priesnitz [Priesnitz 2000, 2004] develop the automata approach further. Niehren et al. [2005] classify the complexity of satisfiability over posets and note that their technique does not extend beyond satisfiability. Our reduction keeps the spirit of this line: the obstructions are periodicity conditions between suffixes of paths. The new features are that two words with a common suffix are compared, that a word may be compared with itself through a self event, and that the separator encoding returns the two words to the one-word readers of Zhao [2026].

Pushdown Saturation Through Internal Variables. The type inference of Noonan et al. [2016] for machine code uses derived type variables, which extend a variable by a word of field labels, and observes that "the full set of constraints entailed by C has a finite encoding by an automaton". It builds a transducer for derivations through uninteresting variables by a modification of the saturation procedure for prefix rewriting [Büchi 1964; Caucal 1992], and normalizes derivations to pops followed by pushes. This is a close precedent for our pure derivations through hidden variables and their lowest-point decomposition (Lemma 5.3). Their entailment is derivability in a proof system, while ours is semantic entailment over ground trees. Completeness for the semantic notion is what needs the leaf-cone characterization and the analysis of two-word holes, which have no counterpart there. The saturation in our setting is the computation of the closure D_χ of the

signed system; no further saturation is needed, because the pushdown summaries are read off from it (Definition 5.1 and Lemma 5.2).

Algebraic and Semantic Subtyping. Algebraic subtyping [Dolan 2016; Dolan and Mycroft 2017] interprets types in an extensible algebra rather than as ground trees and infers principal types by biunification. Subsumption between its polar type schemes is decided in the thesis [Dolan 2016]; the conference paper does not discuss deciding subsumption [Dolan and Mycroft 2017]. Simple-sub [Parreaux 2020] gives a compact account of this inference; MLstruct extends it to a Boolean algebra of structural types and does not formally describe its subsumption check [Parreaux and Chau 2022]; and Parreaux et al. [2024] keep subtyping constraints in first-class polymorphic types. Because the semantics differ, decidability there does not decide the ground-tree problem; Appendix G.1 gives a correct closed-tree simplification that is not valid in an extensible algebra. Semantic subtyping interprets types as sets of values. Castagna and Xu [2011] decide subtyping between polymorphic regular types in EXPTIME, and Gesbert et al. [2015] decide semantic subtyping through a logic. For polymorphic application, Castagna et al. [2015] decide tallying, but inferring the type substitutions in general uses sound and complete semi-decision procedures whose termination is left open. These are questions about types, not about the solutions of constraint systems over ground trees.

Representations of Solution Spaces. Kozen [1995] represents the solution spaces of set constraints as rational spaces and relates logical entailment to rational embeddings between these spaces. Our leaf representation (Theorem 3.2) is a representation theorem of a different kind, in the tradition of rings of sets [Birkhoff 1937] and Alexandrov topologies [Alexandroff 1937]: a tree of any variance is a pair of disjoint sets of words that are closed under extension, and the order of trees is inclusion of the first set and reverse inclusion of the second. This elementary fact becomes useful because every literal then becomes clauses with a single premise. For a covariant constructor over $\{\bot, f\}$, solution sets are closed under unions and intersections (Remark 4.6); in general they are closed under intersections of leaf cones, which in the order of trees is the median with Ω (Proposition 10.5).

Decidability of Plain Entailment. Zhao [2026] decides plain entailment $\varphi \models x \leq y$ for one constructor of any arity and variance, by reducing non-entailment to holes of finite readers and bounding a shortest hole through its rejected tail, its depth, and a compression argument. It lists existential entailment among the natural extensions that are not decided by testing component inequalities independently. We use the results recalled in Section 2 and Appendix A. Condition (U) is a hole problem of exactly the kind solved there, decided by its three bounds, applied to a reduced reader whose depth is bounded through a clock on the antecedent (Section 6); the readers of that paper already compare a root with its own signed copy, the plain counterpart of our self events. For the two-word holes of condition (II), the rejected tail of a mortal hole is short by definition, so its depth theorem already bounds the depth of the separator word (Lemma 9.6). Our main technical additions to that machinery are the pump for holes that are not mortal (Theorem 8.14) and the extension of the bounded-depth search to holes that must contain an anchored comparison (Theorem 9.14).

16 Conclusion

We have shown that existential subtype entailment $\varphi \models \exists \bar{z}\, \psi$ for one constructor is decidable over arbitrary, regular, and finite trees, for every arity and every variance over $\{\bot, f, \top\}$, and over $\{\bot, f\}$ when the constructor is covariant (Theorem 11.2). Signed leaf cones turn every system into a Horn system with unary rules, and collision return reduces the entailment to three conditions on single seeds. Two of them become hole problems for finite readers. Condition (U) is a hole problem of the kind solved for plain entailment [Zhao 2026]; condition (II) is a hole problem for two words

that share a suffix, in which a word can also obstruct itself through a self event. Holes that are not mortal are found through breakers, lassos and a pump, and are singly exponential. Mortal holes are packed around a separator into one word of bounded depth, and a compression that carries a positive channel up the hierarchy keeps the anchored comparison that a mortal hole must contain. The transfers to regular and finite trees hold for every variance, and a gadget reduces covariant $\{\bot, f\}$ to $\{\bot, f, \top\}$.

The Signature without Top and a Contravariant Argument. We do not know whether existential entailment over $\{\bot, f\}$ is decidable when f has a contravariant argument. Neither omitting the literals $\top \le u$ nor the gadget of Theorem 10.16 reduces this case to $\{\bot, f, \top\}$ (Remark 10.18): the gadget excludes leaves $\top$ only because each covariant child of its variable lies below the variable itself. A decision for this case needs either a gadget that bounds the leaves of both polarities without $\top$ literals, or a direct treatment of the cones of trees without $\top$, in which every leaf is $\bot$ but its sign still depends on the polarity of its path.

Complexity. The problem is PSPACE-hard for every fixed arity $n \ge 2$ with f covariant, and in 2-EXPSPACE (Theorems 13.2 and 13.9). For $n \le 1$ it is in PSPACE (Theorem 13.4 and Remark 13.5), but we know no lower bound for $n = 1$: the reductions of Theorem 13.9 need two independent directions, and a unary constructor offers only one. Closing the gap between PSPACE and 2-EXPSPACE for $n \ge 2$ is open, as is the complexity for a fixed variance with a contravariant argument, for which we claim no lower bound.

Witness Lengths. Shortest counterexample words can be exponentially long (Proposition 13.6), and they are at most triply exponential (Theorem 11.2). Words for (I0) and holes that are not mortal are singly exponential, and the words for (U) stay below the mortal bound (Theorem 6.11), so the bound of Theorem 11.2 is that of the mortal holes. Whether mortal holes have a doubly exponential bound is open. The length bound of anchored compression is exponential in the depth and in the square of the dictionary size, and the dictionaries are exponential in the number of states and the depth; dictionaries of polynomial size would give a doubly exponential bound. Bounded depth and the subset structure of the automata do not give such dictionaries by themselves, and a different compression, or a direct analysis of the long common suffix of a mortal hole, may be needed. The words for (U) come from the compression of Zhao [2026], whose bound has the same form, so an improvement of the overall bound would need it there as well. The exponential lower bound holds over arbitrary and regular trees; we have no superpolynomial lower bound over finite trees.

Richer Signatures and Conclusions. Several constructors, possibly together with nullary base types, change the order on labels: incomparable constructors clash, and the leaf cones would have to record which constructor occurs at each position, while the polarity of a path would depend on the constructors along it. Entailment of disjunctions, $\varphi \models \psi_1 \vee \psi_2$, is not covered either: a disjunction is not a Horn clause, and the single-seed characterization of Section 4 rests on the Horn form.

Mechanization. The mechanized proof covers decidability on flat input in all six cases, with anchored compression and all search bounds as explicit functions of automaton state counts (Section 14). Mechanizing the flattening of Lemma 2.6, the bounds of Proposition 11.5 in terms of the number of variables, the countermodels of Lemma 11.3, and the summary search of Proposition 13.1 would make the quantitative and certifying statements machine checked as well.

Implementation in Type Inference. The procedure of Theorem 11.2 is not meant to be run as stated. A practical subsumption or simplification checker could first apply the classes of Section 12.3, which reduce to plain entailment, then test satisfiability and condition (I0), both in polynomial space, and then search for short holes of the unit readers and of the extracted instances. Each counterexample

found can be checked directly and yields a regular countermodel (Lemma 11.3); only the absence of counterexamples depends on the bounds. Which of these cases arise in the constrained types produced by inference, and how often mortal holes or long words for (U) are needed, are empirical questions that we leave open.

Use of LLM-Based Tools

Tools based on large language models (LLMs) assisted proof attempts, checks of arguments, exploratory programs, and the formalization. Kernel-checked proofs support the decision procedures (Section 14). Search results were treated as evidence, not as proofs of universal claims.

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A Results Used from the Plain-Entailment Proof

This appendix collects what we use from the decidability proof for plain entailment of Zhao [2026] beyond Section 2: the satisfiability criteria and their extension to copy literals (Appendix A.1); the cores of the canonical hierarchy (Appendix A.2); the ordinary and symbolic states of derived monitors (Appendix A.3); the structure of the proof of Theorem 2.20 in the precise terms that later sections extend (Appendix A.4); and the constants of its length bound (Appendix A.5). The results are quoted with the hypotheses of that paper and proved there, for every arity and every variance; we give proofs only where we extend a result. Except in Appendix A.1, nothing here uses a property of subtype constraints, and every alphabet may be infinite unless it is said to be finite.

A.1 Satisfiability

In the syntax of Zhao [2026] a system has no copy literals (Section 2.4). For a covariant system χ without copy literals over a finite set V, the *closure* D^{cov}_{χ} is the least reflexive and transitive relation on V such that, for every $i \in \Sigma$,

$$\frac{f(a_j)_j \le u \in \chi \qquad D^{\mathrm{cov}}_{\chi}(u,v) \qquad v \le f(b_j)_j \in \chi}{D^{\mathrm{cov}}_{\chi}(a_i,b_i)}. \tag{10}$$

It is computed by saturation in time polynomial in the size of χ, since there are $|V|^2$ pairs. The *path bounds* $\mathrm{Up}(\eta,u,v)$ and $\mathrm{Lo}(\eta,u,v)$, for words $\eta \in \Sigma^*$ and $u,v \in V$, are defined by $\mathrm{Up}(\varepsilon,u,v) \iff \mathrm{Lo}(\varepsilon,u,v) \iff D^{\mathrm{cov}}_{\chi}(u,v)$ and

$$\begin{aligned} \mathrm{Up}(i\eta,u,v) &\iff \exists z,\bar{b}\,\left[D^{\mathrm{cov}}_{\chi}(u,z) \wedge (z \le f(b_j)_j) \in \chi \wedge \mathrm{Up}(\eta,b_i,v)\right], \\ \mathrm{Lo}(i\eta,u,v) &\iff \exists \bar{a},z\,\left[(f(a_j)_j \le z) \in \chi \wedge D^{\mathrm{cov}}_{\chi}(z,v) \wedge \mathrm{Lo}(\eta,u,a_i)\right]. \end{aligned} \tag{11}$$

In every solution, $\mathrm{Up}(\eta,u,v)$ says that the subtree of u at η, when it exists, is below v, and $\mathrm{Lo}(\eta,u,v)$ says that u is below the subtree of v at η, when it exists [Zhao 2026, Lemma "Path rules"]. A *label clash* is a literal $f(\bar{a}) \le u$ or $\top \le u$ of χ, with *root label* f or $\top$, and a literal $v \le f(\bar{b})$ or $v = \bot$, with root label f or $\bot$, such that $D^{\mathrm{cov}}_{\chi}(u,v)$ and the first root label is greater than the second; that is, every such pair except two constructor literals. A *cycle clash* is a nonempty word η and variables a,b with

$$\mathrm{Lo}(\eta,a,a), \qquad D^{\mathrm{cov}}_{\chi}(a,b), \qquad \mathrm{Up}(\eta,b,b). \tag{12}$$

Theorem A.1 (Satisfiability criteria [Zhao 2026, Sect. "Closure and Supported Path Bounds"; Theorems "Least shape and satisfiability", "Covariant finite satisfiability" and "Regular countermodel transfer"; Lemma "Common shape and selection"; Lemma "Finite signed correspondence"]). *For every arity and every variance:*

(1) Let χ be a covariant system without copy literals. In every solution σ of χ, $D^{\mathrm{cov}}_{\chi}(u,v)$ implies $\sigma(u) \le \sigma(v)$. The system χ has a solution iff it has no label clash, and a finite solution iff it has neither a label clash nor a cycle clash. A cycle clash exists iff the finite graph on the pairs (a,b) with $D^{\mathrm{cov}}_{\chi}(a,b)$, with an edge from (a,b) to (a',b') whenever $\mathrm{Lo}(i,a',a)$ and $\mathrm{Up}(i,b,b')$ for some letter i, has a cycle; it suffices to test words η of lengths 1 through $|V|^2$.

(2) For every system χ without copy literals and every variance c, $\mathrm{Sat}_c(\chi) \iff \mathrm{Sat}(\chi^{\pm})$ and $\mathrm{Sat}_{\mathrm{fin},c}(\chi) \iff \mathrm{Sat}_{\mathrm{fin}}(\chi^{\pm})$.

(3) Every system without copy literals that has a solution has a regular solution.

The signed system of a system without copy literals again has no copy literals, so part (1) decides both satisfiability problems of part (2). The following lemma extends parts (2) and (3) to copy literals and to all three tree classes.

Lemma A.2 (Signed satisfiability). *For every arity $n \geq 0$, every variance c, every system χ over $\{\bot, f, \top\}$, copy literals included, and every tree class $\mathcal{D}$,*

$$\mathrm{Sat}_{\mathcal{D},c}(\chi) \iff \mathrm{Sat}_{\mathcal{D}}(\chi^{\pm}), \qquad \textit{and} \qquad \mathrm{Sat}_{c}(\chi) \iff \mathrm{Sat}_{\mathrm{reg},c}(\chi).$$

Proof. If ρ is a $\mathcal{D}$-solution of χ, then $\rho^{\pm}$ is a $\mathcal{D}$-solution of $\chi^{\pm}$ by Lemma 2.5. For the converse let σ be a $\mathcal{D}$-solution of $\chi^{\pm}$.

For $n = 0$ there are no children, so the literals of $\chi^{\pm}$ of sign + in Table 2 are the literals of χ with every variable v replaced by v^{+}, and the order is the same chain for every variance. Hence $v \mapsto \sigma(v^{+})$ solves χ, and the three classes coincide.

Let $n \geq 1$, and let $\chi^{\flat}$ be obtained by Lemma 2.7, with fresh variables y, o for each copy literal $u \leq v$ and the literals $f(a_i)_i \leq y$ and $y \leq f(b_i)_i$. We extend σ to a $\mathcal{D}$-solution of $(\chi^{\flat})^{\pm}$. The four signed literals of the two new literals are

$$f(a_i^{c(i)})_i \leq y^{+}, \quad y^{-} \leq f(a_i^{\bar{c}(i)})_i, \quad y^{+} \leq f(b_i^{c(i)})_i, \quad f(b_i^{\bar{c}(i)})_i \leq y^{-}.$$

Put $\sigma(o^{+}) = \sigma(o^{-}) = \bot$, $\sigma(y^{+}) = f(\sigma(a_i^{c(i)}))_i$, and $\sigma(y^{-}) = f(\sigma(b_i^{\bar{c}(i)}))_i$. The first and fourth literals hold with equality. The other two hold iff $\sigma(a_i^{c(i)}) \leq \sigma(b_i^{c(i)})$ and $\sigma(b_i^{\bar{c}(i)}) \leq \sigma(a_i^{\bar{c}(i)})$ for every i. For $i \geq 1$ both sides are $\sigma(o^{\pm})$. For $i = 0$ these are the two inequalities $\sigma(u^{+}) \leq \sigma(v^{+})$ and $\sigma(v^{-}) \leq \sigma(u^{-})$, in either case of $c(0)$, and they hold because $u^{+} \leq v^{+}$ and $v^{-} \leq u^{-}$ are the signed literals of $u \leq v$. The new values lie in $\mathcal{D}$.

So $(\chi^{\flat})^{\pm}$ is $\mathcal{D}$-satisfiable, and in particular satisfiable. By Theorem A.1(2), $\chi^{\flat}$ has a solution, and a finite one if $\mathcal{D} = \mathcal{T}_{\mathrm{fin}}$; by Theorem A.1(3) it has a regular one. In each case $\chi^{\flat}$ has a $\mathcal{D}$-solution for the variance c, whose restriction to the variables of χ is a $\mathcal{D}$-solution of χ by Lemma 2.7.

Finally, if χ has a solution, then so has $\chi^{\flat}$ by Lemma 2.7, hence a regular one by Theorem A.1(3), and its restriction is a regular solution of χ. □

A.2 Cores

Let w be a nonempty word with canonical hierarchy (w_j, ℓ_j). Every position of w_j carries a *core*, which is either a cut of w, an *ordinary core*, or a symbolic *start core*. At level 0, position i has core i. At level $j+1$, the first position has a new start core $\star_{j+1}$, and position $i \geq 1$ inherits the core of the $(i-1)$st occurrence of ℓ_j in w_j, counting occurrences from 0; this is the marker that closes the block X_{i-1}, and the block X_i is read from it. Let C_j be the set of ordinary cores of w_j.

Lemma A.3 (Cores [Zhao 2026, Lemma "Cores"]).

(1) $C_{j+1} \subseteq C_j$. The start cores of w_j occupy an initial segment of its positions, are pairwise distinct, and number at most j; the ordinary cores increase from left to right.

(2) If $C_j \neq \emptyset$ and $e_j = \max C_j$, then

$$C_{j+1} = \{p : p < e_j \textit{ and } w[e_j{:}] \preceq w[p{:}]\}.$$

In particular, (p, e_j) is a comparison of w for every $p \in C_{j+1}$. If $C_j = \emptyset$, then $C_{j+1} = \emptyset$.

The cut e_j is the *terminal core* of level j, the core of the last position of w_j. Thus each level collects the left ends of the comparisons whose right end is the terminal core of the level below. The proof of Theorem 2.19 combines this lemma with Lemma 2.18. Both lemmas hold for every word, respectively every rejected word, and every reader.

A.3 Lettered Monitors and Derivatives

Lettered monitors (Definition 2.22), the lettered monitor of a reader, which has the same nonempty holes as the reader, shifted blocks, derivatives (Definition 2.23), the label facts Eq. (4) and the

desubstitution lemma (Lemma 2.24) are stated in Section 2.6. We add the structure of the states along a hierarchy.

Ordinary and symbolic states. Along a hierarchy, the states of $\mathcal{M}_i$ are the N states of $\mathcal{M}_0$, the *ordinary* states, and the i *symbolic* states $\star_1, \ldots, \star_i$ added by the derivatives. The ordinary states form a closed copy: a derived transition from an ordinary state runs the previous monitor over ℓX, so it stays ordinary. The *entry state* of a position x of w_i is the state of $\mathcal{M}_i$ at the cut x, from which its letter is read. By the label facts and the core recursion, a position with an ordinary core p has as entry state the state of $\mathcal{M}_0$ at the cut p of w, and a position with a start core has a symbolic entry state [Zhao 2026, Appendix "Details of the Bounded-Depth Compression"].

A.4 Structure of the Bounded-Depth Search

Let $\mathcal{M}_0$ be a lettered monitor with $N \geq 1$ states, for instance the monitor of a reader, and let w be a nonempty hole of depth $D \leq d$, with hierarchy (w_i, ℓ_i) and derived monitors $\mathcal{M}_i$. The proof of Theorem 2.20 has the five steps sketched in Section 2.6.

Step 1: derivatives. By Lemma 2.24, w_i is a hole of $\mathcal{M}_i$, which has $N + i$ states.

Step 2: dictionary normal form. For a word v let $\mathrm{alph}(v)$ be its set of letters. The level-i *dictionary* is $\mathcal{A}_i = \mathrm{alph}(w_i)$, and $g_i = |\mathcal{A}_i|$. The *start count* $s_i \leq i$ is the number of positions of w_i with a start core; by Lemma A.3 they form the *start prefix* $w_i[:s_i]$. The set U_i of *entry states* is the set of ordinary states together with the symbolic states that are entry states of positions in the start prefix, so $|U_i| \leq N + s_i$. Every entry state occurring in w_i lies in U_i. The *protected* letters are

$$P_i = \{\ell_i\} \cup \mathrm{alph}(w_i[:s_i]), \qquad |P_i| \leq s_i + 1.$$

An unprotected letter occurs only outside the start prefix, so all its occurrences have ordinary entry states. A comparison (s, e) of w_i is *left maximal* if $s \geq 1$ and its predecessor letters $w_i[s-1]$ and $w_i[e-1]$ differ. The *boundary pairs* BP_i are the ordered pairs $(w_i[s-1], w_i[e-1])$ over the left-maximal comparisons with $e < |w_i|$, and the *endpoint pairs* EP_i the same pairs over those with $e = |w_i|$; each pair is counted once, and $b_i = |\mathrm{BP}_i|$. Endpoint pairs have ℓ_i as second component. A letter $a \in \mathcal{A}_i \setminus P_i$ is *isolated* if no pair in BP_i has a as one component and a letter outside P_i as the other; then all unequal comparison partners of a, endpoint partners included, are protected. The *exchange type* of an unprotected letter a consists of the following tables of $\mathcal{M}_i$:

- the transition $\zeta \mapsto \delta_i(\zeta, a)$ and the target $T_i^{\mathrm{fin}}(\zeta, a)$ for every ordinary state ζ;
- for each $R \in \{\Lambda_i, \Lambda_i^{\mathrm{fin}}\}$, the *self table* $R((\zeta, a), (\zeta', a))$ for ordinary ζ, ζ';
- for each such R, the *protected rows* $R((\zeta, a), (\vartheta, z))$ and $R((\vartheta, z), (\zeta, a))$ for ordinary ζ, $\vartheta \in U_i$, and $z \in P_i$;
- for each such R, the *initial row* $R(\triangleright, (\zeta, a))$ for ordinary ζ, which is false at every level $i \geq 1$.

Lemma A.4 (Exchange types [Zhao 2026, Lemma "Exchange types"]). *At every level $1 \leq i < D \leq d$ there are at most $\mathsf{C}(N, d)$ exchange types, with C from Eq. (14).*

Lemma A.5 (Comparison-preserving maps [Zhao 2026, Lemma "Comparison-preserving maps"]). *Let v be a nonempty word with last letter ℓ, and h a letter map that* identifies no unequal predecessor pair: *for every comparison (s, e) of v with $s \geq 1$, terminal ones included, $h(v[s-1]) = h(v[e-1])$ implies $v[s-1] = v[e-1]$. Then v and $h(v)$ have exactly the same comparisons, and on the letters of v, $h(a) = h(\ell)$ implies $a = \ell$.*

A *level map* at level i is a map h from $\mathcal{A}_i$ into itself. It acts letterwise on w_i, and $h(w_i)$ is expanded with the original lower markers $\ell_{i-1}, \ldots, \ell_0$ to a new word w'. The induced maps on the higher levels are $h_0 = h$ and $h_{r+1}([X]) = [h_r^*(X)]$, where h_r^* applies h_r letterwise to a block.

Lemma A.6 (Shape under a level map [Zhao 2026, Lemma "Shape under a level map"]). *Let h be a level map at level i that identifies no unequal predecessor pair of w_i, fixes the start prefix $w_i[:s_i]$ letterwise, preserves the level-i state run, and makes $h(w_i)$ a hole of $\mathcal{M}_i$. Then w' is a hole of $\mathcal{M}_0$, its markers below i are unchanged, and its dictionaries below i are subsets of the old ones. For every $r \geq 0$, $w'_{i+r} = h_r(w_{i+r})$, $\ell'_{i+r} = h_r(\ell_{i+r})$, and h_r identifies no unequal predecessor pair of w_{i+r}. Every start count is unchanged, and exactly the same levels are unary; in particular w' has depth D and no dictionary of w' is larger than the corresponding one of w.*

Lemma A.7 (Isolated exchange [Zhao 2026, Lemma "Isolated exchange"]). *Let $1 \leq i < D$, and let distinct isolated letters a, b of w_i have the same exchange type. The level map that replaces a by b satisfies the hypotheses of Lemma A.6, so the resulting word is a hole of depth D with every start count unchanged and no dictionary larger; its level-i dictionary is strictly smaller. Choosing, between the two directions, the letter with the shorter level-zero expansion as the image also makes the word no longer.*

A hole is *normal* if, at every level $1 \leq i < D$, distinct isolated letters have distinct exchange types, computed in its own hierarchy.

Lemma A.8 (Dictionary normal form and bounded dictionaries [Zhao 2026, Lemmas "Dictionary normal form", "Dictionary counting and propagation" and "Bounded dictionaries"]).

(1) *Every nonempty hole has a normal hole of the same depth, no greater length, the same start counts, and no larger dictionary at any level.*
(2) *For a normal hole of depth $D \leq d$ and $1 \leq i < D$, $g_i \leq \mathrm{A}(N,d) + 2b_i$. For every nonempty word and every level i, $b_i \leq b_{i+1} + g_{i+1} - 1$.*
(3) *For a normal hole of depth $D \leq d$, with $\mathrm{A} = \mathrm{A}(N,d)$, at every $1 \leq i < D$,*

$$2b_i \leq (\mathrm{A}-1)(3^{D-1-i}-1), \qquad g_i \leq 1 + (\mathrm{A}-1)3^{D-1-i}.$$

Consequently every nonempty hole of depth $D \leq d$ has a hole of depth D and no greater length with $g_i \leq \mathrm{G}(N,d)$ for every $1 \leq i \leq D$.

The counting in part (2) partitions $\mathcal{A}_i$ into the at most $d+1$ protected letters, the isolated letters, at most one per exchange type, and the remaining letters, each of which is a component of a boundary pair with both components unprotected. For the propagation, Lemma 2.17 shows that every boundary pair of level i consists of the first unequal letters, scanning from the right, of the two blocks of a boundary or endpoint pair of level $i+1$, with ℓ_i standing for an exhausted block. No bound on g_0 is needed.

Step 3: the top level.

Lemma A.9 (Unary words [Zhao 2026, Appendix "Details of the Bounded-Depth Compression"]). *Let z^m be a hole of a lettered monitor with state set Q. If two of the nonfinal cuts $i < i' \leq m-1$ have equal states, deleting the letters between them gives a hole whose labels are a subsequence of those of z^m with the same last label. Repeating this gives a unary hole $z^{m'}$ with $1 \leq m' \leq |Q|$.*

Every pair of cuts of a unary word is a comparison, so this deletion creates no new kind of label pair. Applied to w_D in $\mathcal{M}_D$, it gives at most $N + D$ letters.

Step 4: simultaneous compression. Let $\mathcal{M}$ be a lettered monitor with state set Q and start ζ_0, and ℓ a marker. Write $lab_0(\zeta)$ for the initial shifted-block label: (ζ, ℓ) for $\zeta \in Q$ and $\triangleright$ for $\star$. For blocks

$X_1, \dots, X_k$ avoiding ℓ, their *summary* is the tuple of the four functions

$$\begin{aligned} exit_a(\zeta) &\in Q, && \text{the exit of the shifted block of } X_a \text{ read from } \zeta,\\ term_a(\zeta,\zeta'') &\in \{0,1\}, && \text{whether some label of it is related by } \Lambda^{\mathrm{fin}} \text{ to } (\zeta'',\ell),\\ eq(a,b) &\in \{0,1\}, && \text{whether } X_a = X_b,\\ adm(a,b,\zeta,\zeta') &\in \{0,1\}, && \text{whether } \Lambda_\ell \text{ relates } (\zeta, X_a) \text{ and } (\zeta', X_b), \end{aligned}$$

for blocks a, b, entry states $\zeta, \zeta' \in Q \sqcup \{\star\}$, and states $\zeta'' \in Q$. It determines the restriction of $\mathcal{M}_\ell$ to the letters $X_1, \dots, X_k$, together with their equalities, and it has at most $\mathrm{col}(|Q|, k)$ values Eq. (14).

The *column word* of the tuple aligns the blocks at their right ends, pads them on the left, and reads the resulting word of k-tuples of optional letters, the *columns*, from left to right. It is *valid* if every track consists of absent entries followed by letters; a subsequence of a valid column word is valid. A deterministic automaton computes the summary along the column word. The tuple of k empty blocks has $exit_a(\zeta) = \delta(\zeta, \ell)$ for $\zeta \in Q$, $exit_a(\star) = \zeta_0$, $term_a(\zeta,\zeta'') = \Lambda^{\mathrm{fin}}(lab_0(\zeta), (\zeta'', \ell))$, all blocks equal, and $adm(a,b,\zeta,\zeta') = \Lambda(lab_0(\zeta), lab_0(\zeta'))$. Reading a column γ updates all components simultaneously, from the old values. If γ_a is absent, the components of block a are unchanged. If $\gamma_a = z$, then

$$exit'_a(\zeta) = \delta(exit_a(\zeta), z), \qquad term'_a(\zeta,\zeta'') = term_a(\zeta,\zeta'') \vee \Lambda^{\mathrm{fin}}\big((exit_a(\zeta), z), (\zeta'', \ell)\big).$$

Equality becomes $eq(a,b) \wedge \gamma_a = \gamma_b$. If $\gamma_a = z$ and $\gamma_b = z'$ are both present,

$$adm'(a,b,\zeta,\zeta') = \Lambda\big((exit_a(\zeta), z), (exit_b(\zeta'), z')\big) \vee \big(z = z' \wedge adm(a,b,\zeta,\zeta')\big);$$

if only $\gamma_a = z$ is present, $adm'(a,b,\zeta,\zeta') = \Lambda((exit_a(\zeta), z), lab_0(\zeta'))$; if only $\gamma_b = z'$ is present, $adm'(a,b,\zeta,\zeta') = \Lambda(lab_0(\zeta), (exit_b(\zeta'), z'))$; and if both are absent, the bit is unchanged. The update is correct because Λ_ℓ compares positions at the same distance from the ends of two blocks, with equal letters after them: appending a column either relates the two new last positions, or extends the common suffix of a pair related earlier by an equal letter.

Lemma A.10 (Column shortening [Zhao 2026, Appendix "Details of the Bounded-Depth Compression"]). *Deleting the columns between two cuts of a valid column word with equal summaries gives a valid column word with the same final summary. Hence every tuple of k blocks avoiding ℓ can be replaced by subsequences, one per block, with the same summary and all lengths below $\mathrm{col}(|Q|, k)$. The replacement is injective on the letters, preserves marker avoidance, maps holes of $\mathcal{M}_\ell$ over these letters to holes, and keeps a unary level unary.*

For lettered monitors $\mathcal{U}, \mathcal{V}$ with the same state set, a finite set E of letters, and a map ρ from E into the alphabet of $\mathcal{V}$, *agreement* $\mathcal{U} \equiv_{E,\rho} \mathcal{V}$ means: their start states coincide, ρ is injective on E, and for every $a \in E$ and every state ζ,

$$\delta_{\mathcal{V}}(\zeta, \rho(a)) = \delta_{\mathcal{U}}(\zeta, a), \quad T^{\mathrm{fin}}_{\mathcal{V}}(\zeta, \rho(a)) \iff T^{\mathrm{fin}}_{\mathcal{U}}(\zeta, a), \quad R_{\mathcal{V}}(\widehat{\rho}L, \widehat{\rho}L') \iff R_{\mathcal{U}}(L, L') \tag{13}$$

for $R \in \{\Lambda, \Lambda^{\mathrm{fin}}\}$ and all labels L, L' whose letters lie in E, the start label included; here $\widehat{\rho}(\triangleright) = \triangleright$ and $\widehat{\rho}(\zeta, a) = (\zeta, \rho(a))$, and states are not renamed. Agreement preserves runs, labels up to $\widehat{\rho}$, and holes of words over E, and agreements compose.

Lemma A.11 (Agreement under derivation [Zhao 2026, Lemma "Agreement under derivation"]). *If $\mathcal{U} \equiv_{E,\rho} \mathcal{V}$ and $\ell \in E$, then for every finite family F of blocks over E avoiding ℓ, the letterwise map ρ^* gives $\mathcal{U}_\ell \equiv_{F,\rho^*} \mathcal{V}_{\rho(\ell)}$, and the image blocks avoid $\rho(\ell)$.*

For markers $\boldsymbol{\ell} = (\ell_0, \dots, \ell_{j-1})$ write $\mathcal{M}^{\boldsymbol{\ell}}_0 = \mathcal{M}_0$ and $\mathcal{M}^{\boldsymbol{\ell}}_{h+1} = (\mathcal{M}^{\boldsymbol{\ell}}_h)_{\ell_h}$. A finite family of level-j letters is *valid* for the markers if every block avoids the marker of its level and its letters are,

recursively, valid letters of the level below; $\mathrm{Exp}_{\ell,j}(a)$ is the level-zero word obtained by expanding a valid level-j letter a, each block followed by its marker.

Lemma A.12 (Supported simultaneous compression [Zhao 2026, Lemma "Supported simultaneous compression"]). *Fix a nonempty word w with hierarchy markers $\boldsymbol{\ell}$ and dictionaries $\mathcal{A}_h$, and $j \le d$. Suppose $|\mathcal{A}_h| \le g$ for $1 \le h \le j$, and put $\kappa = \mathrm{col}(N+d+1, g)$. For every set $E_j \subseteq \mathcal{A}_j$ there are new markers $\widetilde{\boldsymbol{\ell}}$ and a map ρ_j, injective on E_j, such that*

$$\mathcal{M}_j^{\boldsymbol{\ell}} \equiv_{E_j,\rho_j} \mathcal{M}_j^{\widetilde{\boldsymbol{\ell}}}, \qquad |\mathrm{Exp}_{\widetilde{\ell},j}(\rho_j(a))| \le \kappa^j \quad (a \in E_j),$$

and the image family is valid for the new markers. The recursion keeps at each level $h < j$ the support

$$E_h = \{\ell_h\} \cup \bigcup_{X \in E_{h+1}} \mathrm{alph}(\tau_h(X)) \;\subseteq\; \mathcal{A}_h,$$

where $\tau_h(X)$ is the subsequence of X produced by Lemma A.10; at level zero no cardinality bound is needed. Agreement holds at every state, not only at the states visited by w.

Step 5: the bound. Normalize the given hole by Lemma A.8, so that $g_i \le \mathrm{G} = \mathrm{G}(N,d)$ for $1 \le i \le D$, and keep it fixed as the original hierarchy; put $\kappa = \mathrm{col}(N+d+1, \mathrm{G})$. Lemma A.9 gives a hole z^m of $\mathcal{M}_D$ with $1 \le m \le N+D$. Compressing $E_D = \{z\}$ by Lemma A.12, agreement makes $\rho_D(z)^m$ a hole of the new level-D monitor, and validity allows its exact expansion by Lemma 2.24 to a nonempty hole of $\mathcal{M}_0$. Cutting that expansion recovers the unary word $\rho_D(z)^m$ at level D, so the depth is at most D, and the length is at most

$$m\kappa^D \le (N+D)\kappa^D \le (N+d+1)\kappa^d = \mathsf{Len}(N,d).$$

No step uses finiteness of the alphabet, and the alphabet size does not enter the bound.

A.5 Constants

For integers $N \ge 1$, $d \ge 0$, and $k \ge 0$ put

$$\begin{aligned} \mathrm{col}(N,k) &= N^{(N+1)k} \cdot 2^{N(N+1)k} \cdot 2^{k^2} \cdot 2^{(N+1)^2k^2}, & \mathrm{C}(N,d) &= N^N\, 2^{2N^2+4N(N+d)(d+1)+3N}, \\ \mathrm{A}(N,d) &= d+1+\mathrm{C}(N,d), & \mathrm{G}(N,d) &= 2+\mathrm{A}(N,d)\,3^d, \end{aligned} \tag{14}$$

and

$$\mathsf{Len}(N,d) = (N+d+1)\cdot \mathrm{col}\big(N+d+1, \mathrm{G}(N,d)\big)^d. \tag{15}$$

All five functions are nondecreasing in each argument. Their meaning is as follows; $D \le d$ is the depth of the hole being shortened.

- $\mathrm{col}(N,k)$ bounds the number of summary values of a k-tuple of blocks for a lettered monitor with N states. It is the product of the numbers of values of the summary fields:

Factor	Summary field
$N^{(N+1)k}$	exits: k blocks, $N+1$ entry states
$2^{N(N+1)k}$	terminal-comparison bits, also for N final-marker states
2^{k^2}	equality bits for ordered pairs of blocks
$2^{(N+1)^2k^2}$	derived-admission bits for pairs of blocks and entry states

- $\mathrm{C}(N,d)$ bounds the number of exchange types at a level $1 \le i < D$ (Lemma A.4). The transition table of the ordinary states has N^N values; the target table uses N bits, the two self tables $2N^2$ bits, the protected rows $4N|U_i||P_i|$ bits, where $|P_i| \le d+1$ and $|U_i| \le N+d$, and the initial rows $2N$ bits.

- $\mathsf{A}(N,d)$ bounds the protected letters, at most $d+1$, together with one isolated letter for each exchange type.
- $\mathsf{G}(N,d)$ bounds the dictionary at every level $1 \le i \le D$ of a suitable normal hole (Lemma A.8).
- $\mathsf{Len}(N,d)$ is the length bound. The top level of the shortened hole has at most $N+D \le N+d+1$ letters, and each of the at most d expansion steps multiplies the length by at most $\mathrm{col}(N+d+1, \mathsf{G}(N,d))$, which bounds the summary values of the level-j monitors, with $N+j$ states, for every $j \le d$. For $d=0$ the bound is $N+1$: a hole of depth 0 is unary and is its own top level.

Lemma A.13 (Elementary estimate [Zhao 2026, Lemma "Elementary estimate"]). *Put* $\Xi(N,d) = 3N^2+4N(N+d)(d+1)+3N+3d+4$. *Then* $\mathsf{G}(N,d) \le 2^{\Xi(N,d)}$ *and* $\mathsf{Len}(N,d) \le 2^{2^{5\Xi(N,d)+3}}$. *For* $N \ge 1$ *and* $d_N = 2N(4N-1)-1$,

$$\mathsf{Len}(N,d_N) \le 2^{2^{1280(N+1)^5}}.$$

Proof. Write $\Xi = \Xi(N,d)$ and $\Xi_0 = 3N^2+4N(N+d)(d+1)+3N$. From $N \le 2^N$ we get $\mathsf{C}(N,d) \le 2^{\Xi_0}$; a sum of two quantities bounded by 2^p and 2^q is at most 2^{p+q+1}, and $d+1 \le 2^{d+1}$, so $\mathsf{A}(N,d) \le 2^{\Xi_0+d+2}$; with $3^d \le 2^{2d}$ this gives $\mathsf{G}(N,d) \le 2^{\Xi_0+3d+3} \le 2^{\Xi}$. Put $M = N+d+1$, $g = \mathsf{G}(N,d)$, and $P = 2^{\Xi}$. Then $M+1 \le \Xi$, so $M, M+1, d, g \le P$. Replacing M by 2^M in the first factor of Eq. (14) yields $\mathrm{col}(M,g) \le 2^{2M(M+1)g+g^2+(M+1)^2g^2} \le 2^{4P^4} = 2^{2^{4\Xi+2}}$. Therefore $\mathsf{Len}(N,d) \le 2^{\Xi+d2^{4\Xi+2}} \le 2^{2^{5\Xi+3}}$, because $d \le 2^{\Xi}$ and $\Xi \le 2^{5\Xi+2}$. For $d_N = 8N^2-2N-1$ we have d_N, $N+d_N$, $d_N+1 \le 8N^2$, so $4N(N+d_N)(d_N+1) \le 256N^5$ and $\Xi(N,d_N) \le 256N^5+27N^2+3N+4$. Hence $5\Xi(N,d_N)+3 \le 1280N^5+135N^2+15N+23 \le 1280(N+1)^5$. □

B Proofs for Leaf Cones and the Characterization

This appendix proves the results of Sections 3 and 4 whose proofs were omitted there. We keep their notation. For a tree t, a word w *meets* a leaf v of t if $v \preceq w$. By the chain property, a word meets at most one leaf, a word that meets no leaf is a position labeled f, and a proper prefix of a position is never a leaf. We use Eq. (5) throughout: $w \in L_t^s$ iff w meets a leaf of t of sign s.

Proof of Theorem 3.2. (1) A word meets at most one leaf, and every extension wi of w meets the leaf that w meets. By Eq. (5), the two cones are disjoint and extension closed.

Conversely, let (L^0, L^1) be a pair of disjoint extension-closed sets, and put $L = L^0 \cup L^1$. Let D be the set of words none of whose proper prefixes lies in L. For $w \in D$ put $t(w) = f$ if $w \notin L$, and, if $w \in L^s$, put $t(w) = \top$ if $s \oplus \mathrm{par}(w) = 0$ and $t(w) = \bot$ if $s \oplus \mathrm{par}(w) = 1$; this is well defined because $L^0 \cap L^1 = \varnothing$. The set D contains ε and is prefix closed, and $wi \in D$ iff $w \in D$ and $w \notin L$, that is, iff $w \in D$ and $t(w) = f$. So t is a tree. Its leaves are the words of $D \cap L$, and by construction a leaf $w \in L^s$ has sign s. We show $L_t^s = L^s$. If $w \in L_t^s$, then w meets a leaf v of sign s, so $v \in L^s$, and $w \in L^s$ by extension closure. If $w \in L^s$, let v be the shortest prefix of w that lies in L. Then $v \in D$, so v is a leaf of t. If $v \in L^{s'}$, then $w \in L^{s'}$ by extension closure, so $s' = s$ by disjointness; thus w meets the leaf v of sign s, and $w \in L_t^s$.

For uniqueness, we show that the leaf pair determines the tree. A word w is a position of t iff no proper prefix of w is a leaf, iff no proper prefix of w lies in $L_t^0 \cup L_t^1$: a proper prefix in this union meets a leaf, which is again a proper prefix of w. A position w is labeled f iff it is not a leaf, iff $w \notin L_t^0 \cup L_t^1$, since the only leaf that a position can meet is itself. A leaf $w \in L_t^s$ has sign s, so its label is $\top$ if $s \oplus \mathrm{par}(w) = 0$ and $\bot$ otherwise.

(2) For trees t and u put $T = v_0 t$ and $U = v_0 u$; they have the domains of t and u. By Lemma 2.2(2), $t \le_c u$ iff $T \le U$ in the covariant order, that is, by Eq. (3) for the covariant variance,

$$t \le_c u \iff T(w) \le U(w) \text{ at every common position } w \text{ of } t \text{ and } u.$$

Moreover, a leaf v of t has sign 0 iff $T(v) = \top$, and sign 1 iff $T(v) = \bot$, because ν_0 exchanges the label of v iff $\mathrm{par}(v) = 1$.

Assume $t \leq_c u$, and let $w \in L^0_t$. Then w meets a leaf v of t with $T(v) = \top$, and every proper prefix of v is a position of t labeled f. If u has a leaf at a proper prefix v' of v, then v' is a common position, $f = T(v') \leq U(v')$ forces $U(v') = \top$, and w meets the leaf v' of u of sign 0. Otherwise v is a position of u, by the chain property, and $\top = T(v) \leq U(v)$ forces $U(v) = \top$; again $w \in L^0_u$. Symmetrically, let $w \in L^1_u$, so that w meets a leaf v of u with $U(v) = \bot$. If t has a leaf at a proper prefix v' of v, then $T(v') \leq U(v') = f$ forces $T(v') = \bot$; otherwise v is a position of t, and $T(v) \leq \bot$ forces $T(v) = \bot$. In both cases $w \in L^1_t$.

Conversely, assume $L^0_t \subseteq L^0_u$ and $L^1_u \subseteq L^1_t$, and let w be a common position. If $T(w) = \top$, then w is a leaf of t of sign 0, so $w \in L^0_u$. As w is a position of u, the leaf of u that w meets is w itself, so $U(w) = \top$. If $U(w) = \bot$, then likewise $T(w) = \bot$. In all remaining cases $T(w) \in \{\bot, f\}$ and $U(w) \in \{f, \top\}$. So $T(w) \leq U(w)$ in every case, and $t \leq_c u$.

(3) The tree $\top$ consists of the leaf ε, of polarity 0 and hence of sign 0; every word meets it, so $L^0_\top = \Sigma^*$ and $L^1_\top = \varnothing$. Likewise the leaf ε of $\bot$ has sign 1. The dual t^∂ has the leaves of t, with $\bot$ and $\top$ exchanged, so every leaf changes its sign, and $L^s_{t^\partial} = L^{1-s}_t$; equivalently, $\nu_s t^\partial = \nu_{1-s} t$ by Lemma 2.2(1). Let $t = f(t_i)_i$. Its root is labeled f, so ε meets no leaf and lies in neither cone. The positions of t other than ε are the words iv with v a position of t_i, with $t(iv) = t_i(v)$, and $\mathrm{par}(iv) = c(i) \oplus \mathrm{par}(v)$. Hence iv is a leaf of t iff v is a leaf of t_i, and then the sign of iv in t is the sign of v in t_i plus $c(i)$. So iw meets a leaf of t of sign s iff w meets a leaf of t_i of sign $s \oplus c(i)$. Finally, ε lies in a cone iff it meets a leaf, that is, iff ε is a leaf, and the leaf ε has sign 0 iff it is labeled $\top$.

(4) A word w is a position labeled f iff it meets no leaf, that is, iff $w \notin L^0_t \cup L^1_t$. The positions of t are ε and the children of its positions labeled f, so t has at most $1 + nk$ positions if it has k positions labeled f; hence t is finite iff $\Sigma^* \setminus (L^0_t \cup L^1_t)$ is finite.

For regularity, let w be a position of t and $w^{-1}t$ the subtree of t at w. The proper prefixes of w are not leaves, so wv meets a leaf of t iff it meets a leaf wv' with v' a leaf of $w^{-1}t$, and the sign of wv' in t is the sign of v' in $w^{-1}t$ plus $\mathrm{par}(w)$. Hence

$$L^s_{w^{-1}t} = w^{-1} L^{s \oplus \mathrm{par}(w)}_t \qquad \text{for every position } w \text{ of } t \text{ and every sign } s. \tag{16}$$

If w is not a position of t, then a proper prefix of w is a leaf, of some sign s, and $w^{-1}L^s_t = \Sigma^*$ and $w^{-1}L^{1-s}_t = \varnothing$. Suppose L^0_t and L^1_t are regular. By the Myhill–Nerode theorem they have finitely many left quotients, so by Eq. (16) the subtrees of t have finitely many leaf pairs, and by (1) t has finitely many distinct subtrees. Conversely, if t has finitely many distinct subtrees, then by Eq. (16) and the case of non-positions, the pair $(w^{-1}L^0_t, w^{-1}L^1_t)$ takes finitely many values as w ranges over all words, so both languages have finitely many left quotients and are regular. □

Proof of Lemma 3.5. Let ρ be an assignment and write $t_v = \rho(v)$. By Theorem 3.2(2), $u \leq v$ holds iff $L^0_{t_u} \subseteq L^0_{t_v}$ and $L^1_{t_v} \subseteq L^1_{t_u}$, that is, iff S_ρ is closed under the two copies of $u \leq v$.

Let $t = f(t_{a_i})_i$. Every nonempty word has the form iw, so by Theorem 3.2(3), $L^s_t = \bigcup_{i \in \Sigma} i L^{s \oplus c(i)}_{t_{a_i}}$. The literal $f(a_i)_i \leq u$ holds iff $t \leq_c t_u$, iff $L^0_t \subseteq L^0_{t_u}$ and $L^1_{t_u} \subseteq L^1_t$. The first inclusion says that $w \in L^{c(i)}_{t_{a_i}}$ implies $iw \in L^0_{t_u}$, for all i and w, which is closure under push_i. The second says that $\varepsilon \notin L^1_{t_u}$, which is the avoidance of the forbidden root (u^1, ε), and that $iw \in L^1_{t_u}$ implies $w \in L^{\bar{c}(i)}_{t_{a_i}}$, which is closure under pop_i. The literal $u \leq f(a_i)_i$ holds iff $L^0_{t_u} \subseteq L^0_t$ and $L^1_t \subseteq L^1_{t_u}$. The first inclusion says that $\varepsilon \notin L^0_{t_u}$ and that $iw \in L^0_{t_u}$ implies $w \in L^{c(i)}_{t_{a_i}}$, which are the forbidden root (u^0, ε) and closure under pop_i; the second says that $w \in L^{\bar{c}(i)}_{t_{a_i}}$ implies $iw \in L^1_{t_u}$, which is closure under push_i.

The literal $u = \bot$ holds iff $t_u = \bot$, that is, by Theorem 3.2(1,3), iff the leaf pair of t_u is $(\varnothing, \Sigma^*)$. This holds iff $\varepsilon \in L^1_{t_u}$: if so, then $L^1_{t_u} = \Sigma^*$ by extension closure, and $L^0_{t_u} = \varnothing$ by disjointness. So $u = \bot$ holds iff S_ρ contains the fact (u^1, ε). Likewise $\top \le u$ holds iff $t_u = \top$, iff $(u^0, \varepsilon) \in S_\rho$. This proves the first claim.

For the second, $\rho \mapsto S_\rho$ is a bijection from the assignments onto the extension-closed sets without exclusion pairs, as noted before Table 3. Every model has these two properties, and by the first claim ρ is a solution of χ iff S_ρ is a model of χ. So the bijection restricts to a bijection from the solutions onto $\mathrm{Mod}(\chi)$. □

Proof of Lemma 3.6. (1) Every scheme of Table 3 holds for all words w; replacing w by $w\sigma$ turns the move $(p, u) \to (q, t)$ into $(p, u\sigma) \to (q, t\sigma)$.

(2) For a copy, $t = u$ and $\mathrm{sgn}(q) = \mathrm{sgn}(p)$. For the other moves we use $\mathrm{par}(iw) = c(i) \oplus \mathrm{par}(w)$. The move $(a_i^{c(i)}, w) \to (u^0, iw)$ has $c(i) \oplus \mathrm{par}(w) = 0 \oplus \mathrm{par}(iw)$; the move $(u^1, iw) \to (a_i^{\bar{c}(i)}, w)$ has $1 \oplus \mathrm{par}(iw) = 1 \oplus c(i) \oplus \mathrm{par}(w)$; the move $(u^0, iw) \to (a_i^{c(i)}, w)$ has $\mathrm{par}(iw) = c(i) \oplus \mathrm{par}(w)$; and the move $(a_i^{\bar{c}(i)}, w) \to (u^1, iw)$ has $1 \oplus c(i) \oplus \mathrm{par}(w) = 1 \oplus \mathrm{par}(iw)$. □

Proof of Lemma 3.8. (1) Every set that contains $\mathrm{Fact}_\chi \cup A$ and is closed under the rules contains $H_\chi(A)$, by induction along derivations. Conversely, $H_\chi(A)$ contains $\mathrm{Fact}_\chi \cup A$, since derivations may have length zero, and it is closed under the rules: each rule has a single premise, so applying it to an atom of $\mathrm{Reach}_\chi(x)$ yields an atom of $\mathrm{Reach}_\chi(x)$.

(2) A model M that contains A contains Fact_χ and is closed under the rules, so $H_\chi(A) \subseteq M$ by (1); since M is consistent, so is every subset of M. Conversely, if $H_\chi(A)$ is consistent, then it is a model by (1), and by the first part it is the least model containing A. The remaining claims are the cases $A = \varnothing$ and $A = \{a\}$; for the first, χ is satisfiable iff it has a model, by Lemma 3.5.

(3) An inconsistent set contains a forbidden root or an exclusion pair, and every atom of $H_\chi(A)$ has a seed in $\mathrm{Fact}_\chi \cup A$.

(4) The extensions by the letters of α derive $(p, u\alpha)$ from (p, u), so $\mathrm{Reach}_\chi(p, u\alpha) \subseteq \mathrm{Reach}_\chi(p, u)$ and $\mathrm{Cl}_\chi(p, u\alpha) \subseteq \mathrm{Cl}_\chi(p, u)$. A model that contains (p, u) is closed under extensions, so it contains $(p, u\alpha)$.

(5) If $S = T|_X$ for a model T of χ, then T contains S and Fact_χ and is closed under the rules, so $H_\chi(S) \subseteq T$ by (1). Hence $H_\chi(S)$ is consistent, and $S \subseteq H_\chi(S)|_X \subseteq T|_X = S$. Conversely, if $H_\chi(S)$ is consistent and $H_\chi(S)|_X = S$, then $H_\chi(S)$ is a model by (1), with visible part S, and it is contained in every model that contains S. The hypothesis on S only ensures that S is the set of atoms of an assignment of X. □

Proof of Lemma 3.11. We show by induction on k that if a derivation of length k leads from (p, u) to b, then $(p, u\alpha) \to^* b$ for some α. For $k = 0$ take $\alpha = \varepsilon$. For $k > 0$, let the derivation end with a rule from b' to b, and by induction let $(p, u\alpha') \to^* b'$. If the last rule is a pure move, append it, with $\alpha = \alpha'$. If it is an extension from $b' = (q, t)$ to $b = (q, ti)$, apply Lemma 3.6(1) with $\sigma = i$ to each move of the pure derivation from $(p, u\alpha')$ to (q, t); this gives $(p, u\alpha' i) \to^* (q, ti)$, and $\alpha = \alpha' i$. Conversely, if $(p, u\alpha) \to^* b$, then the extensions by the letters of α, followed by this pure derivation, form a derivation from (p, u) to b. □

Proof of Corollary 3.13. Put $a = (p, u)$. Suppose that (a), (b) or (c) holds. In case (a), $\mathrm{Cl}_\chi(a) \supseteq \mathrm{Reach}_\chi(a)$ contains a forbidden root. In cases (b) and (c), $\mathrm{Cl}_\chi(a)$ contains $(\bar{p}, u\delta)$, and it contains $(p, u\delta)$ by Lemma 3.8(4), an exclusion pair. In each case $\mathrm{Cl}_\chi(a)$ is inconsistent, and a is dead by Lemma 3.8(2).

Conversely, let a be dead. Then $\mathrm{Cl}_\chi(a)$ is inconsistent; take seeds $x, y \in \mathrm{Fact}_\chi \cup \{a\}$ as in Lemma 3.8(3). Since χ is satisfiable, μ_χ is consistent. If $\mathrm{Reach}_\chi(x)$ contains a forbidden root, then

$x \notin \mathrm{Fact}_\chi$, so $x = a$, and (a) holds. Otherwise $b \in \mathrm{Reach}_\chi(x)$ and $\bar{b} \in \mathrm{Reach}_\chi(y)$ for some atom b, and x and y are not both facts. Since $\{b, \bar{b}\}$ is unordered, we may assume $y = a$. If x is a fact e, then Lemma 3.12 gives $(\bar{p}, u\delta) \in \mathrm{Reach}_\chi(e) \subseteq \mu_\chi$ for some δ, which is (b). If $x = a$, it gives $(\bar{p}, u\delta) \in \mathrm{Reach}_\chi(a)$, which is (c). The last two claims follow from Lemma 3.11. □

C Proofs for Summaries, Condition (U), and Extracted Instances

C.1 Summaries and the Lowest Point

Throughout, χ is a system over V, pure derivations are those of χ, that is, of the copy and constructor literals of $\chi^\pm$ (Section 5.1), and the automata are those of Definition 5.1.

Lemma C.1 (Lifting).

(1) If $(a, \eta) \to_\chi^* (b, \eta')$, then $(a, \eta\sigma) \to_\chi^* (b, \eta'\sigma)$ by a pure derivation above σ, for every word σ.
(2) A pure derivation that starts at an atom $(a, \eta\sigma)$ and whose atoms all have height at least $|\sigma|$ is above σ, and deleting the suffix σ from all its words yields a pure derivation.

Proof. (1) is Lemma 3.6(1), applied move by move; every word of the new derivation ends with σ. (2) Suppose that an atom $(r, \eta\sigma)$ is followed by a move to an atom of height at least $|\sigma|$. A copy or a push yields an atom $(r', \eta'\sigma)$ with $(r, \eta) \to_\chi (r', \eta')$. A pop must have $\eta \neq \varepsilon$, since popping the first letter of σ would lower the height below $|\sigma|$; so again the result is $(r', \eta'\sigma)$ with $(r, \eta) \to_\chi (r', \eta')$. Induction along the derivation proves both claims. □

Proof of Lemma 5.2. (a) For the forward implication, induct on the derivation of $D_\chi(a, b)$ by the rules of Definition 5.1, constructing a pure derivation for every w at once. Reflexivity gives the empty derivation, a copy literal $a \le b$ of $\chi^\pm$ one copy, and transitivity a concatenation. For the decomposition rule, which concludes $D_\chi(a_i, b_i)$ from $f(a_j)_j \le u$, $D_\chi(u, v)$, and $v \le f(b_j)_j$: push i from (a_i, w) to (u, iw), follow the derivation $(u, iw) \to_\chi^* (v, iw)$ above iw given by the induction hypothesis, and pop i to (b_i, w). The result is above w.

For the reverse implication, induct on the length of the derivation. If it is empty, $a = b$. Otherwise its first move, from (a, w), is a copy or a push, since a pop would produce a word shorter than w. If it is a copy to (a', w) by a literal $a \le a'$, then $D_\chi(a, a')$, the induction hypothesis gives $D_\chi(a', b)$ for the rest, and transitivity concludes. If it is a push of i to (u, iw) by a literal $f(a_j)_j \le u$ with $a = a_i$, let (b', w) be the first later atom of height $|w|$; it exists because the derivation ends at height $|w|$. The atoms from (u, iw) up to the one just before (b', w) have height at least $|w| + 1$, so by Lemma C.1(2) they form a derivation above iw, which ends at an atom (v, iw). The next move lowers the height, so it is a pop of i by a literal $v \le f(b_j)_j$, and $b' = b_i$. The induction hypothesis gives $D_\chi(u, v)$, the decomposition rule gives $D_\chi(a, b')$, and the induction hypothesis applied to the rest, from (b', w) to (b, w), gives $D_\chi(b', b)$. Transitivity concludes.

(b) Induct on $|\gamma|$. For $\gamma = \varepsilon$, $P_p^\chi(\varepsilon) = \uparrow_\chi p$, and (a) applies. Let $\gamma = \gamma' i$. By definition, $b \in P_p^\chi(\gamma' i)$ iff there are $r \in P_p^\chi(\gamma')$ and a literal $r \le f(b_j)_j$ of $\chi^\pm$ with $D_\chi(b_i, b)$. If so, the induction hypothesis with $i\sigma$ in place of σ, a pop of i, and (a) compose to the required derivation. Conversely, given a derivation from $(p, \gamma' i\sigma)$ to (b, σ) above σ, let (b', σ) be its first atom of height $|\sigma|$. The earlier atoms have height at least $|\sigma| + 1$, so by Lemma C.1(2) they form a derivation above $i\sigma$, which ends at some $(r, i\sigma)$; the next move is a pop of i by a literal $r \le f(b_j)_j$, and $b' = b_i$. The induction hypothesis gives $r \in P_p^\chi(\gamma')$, and (a) applied to the rest gives $D_\chi(b_i, b)$. Hence $b \in \delta_i^\uparrow(P_p^\chi(\gamma')) = P_p^\chi(\gamma)$.

(c) Induct on $|\beta|$. For $\beta = \varepsilon$, $Q_q^\chi(\varepsilon) = \downarrow_\chi q$, and (a) applies. Let $\beta = \beta' i$. By definition, $a \in Q_q^\chi(\beta' i)$ iff there are $r \in Q_q^\chi(\beta')$ and a literal $f(a_j)_j \le r$ of $\chi^\pm$ with $D_\chi(a, a_i)$. If so, (a), a push of i from (a_i, σ) to $(r, i\sigma)$, and the induction hypothesis with $i\sigma$ in place of σ compose. Conversely, given a derivation from (a, σ) to $(q, \beta' i\sigma)$ above σ, let (a', σ) be its last atom of height $|\sigma|$. It is not the

final atom, and all later atoms have height at least $|\sigma| + 1$. So the next move is a push from (a', σ) to some $(r, i'\sigma)$ by a literal $f(a_j)_j \le r$ with $a' = a_{i'}$, and by Lemma C.1(2) the rest is a derivation above $i'\sigma$. Its final word $\beta' i \sigma$ therefore ends with $i'\sigma$, so $i' = i$. The induction hypothesis gives $r \in Q^{\chi}_{q}(\beta')$, and (a) applied to the part before (a', σ) gives $D_\chi(a, a_i)$.

The last claim holds because the conditions on the left do not mention w or σ. □

Proof of Lemma 5.3. Given the derivation, let h be its least height and σ the suffix of w of length h. Every atom has height at least $|\sigma|$, so the derivation is above σ by Lemma C.1(2), and its first atom of height h is some (r, σ). Then $v = \beta\sigma$ for a word β; put $i = |w| - h$ and $e = |v| - h$, so that $w[i:] = \sigma = v[e:]$. The part of the derivation up to (r, σ) leads from $(p, w[:i]\sigma)$ to (r, σ) above σ, so $r \in P^{\chi}_{p,i}(w)$ by Lemma 5.2(b). The rest leads from (r, σ) to $(q, v[:e]\sigma)$ above σ, so $r \in Q^{\chi}_{q,e}(v)$ by Lemma 5.2(c). Conversely, given i, e, and r in the intersection, put $\sigma = w[i:] = v[e:]$; Lemma 5.2(b,c) give pure derivations $(p, w) \to^*_\chi (r, \sigma) \to^*_\chi (q, v)$. □

Proof of Lemma 5.4. By Lemma 3.10, reversing a pure derivation and flipping every signed variable gives a pure derivation, and the dual of an atom has the same word. So the reversed derivation visits the same words, and it is above σ iff the original one is. (1) By Lemma 5.2(a), $D_\chi(a, b)$ iff $(a, w) \to^*_\chi (b, w)$ above w, iff $(\bar{b}, w) \to^*_\chi (\bar{a}, w)$ above w, iff $D_\chi(\bar{b}, \bar{a})$. Hence $r \in \downarrow_\chi \bar{q}$ iff $D_\chi(r, \bar{q})$ iff $D_\chi(q, \bar{r})$ iff $r \in \overline{\uparrow_\chi q}$. (2) By Lemma 5.2(c,b), $r \in Q^{\chi}_{\bar{q}}(w)$ iff $(r, \sigma) \to^*_\chi (\bar{q}, w\sigma)$ above σ, iff $(q, w\sigma) \to^*_\chi (\bar{r}, \sigma)$ above σ, iff $\bar{r} \in P^{\chi}_{q}(w)$. (3) By Lemma 5.2, $D_\chi(a, b)$ gives $(a, \varepsilon) \to^*_\chi (b, \varepsilon)$, $r \in P^{\chi}_{p}(w)$ gives $(p, w) \to^*_\chi (r, \varepsilon)$, and $r \in Q^{\chi}_{q}(w)$ gives $(r, \varepsilon) \to^*_\chi (q, w)$. The sign invariant of Lemma 3.6(2), extended to pure derivations, gives the three claims, since $\mathrm{par}(\varepsilon) = 0$. □

C.2 Least Closures and Deadness

Proof of Lemma 5.6. By definition, $(q, v) \in \mu_\chi$ iff $(q, v) \in \mathrm{Reach}_\chi(p, \varepsilon)$ for some fact (p, ε), iff, by Lemma 3.11, $(p, \alpha) \to^*_\chi (q, v)$ for some α. By Lemma 5.3, this holds iff there are cuts $i \le |\alpha|$ and $e \le |v|$ with $\alpha[i:] = v[e:]$ and $P^{\chi}_{p}(\alpha[:i]) \cap Q^{\chi}_{q,e}(v) \neq \emptyset$. If so, then $Q^{\chi}_{q,e}(v)$ meets $P^{\chi}_{p}(\gamma) \subseteq \mathrm{Fact}^{\uparrow}_{\chi}$ for $\gamma = \alpha[:i]$. Conversely, if $r \in P^{\chi}_{p}(\gamma) \cap Q^{\chi}_{q,e}(v)$ for a fact (p, ε), a word γ, and a cut e, take $\alpha = \gamma\, v[e:]$ and $i = |\gamma|$.

Let Y be the least set that contains $\uparrow_\chi p$ for every fact (p, ε) and contains $\delta^{\uparrow}_i(\{r\})$ for each of its elements r. Since $\delta^{\uparrow}_i(P) = \bigcup_{r \in P} \delta^{\uparrow}_i(\{r\})$, induction on γ gives $P^{\chi}_{p}(\gamma) \subseteq Y$, so $\mathrm{Fact}^{\uparrow}_{\chi} \subseteq Y$. Conversely, $\uparrow_\chi p = P^{\chi}_{p}(\varepsilon) \subseteq \mathrm{Fact}^{\uparrow}_{\chi}$, and if $r \in P^{\chi}_{p}(\gamma)$, then $\delta^{\uparrow}_i(\{r\}) \subseteq \delta^{\uparrow}_i(P^{\chi}_{p}(\gamma)) = P^{\chi}_{p}(\gamma i)$; so $\mathrm{Fact}^{\uparrow}_{\chi}$ has the closure properties of Y, and $Y \subseteq \mathrm{Fact}^{\uparrow}_{\chi}$. The set Y is computed by adding elements of $V^{\pm}$ until nothing changes, which takes at most $2|V|$ rounds of polynomial time each. □

Proof of Lemma 5.7. By definition, $H_\chi(S) = \mu_\chi \cup \bigcup_{a \in S} \mathrm{Reach}_\chi(a)$. By Lemma 5.6, $(q, v) \in \mu_\chi$ iff $v[:e] \in V_{r,q}$ for some cut e and some $r \in \mathrm{Fact}^{\uparrow}_{\chi}$, that is, iff $v \in \bigcup_{r \in \mathrm{Fact}^{\uparrow}_{\chi}} V_{r,q}\Sigma^*$. Next, let $w \in S_p$. By Lemmas 3.11 and 5.3, $(q, v) \in \mathrm{Reach}_\chi(p, w)$ iff there are a word α, cuts $i \le |w\alpha|$ and $e \le |v|$, and a signed variable r with $(w\alpha)[i:] = v[e:]$ and $r \in P^{\chi}_{p}((w\alpha)[:i]) \cap Q^{\chi}_{q,e}(v)$. Put $\gamma = (w\alpha)[:i]$, $\beta = v[:e]$, and $\sigma = v[e:]$. Then $v = \beta\sigma$, $\beta \in V_{r,q}$, $\gamma \in U_{p,r}$, and $\gamma\sigma = w\alpha \in S_p\Sigma^*$, so $v \in V_{r,q}\mathrm{Res}_{p,r}$. Conversely, if $v = \beta\sigma$ with $\beta \in V_{r,q}$ and $\gamma\sigma = w\alpha$ for some $\gamma \in U_{p,r}$, $w \in S_p$, and word α, then Lemma 5.3 with the cuts $|\gamma|$ and $|\beta|$ gives $(p, w\alpha) \to^*_\chi (q, v)$, hence $(q, v) \in \mathrm{Reach}_\chi(p, w)$ by Lemma 3.11. This proves the formula.

The language $V_{r,q}$ is recognized by the lower automaton of q with the accepting states $\{Q : r \in Q\}$, and $U_{p,r}$ by the upper automaton of p with the accepting states $\{P : r \in P\}$. Let M be a deterministic automaton for $S_p\Sigma^*$, which is regular. The set I of states of M reached by words of $U_{p,r}$ is computable: it consists of the second components of the reachable states (P, m) of the product of the upper

automaton of p with M such that $r \in P$. Then $\mathrm{Res}_{p,r}$ is the union over $m \in I$ of the languages accepted by M from m, which is regular. Concatenation, finite union, and the language Σ^* are effective. Finally, $H_\chi(S)$ is consistent iff $\varepsilon \notin H_\chi(S)_m$ for every forbidden root (m, ε) and $H_\chi(S)_{x^+} \cap H_\chi(S)_{x^-} = \varnothing$ for every variable x, which are emptiness tests for regular languages. □

Proof of Lemma 5.11. We first show that

$$P^\chi_p(wz) = \bigcup_{r \in P^\chi_p(w)} P^\chi_r(z) \qquad \text{for all words } w, z. \tag{17}$$

For $z = \varepsilon$ this holds because $P^\chi_p(w)$ is upward closed under D_χ and $r \in \uparrow_\chi r = P^\chi_r(\varepsilon)$. The induction step from z to zi applies $\delta_i^\uparrow$, which distributes over unions.

(1) By Lemma 5.9(3) and Eq. (17), $\mathsf{DeadU}_\chi(p, w)$ iff some $r \in P^\chi_p(w)$ and some z have $P^\chi_r(z) \cap \mathrm{Neg}^\downarrow_\chi \neq \varnothing$, that is, by Lemma 5.9(3) for the atom (r, ε), iff some $r \in P^\chi_p(w)$ has $\mathsf{DeadU}_\chi(r, \varepsilon)$.

(2) This is Lemma 5.6 for the atom $(\bar{p}, w)$, together with $Q^\chi_p = \overline{P^\chi_p}$ (Lemma 5.4(2)).

(3) Suppose $\mathsf{DeadT}_\chi(p, w)$. If $\mathsf{DeadU}_\chi(p, w)$, then (1) gives some $r \in P^\chi_p(w)$ with $\mathsf{DeadU}_\chi(r, \varepsilon)$, hence $r \in \mathsf{DeadT}^\varepsilon_\chi$. Otherwise $(\bar{p}, w\delta) \in \mu_\chi$ for some δ, and by (2) applied to $w\delta$ there is a cut $e \le |w\delta|$ with $\overline{P^\chi_p((w\delta)[:e])} \cap \mathsf{Fact}^\uparrow_\chi \neq \varnothing$. If $e \le |w|$, then $(\bar{p}, w) \in \mu_\chi$ by (2). If $e > |w|$, write $(w\delta)[:e] = wz$; by Eq. (17), some $r \in P^\chi_p(w)$ has $\overline{P^\chi_r(z)} \cap \mathsf{Fact}^\uparrow_\chi \neq \varnothing$, and (2) for the atom $(\bar{r}, z)$ gives $(\bar{r}, z) \in \mu_\chi$, so $r \in \mathsf{DeadT}^\varepsilon_\chi$.

Conversely, $(\bar{p}, w) \in \mu_\chi$ gives $\mathsf{DeadT}_\chi(p, w)$ with $\delta = \varepsilon$. Let $r \in P^\chi_p(w) \cap \mathsf{DeadT}^\varepsilon_\chi$; then $(p, w) \to^*_\chi (r, \varepsilon)$ by Lemma 5.2(b). If $\mathsf{DeadU}_\chi(r, \varepsilon)$, then $\mathrm{Reach}_\chi(r, \varepsilon) \subseteq \mathrm{Reach}_\chi(p, w)$ contains a forbidden root. If $(\bar{r}, \delta) \in \mu_\chi$, then duality reverses the derivation to $(\bar{r}, \varepsilon) \to^*_\chi (\bar{p}, w)$, Lemma C.1(1) lifts it to $(\bar{r}, \delta) \to^*_\chi (\bar{p}, w\delta)$, and $(\bar{p}, w\delta) \in \mu_\chi$ because μ_χ is closed under the rules (Lemma 3.8(1)).

(4) The first part of DeadT_χ is inherited by prefixes by Lemma 5.9(4), and $(\bar{p}, wv\delta) = (\bar{p}, w(v\delta))$.

(5) By Lemma 5.9(3) and Eq. (17), $r \in \mathsf{DeadU}^\varepsilon_\chi$ iff $r \in \mathrm{Neg}^\downarrow_\chi$, which is the case $z = \varepsilon$, or some letter i and some $b \in P^\chi_r(i)$ have $b \in \mathsf{DeadU}^\varepsilon_\chi$. So $\mathsf{DeadU}^\varepsilon_\chi$ is the least set that contains $\mathrm{Neg}^\downarrow_\chi$ and contains every r with $P^\chi_r(i) \cap \mathsf{DeadU}^\varepsilon_\chi \neq \varnothing$ for some i; it is computed by iteration in polynomial time. Similarly, by (2) and Eq. (17), the set M of signed variables r with $(\bar{r}, \delta) \in \mu_\chi$ for some δ consists of the r with $P^\chi_r(z) \cap \overline{\mathsf{Fact}^\uparrow_\chi} \neq \varnothing$ for some z; it is the least set that contains every r with $\uparrow_\chi r \cap \overline{\mathsf{Fact}^\uparrow_\chi} \neq \varnothing$ and every r with $P^\chi_r(i) \cap M \neq \varnothing$ for some i. Finally $\mathsf{DeadT}^\varepsilon_\chi = \mathsf{DeadU}^\varepsilon_\chi \cup M$. □

Proof of Theorem 5.12. By Corollary 3.13, (p, w) is dead iff (a) $\mathsf{DeadU}_\chi(p, w)$, or (b) $(\bar{p}, w\delta) \in \mu_\chi$ for some δ, or (c) (p, w) has a self-collision, that is, $(p, w\alpha) \to^*_\chi (\bar{p}, w\delta)$ for some words α and δ. Causes (a) and (b) together are $\mathsf{DeadT}_\chi(p, w)$, and the causes may overlap. It remains to show that a self-collision implies $\mathsf{DeadT}_\chi(p, w)$ or a self event, and that a self event implies a self-collision.

Assume a self-collision. By Lemmas 5.3 and 5.4(2), there are cuts $i \le |w\alpha|$ and $j \le |w\delta|$ with $(w\alpha)[i:] = (w\delta)[j:]$ and

$$P^\chi_p((w\alpha)[:i]) \cap \overline{P^\chi_p((w\delta)[:j])} \neq \varnothing.$$

In particular both sets $P^\chi_p((w\alpha)[:i])$ and $P^\chi_p((w\delta)[:j])$ are nonempty. If $i > |w|$, then $(w\alpha)[:i]$ is a proper extension of w with a nonempty upper set, and the pop bound (Lemma 5.9(2) with $k = |w|$) gives $\mathsf{DeadU}_\chi(p, w)$; the case $j > |w|$ is the same. Otherwise $i, j \le |w|$, and the equation reads $w[i:]\,\alpha = w[j:]\,\delta$, so $w[i:]$ and $w[j:]$ are prefixes of one word and hence comparable in the prefix order, and the displayed condition reads $P^\chi_{p,i}(w) \cap \overline{P^\chi_{p,j}(w)} \neq \varnothing$. By sign coherence (Lemma 5.4(3)), the elements of $P^\chi_{p,i}(w)$ have the sign $\mathrm{sgn}(p) \oplus \mathrm{par}(w[:i])$ and those of $\overline{P^\chi_{p,i}(w)}$ the opposite sign,

so $i \neq j$. If $i < j$, then $w[j{:}]$ is the shorter suffix, so $w[j{:}] \preceq w[i{:}]$, and (i, j) is a self event. If $j < i$, flipping both sets gives $P^{\chi}_{p,j}(w) \cap \overline{P^{\chi}_{p,i}(w)} \neq \varnothing$ and $w[i{:}] \preceq w[j{:}]$, so (j, i) is a self event.

Conversely, let (i, j) be a self event of (p, w). Then $w[i{:}] = w[j{:}]\,\delta$ for some word δ, and the guard says, by Lemma 5.4(2), that $P^{\chi}_{p,i}(w) \cap Q^{\chi}_{\bar{p}}(w[{:}j]) \neq \varnothing$. Since $(w\delta)[{:}j] = w[{:}j]$ and $(w\delta)[j{:}] = w[j{:}]\,\delta = w[i{:}]$, Lemma 5.3 gives $(p, w) \to^{*}_{\chi} (\bar{p}, w\delta)$, a self-collision with $\alpha = \varepsilon$. □

C.3 The Reduced Unit Reader and the Rejection Clock

Proof of Lemma 6.9. (1) The bit b never returns from 1 to 0, and if du fails at a cut, it fails at every later cut (Lemma 6.3(2)). So the replaced states are closed under the transitions of $\mathcal{R}_{\mathrm{U}}(p)$, and a run that meets one of them stays among them. None of them is rejected, since a rejected state has $b = 0$ and satisfies du. Hence a word rejected by $\mathcal{R}_{\mathrm{U}}(p)$ meets no replaced state, and a word rejected by $\widehat{\mathcal{R}}_p$ never meets the absorbing state $\bot$. In both cases the run consists of kept states, between which the transitions are unchanged, so the word has the same run in both readers and is rejected by both. Each comparison of the word is tested on states of this run, which carry the same self guard in both readers. So the two readers reject the same words, with the same states at every cut and the same admitted comparisons, and they have the same holes.

(2) If two prefixes of a word are rejected by $\widehat{\mathcal{R}}_p$, they are rejected by $\mathcal{R}_{\mathrm{U}}(p)$ by (1); by betweenness for $\mathcal{R}_{\mathrm{U}}(p)$ (Lemma 6.6), so is every intermediate prefix, and by (1) again it is rejected by $\widehat{\mathcal{R}}_p$. A word rejected by $\widehat{\mathcal{R}}_p$ has, by (1), the same first rejected cut and the same admitted comparisons in both readers, so the admission cone of $\mathcal{R}_{\mathrm{U}}(p)$ transfers.

(3) A kept state after a word u is $\zeta = (P^{\varphi}_{p}(u), 0, P^{\psi}_{p}(u))$ by Lemma 6.3(1), and $\mathrm{du}(\zeta)$ gives $P^{\psi}_{p}(u) \cap \mathrm{DeadU}^{\varepsilon}_{\psi} \neq \varnothing$, so $P^{\psi}_{p}(u) \neq \varnothing$. By Lemma 5.4(3), applied in φ and in ψ to the same seed p and the same word u, all elements of $P^{\varphi}_{p}(u)$ and of $P^{\psi}_{p}(u)$ have the sign $\mathrm{sgn}(p) \oplus \mathrm{par}(u)$, which the nonempty set $P^{\psi}_{p}(u)$ determines. For each sign there are at most 2^{v} sets $P \subseteq X^{\pm}$ and $2^{s} - 1$ nonempty sets $P' \subseteq (X \cup Z)^{\pm}$ of that sign. Adding $\bot$ gives at most $1 + 2^{v+1}(2^{s} - 1) = 2^{v+s+1} - 2^{v+1} + 1 < 2^{v+s+1}$ states. □

Proof of Lemma 6.10. Let w be rejected by $\mathcal{R}_{\mathrm{U}}(p)$, and let $\zeta_j = (P_j, b_j, P'_j)$ be the state of $\mathcal{B}_p$ at the cut j, so that $P_j = P^{\varphi}_{p,j}(w)$ is also the state of $\mathcal{A}_p$ there. At the last cut, $b = 0$ and du holds, so by Lemma 6.3(2) and the monotonicity of the bit, $b_j = 0$ and $\mathrm{du}(\zeta_j)$ hold at every cut j. Hence $\zeta_j \in \mathrm{Rej}_{\mathrm{U}}$ iff $\mathrm{dt}(\zeta_j)$ fails, iff $P_j \cap \mathrm{DeadT}^{\varepsilon}_{\varphi} = \varnothing$, which is rejection by $\mathcal{A}_p$ at the cut j. So w is rejected by $\mathcal{A}_p$, with the same first rejected cut, and $r(w)$ is the same for both readers. By sign coherence (Lemma 5.4(3)), every reachable state of $\mathcal{A}_p$ is empty or a nonempty set of elements of $X^{\pm}$ of one sign, so $\mathcal{A}_p$ has at most $2(2^{v} - 1) + 1 = 2^{v+1} - 1$ states. Theorem 2.19 holds for every reader and every nonempty rejected word, without any condition on the admission relation, so for $w \neq \varepsilon$ it gives $\mathrm{depth}(w) \leq 2(2^{v+1} - 1)(r(w) + 1) - 1$. □

C.4 Extracted Instances

Proof of Theorem 7.7. Write ζ_e for the state of $B = \mathcal{B}_p$ at the cut e of t. By Definition 7.5 and Lemma 6.3, $A_s(u) = Q^{\varphi}_{q,s}(u)$, $A'_s(u) = Q^{\psi}_{q,s}(u)$, $B'_e(t) = P^{\psi}_{p,e}(t)$, and $\zeta_e = (P^{\varphi}_{p,e}(t), [(\bar{p}, t[{:}e]) \in \mu_{\varphi}], P^{\psi}_{p,e}(t))$. We translate the conditions of Definition 7.1.

- Condition (a) says that $Q^{\varphi}_{q,s}(u) \cap \mathrm{Fact}^{\uparrow}_{\varphi} = \varnothing$ for every cut s, that is, $(q, u) \notin \mu_{\varphi}$ (Lemma 5.6).
- By Lemma 6.3(2), a cut that is rejected stays rejected at every later cut, so condition (b) says that the last cut of t is rejected: neither $\mathrm{DeadT}_{\varphi}(p, t)$ nor $\mathrm{DeadU}_{\psi}(p, t)$ holds.

- Condition (d) says that some cuts $s \le |u|$ and $e \le |t|$ have $t[e{:}] = u[s{:}]$ and $P^{\psi}_{p,e}(t) \cap Q^{\psi}_{q,s}(u) \neq \varnothing$; by Lemma 5.3 for ψ, this says $(p,t) \to^*_{\psi} (q,u)$.
- Assume (b). At the cut $J(t)$, dt fails, so $\mathrm{DeadU}_{\varphi}(p, t[{:}J(t)])$ fails, and the pop bound (Lemma 5.9(2)) gives $P^{\varphi}_{p,e}(t) = \varnothing$ for every $e > J(t)$. A guard $Q^{\varphi}_{q,s}(u) \cap P^{\varphi}_{p,e}(t) \neq \varnothing$ or a self guard $P^{\varphi}_{p,i}(t) \cap \overline{P^{\varphi}_{p,j}(t)} \neq \varnothing$ therefore forces $e \le J(t)$, respectively $j \le J(t)$. Hence, under (b), condition (c) says that there are no cuts $e \le |t|$ and $s \le |u|$ with $t[e{:}] \preceq u[s{:}]$ and $P^{\varphi}_{p,e}(t) \cap Q^{\varphi}_{q,s}(u) \neq \varnothing$, and condition (e) says that (p,t) has no self event in φ (Lemma 6.3(3)).

Let (u,t) be a hole. By (b) and (e), (p,t) is not dead by support and has no self event, so it is live by Theorem 5.12, since φ is satisfiable. By (d), $(p,t) \to^*_{\psi} (q,u)$. By (a) and (c), and by Lemma 5.15 for the live seed (p,t) and the atom (q,u), $(q,u) \notin \mathrm{Cl}_{\varphi}(p,t)$.

Conversely, let (p,t) be live in φ, $(p,t) \to^*_{\psi} (q,u)$, and $(q,u) \notin \mathrm{Cl}_{\varphi}(p,t)$. By Theorem 5.12, $\mathrm{DeadT}_{\varphi}(p,t)$ fails and (p,t) has no self event in φ; by (U) and Lemma 6.1, $\mathrm{DeadU}_{\psi}(p,t)$ fails. So (b) and (e) hold. Condition (d) holds by the derivation. Since $(q,u) \notin \mathrm{Cl}_{\varphi}(p,t)$, Lemma 5.15 gives $(q,u) \notin \mu_{\varphi}$, which is (a), and the absence of the cuts described above, which is (c).

(1) By Lemma 4.3(2), (II) fails iff some visible atom (p,t) that is live in φ and some visible atom (q,u) satisfy $(p,t) \to^*_{\psi} (q,u)$ and $(q,u) \notin \mathrm{Cl}_{\varphi}(p,t)$. By the equivalence just proved, this happens iff some I_{pq} has a hole.

(2) Let (u,t) be a hole of I_{pq}. Then (p,t) is live, so $S = \mathrm{Cl}_{\varphi}(p,t)$ is a model of φ (Lemma 3.8(2)). If $S = T|_X$ for a model T of ψ, then T contains (p,t) and is closed under the rules of ψ, so it contains (q,u), which is visible; hence $(q,u) \in S$, a contradiction. By Lemma 3.5, the solution of φ whose set of atoms is S has no extension to a solution of ψ. □

D Proofs for Non-Mortal Holes

This appendix contains the proofs omitted from Section 8: facts on infinite periodic words (Appendix D.1), the continuation identity and the decomposition (Appendix D.2), the breaker and thin states (Appendix D.3), survival (Appendix D.4), the pre-hole graph (Appendix D.5), the pump (Appendix D.6), and the three-case theorem (Appendix D.7). We keep the conventions of Section 8.

D.1 Infinite Periodic Words

Periods of finite words are as in Section 2.1, and the equivalence $\theta \preceq o\theta \iff \theta \preceq o^{\omega}$ for a nonempty word o is Lemma 2.1. We use the theorem of Fine and Wilf [Fine and Wilf 1965; Lothaire 2002] in its word form: a finite word with periods ϖ and ϖ' and length at least $\varpi + \varpi' - \gcd(\varpi, \varpi')$ has period $\gcd(\varpi, \varpi')$. It concerns equality of letters only and holds over every alphabet. The following facts on infinite words are used throughout.

Lemma D.1 (Infinite periodic words).

(a) Let ξ' be an infinite word with a period ϖ', and let $m \ge 0$. If $\xi'[m{:}]$ has a period ϖ, then ξ' has the period ϖ.

(b) Let ξ be purely periodic with least period ϖ. Every period of a suffix of ξ is a period of ξ, and therefore at least ϖ. For $i, j \ge 0$, $\xi[i{:}] = \xi[j{:}]$ if and only if $i \equiv j \pmod{\varpi}$. In particular, if $\xi = \alpha\lambda^{\omega}$ with $\lambda \neq \varepsilon$, then $\varpi \le |\lambda|$.

(c) Let o be a nonempty word. An infinite word whose finite prefixes are all prefixes of o^{ω} equals o^{ω}. If ξ' is an infinite word with $o\xi' = \xi'$, then $\xi' = o^{\omega}$; in particular ξ' has the period $|o|$.

(d) Let $\xi_1 = \alpha_1\lambda_1^{\omega}$ and $\xi_2 = \alpha_2\lambda_2^{\omega}$ with $\lambda_1, \lambda_2 \neq \varepsilon$. If ξ_1 and ξ_2 agree on their first $\max(|\alpha_1|, |\alpha_2|) + |\lambda_1| + |\lambda_2| - 1$ letters, then $\xi_1 = \xi_2$.

Proof. (a) Let $k \geq 0$, and choose $N \geq 0$ with $N\varpi' \geq m$. Using the period ϖ' of ξ' twice and the period ϖ of $\xi'[m{:}]$ once,

$$\begin{aligned}\xi'[k] &= \xi'[k + N\varpi'] = (\xi'[m{:}])[k + N\varpi' - m] \\ &= (\xi'[m{:}])[k + N\varpi' - m + \varpi] = \xi'[k + N\varpi' + \varpi] = \xi'[k + \varpi].\end{aligned}$$

(b) If a suffix $\xi[m{:}]$ has a period ϖ', then (a), applied to $\xi' = \xi$ with its period ϖ, shows that ξ has the period ϖ'; so $\varpi' \geq \varpi$. If $i \equiv j \pmod{\varpi}$, then $\xi[i{:}] = \xi[j{:}]$, since ϖ is a period of ξ. Conversely, let $\xi[i{:}] = \xi[j{:}]$, and let i' and j' be the remainders of i and j modulo ϖ. Then $\xi[i'{:}] = \xi[i{:}]$ and $\xi[j'{:}] = \xi[j{:}]$, so $\xi[i'{:}] = \xi[j'{:}]$. If $i' < j'$, this equation says that $\xi[i'{:}]$ has the period $j' - i'$, which is then a period of ξ smaller than ϖ; this is impossible, and so is $j' < i'$. Hence $i' = j'$. Finally, $\xi[|\alpha|{:}] = \lambda^\omega$ has the period $|\lambda|$, so $\varpi \leq |\lambda|$.

(c) The first claim holds letter by letter. If $o\xi' = \xi'$, then $\xi' = o^k\xi'$ for every k, so ξ' begins with o^k for every k, and $\xi' = o^\omega$.

(d) Let $k = \max(|\alpha_1|, |\alpha_2|)$, $\varpi_1 = |\lambda_1|$, $\varpi_2 = |\lambda_2|$ and $g = \gcd(\varpi_1, \varpi_2)$. The suffixes $\xi_1[k{:}]$ and $\xi_2[k{:}]$ have the periods ϖ_1 and ϖ_2, respectively, and they share the prefix v of length $\varpi_1 + \varpi_2 - 1 \geq \varpi_1 + \varpi_2 - g$, which therefore has both periods and, by the theorem of Fine and Wilf, the period g. As g divides ϖ_1 and $|v| \geq \varpi_1$, the prefix of length ϖ_1 of v is $(v[{:}g])^{\varpi_1/g}$, and so $\xi_1[k{:}] = (v[{:}g])^\omega$. Likewise $\xi_2[k{:}] = (v[{:}g])^\omega$. The first k letters agree by assumption, so $\xi_1 = \xi_2$. □

D.2 The Continuation Identity and the Decomposition

Proof of Lemma 8.1. (a) The automata read the same letters up to these cuts. Since $J(t) \leq |t|$, the cuts of $t\theta$ up to $J(t)$ carry the states of the same cuts of t, so $J(t\theta) = J(t)$.

(b) Let (s, e) be a cross comparison of $(u\theta, t\theta)$ with $e \leq J(t)$. Then $|u\theta| - s \geq |t\theta| - e \geq |\theta|$, because $e \leq |t|$; hence $s \leq |u|$. For $s \leq |u|$ and $e \leq |t|$ we have $(u\theta)[s{:}] = u[s{:}]\,\theta$ and $(t\theta)[e{:}] = t[e{:}]\,\theta$, so (s, e) is a cross comparison of $(u\theta, t\theta)$ if and only if $t[e{:}]\,\theta \preceq u[s{:}]\,\theta$. This relation implies $t[e{:}] \preceq u[s{:}]$: if $|t[e{:}]| \leq |u[s{:}]|$, then $t[e{:}]$ is a prefix of $u[s{:}]\,\theta$ of length at most $|u[s{:}]|$, and if $|t[e{:}]| > |u[s{:}]|$, then $t[e{:}]\,\theta$ is longer than $u[s{:}]\,\theta$. So it holds if and only if (s, e) is a cross comparison of (u, t) that θ keeps. For a self comparison, apply the same argument to the pair (t, t) and cuts $i < j \leq J(t)$.

(c) For a cross comparison (s, e) with overhang o we have $u[s{:}] = t[e{:}]\,o$, so $t[e{:}]\,\theta \preceq u[s{:}]\,\theta$ if and only if $\theta \preceq o\theta$, by canceling the common prefix $t[e{:}]$; the same holds for self comparisons. For $o \neq \varepsilon$ this is equivalent to $\theta \preceq o^\omega$ by Lemma 2.1. For $o = \varepsilon$ it always holds. □

Proof of the continuation facts. (K1) The events of $(u\theta, t\theta)$ are the cross comparisons (s, e) of $(u\theta, t\theta)$ with $e \leq J(t\theta) = J(t)$ whose states lie in G. By Lemma 8.1(b) they are the cross comparisons (s, e) of (u, t) with $e \leq J(t)$ that θ keeps, and by Lemma 8.1(a) their states are those in (u, t). The same argument applies to self events. (K2) is part of Lemma 8.1(c). (K3) An aligned comparison is never broken. If a word $\theta\theta'$ keeps a comparison with nonempty overhang o, then $\theta\theta' \preceq o^\omega$, hence $\theta \preceq o^\omega$, and θ keeps it. (K4) If $u[s{:}] = t[e{:}]$, then $(u\theta)[s{:}] = u[s{:}]\,\theta = t[e{:}]\,\theta = (t\theta)[e{:}]$, and the states of A' and B' at these cuts are unchanged. (K5) The states of A at the cuts $s \leq |u|$ of $u\theta$ are those of u, and its state at the cut $|u| + k$ is $A(A(u), \theta[{:}k])$. □

Proof of Lemma 8.3. Let (u_0, t_0) be a pre-hole completed by θ. By (K5), $u_0\theta$ is safe. By Lemma 8.1(a), $J(t_0\theta) = J(t_0)$ is defined. By (K4), $(u_0\theta, t_0\theta)$ has a head event. By (K1), its events are the events of (u_0, t_0) that θ keeps, and the self events of $t_0\theta$ are the self events of t_0 that θ keeps; there are none. So $(u_0\theta, t_0\theta)$ is a hole.

Conversely, let (u, t) be a hole and $J = J(t)$. It has a head event (s, e). Put $\gamma = t[{:}e]$ and $\sigma = u[s{:}] = t[e{:}]$. By (FH) and Lemma 7.4, $J \geq e = |\gamma|$. Hence $\theta = t[J{:}]$ has length $|t| - J \leq |t| - |\gamma| = |\sigma|$, so θ is a suffix of σ and therefore of u. Write $u = u_0\theta$ and $t = t_0\theta$. Then u_0 is safe, as a prefix of u,

and $J(t_0) = J = |t_0|$, since $B_i(t_0) = B_i(t)$ for $i \le |t_0|$. The head event satisfies $e = |\gamma| \le J = |t_0|$ and $s = |u| - |\sigma| \le |u| - |\theta| = |u_0|$, and $u_0[s{:}] = t_0[e{:}]$ because θ is a common suffix of $u[s{:}] = t[e{:}]$; by Lemma 8.1(a) the states at these cuts are unchanged, so (s, e) is a head event of (u_0, t_0). By (K1), the events of (u, t) are the events of (u_0, t_0) that θ keeps, and the self events of t are the self events of t_0 that θ keeps. Since (u, t) is a hole, θ breaks every obligation of (u_0, t_0), and by (K2) (u_0, t_0) has no aligned event. Thus (u_0, t_0) is a pre-hole, and θ, which is safe from $A(u_0)$ by (K5), completes it. □

D.3 The Breaker and Thin States

Proof of Lemma 8.6. Since q is branching, $\mathcal{G}_A$ has a path from q to a cyclic state r that has two distinct safe letters. Let α be the label of a shortest such path; it visits pairwise distinct states, so $|\alpha| < N_A$. Let λ be the label of a shortest cycle through r, so $1 \le |\lambda| \le N_A$. At least one of the two safe letters at r differs from $\lambda[0]$; call it ℓ. Let k be the least integer with $k|\lambda| \ge W + |\lambda|$, and put $\theta = \alpha\lambda^k\ell$. The word θ is safe from q: α leads to r, λ^k returns to r, and $A(r, \ell)$ is safe. By the choice of k, $k|\lambda| \le W + 2|\lambda| - 1$, so $|\theta| \le (N_A - 1) + (W + 2N_A - 1) + 1 < W + 3N_A$.

Let o be a word with $1 \le \Delta = |o| \le W$, and suppose that $\theta \preceq o^\omega$. Then θ has the period Δ, and so has its suffix $\theta' = \lambda^k\ell$. The prefix λ^k of θ' has the periods Δ and $|\lambda|$ and length $k|\lambda| \ge \Delta + |\lambda|$, so by the theorem of Fine and Wilf it has the period $g = \gcd(\Delta, |\lambda|)$. Since $k|\lambda| \ge \Delta$, the period Δ of θ' gives $\ell = \theta'[k|\lambda|] = \theta'[k|\lambda| - \Delta]$. The position $k|\lambda| - \Delta$ lies in λ^k and is divisible by g, so the period g of λ^k gives $\theta'[k|\lambda| - \Delta] = \theta'[0] = \lambda[0]$. Thus $\ell = \lambda[0]$, a contradiction.

For the second claim, let (u_0, t_0) be a pre-hole with $A(u_0)$ branching, and apply the first claim with $q = A(u_0)$ and $W = \max(|u_0|, |t_0|)$. Every obligation of (u_0, t_0) has a nonempty overhang o with $|o| \le W$ (Section 8.2). Since θ is not a prefix of o^ω, it breaks the obligation by Lemma 8.1(c). So θ completes (u_0, t_0). □

Proof of Lemma 8.8. (a) No state on a path from q in $\mathcal{G}_A$ is cyclic. Hence the $|\theta| + 1$ states visited by a word θ safe from q are pairwise distinct, since a repetition would close a cycle, and $|\theta| + 1 \le N_A$. The endpoint is mortal, since every state reachable from a mortal state is mortal.

(b) A cyclic state r has at least one safe letter, namely the first letter of a cycle through r. If it had two, q would be branching. So the unique safe letter at r is the first letter of every cycle through r, and it leads to the next state of such a cycle, which is cyclic.

(c) Let ξ be an infinite word safe from q, and let $r_0 = q, r_1, \ldots$ be the states it visits. Some state repeats, so some r_i is cyclic; let i be the least such index. By (b) and induction, every r_j with $j \ge i$ is cyclic, and $\xi[j]$ is the unique safe letter at r_j; so from i on the run is determined by its current state. By (b), each state of a cycle through r_i has the edge of that cycle as its only safe edge, so the run follows the cycle and returns to r_i; let $l \ge 1$ be least with $r_{i+l} = r_i$. Then $r_{j+l} = r_j$ for all $j \ge i$, and $\xi[i{:}] = \lambda^\omega$ with $\lambda = \xi[i{:}i + l]$. Put $\alpha = \xi[{:}i]$. The states $r_0, \ldots, r_{i-1}$ are not cyclic, so they are pairwise distinct and differ from the cyclic states $r_i, \ldots, r_{i+l-1}$. These are pairwise distinct as well: if $r_{i+m} = r_{i+m'}$ with $0 \le m < m' < l$, then by determinism the run has the period $m' - m$ from time $i + m$ on, so $r_{i+l-(m'-m)} = r_{i+l} = r_i$ with $1 \le l - (m' - m) < l$, contradicting the minimality of l. Hence $|\alpha| + |\lambda| = i + l \le N_A$, and $\alpha\lambda$ is safe from q with $A(q, \alpha\lambda) = A(q, \alpha)$, so $\xi = \alpha\lambda^\omega$ is a lasso of q. Since q is not mortal, some infinite word is safe from q, so q has a lasso.

(d) Let $r_0 = q, \ldots, r_{|\theta|}$ be the states visited by θ. If some r_i were cyclic, then by (b) and induction every r_m with $m \ge i$ would be cyclic, since $\theta[m]$ is a safe letter at r_m and hence the unique one; then $A(q, \theta) = r_{|\theta|}$ would be cyclic and not mortal. So no r_i is cyclic. Then the r_i are pairwise distinct, since a repetition would make a state cyclic, and $|\theta| < N_A$. □

D.4 Survival

Proof of Lemma 8.10. (a) By Lemma 8.1(c), a finite word keeps a comparison with nonempty overhang o if and only if it is a prefix of o^ω. So the comparison survives ξ if and only if every finite prefix of ξ is a prefix of o^ω, that is, by Lemma D.1(c), if and only if $\xi = o^\omega$. An aligned comparison is kept by every word.

(b) Suppose that $u[s{:}]\,\xi = t[e{:}]\,\xi$. The finite words $t[e{:}]$ and $u[s{:}]$ are prefixes of this infinite word, and $|t[e{:}]| \le |u[s{:}]|$, so $t[e{:}] \preceq u[s{:}]$ and (s, e) is a cross comparison. For every prefix θ of ξ, the words $t[e{:}]\,\theta$ and $u[s{:}]\,\theta$ are prefixes of the same infinite word, and the first is not longer, so $t[e{:}]\,\theta \preceq u[s{:}]\,\theta$: the comparison survives ξ. Conversely, let (s, e) be a cross comparison that survives ξ. For every n, $t[e{:}]\,\xi[{:}n] \preceq u[s{:}]\,\xi[{:}n]$, so the first $|t[e{:}]| + n$ letters of $u[s{:}]\,\xi$ are those of $t[e{:}]\,\xi$. As n is arbitrary, $u[s{:}]\,\xi = t[e{:}]\,\xi$. The claim on self comparisons is the case $u = t$, $s = i$, $e = j$.

(c) By (a), $\xi \ne o^\omega$. By Lemma D.1(d), applied to $\alpha\lambda^\omega$ and $\varepsilon\, o^\omega$, the two words differ at some position $i < |\alpha| + |\lambda| + |o| - 1$. The prefix of ξ of length $i + 1$ is not a prefix of o^ω, so it breaks the comparison by Lemma 8.1(c), and by (K3) so does every longer prefix, in particular the prefix of length $|\alpha| + |\lambda| + |o| - 1$. □

D.5 The Pre-Hole Graph

For words v and z, the *aligned pairs* of (v, z) are the pairs of cuts (s, e), $s \le |v|$ and $e \le |z|$, with $v[s{:}] = z[e{:}]$. Let $\mathrm{h}(v, z) \in \{0, 1\}$ be 1 if and only if some aligned pair (s, e) of (v, z) has $(A'_s(v), B'_e(z)) \in H$, and let $\mathrm{g}(v, z) \in \{0, 1\}$ be 1 if and only if some aligned pair has $(A_s(v), B_e(z)) \in G$.

Lemma D.2 (Aligned pairs under extension). *For words v, z and letters ℓ, ℓ':*

(a) the only aligned pair of (v, ε) is $(|v|, 0)$, and the only aligned pair of (ε, z) is $(0, |z|)$;
(b) the aligned pairs of $(v\ell, z\ell')$ are $(|v\ell|, |z\ell'|)$ and, if $\ell = \ell'$, the aligned pairs of (v, z).

Consequently $\mathrm{h}(v\ell, z\ell') = 1$ if and only if $(A'(v\ell), B'(z\ell')) \in H$, or $\ell = \ell'$ and $\mathrm{h}(v, z) = 1$; and likewise for g*, with A, B and G.*

Proof. (a) The suffixes of ε are empty, and the only empty suffix of v is $v[|v|{:}]$. (b) The pair of last cuts is aligned. Let (s, e) be another aligned pair. If $s = |v\ell|$, then the common suffix is empty and $e = |z\ell'|$; so $s \le |v|$ and, likewise, $e \le |z|$. Then $(v\ell)[s{:}] = v[s{:}]\,\ell$ and $(z\ell')[e{:}] = z[e{:}]\,\ell'$, which are equal if and only if $\ell = \ell'$ and $v[s{:}] = z[e{:}]$. The states at the cuts $s \le |v|$ and $e \le |z|$ are those of v and z. □

The graph $\mathcal{P}_E$. Let E be a set of safe states of A. The nodes of $\mathcal{P}_E$ are the tuples $\kappa = (r_A, r_B, r_{A'}, r_{B'}, h, g, \phi)$ with $r_M \in S_M$ for each automaton M, bits $h, g \in \{0, 1\}$, and a phase $\phi \in \{0, 1, 2\}$; there are $12aba'b'$ of them. Every edge is labeled by a pair of words of length at most one. From a node κ there are the following edges.

- A *spine edge*, labeled (ℓ, ε), if $\phi = 0$ and $A(r_A, \ell)$ is safe. It leads to the node with $A(r_A, \ell)$ and $A'(r_{A'}, \ell)$ in place of r_A and $r_{A'}$, with $h = 1$ if and only if $(A'(r_{A'}, \ell), r_{B'}) \in H$, with $g = 1$ if and only if $(A(r_A, \ell), r_B) \in G$, and with phase 0.
- A *target edge*, labeled (ε, ℓ'), if $\phi = 1$ and $r_B \notin \mathrm{Rej}$. It leads to the node with $B(r_B, \ell')$ and $B'(r_{B'}, \ell')$ in place of r_B and $r_{B'}$, with $h = 1$ if and only if $(r_{A'}, B'(r_{B'}, \ell')) \in H$, with $g = 1$ if and only if $(r_A, B(r_B, \ell')) \in G$, and with phase 1.
- A *joint edge*, labeled (ℓ, ℓ'), if $A(r_A, \ell)$ is safe and $r_B \notin \mathrm{Rej}$. It leads to the node with all four states advanced, with $h = 1$ if and only if $(A'(r_{A'}, \ell), B'(r_{B'}, \ell')) \in H$ or $\ell = \ell'$ and the old h is 1, with g defined in the same way from A, B, G and the old g, and with phase 2.

A path *spells* the pair of words obtained by concatenating the first and the second components of its labels. For words v and z and a phase ϕ, put

$$\mathrm{node}(v, z, \phi) = (A(v), B(z), A'(v), B'(z), \mathrm{h}(v, z), \mathrm{g}(v, z), \phi).$$

The *initial* nodes are $\mathrm{node}(\varepsilon, \varepsilon, \phi)$ for $\phi \in \{0, 1, 2\}$, provided that 0 is safe; the *final* nodes are those with $r_A \in E$, $r_B \in \mathrm{Rej}$, $h = 1$ and $g = 0$.

Lemma D.3 (Invariant). *If a path from an initial node spells (v, z) and ends at the node κ, then $\kappa = \mathrm{node}(v, z, \phi)$ for some phase ϕ, the word v is safe, $B_e(z) \notin \mathrm{Rej}$ for all $e < |z|$, and $z = \varepsilon$ if $\phi = 0$, and $v = \varepsilon$ if $\phi = 1$.*

Proof. By induction on the length of the path. An initial node $\mathrm{node}(\varepsilon, \varepsilon, \phi)$ satisfies the claim, since 0 is safe. Let the claim hold for a path that spells (v, z) and ends at $\kappa = \mathrm{node}(v, z, \phi)$. A spine edge requires $\phi = 0$, so $z = \varepsilon$, and by Lemma D.2(a) it leads to $\mathrm{node}(v\ell, \varepsilon, 0)$; the word $v\ell$ is safe, since the new state of A is safe. A target edge requires $\phi = 1$, so $v = \varepsilon$; it leads to $\mathrm{node}(\varepsilon, z\ell', 1)$ by Lemma D.2(a), and $B_{|z|}(z\ell') = B(z) = r_B$ is not in Rej. A joint edge leads to $\mathrm{node}(v\ell, z\ell', 2)$ by Lemma D.2(b), and safety and non-rejection follow as before. The phase 2 imposes no condition. □

Lemma D.4 (Correctness of the pre-hole graph). *The graph $\mathcal{P}_E$ has a path from an initial node to a final node that spells (v, z) if and only if (v, z) is a pre-hole with $A(v) \in E$.*

Proof. Let such a path end at the final node κ. By Lemma D.3, $\kappa = \mathrm{node}(v, z, \phi)$, the word v is safe, and $B_e(z) \notin \mathrm{Rej}$ for $e < |z|$; as κ is final, $A(v) \in E$, $B(z) \in \mathrm{Rej}$, so that $J(z) = |z|$, $\mathrm{h}(v, z) = 1$, so that (v, z) has a head event, and $\mathrm{g}(v, z) = 0$. Since every cut $e \le |z| = J(z)$ is admissible for an event, $\mathrm{g}(v, z) = 0$ says that (v, z) has no aligned event. So (v, z) is a pre-hole with $A(v) \in E$.

Conversely, let (v, z) be a pre-hole with $A(v) \in E$. Suppose first that $|v| \ge |z|$, and put $m = |v| - |z|$. Start at $\mathrm{node}(\varepsilon, \varepsilon, 0)$, which is initial since v is safe; take spine edges labeled $(v[0], \varepsilon), \ldots, (v[m-1], \varepsilon)$, and then joint edges labeled $(v[m+i], z[i])$ for $i < |z|$. By Lemma D.2, the path is at $\mathrm{node}(v[:m'], z[:e'], \phi)$ after it has spelled $(v[:m'], z[:e'])$. Every edge is allowed: the new states of A are states of the safe word v, and the joint edges start at states $B_i(z)$ with $i < |z| = J(z)$, which are not in Rej. The path ends at $\mathrm{node}(v, z, \phi)$, which is final because (v, z) is a pre-hole with $A(v) \in E$. If $|z| > |v|$, start at $\mathrm{node}(\varepsilon, \varepsilon, 1)$, take target edges for the first $|z| - |v|$ letters of z, which start at states $B_i(z)$ with $i < |z|$, and continue with joint edges as before. □

Proof of Proposition 8.12. Let (u_0, t_0) be a pre-hole with $A(u_0) \in E$. By Lemma D.4, $\mathcal{P}_E$ has a path from an initial node to a final node. A shortest such path visits no node twice, since a repeated node would allow us to cut out a cycle; so it has fewer than $12aba'b'$ edges. Each edge contributes at most one letter to each word, so the path spells a pair (u_1, t_1) with $|u_1|, |t_1| < 12aba'b'$, which is a pre-hole with $A(u_1) \in E$ by Lemma D.4. For decidability, test every pair of words over the finite alphabet Σ of these lengths. □

D.6 The Pump

Throughout this subsection, ξ is a purely periodic infinite word with least period ϖ. We prove Theorem 8.14 in five steps: a reformulation of survivor-freeness, the structure of rests and periodic ends, with a bound on the periodic end of the target, non-self-similar words, a deletion lemma, and four deletions together with a count.

D.6.1 Matched Pairs and Good Pairs.

Definition D.5 (Matched pairs, good pairs). Let u and t be words. A *matched pair* of (u, t) is a pair of cuts (s, e), $s \le |u|$ and $e \le |t|$, with $|t| - e \le |u| - s$ and $u[s:]\,\xi = t[e:]\,\xi$. A *matched self pair* of t is a pair of cuts $i < j \le |t|$ with $t[i:]\,\xi = t[j:]\,\xi$. The pair (u, t) is *good* if

(G1) u is safe;
(G2) $B_e(t) \notin \mathrm{Rej}$ for every $e < |t|$, and $B(t) \in \mathrm{Rej}$;
(G3) (u, t) has a head event;
(G4) $(A_s(u), B_e(t)) \notin G$ for every matched pair (s, e) of (u, t);
(G5) $(B_i(t), B_j(t)) \notin G_\mathrm{s}$ for every matched self pair (i, j) of t.

Lemma D.6 (Good pairs). *A pair (u, t) is good if and only if it is a pre-hole for which ξ is survivor-free.*

Proof. Condition (G2) says that $J(t) = |t|$; so every cut of t is admissible for events and self events. Let (u, t) be good. Every aligned pair of cuts is a matched pair, so by (G4) (u, t) has no aligned event; with (G1)–(G3), it is a pre-hole. If an event (s, e) survived ξ, then, being a cross comparison, it would satisfy $|t| - e \le |u| - s$, and by Lemma 8.10(b) $u[s{:}]\,\xi = t[e{:}]\,\xi$; so it would be a matched pair carrying G, contrary to (G4). Likewise a surviving self event would be a matched self pair carrying G_s, contrary to (G5). So ξ is survivor-free for (u, t).

Conversely, let (u, t) be a pre-hole for which ξ is survivor-free. Conditions (G1)–(G3) hold. By Lemma 8.10(b), a matched pair is a cross comparison that survives ξ; if it carried G, it would be an event, since $e \le |t| = J(t)$, that survives ξ. Likewise a matched self pair carrying G_s would be a surviving self event. So (G4) and (G5) hold. □

D.6.2 Rests and Periodic Ends. Let w be a word. A cut s of w is *periodic* if $w[s{:}]\,\xi$ has the period ϖ. For $s \le s'$ the word $w[s'{:}]\xi$ is a suffix of $w[s{:}]\,\xi$, so every cut after a periodic cut is periodic; the cut $|w|$ is periodic, because ξ has the period ϖ. Let R_w be the least periodic cut. The cuts $s \ge R_w$ are periodic, and the cuts $s < R_w$, the *rest cuts*, are not. Put $T_w = |w| - R_w$, and call $|w| - s$ the *end distance* of the cut s; the periodic cuts are the cuts of end distance at most T_w.

Lemma D.7 (Rests and periodic ends). *Let u and t be words.*

(a) For periodic cuts s of u and e of t, $u[s{:}]\,\xi = t[e{:}]\,\xi$ if and only if $|u| - s \equiv |t| - e \pmod{\varpi}$.
(b) If $s \le |u|$, $e \le |t|$ and $u[s{:}]\,\xi = t[e{:}]\,\xi$, then $\max(R_u - s, 0) = \max(R_t - e, 0)$. In particular, s is periodic if and only if e is, and if s is a rest cut, then e is a rest cut with $R_t - e = R_u - s$.
(c) If $T_u \ne T_t$, then the end distance $|u| - s = |t| - e$ of every aligned pair (s, e) of (u, t) is at most $\min(T_u, T_t)$.

Proof. (a) Let $k = |u| - s$ and $k' = |t| - e$, and choose N with $N\varpi \ge \max(k, k')$. The infinite word $\xi' = u[s{:}]\,\xi$ has the period ϖ, so $\xi' = \xi'[N\varpi{:}] = \xi[N\varpi - k{:}]$. Likewise $t[e{:}]\,\xi = \xi[N\varpi - k'{:}]$. By Lemma D.1(b), these are equal if and only if $N\varpi - k \equiv N\varpi - k'$, that is, $k \equiv k' \pmod{\varpi}$.

(b) Put $m_u = \max(R_u - s, 0)$ and $m_t = \max(R_t - e, 0)$, and suppose that $m_u > m_t$; the other case is symmetric. Then $s + m_t < R_u \le |u|$, and $e \le e + m_t \le |t|$ with $e + m_t \ge R_t$. Removing the first m_t letters of both infinite words gives $u[s + m_t{:}]\xi = t[e + m_t{:}]\xi$. The right side has the period ϖ, because $e + m_t$ is periodic; so $s + m_t$ is a periodic cut, which contradicts $s + m_t < R_u$. The remaining claims follow, because s is a rest cut exactly when $R_u - s > 0$.

(c) Let (s, e) be an aligned pair and $k = |u| - s = |t| - e$. Then $u[s{:}]\,\xi = t[e{:}]\,\xi$. If s were a rest cut, then by (b) so would be e, with $R_u - s = R_t - e$, and $T_u = k - (R_u - s) = k - (R_t - e) = T_t$, which is excluded. So s is periodic, and by (b) so is e; that is, $k \le T_u$ and $k \le T_t$. □

For a rest cut j of u, a cut e of t with $u[j{:}]\,\xi = t[e{:}]\,\xi$ is a *partner* of j; by Lemma D.7(b), it is the rest cut $e = R_t - (R_u - j)$, so j has at most one partner. Partners of rest cuts of t are defined in the same way. A rest cut is *partnered* if it has a partner and *unpartnered* otherwise. Let C_u and U_u be the sets of partnered and unpartnered rest cuts of u, and C_t and U_t those of t.

Lemma D.8 (Periodic target end). *Let t be a word with $B_e(t) \notin \mathrm{Rej}$ for every $e < |t|$ and $B(t) \in \mathrm{Rej}$, as in condition* (G2). *Then $T_t < N_B\,\varpi$.*

Proof. Put $L = |t|$, and suppose that two periodic cuts $i < j$ of t have the same state of B and end distances that agree modulo ϖ. By Lemma D.7(a), $t[i{:}]\,\xi = t[j{:}]\,\xi$, and comparing the first $L - j$ letters gives $t[i{:}i + L - j] = t[j{:}]$. By determinism,

$$B_{L-(j-i)}(t) = B(B_i(t), t[j{:}]) = B(B_j(t), t[j{:}]) = B_L(t) \in \mathsf{Rej},$$

although $L - (j - i) < L$. This contradicts the hypothesis. Hence the $T_t + 1$ periodic cuts e of t have pairwise distinct signatures $(B_e(t), (L - e) \bmod \varpi)$, of which there are at most $N_B\varpi$; so $T_t + 1 \le N_B\varpi$. □

D.6.3 Non-Self-Similar Words. An infinite word ξ' is *non-self-similar* if $z\xi' \neq \xi'$ for every nonempty finite word z.

Lemma D.9 (Non-self-similar words).

(a) If j is a rest cut of a word w, then $w[j{:}]\,\xi$ is non-self-similar.
(b) Let ξ' be non-self-similar. If $z_1\xi' = z_2\xi'$ for finite words z_1 and z_2, then $z_1 = z_2$. If z is a nonempty finite word and $k \ge 0$, then $z\xi' \neq \xi'[k{:}]$.
(c) Let ξ' be non-self-similar and t a word. At most one cut e of t satisfies $t[e{:}]\,\xi = \xi'$.
(d) Let v and v' be words, and let j be a rest cut of v such that $v[j{:}]\,\xi \neq v'[e{:}]\,\xi$ for every cut e of v'. Then $z\,v[j{:}]\,\xi \neq v'[k{:}]\,\xi$ for every finite word z and every cut k of v'.

Proof. (a) Let $\xi' = w[j{:}]\,\xi$, and suppose that $z\xi' = \xi'$ with $z \neq \varepsilon$. Then ξ' has the period $|z|$ by Lemma D.1(c). Its suffix $\xi'[|w| - j{:}] = \xi$ has the period ϖ, so ξ' has the period ϖ by Lemma D.1(a). Then j is a periodic cut, a contradiction.

(b) Let $|z_1| \ge |z_2|$. Both words are prefixes of $z_1\xi' = z_2\xi'$, so $z_1 = z_2 z$ for a word z; canceling z_2 gives $z\xi' = \xi'$, so $z = \varepsilon$. If $z\xi' = \xi'[k{:}]$, then $(\xi'[{:}k]\,z)\xi' = \xi'[{:}k]\,\xi'[k{:}] = \xi'$ with $\xi'[{:}k]\,z$ nonempty, a contradiction.

(c) If $e < e'$ are cuts with $t[e{:}]\,\xi = t[e'{:}]\xi = \xi'$, then $t[e{:}e']\xi' = t[e{:}]\,\xi = \xi'$ with $t[e{:}e']$ nonempty, a contradiction.

(d) Let $\xi' = v[j{:}]\,\xi$, and suppose that $z\xi' = v'[k{:}]\,\xi$. If $|z| \le |v'| - k$, then $z = v'[k{:}k + |z|]$, and canceling z gives $\xi' = v'[k + |z|{:}]\xi$, contrary to the hypothesis. If $|z| > |v'| - k$, then $z = v'[k{:}]\,z'$ with z' nonempty, and canceling $v'[k{:}]$ gives $z'\xi' = \xi$. Then $\xi' = \xi[|z'|{:}]$ has the period ϖ, and j is a periodic cut, a contradiction. □

D.6.4 The Deletion Lemma. Let $u = xwy$ and $t = x'w'y'$ with $w \neq \varepsilon$ or $w' \neq \varepsilon$, and put $\bar{u} = xy$ and $\bar{t} = x'y'$. A cut s of $\bar{u}$ is *deep* if $s < |x|$ and *shallow* otherwise; its *image* is $\hat{s} = s$ if s is deep and $\hat{s} = s + |w|$ if s is shallow. Cuts e of $\bar{t}$ are deep if $e < |x'|$, shallow otherwise, with images $\hat{e}$ defined in the same way with w'. The image map is strictly increasing. For a deep cut s, $\bar{u}[s{:}] = x[s{:}]\,y$ and $u[\hat{s}{:}] = x[s{:}]\,wy$; for a shallow cut $s = |x| + k$, $\bar{u}[s{:}] = u[\hat{s}{:}] = y[k{:}]$, and the end distances agree: $|\bar{u}| - s = |u| - \hat{s}$. The same holds for $\bar{t}$. Put $Y = y\xi$ and $Y' = y'\xi$.

Lemma D.10 (Deletion). *Let (u, t) be good, and suppose:*

(D1) each of A and A' has the same state at the cuts $|x|$ and $|xw|$ of u, and each of B and B' has the same state at the cuts $|x'|$ and $|x'w'|$ of t;
(D2) the image $(\hat{s}, \hat{e})$ of every matched pair (s, e) of $(\bar{u}, \bar{t})$ is a matched pair of (u, t), and the image $(\hat{i}, \hat{j})$ of every matched self pair (i, j) of $\bar{t}$ is a matched self pair of t;
(D3) some head event (s, e) of (u, t) is shallow in both words, that is, $s \ge |xw|$ and $e \ge |x'w'|$, or deep in both words at equal distance from factors of equal length, that is, $s \le |x|$, $e \le |x'|$, $|x| - s = |x'| - e$ and $|w| = |w'|$.

Then $(\bar{u}, \bar{t})$ is good, $A(\bar{u}) = A(u)$, and $|\bar{u}| + |\bar{t}| < |u| + |t|$.

Proof. First, for $M \in \{A, A'\}$ and every cut s of $\bar{u}$, $M_s(\bar{u}) = M_{\hat{s}}(u)$. For $s \le |x|$ the two words have the common prefix $x[:s]$. For a shallow cut $s = |x| + k$, the automaton M reads x and then $y[:k]$ in $\bar{u}$, and x, w and then $y[:k]$ in u; by (D1), $M(0, xw) = M(0, x)$, so both runs end in the same state. Likewise $M_e(\bar{t}) = M_{\hat{e}}(t)$ for $M \in \{B, B'\}$ and every cut e of $\bar{t}$. Moreover, the states of A' and B' at cuts $s \le |x|$ of $\bar{u}$ and $e \le |x'|$ of $\bar{t}$ are those at the same cuts of u and t, because the prefixes coincide.

(G1) Every $A_s(\bar{u}) = A_{\hat{s}}(u)$ is safe. (G2) For $e < |\bar{t}|$ we have $\hat{e} < |t|$, so $B_e(\bar{t}) = B_{\hat{e}}(t) \notin \mathrm{Rej}$. The last cut $|\bar{t}|$ is shallow with image $|t|$, so $B(\bar{t}) = B(t) \in \mathrm{Rej}$; in the same way $A(\bar{u}) = A(u)$.

(G3) Let (s, e) be the head event of (D3). If it is shallow in both words, then $\bar{s} = s - |w|$ and $\bar{e} = e - |w'|$ are shallow cuts with images s and e, so $\bar{u}[\bar{s}:] = u[s:] = t[e:] = \bar{t}[\bar{e}:]$, and the states of A' and B' there are those at s and e; so $(\bar{s}, \bar{e})$ is a head event of $(\bar{u}, \bar{t})$. Otherwise $u[s:] = x[s:]\, wy$ equals $t[e:] = x'[e:]\, w'y'$, where $|x[s:]| = |x'[e:]|$ and $|w| = |w'|$; comparing factors of equal lengths gives $x[s:] = x'[e:]$, $w = w'$ and $y = y'$. Hence $\bar{u}[s:] = x[s:]\, y = x'[e:]\, y' = \bar{t}[e:]$, and since $s \le |x|$ and $e \le |x'|$, the states of A' and B' at these cuts are those in u and t. So (s, e) is a head event of $(\bar{u}, \bar{t})$.

(G4) Let (s, e) be a matched pair of $(\bar{u}, \bar{t})$. By (D2), $(\hat{s}, \hat{e})$ is a matched pair of (u, t), so $(A_s(\bar{u}), B_e(\bar{t})) = (A_{\hat{s}}(u), B_{\hat{e}}(t)) \notin G$. (G5) Let (i, j) be a matched self pair of $\bar{t}$. By (D2), $(\hat{i}, \hat{j})$ is a matched self pair of t, so $(B_i(\bar{t}), B_j(\bar{t})) = (B_{\hat{i}}(t), B_{\hat{j}}(t)) \notin G_s$.

Finally, $|\bar{u}| + |\bar{t}| = |u| + |t| - |w| - |w'|$, and one of the deleted factors is nonempty. □

A pair whose cuts are both shallow satisfies (D2) automatically: its suffixes and end distances are those of its image. Since the image map is increasing, a self pair $i < j$ with i shallow has j shallow. So (D2) follows from the following five implications, which are sufficient conditions for it, for deep cuts $s < |x|$ of $\bar{u}$ and $e < |x'|$ of $\bar{t}$, shallow cuts $|x| + k$ of $\bar{u}$ and $|x'| + k$ of $\bar{t}$, and deep cuts $i < j < |x'|$ of $\bar{t}$.

(i) Both cuts deep: if $|x'[e:]| + |y'| \le |x[s:]| + |y|$ and $x[s:]\, Y = x'[e:]\, Y'$, then $|x'[e:]| + |w'| + |y'| \le |x[s:]| + |w| + |y|$ and $x[s:]\, wY = x'[e:]\, w'Y'$.

(ii) Spine cut deep, target cut shallow: if $|y'| - k \le |x[s:]| + |y|$ and $x[s:]\, Y = y'[k:]\, \xi$, then $x[s:]\, wY = y'[k:]\, \xi$. The length condition of the image, $|y'| - k \le |x[s:]| + |w| + |y|$, then holds, since only the spine side grows.

(iii) Spine cut shallow, target cut deep: if $|x'[e:]| + |y'| \le |y| - k$ and $y[k:]\, \xi = x'[e:]\, Y'$, then $|x'[e:]| + |w'| + |y'| \le |y| - k$ and $y[k:]\, \xi = x'[e:]\, w'Y'$.

(iv) Self pair, both cuts deep: if $x'[i:]\, Y' = x'[j:]\, Y'$, then $x'[i:]\, w'Y' = x'[j:]\, w'Y'$.

(v) Self pair, first cut deep, second cut $|x'| + k$ shallow: if $x'[i:]\, Y' = y'[k:]\, \xi$, then $x'[i:]\, w'Y' = y'[k:]\, \xi$.

Lemma D.11 (Two ways to lift matched pairs).

(a) Absorption. *If* $wY = Y$, *then* (ii) *holds.*

(b) Rigidity. *If* Y' *is non-self-similar, then the premises of* (iv) *and* (v) *never hold, so* (iv) *and* (v) *hold.*

Proof. (a) If $wY = Y$, then $x[s:]\, wY = x[s:]\, Y$, which gives (ii).

(b) In (iv), $x'[i:]\, Y' = x'[j:]\, Y'$ would give $x'[i:] = x'[j:]$ by Lemma D.9(b), although the two words have different lengths. In (v), $y'[k:]\, \xi = Y'[k:]$ because $k \le |y'|$, and $x'[i:]\, Y' = Y'[k:]$ with $x'[i:]$ nonempty contradicts Lemma D.9(b). □

D.6.5 The Four Deletions. Put

$$N_1 = aa'\varnothing, \qquad N_2 = 2aba'b', \qquad N_3 = aa', \qquad N_4 = bb'.$$

For a pair (u, t) and a cut s of u, let the *head bit* $\eta(s)$ be 1 if some head event (s', e') of (u, t) has $s' \ge s$, and 0 otherwise; it does not increase with s.

Lemma D.12 (Four deletions). *Let (u, t) be good, and suppose that one of the following conditions holds:*

(1) $T_u \geq T_t + N_1$;
(2) $|C_u| > N_2$;
(3) $|U_u| > N_3$;
(4) $|U_t| > N_4$.

Then there is a good pair $(\bar{u}, \bar{t})$ with $A(\bar{u}) = A(u)$ and $|\bar{u}| + |\bar{t}| < |u| + |t|$.

Proof. In each case we choose $u = xwy$ and $t = x'w'y'$, check (D1)–(D3), and apply Lemma D.10; for (D2) we check the implications (i)–(v), using Lemma D.11 where possible. When $x' = w' = \varepsilon$ and $y' = t$, the word $\bar{t} = t$ has no deep cut, so only (ii) has to be checked, and (D1) for B and B' is trivial. When $x = w = \varepsilon$ and $y = u$, only (iii), (iv) and (v) have to be checked, and (D1) for A and A' is trivial.

(1) Spine-end deletion. For $0 \leq \delta \leq N_1$ let $s_\delta = |u| - T_t - \delta$; it is a periodic cut of u, since $T_t + \delta \leq T_t + N_1 \leq T_u$. The signature $(A_{s_\delta}(u), A'_{s_\delta}(u), \delta \bmod \varpi)$ takes at most N_1 values, so some $\delta_1 < \delta_2$ have the same signature. Put $x = u[:s_{\delta_2}]$, $w = u[s_{\delta_2}:s_{\delta_1}]$, $y = u[s_{\delta_1}:]$, $x' = w' = \varepsilon$ and $y' = t$. (D1) holds by the signature, and $wY = Y$ by Lemma D.7(a), since the end distances $T_t + \delta_2$ and $T_t + \delta_1$ agree modulo ϖ; so (ii) holds by absorption. For (D3), note that $T_u \neq T_t$, as $N_1 \geq 1$. By Lemma D.7(c), every head event (s, e) has end distance at most T_t, so $s \geq |u| - T_t \geq s_{\delta_1} = |xw|$, and $e \geq 0 = |x'w'|$.

(2) Synchronous rest deletion. For $j \in C_u$ let $e(j)$ be its partner, $e(j) = R_t - (R_u - j)$. The signature $(A_j(u), A'_j(u), B_{e(j)}(t), B'_{e(j)}(t), \eta(j))$ takes at most N_2 values, so some $j_1 < j_2$ in C_u have the same signature. Put $e_1 = e(j_1)$ and $e_2 = e(j_2)$, so that $e_2 - e_1 = j_2 - j_1 > 0$, and let $x = u[:j_1]$, $w = u[j_1:j_2]$, $y = u[j_2:]$, $x' = t[:e_1]$, $w' = t[e_1:e_2]$ and $y' = t[e_2:]$. Then $Y = u[j_2:]\xi = t[e_2:]\xi = Y'$; call this word ξ'. It is non-self-similar by Lemma D.9(a). Moreover $w\xi' = u[j_1:]\xi = t[e_1:]\xi = w'\xi'$ and $|w| = |w'|$, so $w = w'$. (D1) holds by the signature. For (D2), (iv) and (v) hold by rigidity. In (i), the premise $x[s:]\,\xi' = x'[e:]\,\xi'$ gives $x[s:] = x'[e:]$ by Lemma D.9(b), hence $x[s:]\,w\xi' = x'[e:]\,w'\xi'$, and the length condition gains $|w| = |w'|$ on both sides. The premise of (ii) never holds: $x[s:]\,\xi' = y'[k:]\,\xi = \xi'[k:]$ with $x[s:]$ nonempty contradicts Lemma D.9(b). Neither does that of (iii), for the same reason. For (D3), suppose first that $\eta(j_2) = 1$, so some head event (s, e) has $s \geq j_2$. If $e < e_2$, then $u[s:]\,\xi = t[e:]\,\xi = t[e:e_2]\xi'$ with $t[e:e_2]$ nonempty, while $u[s:]\,\xi = \xi'[s - j_2:]$, which contradicts Lemma D.9(b). So $e \geq e_2 = |x'w'|$ and $s \geq j_2 = |xw|$. Otherwise $\eta(j_1) = \eta(j_2) = 0$, so every head event (s, e) has $s < j_1$. Removing the first $j_1 - s$ letters from $u[s:] = t[e:]$ gives $u[j_1:] = t[e + j_1 - s:]$, so $t[e + j_1 - s:]\xi = u[j_1:]\xi = t[e_1:]\xi$. This word is non-self-similar by Lemma D.9(a), so $e + j_1 - s = e_1$ by Lemma D.9(c). Hence $s \leq |x|$, $e \leq |x'|$, $|x| - s = |x'| - e$, and $|w| = |w'|$.

(3) Spine rest deletion. The pair $(A_j(u), A'_j(u))$ takes at most N_3 values on U_u, so some $i < j$ in U_u have the same pair. Put $x = u[:i]$, $w = u[i:j]$, $y = u[j:]$, $x' = w' = \varepsilon$ and $y' = t$. (D1) holds. The premise of (ii), $x[s:]\,(u[j:]\,\xi) = t[k:]\,\xi$, never holds, by Lemma D.9(d) with $v = u$, $v' = t$ and the unpartnered rest cut j. For (D3), every head event (s, e) has $s \geq j$: otherwise removing the first $j - s$ letters from $u[s:] = t[e:]$ gives $u[j:] = t[e + j - s:]$, and j would be partnered. So $s \geq |xw|$, and $e \geq 0 = |x'w'|$.

(4) Target rest deletion. The pair $(B_e(t), B'_e(t))$ takes at most N_4 values on U_t, so some $i < j$ in U_t have the same pair. Put $x = w = \varepsilon$, $y = u$, $x' = t[:i]$, $w' = t[i:j]$ and $y' = t[j:]$. (D1) holds. The word $Y' = t[j:]\,\xi$ is non-self-similar by Lemma D.9(a), so (iv) and (v) hold by rigidity. The premise of (iii), $u[k:]\,\xi = x'[e:]\,(t[j:]\,\xi)$, never holds, by Lemma D.9(d) with $v = t$, $v' = u$ and the unpartnered rest cut j. For (D3), every head event (s, e) has $e \geq j = |x'w'|$, since otherwise $t[j:] = u[s + j - e:]$ and j would be partnered; and $s \geq 0 = |xw|$. □

No deletion inside the periodic end of t is needed: by Lemma D.8, that periodic end is already short in every good pair.

D.6.6 *Counting and the Proof of the Pump.*

Lemma D.13 (Counting). *Let (u,t) be a good pair for which none of the conditions (1)–(4) of Lemma D.12 holds. Then $|u|, |t| < N_1 + N_2 + N_3 + N_4 + b\varpi$.*

Proof. Map each $e \in C_t$ to its partner. By Lemma D.7(b) the partner is the rest cut $R_u - (R_t - e)$ of u, which is partnered; so the map is an injection of C_t into C_u, and $|C_t| \le |C_u|$. Since (2), (3) and (4) fail, $R_u = |C_u| + |U_u| \le N_2 + N_3$ and $R_t = |C_t| + |U_t| \le N_2 + N_4$. By Lemma D.8, $T_t \le b\varpi - 1$, and since (1) fails, $T_u \le T_t + N_1 - 1 \le b\varpi + N_1 - 2$. Hence $|u| = R_u + T_u \le N_1 + N_2 + N_3 + b\varpi - 2$ and $|t| = R_t + T_t \le N_2 + N_4 + b\varpi - 1$, and both are less than $N_1 + N_2 + N_3 + N_4 + b\varpi$. □

Proof of Theorem 8.14. Put $K = N_1 + N_2 + N_3 + N_4 + b\varpi = 2aba'b' + aa'\varpi + b\varpi + aa' + bb'$. By Lemma D.6, the pair (u,t) is good. Among the good pairs (u',t') with $A(u') = A(u)$ choose one with $|u'| + |t'|$ least. If $|u'| \ge K$ or $|t'| \ge K$, then by Lemma D.13 one of the conditions (1)–(4) of Lemma D.12 holds for (u',t'), and Lemma D.12 gives a good pair with the same endpoint and a smaller total length, a contradiction. Hence $|u'|, |t'| < K$, and (u',t') is a pre-hole for which ξ is survivor-free, by Lemma D.6. If $\varpi \le N_A = a$, then $aa'\varpi \le a^2a'$ and $b\varpi \le ab$, so $K \le \mathsf{P}(a,b,a',b')$. □

D.7 The Three-Case Theorem

The bound L_{nm}. Since $a,b,a',b' \ge 1$, we have $ab, aa', bb' \le aba'b'$, so $\mathsf{P} \le 5aba'b' + a^2a' \le 6a^2ba'b'$. Hence $\max(12aba'b', \mathsf{P}) \le 12aba'b' + a^2a'$ and $\max(12aba'b', \mathsf{P}) \le 12a^2ba'b'$, which give

$$\mathsf{L}_{\mathrm{nm}}(a,b,a',b') \le 48aba'b' + 4a^2a' + 6a \qquad \text{and} \qquad \mathsf{L}_{\mathrm{nm}}(a,b,a',b') \le 48a^2ba'b' + 6a \le 54a^2ba'b'.$$

Proof of Theorem 8.17. We first show that (1) or (2) yields a short hole; this part does not use (FH). Put $K_0 = 12aba'b'$ and $K = \max(K_0, \mathsf{P}(a,b,a',b'))$.

Suppose that (1) holds for a pre-hole (u_0,t_0), and let $q = A(u_0)$. By Proposition 8.12 with $E = \{q\}$, there is a pre-hole (u_1,t_1) with $A(u_1) = q$ and $|u_1|, |t_1| \le K_0 - 1$. By Lemma 8.6 it is completed by a word θ with $|\theta| \le K_0 - 1 + 3a$, and by Lemma 8.3 $(u_1\theta, t_1\theta)$ is a hole with $|u_1\theta| + |t_1\theta| = |u_1| + |t_1| + 2|\theta| \le 4(K_0 - 1) + 6a < 4K + 6a$.

Suppose that (2) holds for a pre-hole (u_0,t_0) and an infinite word ξ safe from the thin state $q = A(u_0)$. By Lemma 8.8(c), ξ is a lasso $\alpha\lambda^\omega$ of q. If ξ is not purely periodic, then Proposition 8.12 with $E = \{q\}$ gives a pre-hole (u_1,t_1) with $A(u_1) = q$ and $|u_1|, |t_1| \le K_0 - 1$, and ξ is survivor-free for it by Lemma 8.13. If ξ is purely periodic, its least period is at most $|\lambda| \le a$ by Lemma D.1(b), and Theorem 8.14 gives a pre-hole (u_1,t_1) for which ξ is survivor-free, with $A(u_1) = q$ and $|u_1|, |t_1| \le \mathsf{P}(a,b,a',b') - 1$. In both cases $|u_1|, |t_1| \le K - 1$, and ξ is a lasso of $A(u_1)$ that is survivor-free for (u_1,t_1). By Corollary 8.11, the prefix θ of ξ of length $\max(|u_1|, |t_1|) + a - 1 \le K + a - 2$ completes (u_1,t_1), and by Lemma 8.3 $(u_1\theta, t_1\theta)$ is a hole with $|u_1\theta| + |t_1\theta| \le 2(K-1) + 2(K + a - 2) < 4K + 6a$.

In particular, (1) or (2) implies that I has a hole, and (3) does so trivially. Conversely, let I satisfy (FH), and let (u,t) be a hole. By Lemma 8.3, $(u,t) = (u_0\theta, t_0\theta)$, where $\theta = t[J(t){:}]$ completes the pre-hole (u_0,t_0). Let $q = A(u_0)$, which is safe. Then $A(u) = A(q,\theta)$, and $J(t) = J(t_0) = |t_0|$, so $|t| - J(t) = |\theta|$. If q is branching, (1) holds. If q is mortal, then by Lemma 8.8(a) $|\theta| < N_A$ and $A(u)$ is mortal, so (u,t) is a mortal hole. Let q be thin. If $A(u)$ is not mortal, some infinite word ξ' is safe from $A(u)$, so $\xi = \theta\xi'$ is safe from q. Every obligation of (u_0,t_0) is broken by θ, which is a prefix of ξ, and so does not survive ξ; hence (2) holds with ξ. If $A(u)$ is mortal, then $|\theta| < N_A$ by Lemma 8.8(d), and (u,t) is a mortal hole. □

E Proofs for Mortal Holes

This appendix defines the constants of Theorem 9.14 and proves the results of Section 9 whose proofs were omitted or outlined there. Appendix E.1 proves Lemmas 9.4 and 9.5, and Appendix E.2 proves Lemma 9.10. Appendices E.3 to E.8 prove Theorem 9.14, following the structure of the proof of Theorem 2.20 by Zhao [2026] as recalled in Appendix A.4; each subsection states what it reuses and what it adds. Appendix E.9 proves the closed-form estimates and Corollary 9.16.

What is reused unchanged: the alignment lemma (Lemma 2.17); the label facts Eq. (4) and the desubstitution lemma (Lemma 2.24); the facts on ordinary and symbolic states, start prefixes and entry states (Appendices A.3 and A.4); the comparison-preserving map lemma, the shape of level maps and the isolated exchange for holes (Lemmas A.5 to A.7); the propagation of boundary pairs (Lemma A.8(2)); the shortening of unary holes (Lemma A.9); and the column summaries, column shortening and agreement under derivation (Lemmas A.10 and A.11). What is new is that every step that replaces the witness, namely isolated exchange, the top shortening, column shortening and supported compression, also preserves the positive condition. Nothing below uses finiteness of an alphabet: block alphabets are infinite, and only state sets, supports, type spaces and summary values must be finite.

The constants. The bound $\mathsf{Len}^\sharp(N,d)$ of Theorem 9.14 and the constants behind it are as follows. For integers $N \geq 1$, $d \geq 0$, $M \geq 1$ and $k \geq 0$ put

$$\begin{aligned} \mathsf{C}^\sharp(N,d) &= N^N\, 2^{5N^2+10N(N+d)(d+1)+7N+1}, & \mathsf{A}^\sharp(N,d) &= d+1+\mathsf{C}^\sharp(N,d),\\ \Xi^\sharp(N,d) &= 6N^2+10N(N+d)(d+1)+7N+3d+4, & \mathsf{G}^\sharp(N,d) &= 2+\mathsf{A}^\sharp(N,d)\,3^d,\\ \mathsf{col}^\sharp(M,k) &= M^{(M+1)k}\, 2^{4M(M+1)k+2k+k^2+2(M+1)^2k^2},\\ \mathsf{Len}^\sharp(N,d) &= 3(N+d+1)\,\mathsf{col}^\sharp\big(N+d+1,\mathsf{G}^\sharp(N,d)\big)^d. \end{aligned} \tag{18}$$

They extend the constants Eq. (14) of the plain-entailment proof by the new tables and fields: $\mathsf{C}^\sharp(N,d)$ bounds the anchored exchange types at one level, $\mathsf{A}^\sharp(N,d)$ the protected and isolated letters of a normal hole at one level, $\mathsf{G}^\sharp(N,d)$ its dictionaries, $\mathsf{col}^\sharp(M,k)$ the positive column summaries of k blocks over a monitor with M states, and $\mathsf{Len}^\sharp(N,d)$ is the resulting length bound; $\Xi^\sharp(N,d)$ is the exponent of its closed form. All six functions are increasing in each argument, and the alphabet enters none of them.

E.1 The Separator Word

Proof of Lemma 9.4. (i) Let $y \leq m$. The suffix $w[y{:}]$ contains $\sharp$ at offset $m-y$. The suffix $w[x{:}]$ is longer, and its letter at offset $m-y$ is $w[x+m-y]$ with $x+m-y < m$, a letter of Σ. Hence $w[y{:}] \not\preceq w[x{:}]$.

(ii) Here $w[m+1+e{:}] = t[e{:}]$ is a word over Σ and $w[x{:}] = u[x{:}]\,\sharp\,t$. If $|t[e{:}]| \leq |u[x{:}]|$, the prefix test reads only letters of $u[x{:}]$, so it is the test $t[e{:}] \preceq u[x{:}]$. Otherwise $t[e{:}] \not\preceq u[x{:}]$ by length, and also $t[e{:}] \not\preceq w[x{:}]$, because the letter of $w[x{:}]$ at offset $m-x$ is $\sharp$, while the letter of $t[e{:}]$ at that offset lies in Σ.

(iii) Here $w[m+1+i{:}] = t[i{:}]$ and $w[m+1+j{:}] = t[j{:}]$.

(iv) The letter $\sharp$ occurs in w only at m. The continuation letter of a comparison $(x, m+1+e)$ is $w[x+|w|-m-1-e] = w[x+|t|-e]$, so the comparison is anchored iff $x+|t|-e = m$, that is, iff $|t[e{:}]| = |u[x{:}]|$. By (ii), the pair is a comparison iff $t[e{:}] \preceq u[x{:}]$, which together with equal lengths says $t[e{:}] = u[x{:}]$. Conversely, $t[e{:}] = u[x{:}]$ gives equal lengths and, by (ii), a comparison. □

We prove Lemma 9.5 for the reader $\mathcal{R}_I^{\Pi}$ and the relation Π constructed in Section 9.2. The cap N_A of the counter is the mortal bound of Definition 8.16, not the size of $\mathcal{R}_I^{\Pi}$; for $N_A = 1$ no state $\mathrm{done}(r', k)$ is rejected, as required by $|t| - J(t) < 1$.

Proof of Lemma 9.5. The reader and Π are computable, since mortality is decided by reachability in the safe graph of A (Definition 8.5).

Runs. The states after the separator have no transition on $\sharp$ other than into sink, which is absorbing. So a word without $\sharp$ ends in a state $\mathrm{pre}(\cdot)$ or in sink, and a word with two occurrences of $\sharp$ ends in sink. Let $w = u\sharp t$ with $u, t \in \Sigma^*$ and $m = |u|$. If u is not safe, the run enters sink within u and stays there. If u is safe, then by induction the state at a cut $x \le m$ is $\mathrm{pre}(A_x(u), A'_x(u))$, and the separator leads to act of the start states of B and B' iff $A(u)$ is mortal, and to sink otherwise. In the first case, by induction on e, the state at the cut $m + 1 + e$ is $\mathrm{act}(B_e(t), B'_e(t))$ while no earlier cut of t is rejected by B, that is, for $e \le J(t)$ if $J(t)$ is defined and for all e otherwise; and it is $\mathrm{done}(B'_e(t), \min(e - J(t), N_A))$ for $e > J(t)$.

(i) Every reachable state occurs in the run of some word. By the description of runs, the reachable states of the forms $\mathrm{pre}(q, q')$ and $\mathrm{act}(r, r')$ are pairs of states reachable in $A \times A'$ and in $B \times B'$, so there are at most $r_I + z_I$ of them; there are $N_{B'} N_A$ states $\mathrm{done}(r', k)$ and one sink.

(iii) and rejection. By the description of runs, the last state of a word is rejected iff the word is $u\sharp t$ with $u, t \in \Sigma^*$, u is safe, $A(u)$ is mortal, $J(t)$ is defined, and $|t| - J(t) \le N_A - 1$: if $|t| = J(t)$, the run ends in $\mathrm{act}(B_{J(t)}(t), \cdot)$ with $B_{J(t)}(t) \in \mathrm{Rej}$; if $|t| > J(t)$, it ends in a state done whose counter $\min(|t| - J(t), N_A)$ is at most $N_A - 1$ iff $|t| - J(t) \le N_A - 1$. No state pre is rejected, and the states $\mathrm{act}(B_e(t), \cdot)$ with $e < J(t)$ are not, so the first rejected cut is $m + 1 + J(t)$.

(ii) Let $w = u\sharp t$ be rejected, as just described. By Lemma 9.4, every comparison of w has the form $(x, m + 1 + e)$ with $x \le m$, a cross comparison (x, e) of (u, t), or $(m + 1 + i, m + 1 + j)$ with $i < j$, a comparison (i, j) of t. Admitted pairs join a state pre to a state act, or two states act; the states act occur exactly at the cuts $m+1+e$ with $e \le J(t)$. So a comparison $(x, m+1+e)$ is admitted iff $e \le J(t)$ and $(A_x(u), B_e(t)) \in G$, that is, iff (x, e) is an event of (u, t); and a comparison $(m + 1 + i, m + 1 + j)$ is admitted iff $j \le J(t)$ and $(B_i(t), B_j(t)) \in G_s$, that is, iff (i, j) is a self event of t. Hence w is a hole iff (u, t) has no event and t has no self event. By Lemma 9.4(iv), the anchored comparisons of w are the pairs $(x, m + 1 + e)$ with $u[x:] = t[e:]$. Their first state is $\mathrm{pre}(A_x(u), A'_x(u))$, and their second state is $\mathrm{act}(B_e(t), B'_e(t))$ or $\mathrm{done}(B'_e(t), k)$, with B'-component $B'_e(t)$ in both cases, also when $e > J(t)$. So Π holds on such a pair iff $(A'_x(u), B'_e(t)) \in H$, and an anchored comparison in Π is exactly a head event. Altogether, w is a Π-hole iff $w = u\sharp t$ where (u, t) satisfies conditions (a)–(e) of Definition 7.1, $A(u)$ is mortal and $|t| - J(t) < N_A$, that is, iff (u, t) is a mortal hole. A word that is not rejected is not a hole. □

E.2 Positive Desubstitution

Throughout this subsection, $\mathcal{M}$ is a lettered monitor with a positive channel $\Theta = (F, R, S, S', \Phi)$, ℓ is a marker, and $v = X_0\ell X_1\ell \cdots X_{k-1}\ell$ is the expansion of a nonempty word $v' = [X_0] \cdots [X_{k-1}]$ of blocks avoiding ℓ. We use the notation of the label facts Eq. (4): the run $\zeta'_0 = \star, \ldots, \zeta'_k$ of v' in $\mathcal{M}_\ell$, the cuts $a_i = |X_0\ell \cdots X_{i-1}\ell|$ at which the blocks begin, and $b_i = a_i + |X_i|$; in addition $a_k = |v|$. The letter of v at the position $a_i + g$, for $0 \le g \le |X_i|$, is $(X_i\ell)[g]$. We write L and L' for the labels of v and v'.

Lemma E.1 (Transport of continuation letters). *Let (x, y) be a comparison of v with $y < |v|$, and let $i < i' < k$, $X_i = \hat{x}\hat{y}$ and $X_{i'} = \hat{x}'\hat{y}$ be as in Lemma 2.17, so that $x = a_i + |\hat{x}|$, $y = a_{i'} + |\hat{x}'|$, and $(i + 1, i' + 1)$ is a comparison of v'. Put $r = (i + 1) + k - (i' + 1)$, the position of the continuation letter of $(i + 1, i' + 1)$ in v'. Then $i < r < k$, $x + |v| - y = a_r$, and the continuation letter of (x, y) is $v[a_r] = (X_r\ell)[0]$.*

Proof. Since $\hat{y}$ ends both blocks, $b_i - x = |\hat{y}| = b_{i'} - y$. Hence

$$|v| - y = |\hat{y}| + 1 + \sum_{h=i'+1}^{k-1} (|X_h| + 1) = |\hat{y}| + 1 + (a_k - a_{i'+1}).$$

Since $(i+1, i'+1)$ is a comparison of v', $X_{i+1+m} = X_{i'+1+m}$ for $0 \le m < k - i' - 1$, so the sum equals $a_r - a_{i+1}$, where $r = i + k - i'$. With $x + |\hat{y}| + 1 = b_i + 1 = a_{i+1}$ this gives $x + |v| - y = a_r$. Finally $i < r$ because $i' < k$, and $r < k$ because $i < i'$. □

For $i' + 1 = k$ the comparison $(i+1, k)$ of v' is terminal and $r = i + 1$: the continuation letter is $v'[i+1]$, as in clause (S′).

Proof of Lemma 9.10. The second claim follows from the first and Lemma 2.24. For the first, recall from Eq. (4) that $L_{a_i+g} = lab_g(\zeta'_i, X_i)$ for $g \le |X_i|$, $L_{|v|} = (\zeta'_k, \ell)$, and $L'_{i+1} = (\zeta'_i, X_i)$, with $\zeta'_k = \delta_\ell(\zeta'_{k-1}, X_{k-1})$. Every cut $x < |v|$ lies in exactly one interval $[a_i, b_i]$, and then $x = a_i + g$ with $g \le |X_i|$ and $v[x] = (X_i\ell)[g]$.

($\Rightarrow$) Suppose $\mathrm{Pos}_\Theta(v)$, and consider the clause that holds.

- (F): $F(L_{|v|}) = F(\delta_\ell(\zeta'_{k-1}, X_{k-1}), \ell)$ is the first disjunct of $F_\ell(\zeta'_{k-1}, X_{k-1}) = F_\ell(L'_k)$, which is clause (F) of v'.
- (R): a comparison (x, y) with $y < |v|$, $R(L_x, L_y)$ and $v[x + |v| - y] \in \Phi$. Take $i < i'$, $\hat{x}$, $\hat{x}'$ and $\hat{y}$ as in Lemma 2.17. Then $L_x = lab_{|\hat{x}|}(\zeta'_i, X_i)$ and $L_y = lab_{|\hat{x}'|}(\zeta'_{i'}, X_{i'})$, so $R(L_x, L_y)$ witnesses $R_\ell(L'_{i+1}, L'_{i'+1})$. By Lemma E.1, the continuation letter of (x, y) is $(X_r\ell)[0]$, so $X_r \in \Phi_\ell$, where $v'[r]$ is the continuation letter of the comparison $(i+1, i'+1)$ of v'. If $i' + 1 < k$, this is clause (R) of v'. If $i' + 1 = k$, then $r = i + 1$, and since $S'_\ell = R_\ell$, this is clause (S′) of v' at the cut $i + 1 < k$.
- (S) or (S′): a cut $x < |v|$ with $S(L_x, L_{|v|})$, or with $S'(L_x, L_{|v|})$ and $v[x] \in \Phi$. Write $x = a_i + g$. Then $L_x = lab_g(\zeta'_i, X_i)$, $L_{|v|} = (\zeta'_k, \ell)$ and $v[x] = (X_i\ell)[g]$, so $pt(\zeta'_i, X_i; \zeta'_k)$ holds. If $i < k - 1$, this is $S_\ell((\zeta'_i, X_i), (\zeta'_{k-1}, X_{k-1})) = S_\ell(L'_{i+1}, L'_k)$, clause (S) of v' at the cut $i + 1 < k$. If $i = k - 1$, it is the second disjunct of $F_\ell(L'_k)$, clause (F) of v'.

($\Leftarrow$) Suppose $\mathrm{Pos}_{\Theta_\ell}(v')$. All relations of Θ_ℓ are false at $\triangleright = L'_0$, so a witnessing cut x' of v' satisfies $x' \ge 1$; write $x' = i + 1$.

- (F): $F_\ell(L'_k)$. Either $F(\zeta'_k, \ell) = F(L_{|v|})$, clause (F) of v; or $pt(\zeta'_{k-1}, X_{k-1}; \zeta'_k)$, which gives some $g \le |X_{k-1}|$ with $S(L_x, L_{|v|})$, or with $S'(L_x, L_{|v|})$ and $v[x] \in \Phi$, at the cut $x = a_{k-1} + g \le b_{k-1} < |v|$: clause (S) or (S′) of v.
- (S): $S_\ell(L'_{i+1}, L'_k)$ with $i + 1 < k$, that is, $pt(\zeta'_i, X_i; \zeta'_k)$. As in the previous case this gives clause (S) or (S′) of v at a cut $x = a_i + g \le b_i < |v|$.
- (R) or (S′): a comparison $(i+1, i'+1)$ of v' with $R_\ell(L'_{i+1}, L'_{i'+1})$ and continuation letter $v'[r] \in \Phi_\ell$, where either $i' + 1 < k$ and $r = i + k - i'$ (clause (R)), or $i' + 1 = k$, $r = i + 1$, and the comparison is the terminal pair $(i+1, k)$, which is a comparison because $i + 1 < k$ (clause (S′), since $S'_\ell = R_\ell$). In both cases $i < i' < k$. The relation R_ℓ gives decompositions $X_i = \hat{x}\hat{y}$ and $X_{i'} = \hat{x}'\hat{y}$ with $R(lab_{|\hat{x}|}(\zeta'_i, X_i), lab_{|\hat{x}'|}(\zeta'_{i'}, X_{i'}))$. Put $x = a_i + |\hat{x}|$ and $y = a_{i'} + |\hat{x}'|$. Then $y \le b_{i'} < |v|$, and by Lemma 2.17 the pair (x, y) is a comparison of v; by the label facts $R(L_x, L_y)$ holds; and by Lemma E.1 its continuation letter is $(X_r\ell)[0]$, which lies in Φ because $X_r \in \Phi_\ell$. This is clause (R) of v.

This proves the equivalence. □

E.3 Anchored Exchange and Bounded Dictionaries

Fix a lettered monitor $\mathcal{M}_0$ with $N \ge 1$ states and a positive channel Θ_0 on it, and a positive hole w of depth $D \le d$ with hierarchy (w_i, ℓ_i), monitors $\mathcal{M}_i$ and channels $\Theta_i = (F_i, R_i, S_i, S'_i, \Phi_i)$ as in Corollary 9.11. We use the notions of step 2 of Appendix A.4: the dictionaries $\mathcal{A}_i$ and $g_i = |\mathcal{A}_i|$,

the start counts s_i and start prefixes, the entry states U_i with $|U_i| \le N + s_i \le N + d$, the protected letters P_i with $|P_i| \le s_i + 1 \le d + 1$, the boundary pairs BP_i with $b_i = |\mathrm{BP}_i|$, and isolated letters. Recall that every occurrence of an unprotected letter has an ordinary entry state.

Definition E.2 (Anchored exchange type). Let $1 \le i < D$. The *anchored exchange type* of an unprotected letter a of w_i consists of its exchange type (Appendix A.4) together with the following tables of $(\mathcal{M}_i, \Theta_i)$:

- the row $F_i(\zeta, a)$ for every ordinary state ζ;
- for each $Y \in \{R_i, S_i, S_i'\}$, the self table $Y((\zeta, a), (\zeta', a))$ for ordinary ζ, ζ';
- for each such Y, the protected rows $Y((\zeta, a), (\vartheta, z))$ and $Y((\vartheta, z), (\zeta, a))$ for ordinary $\zeta, \vartheta \in U_i$ and $z \in P_i$;
- for each such Y, the initial row $Y(\triangleright, (\zeta, a))$ for ordinary ζ, which is false at every level $i \ge 1$;
- the bit $[a \in \Phi_i]$.

Lemma E.3 (Anchored exchange types). *At every level $1 \le i < D \le d$ there are at most $\mathrm{C}^\sharp(N, d)$ anchored exchange types.*

Proof. Ordinary states transition to ordinary states, so the transition table has at most N^N values. The rows of T_i^{fin} and F_i take $2N$ bits; the self tables of the five relations $\Lambda_i, \Lambda_i^{\mathrm{fin}}, R_i, S_i, S_i'$ take $5N^2$ bits, their protected rows $10N|U_i||P_i| \le 10N(N + d)(d + 1)$ bits, and their initial rows $5N$ bits; and the bit of Φ_i is one more. Hence there are at most $N^N 2^{5N^2 + 10N(N+d)(d+1) + 7N + 1} = \mathrm{C}^\sharp(N, d)$ types. □

Lemma E.4 (Anchored isolated exchange). *Let $1 \le i < D$, let $a \ne b$ be isolated letters of w_i with the same anchored exchange type, let h replace a by b and fix every other letter, and let w' be the expansion of $h(w_i)$ with the original lower markers. Then w' is a positive hole of $(\mathcal{M}_0, \Theta_0)$ of depth D, with every start count unchanged, no dictionary larger than the corresponding one of w, and level-i dictionary $\mathcal{A}_i \setminus \{a\}$. If the level-zero expansion of b is not longer than that of a, then $|w'| \le |w|$.*

Proof. The anchored type contains the exchange type, so a and b have the same exchange type, and by Lemma A.7 the map h satisfies the hypotheses of Lemma A.6: it identifies no unequal predecessor pair of $v = w_i$, so v and $h(v)$ have exactly the same comparisons (Lemma A.5); it fixes the start prefix; it preserves the state run of v in $\mathcal{M}_i$; and $h(v)$ is a hole of $\mathcal{M}_i$. Hence w' is a hole of $\mathcal{M}_0$ of depth D with the stated start counts, dictionaries and length, and its markers below i are $\ell_0, \dots, \ell_{i-1}$, with $w'_i = h(v)$. So the monitors and channels of w' through level i are $(\mathcal{M}_j, \Theta_j)_{j \le i}$, and by Lemma 9.10, applied at the levels $i - 1, \dots, 0$, it remains to show $\mathrm{Pos}_{\Theta_i}(h(v))$.

Write L and L^h for the labels of v and of $h(v)$ in $\mathcal{M}_i$. The runs coincide, so L^h_c is L_c with its letter mapped by h.

Claim: for every comparison (x, y) of v and each $Y \in \{R_i, S_i, S_i'\}$, $Y(L^h_x, L^h_y) \iff Y(L_x, L_y)$. If $x = 0$, then $L_x = L^h_x = \triangleright$, and L_y is unchanged unless $v[y - 1] = a$, when the initial row applies. Let $x \ge 1$. If neither $v[x - 1]$ nor $v[y - 1]$ is a, both labels are unchanged. If both are a, the self table applies, both entry states being ordinary. If exactly one of them is a, the other letter c differs from a, so (x, y) is left maximal. If $y < |v|$, the ordered pair of predecessor letters lies in BP_i, and isolation of a makes c protected. If $y = |v|$, then $v[y - 1] = \ell_i$ is protected and differs from a, so $v[x - 1] = a$ and $c = \ell_i$. In both cases $c \in P_i$ is read from a state in U_i, $c \ne b$ because b is unprotected, and the protected row of Y in the appropriate orientation applies. This proves the claim.

Now fix a clause witnessing $\mathrm{Pos}_{\Theta_i}(v)$.

- (F): the final label $L_{|v|}$ has the protected letter ℓ_i, so $L^h_{|v|} = L_{|v|}$.

- (R): a comparison (x, y) with $y < |v|$. It is a comparison of $h(v)$, the claim keeps $R_i(L_x, L_y)$, and the continuation letter is at the same position $x + |v| - y$, since h preserves lengths. That letter is either unchanged or changed from a to b, and $[a \in \Phi_i] = [b \in \Phi_i]$.
- (S) or (S′): a cut $x < |v|$. The pair $(x, |v|)$ is a terminal comparison of both words, the claim keeps the value of S_i or S'_i, and for (S′) the letter $h(v)[x]$ lies in Φ_i iff $v[x]$ does.

Hence $\mathrm{Pos}_{\Theta_i}(h(v))$. □

The merge is not injective, so it gives no agreement of the derived data above level i. After every merge we recompute the hierarchy, the towers, the protected sets and the anchored types of the new word; its positivity in the new towers follows from Corollary 9.11.

A positive hole is *anchored-normal* if at every level $1 \le i < D$ distinct isolated letters have distinct anchored exchange types, computed in its own hierarchy and towers.

Lemma E.5 (Normal form and bounded dictionaries).

(1) Every positive hole of depth D has an anchored-normal positive hole of depth D, no greater length, the same start counts, and no larger dictionary at any level.

(2) Let w be an anchored-normal positive hole of depth $D \le d$, and put $\mathrm{A}^\sharp = \mathrm{A}^\sharp(N, d)$. Then $g_D = 1$, and for $1 \le i < D$,

$$g_i \le \mathrm{A}^\sharp + 2b_i, \qquad 2b_i \le (\mathrm{A}^\sharp - 1)(3^{D-1-i} - 1), \qquad g_i \le 1 + (\mathrm{A}^\sharp - 1)3^{D-1-i} \le \mathrm{G}^\sharp(N, d).$$

Consequently, every positive hole of depth $D \le d$ has a positive hole of depth D and no greater length with $g_i \le \mathrm{G}^\sharp(N, d)$ for every $1 \le i \le D$.

Proof. (1) Use the potential $\sum_{j=0}^{D} g_j$. While anchored normality fails at some level $1 \le i < D$, merge an offending pair by Lemma E.4, in the direction of the shorter level-zero expansion. The result is a positive hole of depth D, no longer, with the same start counts; no dictionary grows and the level-i dictionary shrinks, so the potential decreases strictly. The process terminates in an anchored-normal positive hole.

(2) The top level w_D is unary, so $g_D = 1$, and $b_D = 0$ because a unary word has no left-maximal comparison. Let $1 \le i < D$. Partition $\mathcal{A}_i$ into the protected letters, at most $d + 1$; the isolated letters, at most one per anchored type by normality, hence at most $\mathrm{C}^\sharp(N, d)$ by Lemma E.3; and the remaining letters, each of which is unprotected and not isolated, hence a component of a pair in BP_i. The two projections of BP_i have at most $2b_i$ elements, so $g_i \le \mathrm{A}^\sharp + 2b_i$. By the propagation in Lemma A.8(2), which holds for every nonempty word, $b_j \le b_{j+1} + g_{j+1} - 1$ for every j. Hence $b_{D-1} \le b_D + g_D - 1 = 0$, and for $1 \le i < D - 1$, $b_i \le b_{i+1} + \mathrm{A}^\sharp + 2b_{i+1} - 1 = 3b_{i+1} + \mathrm{A}^\sharp - 1$. By downward induction on i, with $r = D - 1 - i$,

$$2b_i \le 3(\mathrm{A}^\sharp - 1)(3^{r-1} - 1) + 2(\mathrm{A}^\sharp - 1) = (\mathrm{A}^\sharp - 1)(3^r - 1),$$

the base case $r = 0$ having both sides zero. Then $g_i \le \mathrm{A}^\sharp + 2b_i \le 1 + (\mathrm{A}^\sharp - 1)3^r \le 2 + \mathrm{A}^\sharp 3^d = \mathrm{G}^\sharp(N, d)$.

The last assertion follows from (1) and (2), since $g_D = 1 \le \mathrm{G}^\sharp(N, d)$. □

E.4 The Unary Top Level

Lemma E.6 (Positive unary words). *Let $\mathcal{M}$ be a lettered monitor with state set Q, Θ a positive channel on it, and z^m a positive hole of $(\mathcal{M}, \Theta)$. Then $z^{m'}$ is a positive hole of $(\mathcal{M}, \Theta)$ for some $1 \le m' \le 3|Q|$.*

Proof. Every pair of cuts $x < y$ of a unary word is a comparison, and its continuation letter is z. Fix a clause witnessing $\mathrm{Pos}_\Theta(z^m)$, and let W be the set consisting of m and the cuts of the witness other than 0: the cuts x and y for (R), the cut x for (S) and (S′), and none for (F); so $W \cup \{0\}$ has at most four elements.

Let $i < i' \le m-1$ be cuts with equal states such that no cut of W lies in $\{i+1,\dots,i'\}$. Deleting the letters at the positions $i,\dots,i'-1$ gives $z^{m-(i'-i)}$, a hole by Lemma A.9. Its cuts correspond to the cuts $c \le i$ and $c > i'$ of z^m, by $c \mapsto c$ and $c \mapsto c-(i'-i)$, an order-preserving map that keeps the labels: the state at the new cut i is the state at the old cut i'. The cuts of W are kept, with their labels and their order, and the last cut maps to the last cut. Since every pair of cuts is a comparison with continuation letter z, the witnessing clause holds for the images of its cuts. So the shorter word is again a positive hole, and W maps onto the corresponding set of the new witness.

Repeat such deletions while possible; the length decreases, so this terminates, in a positive hole $z^{m'}$ with $m' \ge 1$, because the last letter is never deleted. Let $c < c'$ be consecutive elements of $W \cup \{0\}$. The states at the cuts $c,\dots,c'-1$ are pairwise distinct: two equal ones at $i < i'$ would have $i' \le c'-1 \le m'-1$ and no cut of W in $\{i+1,\dots,i'\}$, so a further deletion would be possible. Hence $c'-c \le |Q|$, and, as there are at most three such gaps and they cover $[0,m']$, $m' \le 3|Q|$. □

E.5 Agreement with Channels

We extend agreement Eq. (13) to channels. Let $\mathcal{U}$ and $\mathcal{V}$ be lettered monitors with positive channels $\Theta^{\mathcal{U}} = (F^{\mathcal{U}}, R^{\mathcal{U}}, S^{\mathcal{U}}, S'^{\mathcal{U}}, \Phi^{\mathcal{U}})$ and $\Theta^{\mathcal{V}}$, let E be a finite set of letters of $\mathcal{U}$, and ρ a map from E into the alphabet of $\mathcal{V}$. We write $(\mathcal{U},\Theta^{\mathcal{U}}) \equiv_{E,\rho} (\mathcal{V},\Theta^{\mathcal{V}})$ if $\mathcal{U} \equiv_{E,\rho} \mathcal{V}$ and, for every $a \in E$, every state ζ, and all labels L, L' whose letters lie in E, the start label included,

$$F^{\mathcal{V}}(\zeta,\rho(a)) \iff F^{\mathcal{U}}(\zeta,a), \qquad \rho(a) \in \Phi^{\mathcal{V}} \iff a \in \Phi^{\mathcal{U}},$$

and $Y^{\mathcal{V}}(\widehat{\rho}L, \widehat{\rho}L') \iff Y^{\mathcal{U}}(L,L')$ for $Y \in \{R,S,S'\}$.

Agreement preserves positive holes. Let v be a nonempty word over E. Agreement of the monitors gives the same states along v and $\rho(v)$, labels that correspond under $\widehat{\rho}$, and holes that correspond. Since ρ is injective on E, the words v and $\rho(v)$ have the same letter equalities, hence the same comparisons, and the continuation letter of a comparison of $\rho(v)$ is the image of that of v. Each clause of the positive condition therefore holds for v iff it holds for $\rho(v)$, and v is a positive hole of $(\mathcal{U},\Theta^{\mathcal{U}})$ iff $\rho(v)$ is a positive hole of $(\mathcal{V},\Theta^{\mathcal{V}})$. *Agreements compose*, by composing the maps.

Lemma E.7 (Agreement under derivation). *Suppose $(\mathcal{U},\Theta^{\mathcal{U}}) \equiv_{E,\rho} (\mathcal{V},\Theta^{\mathcal{V}})$ and $\ell \in E$. For every finite family $\mathcal{F}$ of blocks over E avoiding ℓ, the letterwise map ρ^* gives*

$$(\mathcal{U}_\ell,\Theta^{\mathcal{U}}_\ell) \equiv_{\mathcal{F},\rho^*} (\mathcal{V}_{\rho(\ell)},\Theta^{\mathcal{V}}_{\rho(\ell)}),$$

and the image blocks avoid $\rho(\ell)$.

Proof. The agreement of the derived monitors and the avoidance of $\rho(\ell)$ are Lemma A.11. Its proof shows, by induction along a block, that for $X \in \mathcal{F}$ and $\zeta \in Q \sqcup \{\star\}$ the shifted blocks of X and $\rho^*(X)$ have the same states, so that $lab_g(\zeta,\rho^*X) = \widehat{\rho}\, lab_g(\zeta,X)$ for every g and the exits coincide. The letters $(X\ell)[g]$ lie in E, and $(\rho^*(X)\rho(\ell))[g] = \rho((X\ell)[g])$. Hence:

- $\rho^*(X) \in \Phi^{\mathcal{V}}_{\rho(\ell)}$ iff $\rho((X\ell)[0]) \in \Phi^{\mathcal{V}}$ iff $(X\ell)[0] \in \Phi^{\mathcal{U}}$ iff $X \in \Phi^{\mathcal{U}}_\ell$;
- by injectivity, the decompositions $X = xy$, $X' = x'y$ correspond exactly to the decompositions of ρ^*X and ρ^*X' with a common suffix, and R agrees at the corresponding labels, whose letters lie in E or which are $\triangleright$; so R_ℓ, and hence S'_ℓ, agree;
- the positive terminal test $pt(\zeta,X;\zeta'')$ reads S and S' at the labels $lab_g(\zeta,X)$ and (ζ'',ℓ), which correspond to $lab_g(\zeta,\rho^*X)$ and $(\zeta'',\rho(\ell))$, and the letters $(X\ell)[g]$; so it agrees, and with equal exits so do S_ℓ and, using agreement of F at the letter $\ell \in E$, F_ℓ.

All relations at the new start label are false on both sides. □

E.6 Positive Column Summaries

Fix a lettered monitor $\mathcal{M}$ with state set Q, $M = |Q|$, start ζ_0, a positive channel $\Theta = (F, R, S, S', \Phi)$, a marker ℓ, and blocks $X_1, \dots, X_k$ avoiding ℓ. We use the summary, the column word and the notation $lab_0(\zeta)$ of step 4 of Appendix A.4; block indices are a, b, entry states ζ, ζ' range over $Q \sqcup \{\star\}$, and final states ζ'' over Q.

Fields. The *positive summary* of $(X_1, \dots, X_k)$ consists of the four fields *exit*, *term*, *eq* and *adm* of the summary of Appendix A.4 and the following fields:

Field	Meaning
$tS_a(\zeta, \zeta'')$	some $g \le \lvert X_a\rvert$ has $S(lab_g(\zeta, X_a), (\zeta'', \ell))$
$tSL_a(\zeta, \zeta'')$	some $g < \lvert X_a\rvert$ has $S'(lab_g(\zeta, X_a), (\zeta'', \ell))$ and $X_a[g] \in \Phi$
$pend_a(\zeta, \zeta'')$	$S'(lab_{\lvert X_a\rvert}(\zeta, X_a), (\zeta'', \ell))$
ne_a	$X_a \neq \varepsilon$
phi_a	$X_a \neq \varepsilon$ and $X_a[0] \in \Phi$
$rel(a, b, \zeta, \zeta')$	some $X_a = xy$, $X_b = x'y$ has $R(lab_{\lvert x\rvert}(\zeta, X_a), lab_{\lvert x'\rvert}(\zeta', X_b))$

The field *pend* records an S'-test at the last position of the shifted block, whose letter is not known yet: it is the next letter of the block, or the marker if the block ends there. Counting assignments, the positive summary takes at most $\mathrm{col}^{\sharp}(M, k)$ values: the four fields of Appendix A.4 give $M^{(M+1)k} 2^{M(M+1)k + k^2 + (M+1)^2 k^2}$ (Appendix A.5); the fields *tS*, *tSL* and *pend* give $2^{3M(M+1)k}$; *ne* and *phi* give 2^{2k}; and *rel* gives $2^{(M+1)^2 k^2}$.

The automaton. The tuple of empty blocks has the initial summary of Appendix A.4 and

$$tS_a(\zeta, \zeta'') = S(lab_0(\zeta), (\zeta'', \ell)), \qquad pend_a(\zeta, \zeta'') = S'(lab_0(\zeta), (\zeta'', \ell)),$$

tSL_a, ne_a and phi_a false, and $rel(a, b, \zeta, \zeta') = R(lab_0(\zeta), lab_0(\zeta'))$. Reading a column γ updates the four fields as in Appendix A.4 and the new fields as follows, simultaneously and from the old values. If γ_a is absent, the fields of block a are unchanged. If $\gamma_a = z$, then, with $e = exit_a(\zeta)$ the old exit,

$$\begin{aligned} tS'_a(\zeta, \zeta'') &= tS_a(\zeta, \zeta'') \vee S((e, z), (\zeta'', \ell)), \\ tSL'_a(\zeta, \zeta'') &= tSL_a(\zeta, \zeta'') \vee \big(pend_a(\zeta, \zeta'') \wedge z \in \Phi\big), \\ pend'_a(\zeta, \zeta'') &= S'((e, z), (\zeta'', \ell)), \\ ne'_a &= \text{true}, \qquad phi'_a = phi_a \text{ if } ne_a, \text{ and } [z \in \Phi] \text{ otherwise.} \end{aligned}$$

The field *rel* is updated exactly as *adm*, with R in place of Λ.

Lemma E.8 (Positive summaries).

(1) After every prefix of a valid column word, every field equals its meaning for the current tuple.

(2) The final positive summary of $(X_1, \dots, X_k)$ determines the derived channel Θ_ℓ on the letters $X_1, \dots, X_k$, at every state of $\mathcal{M}_\ell$. With

$$T_a(\zeta, \zeta'') = tS_a(\zeta, \zeta'') \vee tSL_a(\zeta, \zeta'') \vee \big(pend_a(\zeta, \zeta'') \wedge \ell \in \Phi\big),$$

it is given by

$$\begin{aligned} X_a \in \Phi_\ell &\iff phi_a \vee (\neg ne_a \wedge \ell \in \Phi), \\ R_\ell((\zeta, X_a), (\zeta', X_b)) \iff S'_\ell((\zeta, X_a), (\zeta', X_b)) &\iff rel(a, b, \zeta, \zeta'), \\ S_\ell((\zeta, X_a), (\zeta', X_b)) &\iff T_a(\zeta, exit_b(\zeta')), \\ F_\ell(\zeta, X_a) &\iff F(exit_a(\zeta), \ell) \vee T_a(\zeta, exit_a(\zeta)). \end{aligned}$$

In particular S_ℓ has no F-term. All relations at the start label of $\mathcal{M}_\ell$ are false by definition.

Proof. (1) Induction on the number of columns read. The initial values are the meanings for empty blocks: the shifted block of ε read from ζ has the single label $lab_0(\zeta)$, no position $g < 0$ exists, and the only decomposition of two empty blocks is $x = x' = y = \varepsilon$. An absent entry leaves the prefix of its track unchanged. Let track a read z, so that its prefix X becomes Xz. The labels of the shifted block of Xz are those of X at the positions $g \le |X|$, and the new last label $lab_{|X|+1}(\zeta, Xz) = (exit_a(\zeta), z)$. The letter at position $|X|$ of Xz is z, and the letters at the positions $g < |X|$ are unchanged. This proves the updates of *tS*, *tSL* and *pend*: the new position $|X|$ of *tSL* carries the old pending test *pend* and the letter z. By validity, letters are appended to a track in order, so the first letter of a nonempty track is the letter read when *ne* changes, which proves the updates of *ne* and *phi*. The field *rel* is the defining formula of R_ℓ and is updated as *adm* (Appendix A.4), by the same argument.

(2) By (1), the final fields are their meanings for $(X_1, \dots, X_k)$. In Definition 9.9, $(X_a\ell)[0]$ is $X_a[0]$ if $X_a \neq \varepsilon$ and ℓ otherwise, which gives Φ_ℓ. The field *rel* is the defining formula of $R_\ell = S'_\ell$. The positive terminal test is $pt(\zeta, X_a; \zeta'') = T_a(\zeta, \zeta'')$, because $(X_a\ell)[g]$ is $X_a[g]$ for $g < |X_a|$ and ℓ for $g = |X_a|$. Finally $\delta_\ell(\zeta', X_b) = exit_b(\zeta')$. □

The four fields of Appendix A.4 alone do not suffice, as Example 9.13 shows. We now give the positive fields in that example.

Example E.9 (Example 9.13 continued: the positive fields). In Example 9.13 the top letter $[ab]$ is shortened as the single track ab over the level-1 monitor $\mathcal{M}_1$ and channel Θ_1, with the marker $c = [\sharp 1]$. For every state ζ'' of $\mathcal{M}_1$, the shifted block of c read from ζ'' reads $0\sharp 1$ from an ordinary state, or $\sharp 1$ from the start, and in both cases the state before the final 1 is 4. After the prefix a, the last label of the shifted track is $(2, a)$ from an ordinary entry, or $(\star_1, a)$ from the fresh entry, and the only common suffix of the blocks ε and $\sharp 1$ is empty. So *pend*, the value of $S'_1 = R_1$ at that label and the final label (ζ'', c), compares the states $\delta(2, 0) = 0$, or the start state 0, with $\delta(4, 1) = 2$; as $(0, 2) \notin \Pi$, *pend* is false everywhere. After the prefix ab the last label is $(0, b)$, and the common suffix 1 of the blocks 1 and $\sharp 1$ compares the states $\delta(0, 0) = 3$ and 4; as $(3, 4) \in \Pi$, *pend* is true everywhere. This is the original witness $(1, 4)$, and it enters F_2 through the disjunct $pend \wedge c \in \Phi_1$ of Lemma E.8(2), since the expansion of c starts with $\sharp$. Hence the positive summaries after a and after ab differ, and the column holding b is kept. At level 0 the tracks ε, 1 and $\sharp 1$ admit no deletion either: their column word has two columns, each of which is the first column of a nonempty track, where *ne* changes. The compression returns w itself.

Lemma E.10 (Positive column shortening). *Let $\mathcal{F} = \{X_1, \dots, X_k\}$ be a set of k distinct blocks avoiding ℓ. There are subsequences $\tau(X_a)$ of the blocks X_a, obtained by deleting the same set of columns, such that $|\tau(X_a)| < \mathrm{col}^\sharp(M, k)$ for every a; $\tau(X_a) = \varepsilon$ iff $X_a = \varepsilon$, and $\tau(X_a)[0] = X_a[0]$ whenever $X_a \neq \varepsilon$; and*

$$(\mathcal{M}_\ell, \Theta_\ell) \equiv_{\mathcal{F},\tau} (\mathcal{M}_\ell, \Theta_\ell).$$

Proof. The new positive summary is a function of the old one and the column, and there are at most $\mathrm{col}^\sharp(M, k)$ positive summaries. If the column word has at least $\mathrm{col}^\sharp(M, k)$ columns, two of its cuts carry the same positive summary. Deleting the columns between them leaves a valid column word, and the same final summary, because the remaining columns are read from the same summary. On each track, *ne* has the same value at the two cuts, so either the track is still empty at the second cut and no letter of it is deleted, or it is already nonempty at the first cut and its first letter lies before the deleted columns. Repeating, we reach fewer than $\mathrm{col}^\sharp(M, k)$ columns. Equal final summaries give equal equality bits, so τ preserves equality and inequality of blocks and is injective on $\mathcal{F}$; by Lemma A.10 and Appendix A.4, the four fields of the summary determine $\mathcal{M}_\ell$ on the letters, and by Lemma E.8(2) the positive fields determine Θ_ℓ on them, at every state and at

labels with letters in $\mathcal{F}$ or the start label. Both sides are the same monitor and channel, with the same states and start. □

E.7 Supported Compression

For markers $\boldsymbol{\ell} = (\ell_0, \dots, \ell_{j-1})$ write $(\mathcal{M}_0^{\boldsymbol{\ell}}, \Theta_0^{\boldsymbol{\ell}}) = (\mathcal{M}_0, \Theta_0)$ and $(\mathcal{M}_{h+1}^{\boldsymbol{\ell}}, \Theta_{h+1}^{\boldsymbol{\ell}}) = ((\mathcal{M}_h^{\boldsymbol{\ell}})_{\ell_h}, (\Theta_h^{\boldsymbol{\ell}})_{\ell_h})$, and use valid families and the expansions $\mathrm{Exp}_{\boldsymbol{\ell},j}$ as in Appendix A.4.

Lemma E.11 (Supported compression with channels). *Fix a nonempty word w with hierarchy markers $\boldsymbol{\ell}$ and dictionaries $\mathcal{A}_h$, and $j \le d$. Suppose $|\mathcal{A}_h| \le g$ for $1 \le h \le j$, and put $\kappa_\sharp = \mathrm{col}^\sharp(N + d + 1, g)$. For every set $E_j \subseteq \mathcal{A}_j$ there are new markers $\widetilde{\boldsymbol{\ell}} = (\widetilde{\ell}_0, \dots, \widetilde{\ell}_{j-1})$ and a map ρ_j, injective on E_j, such that*

$$(\mathcal{M}_j^{\boldsymbol{\ell}}, \Theta_j^{\boldsymbol{\ell}}) \equiv_{E_j,\rho_j} (\mathcal{M}_j^{\widetilde{\boldsymbol{\ell}}}, \Theta_j^{\widetilde{\boldsymbol{\ell}}}), \qquad |\mathrm{Exp}_{\widetilde{\boldsymbol{\ell}},j}(\rho_j(a))| \le \kappa_\sharp^j \quad (a \in E_j),$$

and the image family is valid for the new markers.

Proof. This is the induction of Lemma A.12, with positive summaries and agreement with channels. For $j = 0$ take the identity: a letter expands to itself, of length $1 = \kappa_\sharp^0$, whatever the size of E_0.

For the step from j to $j + 1 \le d$, let $E_{j+1} \subseteq \mathcal{A}_{j+1}$, so $|E_{j+1}| \le g$. Put $(\mathcal{U}, \Theta^{\mathcal{U}}) = (\mathcal{M}_j^{\boldsymbol{\ell}}, \Theta_j^{\boldsymbol{\ell}})$, which has $N + j$ states. The blocks in E_{j+1} are distinct words over $\mathcal{A}_j$ avoiding ℓ_j. Lemma E.10 gives subsequences $\tau(X)$ with $|\tau(X)| < \mathrm{col}^\sharp(N + j, |E_{j+1}|) \le \kappa_\sharp$, by monotonicity, and $(\mathcal{U}_{\ell_j}, \Theta_{\ell_j}^{\mathcal{U}}) \equiv_{E_{j+1},\tau} (\mathcal{U}_{\ell_j}, \Theta_{\ell_j}^{\mathcal{U}})$. Put

$$E_j = \{\ell_j\} \cup \bigcup_{X \in E_{j+1}} \mathrm{alph}(\tau(X)) \;\subseteq\; \mathcal{A}_j.$$

The induction hypothesis gives markers $\widetilde{\ell}_0, \dots, \widetilde{\ell}_{j-1}$, the monitor and channel $(\mathcal{V}, \Theta^{\mathcal{V}}) = (\mathcal{M}_j^{\widetilde{\boldsymbol{\ell}}}, \Theta_j^{\widetilde{\boldsymbol{\ell}}})$, and ρ_j with $(\mathcal{U}, \Theta^{\mathcal{U}}) \equiv_{E_j,\rho_j} (\mathcal{V}, \Theta^{\mathcal{V}})$ and expansions of length at most $\kappa_\sharp^j$. Set $\widetilde{\ell}_j = \rho_j(\ell_j)$ and $\rho_{j+1}(X) = \rho_j^*(\tau(X))$ for $X \in E_{j+1}$. By Lemma E.7, applied to the family $\tau(E_{j+1})$ of blocks over E_j avoiding ℓ_j, $(\mathcal{U}_{\ell_j}, \Theta_{\ell_j}^{\mathcal{U}}) \equiv_{\tau(E_{j+1}),\rho_j^*} (\mathcal{V}_{\widetilde{\ell}_j}, \Theta_{\widetilde{\ell}_j}^{\mathcal{V}})$, and composition with the column agreement gives $(\mathcal{M}_{j+1}^{\boldsymbol{\ell}}, \Theta_{j+1}^{\boldsymbol{\ell}}) \equiv_{E_{j+1},\rho_{j+1}} (\mathcal{M}_{j+1}^{\widetilde{\boldsymbol{\ell}}}, \Theta_{j+1}^{\widetilde{\boldsymbol{\ell}}})$. The image blocks avoid $\widetilde{\ell}_j$ and their letters lie in the valid family $\rho_j(E_j)$, so the new family is valid. Each new block followed by its marker has at most $\kappa_\sharp$ letters, each expanding to at most $\kappa_\sharp^j$ letters, so every new letter expands to at most $\kappa_\sharp^{j+1}$ letters. No dictionary or normality property of an intermediate compressed word is used. □

E.8 Proof of the Anchored Compression Theorem

Let $\mathcal{M}_0$ be a lettered monitor with $N \ge 1$ states and Θ_0 a positive channel on it, and suppose that $(\mathcal{M}_0, \Theta_0)$ has a positive hole of depth at most d. By Lemma E.5, it has a positive hole w of some depth $D \le d$ with $g_i \le \mathrm{G}^\sharp(N, d)$ for $1 \le i \le D$. Keep w, its hierarchy (w_i, ℓ_i) and its dictionaries fixed, and put $\kappa_\sharp = \mathrm{col}^\sharp(N + d + 1, \mathrm{G}^\sharp(N, d)) \ge 1$.

The top level. By Corollary 9.11, the unary word $w_D = z^m$ is a positive hole of $(\mathcal{M}_D, \Theta_D)$, and $\mathcal{M}_D$ has $N + D$ states. By Lemma E.6, $z^{m'}$ is a positive hole of $(\mathcal{M}_D, \Theta_D)$ for some $1 \le m' \le 3(N + D)$.

Compression. Apply Lemma E.11 with $j = D$, $g = \mathrm{G}^\sharp(N, d)$ and $E_D = \{z\}$. It gives markers $\widetilde{\boldsymbol{\ell}}$ and ρ_D with $(\mathcal{M}_D, \Theta_D) \equiv_{\{z\},\rho_D} (\mathcal{M}_D^{\widetilde{\boldsymbol{\ell}}}, \Theta_D^{\widetilde{\boldsymbol{\ell}}})$. Agreement preserves positive holes, so $\widetilde{w}_D = \rho_D(z)^{m'}$ is a positive hole of $(\mathcal{M}_D^{\widetilde{\boldsymbol{\ell}}}, \Theta_D^{\widetilde{\boldsymbol{\ell}}})$.

Expansion. For $i < D$ let $\widetilde{w}_i$ be the expansion of $\widetilde{w}_{i+1}$ with the marker $\widetilde{\ell}_i$. By validity, $\widetilde{w}_{i+1}$ is a nonempty word of blocks avoiding $\widetilde{\ell}_i$, so Lemma 9.10 shows, by downward induction, that $\widetilde{w}_i$ is a

positive hole of $(\mathcal{M}_i^{\widetilde{\ell}}, \Theta_i^{\widetilde{\ell}})$. At $i = 0$, $\widetilde{w} = \widetilde{w}_0$ is a positive hole of $(\mathcal{M}_0, \Theta_0)$. Its length is

$$m' \, |\mathrm{Exp}_{\widetilde{\ell},D}(\rho_D(z))| \le 3(N+D)\,\kappa_\sharp^D \le 3(N+d+1)\,\kappa_\sharp^d = \mathsf{Len}^\sharp(N,d).$$

Depth. Each $\widetilde{w}_i$ with $i < D$ ends with its marker $\widetilde{\ell}_i$, and cutting it at that marker returns $\widetilde{w}_{i+1}$. So the canonical hierarchy of $\widetilde{w}$ is $\widetilde{w}_0, \ldots, \widetilde{w}_D$ through level D, and $\widetilde{w}_D$ is unary. Hence $\mathrm{depth}(\widetilde{w}) \le D \le d$.

If $D = 0$, there is no compression: w is unary and Lemma E.6 gives a positive hole of length at most $3N \le \mathsf{Len}^\sharp(N,0)$ and depth 0.

This proves the general statement of Theorem 9.14. For a reader with a relation Π, apply it to the lettered monitor of the reader and the base channel Θ_Π, whose positive holes are the Π-holes (Lemma 9.8); the monitor has the N states of the reader, and a nonempty word has the same depth whichever monitor reads it. The estimates are Lemma E.12 below.

E.9 Estimates and Decision Procedures

Lemma E.12 (Closed form). *For $N \ge 1$ and $d \ge 0$, with $\Xi^\sharp = \Xi^\sharp(N,d)$,*

$$\mathsf{G}^\sharp(N,d) \le 2^{\Xi^\sharp} \qquad \text{and} \qquad \mathsf{Len}^\sharp(N,d) \le 2^{2^{5\Xi^\sharp}}.$$

Proof. Put $\Xi_0 = \Xi^\sharp - 3d - 3 = 6N^2 + 10N(N+d)(d+1) + 7N + 1$. Since $N \le 2^N$, we have $N^N \le 2^{N^2}$, so $\mathsf{C}^\sharp(N,d) \le 2^{\Xi_0}$. For nonnegative x and y, $2^x + 2^y \le 2^{x+y+1}$; with $d+1 \le 2^{d+1}$ this gives $\mathsf{A}^\sharp(N,d) \le 2^{\Xi_0+d+2}$. Using $3^d \le 2^{2d}$,

$$\mathsf{G}^\sharp(N,d) = 2 + \mathsf{A}^\sharp(N,d)\,3^d \le 2 + 2^{\Xi_0+3d+2} \le 2^{\Xi_0+3d+3} = 2^{\Xi^\sharp}.$$

If $d = 0$, then $\mathsf{Len}^\sharp(N,0) = 3(N+1) \le 2^{N+2} \le 2^{\Xi^\sharp}$. Let $d \ge 1$, so that $\Xi^\sharp \ge \Xi^\sharp(1,1) = 60$, and put $M = N + d + 1$, $g = \mathsf{G}^\sharp(N,d)$ and $P = 2^{\Xi^\sharp}$. Then $M + 1 \le \Xi^\sharp \le P$ and $g \le P$. Using $\log_2 M \le M$,

$$\log_2 \mathsf{col}^\sharp(M,g) \le (M+1)Mg + 4M(M+1)g + 2g + g^2 + 2(M+1)^2 g^2 \le 5P^3 + 2P + P^2 + 2P^4 \le 3P^4 \le 2^{4\Xi^\sharp+2}.$$

Therefore, since $d \le \Xi^\sharp \le 2^{\Xi^\sharp - 3}$ and $\log_2(3M) \le 2M \le 2^{5\Xi^\sharp - 1}$,

$$\log_2 \mathsf{Len}^\sharp(N,d) = \log_2(3M) + d \log_2 \mathsf{col}^\sharp(M,g) \le 2^{5\Xi^\sharp-1} + 2^{\Xi^\sharp-3}\,2^{4\Xi^\sharp+2} \le 2^{5\Xi^\sharp}. \qquad \square$$

Proof of Lemma 9.18. Put $N_0 = 2^\alpha$ and $D_0 = 2^{\alpha+\beta+1}$. Then $\bar{N}_I \le N_0$ and $d_I = 2\bar{N}_I N_A - 1 < D_0$, so, by monotonicity and Lemma E.12, $\mathsf{Len}_I^\sharp \le \mathsf{Len}^\sharp(N_0, D_0) \le 2^{2^{5\Xi^\sharp(N_0,D_0)}}$. Since $\alpha \ge 3$ and $\beta \ge 0$, we have $N_0 \ge 8$, $D_0 \ge 16$ and $N_0 \le D_0/2$. Hence $10N_0(N_0+D_0)(D_0+1) \le 10N_0 \cdot \frac{3}{2}D_0 \cdot \frac{17}{16}D_0 < 16N_0D_0^2$, $6N_0^2 \le 3N_0D_0 \le \frac{3}{16}N_0D_0^2$, and $7N_0 + 3D_0 + 4 \le N_0D_0 \le \frac{1}{16}N_0D_0^2$, so that

$$\Xi^\sharp(N_0,D_0) \le 17N_0D_0^2 = 17 \cdot 2^{3\alpha+2\beta+2} \qquad \text{and} \qquad 5\Xi^\sharp(N_0,D_0) < 2^{3\alpha+2\beta+9}.$$

Therefore $\mathsf{Len}_I^\sharp \le 2^{2^{2^{3\alpha+2\beta+9}}} = E_3(3\alpha+2\beta+9)$. $\square$

Proof of Corollary 9.16. Compute $\mathsf{Len}_I^\sharp$, and test every pair (u,t) of words over Σ with $|u| + |t| < \mathsf{Len}_I^\sharp$: check conditions (a)–(e) of Definition 7.1 over the finitely many cuts, decide whether $A(u)$ is mortal by reachability in the safe graph, and check $|t| - J(t) < N_A$. A pair that passes all tests is a mortal hole. If a mortal hole exists, Theorem 9.15 supplies one among the tested pairs. $\square$

F Proofs for Tree Classes

This appendix proves the results of Section 10 whose proofs were deferred, in its notation. For a covariant tree t, $\nu_0 t = t$ and $\nu_1 t = t^\partial$, so L_t^0 is the set of words with a prefix labeled $\top$ in t, and L_t^1 the set of words with a prefix labeled $\bot$. By Lemma 2.2(3), duality reverses the covariant order, and it commutes with the constructor, $f(t_i)_i^\partial = f(t_i^\partial)_i$, since it exchanges labels position by position.

F.1 Observation Pins

Proof of Lemma 10.4. Let $\mathrm{Pref}(w)$ be the set of prefixes of w, including w, and let $\mathrm{Off}(w) = \Sigma^* \setminus (\mathrm{Pref}(w) \cup w\Sigma^*)$ be the set of words that leave the pin path, that is, the words $w[{:}k]\ j\ v$ with $k < |w|$ and $j \neq w[k]$. The positions of $\mathrm{emb}_b(w, g)$ are the words of $\mathrm{Pref}(w)$, of which the proper prefixes of w are labeled f; the children $w[{:}k]\ j$ off the pin path, labeled b; and w followed by the positions of g. A word of $\mathrm{Off}(w)$ passes a child off the pin path, a proper prefix of w passes no leaf, and an extension of w follows g. Hence the four trees of Eq. (7) have the following cones:

tree	L^0	L^1
$g_1 = \mathrm{emb}_\bot(w, \top)$	$w\Sigma^*$	$\mathrm{Off}(w)$
$g_2 = \mathrm{emb}_\top(w, f(\top^n))$	$\Sigma^* \setminus \mathrm{Pref}(w)$	$\varnothing$
$g_3 = \mathrm{emb}_\top(w, \bot)$	$\mathrm{Off}(w)$	$w\Sigma^*$
$g_4 = \mathrm{emb}_\bot(w, f(\bot^n))$	$\varnothing$	$\Sigma^* \setminus \mathrm{Pref}(w)$

We use Theorem 3.2(1,2): the cones of t are extension closed and disjoint, and $t \le u$ iff $L^0_t \subseteq L^0_u$ and $L^1_u \subseteq L^1_t$.

- $g_1 \le t$ iff $w\Sigma^* \subseteq L^0_t$ and $L^1_t \subseteq \mathrm{Off}(w)$. If $w \in L^0_t$, then $w\Sigma^* \subseteq L^0_t$ by extension closure; L^1_t contains no extension of w, by disjointness, and no proper prefix of w, since otherwise it would contain w; so $L^1_t \subseteq \mathrm{Off}(w)$. Conversely, $g_1 \le t$ gives $w \in L^0_t$.
- $t \le g_2$ iff $L^0_t \subseteq \Sigma^* \setminus \mathrm{Pref}(w)$, since the condition on the cones of sign 1 is vacuous. By extension closure, L^0_t meets $\mathrm{Pref}(w)$ iff $w \in L^0_t$.
- $t \le g_3$ iff $L^0_t \subseteq \mathrm{Off}(w)$ and $w\Sigma^* \subseteq L^1_t$; this is the first case with the two cones exchanged.
- $g_4 \le t$ iff $L^1_t \subseteq \Sigma^* \setminus \mathrm{Pref}(w)$, iff $w \notin L^1_t$, as in the second case.

The pin path has $|w| + 1$ positions, there are $(n-1)|w|$ children off the pin path, and the end of the pin path has at most n children. So each tree has at most $1 + n(|w| + 1)$ positions.

For the consequence, let x be a visible variable and ρ an assignment. The cones of $\rho(x)$ are the covariant cones of $t = \nu_0\rho(x)$, since $\nu_1\rho(x) = t^\partial$ by Lemma 2.2(1). So an observation with variable x and word w is equivalent to one of the four inequalities between t and a tree g_k of Eq. (7). By Lemma 2.2(1,2), $g_k \le \nu_0\rho(x)$ iff $\nu_0 g_k \le_c \rho(x)$, and $\nu_0\rho(x) \le g_k$ iff $\rho(x) \le_c \nu_0 g_k$. The tree $\nu_0 g_k$ has the domain of g_k. □

F.2 Signed Bounds and Medians

Proof of Lemma 10.7. (1) By Lemma 2.5, ρ solves χ iff $\rho^\pm$ solves $\chi^\pm$. For a lower bound, Lemma 2.2(2) gives $g \le_c \rho(x)$ iff $\nu_+ g \le \nu_+\rho(x)$ iff $\nu_-\rho(x) \le \nu_- g$; these are the two bounds of Eq. (8) at $\rho^\pm$. Upper bounds are symmetric.

(2) Since duality is an involution, so is ι. Let σ be an assignment of $V^\pm$. In every row of Table 2, and for the two bounds of each pair in Eq. (8), $\iota\sigma$ satisfies one entry iff σ satisfies the other, because duality reverses the covariant order, commutes with the constructor, exchanges $\bot$ and $\top$, and maps $\nu_+ g$ to $\nu_- g$. For instance, $\iota\sigma$ satisfies $u^+ \le v^+$ iff $\sigma(u^-)^\partial \le \sigma(v^-)^\partial$, iff $\sigma(v^-) \le \sigma(u^-)$; it satisfies $f(a_i^{c(i)})_i \le u^+$ iff $f(\sigma(a_i^{\bar c(i)})^\partial)_i \le \sigma(u^-)^\partial$, iff $\sigma(u^-) \le f(\sigma(a_i^{\bar c(i)}))_i$; it satisfies $u^+ = \bot$ iff $\sigma(u^-) = \top$; and it satisfies $\nu_+ g \le x^+$ iff $\nu_+ g \le \sigma(x^-)^\partial$, iff $\sigma(x^-) \le \nu_- g$. The remaining entries are the same computation. Both entries of every row and both bounds of every pair belong to $\chi^\pm \cup B^\pm$, so ι preserves its solutions. Finally, $(\iota\rho^\pm)(v^\pm) = (\nu_\mp\rho(v))^\partial = \nu_\pm\rho(v)$ by Lemma 2.2(1).

(3) If $\iota\sigma = \sigma$, then $\sigma(v^-) = \sigma(v^+)^\partial$ for every v, and Lemma 2.5 gives $\sigma = \rho^\pm$ for $\rho(v) = \nu_+\sigma(v^+)$, and the preservation of finiteness. □

Proof of Lemma 10.8. Write $\mathrm{maj}(L_1, L_2, L_3)$ for the set of words that lie in at least two of the sets L_j. It is symmetric and monotone for inclusion, and $\mathrm{maj}(L, L, L') = L$.

(1) Symmetry is clear. If $t_j \le t'_j$ for $j = 1, 2, 3$, then $L^0_{t_j} \subseteq L^0_{t'_j}$ and $L^1_{t'_j} \subseteq L^1_{t_j}$ by Theorem 3.2(2), so the same inclusions hold between the cones of the medians, and $\mathrm{med}(t_1, t_2, t_3) \le \mathrm{med}(t'_1, t'_2, t'_3)$. The identity $\mathrm{med}(t, t, u) = t$ follows from $\mathrm{maj}(L, L, L') = L$.

(2) By Theorem 3.2(3), with f covariant, a word iv lies in $L^s_{f(a_i^j)_i}$ iff $v \in L^s_{a_i^j}$, and ε lies in no cone of a constructor tree. So the majority of the three cones $L^s_{f(a_i^j)_i}$ is the union over i of the sets $i\,\mathrm{maj}(L^s_{a_i^1}, L^s_{a_i^2}, L^s_{a_i^3})$, which is $L^s_{f(m_i)_i}$ for $m_i = \mathrm{med}(a_i^1, a_i^2, a_i^3)$; by Theorem 3.2(1) the two trees are equal. For duality, $L^s_{t^\partial} = L^{1-s}_t$, and maj is applied to each sign separately. For assignments of signed variables, $(\iota\,\mathrm{med}(\alpha, \beta, \gamma))(v^\pm) = \mathrm{med}(\alpha(v^\mp), \beta(v^\mp), \gamma(v^\mp))^\partial = \mathrm{med}((\iota\alpha)(v^\pm), (\iota\beta)(v^\pm), (\iota\gamma)(v^\pm))$.

(3) For $n = 0$ every tree is finite. Let $n \ge 1$, let t_1, t_2, t_3 be finite, and let h bound the lengths of their positions. A word w of length $h + 1$ is a position of no t_j, so its longest prefix in the domain of t_j is a leaf, labeled $\bot$ or $\top$; thus $w \in L^0_{t_j} \cup L^1_{t_j}$ for each j. Two of the three trees put w in the same cone L^s, so w lies in the cone L^s of the median, and some prefix of w is a leaf of the median. Every proper prefix of a position is labeled f, so no word of length $h + 2$ is a position of the median, which is therefore finite.

(4) Let α, β, γ solve a covariant system together with covariant ground bounds. A literal $u \le v$ holds in their median $\mathrm{med}(\alpha, \beta, \gamma)$ by monotonicity. For $f(a_i)_i \le u$, monotonicity gives $\mathrm{med}(f(\alpha(a_i))_i, f(\beta(a_i))_i, f(\gamma(a_i))_i) \le \mathrm{med}(\alpha, \beta, \gamma)(u)$, and the left side is $f(\mathrm{med}(\alpha, \beta, \gamma)(a_i))_i$ by (2); the upper constructor literals are symmetric. The extreme literals hold since $\mathrm{med}(\bot, \bot, \bot) = \bot$ and $\mathrm{med}(\top, \top, \top) = \top$. For a lower bound $G \le p$, $G = \mathrm{med}(G, G, G) \le \mathrm{med}(\alpha, \beta, \gamma)(p)$, and upper bounds are symmetric. □

F.3 Covariant Finite-Bound Pinning

In this subsection f is covariant and $n \ge 1$; for $n = 0$, Lemma 10.9 is trivial because every tree is finite. We use the closure D^{cov}_χ, the path bounds Lo and Up, label clashes and cycle clashes of a covariant system without copy literals, as defined in Appendix A.1, and Theorem A.1(1): such a system has a solution iff it has no label clash, and a finite solution iff it has neither a label clash nor a cycle clash.

Proof of Lemma 10.9. Let F be a covariant system over V with a finite solution δ, let B be a finite set of ground bounds on V, and let ρ be a solution of $F \cup B$.

Copy literals. Let $F_0 = F^\flat$ be obtained by Lemma 2.7, applied to every copy literal of F. The variables of F_0 are called *old*. By Lemma 2.7, δ extends to a finite solution of F_0, ρ extends to a solution of $F_0 \cup B$, and every solution of $F_0 \cup B$ restricts to a solution of $F \cup B$.

Encoding the bounds. For a finite ground tree g and a variable x define systems $L(g, x)$, expressing $g \le x$, and $U(x, g)$, expressing $x \le g$, with fresh variables:

$$L(\bot, x) = \varnothing, \quad L(\top, x) = \{\top \le x\}, \quad L(f(g_i)_i, x) = \{f(y_i)_i \le x\} \cup \textstyle\bigcup_i L(g_i, y_i),$$
$$U(x, \top) = \varnothing, \quad U(x, \bot) = \{x = \bot\}, \quad U(x, f(g_i)_i) = \{x \le f(y_i)_i\} \cup \textstyle\bigcup_i U(y_i, g_i).$$

Each fresh variable stands for a nonempty position π of g, and its *rank* is $|\pi|$; old variables have rank 0. By induction on g, an assignment that satisfies $L(g, x)$ satisfies $g \le x$, since $f(g_i)_i \le f(y_i)_i \le x$ by monotonicity of the covariant constructor; conversely, an assignment that satisfies $g \le x$ extends to $L(g, x)$ by giving each fresh variable the subtree of g at its position. The same holds for $U(x, g)$. Let F'' be F_0 together with the encodings of all bounds of B, with distinct fresh variables. Then F'' has a solution, which extends one of $F_0 \cup B$, and every solution of F'' restricts to a solution of $F_0 \cup B$. The fresh variables of the systems L are *sources*, those of the systems U are *sinks*, and the literals of F'' outside F_0 are *new*. The system F'' has no copy literals, and by construction:

(O1) the literals of F_0 contain only old variables;
(O2) the new lower literals are $f(y_i)_i \le x$ and $\top \le x$, where x is old or a source and every y_i is a source of rank one more than x; the new upper literals are $x \le f(y_i)_i$ and $x = \bot$, where x is old or a sink and every y_i is a sink of rank one more than x.

In particular, the children of a lower constructor literal are all old or all sources, and the children of an upper constructor literal are all old or all sinks.

Separation. Write $D = D^{\mathrm{cov}}_{F''}$, Lo, Up for the closure and the path bounds of F'', and $D_0 = D^{\mathrm{cov}}_{F_0}$, Lo_0, Up_0 for those of F_0. We show by induction on the derivation of $D(a,b)$: (a) if b is a source, then $a = b$; (b) if a is a sink, then $b = a$; (c) if a and b are old, then $D_0(a,b)$. Reflexivity is clear. Let $D(a,b)$ follow from $D(a,m)$ and $D(m,b)$. For (a), the induction hypothesis gives $m = b$ and then $a = m$; (b) is symmetric. For (c), m is not a source, since (a) would give $a = m$, and not a sink, since (b) would give $b = m$; so m is old, and $D_0(a,m)$ and $D_0(m,b)$. Let $D(a_i,b_i)$ follow by Eq. (10) from literals $f(a_j)_j \le u$ and $v \le f(b_j)_j$ and $D(u,v)$. Then b_i is old or a sink and a_i is old or a source, so (a) and (b) hold vacuously. If a_i and b_i are old, the two literals have old children; by (O2) they are not new, so they are literals of F_0, and u, v are old by (O1). By the induction hypothesis $D_0(u,v)$, hence $D_0(a_i,b_i)$. Conversely $D_0 \subseteq D$, so D and D_0 agree on old variables.

Paths. We show by induction on η: (d) if $\mathrm{Lo}(\eta,a,s)$ and s is a source, then a is a source of rank $\mathrm{rank}(s) + |\eta|$; (e) if $\mathrm{Up}(\eta,t,b)$ and t is a sink, then b is a sink of rank $\mathrm{rank}(t) + |\eta|$; (f) if a, b are old, then $\mathrm{Lo}(\eta,a,b)$ implies $\mathrm{Lo}_0(\eta,a,b)$ and $\mathrm{Up}(\eta,a,b)$ implies $\mathrm{Up}_0(\eta,a,b)$. For $\eta = \varepsilon$ use (a)–(c). For (d), by Eq. (11), $\mathrm{Lo}(i\eta,a,s)$ gives a literal $f(a'_j)_j \le z$ with $D(z,s)$ and $\mathrm{Lo}(\eta,a,a'_i)$. By (a), $z = s$, so the literal has a source as its root; by (O1) it is new, and by (O2) a'_i is a source of rank $\mathrm{rank}(s) + 1$. The induction hypothesis applies to $\mathrm{Lo}(\eta,a,a'_i)$. (e) is symmetric, using (b). For (f), $\mathrm{Lo}(i\eta,a,b)$ gives $f(a'_j)_j \le z$ with $D(z,b)$ and $\mathrm{Lo}(\eta,a,a'_i)$. If the literal were new, a'_i would be a source, and (d) would make a a source. So it is a literal of F_0, z and a'_i are old, $D_0(z,b)$ by (c), and $\mathrm{Lo}_0(\eta,a,a'_i)$ by the induction hypothesis. The case of Up is symmetric, using (e).

No cycle clash. Suppose that F'' had a cycle clash Eq. (12): $\eta \neq \varepsilon$, $\mathrm{Lo}(\eta,a,a)$, $D(a,b)$ and $\mathrm{Up}(\eta,b,b)$. If a were a source, (d) would give $\mathrm{rank}(a) = \mathrm{rank}(a) + |\eta|$. If b were a sink, (e) would give the same contradiction for b. If a were a sink, (b) would give $b = a$, a sink; if b were a source, (a) would give $a = b$, a source. So a and b are old, and by (c) and (f) the clash is a cycle clash of F_0. But F_0 has a finite solution, so it has no cycle clash.

Conclusion. The system F'' has a solution, so it has no label clash, and it has no cycle clash. Hence it has a finite solution, whose restriction is a finite solution of $F_0 \cup B$ and then of $F \cup B$. □

The proof extends the argument of [Zhao 2026, Theorem "Finite countermodel transfer"], where the bounds are the pin paths of a single word, to arbitrary finite ground bounds. The finite solution of F is used only to exclude cycle clashes among old variables; label clashes are excluded by the solution of $F \cup B$.

F.4 Closure under Intersection

Proof of Proposition 10.5. (1) Let $\mathcal{M}$ be a nonempty family of models of χ and $M = \bigcap \mathcal{M}$. If the premise of a rule lies in M, it lies in every member of $\mathcal{M}$, hence so does its conclusion; so M is closed under the rules. It contains Fact_χ, since every member does, and it is consistent, since it is contained in a member. For the projected family, let $S_j = U_j|_X$ with $U_j \in \mathrm{Mod}(\psi)$ for j in a nonempty index set; then $\bigcap_j U_j$ is a model of ψ with visible part $\bigcap_j S_j$. For assignments, $S_{\rho \sqcap \tau} = S_\rho \cap S_\tau$ by definition, so Lemma 3.5 turns the first two claims into the closure of solutions and of extensible visible assignments under $\sqcap$: if ρ and τ extend to solutions ρ' and τ' of ψ, then $\rho' \sqcap \tau'$ is a solution of ψ that extends $\rho \sqcap \tau$.

(2) Commutativity, associativity and idempotence hold coordinatewise. If $t \le_c t'$, then $L^0_t \cap L^0_u \subseteq L^0_{t'} \cap L^0_u$ and $L^1_t \cap L^1_u \subseteq L^1_{t'} \cap L^1_u$, so $t \sqcap u \le_c t' \sqcap u$ by Theorem 3.2(2). For the constructor, Theorem 3.2(3) gives $L^s_{f(t_i)_i} = \bigcup_i i L^{s \oplus c(i)}_{t_i}$, a union of sets of words with distinct first letters. Intersection therefore distributes over these unions, and the leaf pairs of both sides of the identity agree; by Theorem 3.2(1) the trees agree.

For the median, observe that the pairs $(L^0_t \cup L^0_u,\ L^1_t \cap L^1_u)$ and $(L^0_t \cap L^0_u,\ L^1_t \cup L^1_u)$ are disjoint and extension closed: for instance, a word of L^0_t in $L^1_t \cap L^1_u$ would lie in both cones of t. By Theorem 3.2(1,2) they are the leaf pairs of the least upper bound $t \vee u$ and the greatest lower bound $t \wedge u$ of t and u for $\le_c$. So the trees form a lattice whose joins and meets act on the cone of sign 0 by union and intersection and on the cone of sign 1 by intersection and union; in particular this lattice is distributive. The median $(t \wedge u) \vee (u \wedge v) \vee (v \wedge t)$ of three trees therefore has, as each of its two cones, the set of words that lie in the corresponding cones of at least two of t, u and v. For $v = \Omega$, whose cones are empty, these are the words that lie in the cones of both t and u, so the median is $t \sqcap u$.

(3) Regular languages are closed under intersection, so the first claim follows from Theorem 3.2(4). The cones of $\bot \sqcap \top$ are $\varnothing \cap \Sigma^*$ and $\Sigma^* \cap \varnothing$, so it is Ω, all of whose words are positions labeled f; for $n \ge 1$ there are infinitely many. □

G Proofs for Constrained Types and Simplification

The results of this appendix hold for $\{\bot, f, \top\}$ with any arity n and variance c, and for $\{\bot, f\}$ with f covariant, in each tree class $\mathcal{D}$, with the caveats stated. Terms are finite and built from variables, $\bot$, f and, over $\{\bot, f, \top\}$, $\top$; a term s denotes the tree $\rho(s)$ under an assignment ρ, and $s \le t$ means $\rho(s) \le_c \rho(t)$.

Lemma G.1 (Joins). *Any two trees t, u have a least upper bound $t \vee u$ for $\le_c$, and $\bot$ is the least tree. If t and u are finite, or regular, then so is $t \vee u$. If f is covariant and t and u are free of $\top$, then so is $t \vee u$.*

Proof. The sets $L^0_t \cup L^0_u$ and $L^1_t \cap L^1_u$ are extension closed and disjoint, since $L^0_t \cap L^1_t = L^0_u \cap L^1_u = \varnothing$, so by Theorem 3.2(1) they are the leaf pair of a tree $t \vee u$. By Theorem 3.2(2), a tree lies above t and u iff its first cone contains $L^0_t \cup L^0_u$ and its second cone is contained in $L^1_t \cap L^1_u$, so $t \vee u$ is the least upper bound. The leaf pair of $\bot$ is $(\varnothing, \Sigma^*)$ (Theorem 3.2(3)), so $\bot$ is least. A word outside both cones of $t \vee u$ lies outside $L^0_t \cup L^0_u$ and outside L^1_t or L^1_u, so it is a position of t or of u labeled f; by Theorem 3.2(4) the join of finite trees is finite. Unions and intersections of regular languages are regular. For covariant f, ν_0 is the identity, so a tree is free of $\top$ iff its first cone is empty, which the union preserves. □

Proposition G.2 (Flat bound witnesses). *Let*

$$\psi = \psi_0 \wedge \bigwedge_{(i,j) \in E} z_i \le z_j \wedge \bigwedge_{a \in L} s_a \le z_{i(a)} \wedge \bigwedge_{b \in U} z_{j(b)} \le t_b,$$

where ψ_0 is a conjunction of inequalities between terms over X, the s_a and t_b are terms over X, and E is a set of pairs of hidden indices. Write $i \to^* j$ for reachability in the directed graph E. Then an assignment $\rho : X \to \mathcal{D}$ extends to a $\mathcal{D}$-valued solution of ψ iff ρ satisfies ψ_0 and $s_a \le t_b$ for all $a \in L$ and $b \in U$ with $i(a) \to^* j(b)$. Hence $\varphi \models_{\mathcal{D}} \exists \bar{z}\, \psi$ iff φ entails, in $\mathcal{D}$, each literal of ψ_0 and each of these inequalities.

Proof. If ρ' is a solution of ψ extending ρ, then ρ satisfies ψ_0, and for $i(a) \to^* j(b)$ the chain $s_a \le z_{i(a)} \le \cdots \le z_{j(b)} \le t_b$ gives $s_a \le t_b$. Conversely, put $\rho'(z_j) = \bigvee \{\rho(s_a) : i(a) \to^* j\}$, with $\bot$ for the empty join; it lies in $\mathcal{D}$ by Lemma G.1. Each lower bound $s_a \le z_{i(a)}$ holds since $i(a) \to^* i(a)$.

For $(i, j) \in E$, every a with $i(a) \to^* i$ has $i(a) \to^* j$, so $\rho'(z_i) \leq_c \rho'(z_j)$. Each joinand of $\rho'(z_{j(b)})$ is below $\rho(t_b)$ by hypothesis, hence so is the join. The last statement follows because $\varphi \models_{\mathcal{D}} \exists \bar{z}\, \psi$ says that every solution of φ extends. □

With q hidden variables, reachability costs $O(q^3)$, and at most $|L| \cdot |U|$ plain entailments are added to those for ψ_0. The *occurrence sign* of a variable occurrence in an inequality $s \leq t$ between terms is negative if the occurrence is in s and positive if it is in t, exchanged when the polarity $\mathrm{par}(w)$ of its position w in that term is 1.

Proposition G.3 (Unipolar witnesses). *Let ψ be a conjunction of inequalities between terms over $X \cup Z$. Let Z_- be the hidden variables all of whose occurrences in ψ are negative, and Z_+ those, outside Z_-, all of whose occurrences are positive. Let ψ' be ψ with every $z \in Z_-$ replaced by $\bot$ and every $z \in Z_+$ replaced by $\top$. Over $\{\bot, f, \top\}$, in each tree class, $\varphi \models \exists \bar{z}\, \psi$ iff $\varphi \models \exists \bar{z}'\, \psi'$, where $\bar{z}'$ are the remaining hidden variables. Over $\{\bot, f\}$ the same holds with $\top$ replaced by the tree Ω, whose constructor positions are all words, over arbitrary and regular trees, and also over finite trees if $Z_+ = \varnothing$ or $n = 0$.*

Proof. By induction on terms, a term is monotone in a variable all of whose occurrences in it have polarity 0, and antitone in one all of whose occurrences have polarity 1: an occurrence at the position iw of $f(s_j)_j$ has polarity $c(i) \oplus \mathrm{par}(w)$, and f is monotone in a covariant and antitone in a contravariant argument (Section 2.2). Hence increasing a variable whose occurrences in $s \leq t$ are all positive cannot falsify $s \leq t$, and decreasing one whose occurrences are all negative cannot either. Given a solution of ψ extending a visible assignment, change the values of the variables of Z_- to $\bot$ and those of Z_+ to $\top$, one at a time. Each change preserves every inequality, since occurrence signs do not depend on the values of other variables, and $\bot$ and $\top$ lie in every tree class. The result solves ψ' on the remaining variables. Conversely, a solution of ψ' gives a solution of ψ with the substituted values. Over $\{\bot, f\}$ the same argument uses Ω, the greatest tree over $\{\bot, f\}$, which is regular; for $n \geq 1$ it is not finite, and for $n = 0$ it is the finite tree $f()$. □

When every hidden variable is unipolar, ψ' has no hidden variable, and the entailment is a conjunction of plain entailments. Over $\{\bot, f\}$ the tree Ω is named in a plain query by a variable o with $f(o, \ldots, o) \leq o$, which forces Ω over arbitrary and regular trees; for $n \geq 1$, finite trees over $\{\bot, f\}$ have no greatest element, and this elimination is not available there. The occurrence signs are taken in the whole consequent: in a subsumption formula the constraints C_1 and the context comparisons of R_{12} count, not only the result type.

Proposition G.4 (Functional witnesses). *Let $\psi = D \wedge R$, where D consists of one equation $z = t_z$ for each hidden variable z, with t_z a term over $X \cup Z$, and R is a conjunction of inequalities between terms. The* dependency graph *has an edge from z to z' when z' occurs in t_z; the edge is* guarded *unless $t_z = z'$. Suppose that every cycle of the dependency graph contains a guarded edge. Over arbitrary and over regular trees,*

$$\varphi \models \exists \bar{z}\, (D \wedge R) \iff \varphi \wedge D \models R,$$

where the right side is a conjunction of plain entailments with $\bar{z}$ universally quantified. Over finite trees the same holds if the dependency graph is acyclic. If it has a cycle, then $\varphi \models_{\mathrm{fin}} \exists \bar{z}\, \psi$ iff φ has no finite solution.

A guarded cycle forces infinite trees, so over finite trees the premise $\varphi \wedge D$ of the plain entailments has no solution, and they would hold vacuously; this is why the finite case needs its own dispatch.

Proof. *Unique solutions.* An unguarded edge goes from z to the variable t_z. Since every cycle contains a guarded edge, the unguarded edges form an acyclic graph in which each vertex has at

most one outgoing edge. For a hidden z let z^* be the end of its unguarded chain, so that t_{z^*} is not a hidden variable. Fix values $\rho(x)$ of the visible variables. For a term s over $X \cup Z$ and a word w define a label $\lambda(s, w)$, possibly undefined: for $s = x$ visible, the label of $\rho(x)$ at w; for a constant, its label at $w = \varepsilon$; for $s = f(s_j)_j$, the label f at ε and $\lambda(s_i, w')$ at $w = iw'$; and for $s = z$ hidden, $\lambda(t_{z^*}, w)$. Each step either ends, shortens w, or passes from a hidden variable to a term that is not a hidden variable, so the recursion terminates. By induction on the length of w, the labels $\lambda(s, \cdot)$ define trees $T(s)$ with $T(f(s_j)_j) = f(T(s_j))_j$ and $T(z) = T(t_z)$, so the values $T(z)$ solve D, and every solution of D with these visible values agrees with T at every position, by the same induction. Every subtree of $T(z)$ is $T(s)$ for a subterm s of some $t_{z'}$, a subtree of some $\rho(x)$, or a constant, so $T(z)$ is regular when every $\rho(x)$ is. If the dependency graph is acyclic, then $T(z)$ is the value of the finite term obtained from z by substituting definitions finitely often, which is finite when every $\rho(x)$ is. If the graph has a cycle $z_1 \to z_2 \to \cdots \to z_k = z_1$, let w_i be a position of z_{i+1} in t_{z_i}. In every solution the subtree of z_i at w_i is the value of z_{i+1}, since the path to an occurrence in a term passes through constructors. Some w_i is nonempty, so z_1 is its own subtree at the nonempty position $w = w_1 \cdots w_{k-1}$, has all positions w^j, and is infinite.

The reduction. By the above, each assignment ρ of X has exactly one extension to a solution of D, and it lies in the tree class of ρ in the stated cases. So ρ extends to a solution of $D \wedge R$ iff this unique extension satisfies R, and the solutions of $\varphi \wedge D$ are exactly the solutions of φ with their unique extensions. This proves the equivalence over arbitrary and regular trees, and over finite trees when the graph is acyclic. If it has a cycle, D has no finite solution for any visible values, so finite entailment holds iff φ has no finite solution. □

Proposition G.5 (Mixed sandwich). *For arrow types, $\varphi \models_{\mathcal{D}} \exists z\, [\, x \le (z \to z) \;\wedge\; (z \to z) \le y\,]$ iff $\varphi \wedge x = \top$ and $\varphi \wedge y = \bot$ have no solution in $\mathcal{D}$ and the following plain entailments hold in $\mathcal{D}$, where a, b, c', d' are fresh visible variables:*

$$\varphi \wedge x = (a \to b) \wedge y = \top \models b \le a, \qquad \varphi \wedge x = \bot \wedge y = (c' \to d') \models c' \le d',$$
$$\varphi \wedge x = (a \to b) \wedge y = (c' \to d') \models b \le a \wedge b \le d' \wedge c' \le a \wedge c' \le d'.$$

The hidden variable z occurs at both polarities of $z \to z$, so the consequent has no parity assignment and z is not unipolar; still, two satisfiability tests and six plain entailments decide the instance.

Proof. Let ρ be a solution of φ. If $\rho(x) = \top$ or $\rho(y) = \bot$, no z exists, because $z \to z$ has the root f. Otherwise $\rho(x)$ is $\bot$ or an arrow $a \to b$, and $\rho(y)$ is $\top$ or an arrow $c' \to d'$. By the order on arrows, $a \to b \le_c z \to z$ iff $z \le_c a$ and $b \le_c z$, and $z \to z \le_c c' \to d'$ iff $c' \le_c z$ and $z \le_c d'$, while the comparisons with $\bot$ and $\top$ always hold. So the witnesses form the set of z above the lower bounds and below the upper bounds of the four cases, and by Proposition G.2, with one hidden variable, it is nonempty iff every lower bound lies below every upper bound: no condition, $b \le a$, $c' \le d'$, or the four comparisons. The case premises are mutually exclusive and cover the solutions of φ with $\rho(x) \ne \top$ and $\rho(y) \ne \bot$, and each solution of φ extends in exactly one way to the fresh variables of its case, which then denote the children. Hence the case conditions hold in every solution of φ iff the plain entailments hold, and the two satisfiability tests say that no solution has $\rho(x) = \top$ or $\rho(y) = \bot$. □

G.1 Closed Trees and Extensible Algebras

Let f be binary and covariant over $\{\bot, f, \top\}$, and

$$\Phi = \{x \le f(a, e),\ a \le y,\ f(m, e) \le y,\ f(e, e) \le m,\ \top \le e\}.$$

Then $\Phi \models x \le y$ over trees. Indeed $e = \top$. If $y = \top$ there is nothing to prove. Otherwise $f(m, \top) \le y$ forces $y = f(y_0, \top)$ with $m \le y_0$, and $a \le y$ forces $a \neq \top$, so $a \le f(\top, \top) \le m \le y_0$. Hence $x \le f(a, \top) \le f(y_0, \top) = y$. Thus deleting $x \le y$ from $\Phi \wedge x \le y$ is an exact simplification, decided by one plain entailment, or by Proposition G.2 with the witness $z = x$ for $\exists z\,(x \le z \le y)$.

Now consider the functions from $\{0,1\}^*$ to the subsets of $\{\mathrm{a}, \mathrm{b}\}$, ordered pointwise by inclusion, with $\bot$ the constant $\varnothing$, $\top$ the constant $\{\mathrm{a}, \mathrm{b}\}$, and $f(p, p')$ the function with the value $\{\mathrm{a}\}$ at ε, $p(w)$ at $0w$ and $p'(w)$ at $1w$. Pointwise unions and intersections make this a distributive lattice, and f preserves binary joins and meets. A tree t embeds as the function whose value at w contains a iff $w \notin L^1_t$ and contains b iff $w \in L^0_t$: the value is $\{\mathrm{a}\}$ at the positions labeled f, $\{\mathrm{a}, \mathrm{b}\}$ below a leaf $\top$, and $\varnothing$ below a leaf $\bot$. By Theorem 3.2(2,3) this embedding preserves and reflects the order and commutes with f, $\bot$ and $\top$. Let q have the value $\{\mathrm{b}\}$ at ε and $\varnothing$ elsewhere, and put $e = \top$, $m = f(\top, \top)$, $a = q$, $y = q \vee f(m, \top)$ and $x = f(q, \top)$. All literals of Φ hold, but $x(0) = q(\varepsilon) = \{\mathrm{b}\}$ and $y(0) = q(0) \cup m(\varepsilon) = \{\mathrm{a}\}$, so $x \not\le y$. Hence a procedure that is sound for this extensible algebra cannot certify the simplification; this separates the two semantics and does not indicate an error in a procedure specified for the extensible one.

H The Mechanized Proof

This appendix gives the formal statement, the module layout, and the correspondence between the results of this paper and the declarations that prove them (Section 14). Names are relative to the namespace `DeciENSSE`; names that begin with `DeciNSSE` belong to the library shared with the artifact of Zhao [2026]. Listings open both namespaces and suppress proof bodies and implicit arguments. Paths are lists over `Fin n`, and `V k` abbreviates `Fin k`.

H.1 Statement and Public Interface

Trees, the variance order, finite terms, and regular graphs are those of Zhao [2026]: `DeciNSSE.Tree` labels the paths of a tree by `DeciNSSE.Sym`, `DeciNSSE.Tree.Le` compares the labels at every common path, reversed at odd `DeciNSSE.polarity`, and `DeciNSSE.RGraph.unfold` and `DeciNSSE.FTree.toTree` interpret regular graphs and finite terms. `DeciNSSE.Characterization.treeLe_eq_gfp` identifies the order with the greatest fixed point of the constructor rule, and `DeciNSSE.Characterization.unfold_range_iff` and `DeciNSSE.Characterization.toTree_range_iff` show that the two interpretations have as images the trees with finitely many subtrees and the trees with finite domain. The flat literals of Section 2.3 and existential entailment over arbitrary trees are

```
inductive ELit (n k : ℕ) where
  | le (x y : V k)
  | leF (x : V k) (a : Fin n → V k)
  | fLe (a : Fin n → V k) (x : V k)
  | eqBot (x : V k)
  | eqTop (x : V k)

def ELit.holds (c : Fin n → Bool) (ρ : V k → Tree n) : ELit n k → Prop
  | .le x y => Tree.Le c (ρ x) (ρ y)
  | .leF x a => Tree.Le c (ρ x) (Tree.node (ρ ∘ a))
  | .fLe a x => Tree.Le c (Tree.node (ρ ∘ a)) (ρ x)
  | .eqBot x => ρ x = Tree.bot
  | .eqTop x => ρ x = Tree.top
```

```
def ExistsEntails (c : Fin n → Bool)
    (φ : EConstraint n k) (ψ : EConstraint n (k + h)) : Prop :=
  ∀ ρ, ESat c ρ φ → ∃ σ : V h → Tree n, ESat c (join ρ σ) ψ
```

where `EConstraint` is a list of literals, `ESat` requires every literal to hold, and `join` assigns the visible variables first and the h hidden ones after them. The literal $\top \le u$ of the paper is `eqTop`. In `ExistsEntailsReg` and `ExistsEntailsFin` both assignments range over regular graphs and finite terms, composed with their interpretations. Over $\{\bot, f\}$ the literals `BFLit` omit `eqTop`, `BFSat` reads them covariantly through `BFLit.toELit`, and

```
def TopFree (t : Tree n) : Prop := ∀ w, t.fn w ≠ some Sym.top

def ExistsEntailsTopFree
    (φ : BFConstraint n k) (ψ : BFConstraint n (k + h)) : Prop :=
  ∀ ρ, (∀ x, TopFree (ρ x)) → BFSat ρ φ →
    ∃ σ : V h → Tree n, (∀ z, TopFree (σ z)) ∧ BFSat (join ρ σ) ψ
```

with the regular and finite versions `ExistsEntailsTopFreeReg` and `ExistsEntailsTopFreeFin`. These definitions, together with those of Zhao [2026] that they use, form the trusted statement.

Importing `DeciENSSE` exposes the six procedures

```
def decideExistsEntails (c : Fin n → Bool)
    (φ : EConstraint n k) (ψ : EConstraint n (k + h)) :
    Decidable (ExistsEntails c φ ψ)
def decideExistsEntailsTopFree
    (φ : BFConstraint n k) (ψ : BFConstraint n (k + h)) :
    Decidable (ExistsEntailsTopFree φ ψ)

theorem decideExistsEntails_correct (c : Fin n → Bool)
    (φ : EConstraint n k) (ψ : EConstraint n (k + h)) :
    @decide _ (decideExistsEntails c φ ψ) = true ↔ ExistsEntails c φ ψ
```

and `decideExistsEntailsReg`, `decideExistsEntailsFin`, `decideExistsEntailsTopFreeReg`, and `decideExistsEntailsTopFreeFin`, of the same types for the other relations. Each has a correctness theorem of the same form, named by appending `_correct`, without premises. The procedures are definitions rather than instances, since several of them decide related propositions. The plain-entailment relations `DeciNSSE.Entails`, `DeciNSSE.EntailsReg`, and `DeciNSSE.EntailsFin` of Zhao [2026] take a list `DeciNSSE.Constraint` of literals without copy literals. With `liftConstraint` embedding it and `query` the consequent $x \le y$,

```
theorem nsse_iff_ensse (c : Fin n → Bool)
    (φ : Constraint n k) (x y : V k) :
    Entails c φ x y ↔
      ExistsEntails (h := 0) c (liftConstraint φ) (query x y)
```

and its regular and finite versions, `nsseReg_iff_ensseReg` and `nsseFin_iff_ensseFin`, give the procedures `decideEntails`, `decideEntailsReg`, and `decideEntailsFin`. They have the same types as the procedures of Zhao [2026], and their correctness theorems are named by appending `_correct` as well. With the consequent `bfQuery`, the reduction `topFree_nsse_iff_ensse` and its versions give `decideEntailsTopFree` and its two versions, which decide covariant plain entailment over $\{\bot, f\}$.

The files `lean-toolchain` and `lakefile.toml` pin Lean and Mathlib to `v4.35.0`. The commands

```
lake exe cache get
lake build
```

download compiled dependencies and check both libraries. The axioms of the twelve correctness theorems, of the six existential procedures and the six plain ones, can be listed with `#print axioms`; they are `propext`, `Classical.choice`, and `Quot.sound`.

H.2 Module Overview

The existential proof has 86 files under `DeciENSSE/`, and the shared library 54 files under `DeciNSSE/`. Below the public interface the procedures work over arbitrary finite types `X` of visible and `X ⊕ Z` of all variables; `toCoreFull`, `splitFull`, `toCoreBF`, and `splitBF` translate the public syntax, and `existsEntails_iff_core`, `existsEntailsTopFree_iff_core`, and their versions transfer the semantics.

Module group	Contents	Paper
`Semantics`, `Main`, `NSSE`	public syntax and relations, the six procedures, plain entailment	Sections 2.3 and 11
`Core/Main`, `Core/Nullary`	the arity dispatch; enumeration of root labels at arity zero	Remark 10.19
`Core/Full`	syntax over arbitrary carriers, signed systems, sandwich elimination, observations and ground bounds, path pins, graph equations, $\top$-free trees	Sections 2.3 and 10
`Core/Mixed`: Leaf, LeafHorn, Duality, Conditions	leaf cones, clauses, least closures, duality, collision return, deadness, the three conditions	Sections 3 and 4
`Core/Mixed`: Signed, Membership, Sat, CondI0, Units	summaries, closure membership, self events, satisfiability, (I0), (U)	Sections 5 and 6
`Core/Mixed`: Extract, Decide, Final	extracted instances, bounded search, arbitrary trees	Sections 7 and 11.2
`Core/Mixed`: ClassTransfer, Regular, Finite, FinitePinning*; `Support`	two observations, regular and finite transfer, signed bounds, medians, ranked chains	Section 10
`Core/Words`	automata over finite alphabets, infinite words, two-word instances, kinds of state, lassos, the pre-hole graph, deletions	Sections 7.1 and 8
`Core/Words/Self`	self events, three cases, the pump, mortal holes, the short-hole theorem	Sections 8 and 9.6
`Core/Words/Mortal`	separator words, the Π-reader, its depth, the mortal bound	Sections 9.2 and 9.3
`Anchored`	positive channels and towers, agreement, the positive column automaton, positive unary words, anchored normal forms, the bounds	Sections 9.4 and 9.5 and Appendix E
`Core`: Basic, Horn, Automata*, Reduction, Decide, Final, Classes*	the direct core for $\{\bot, f\}$ with f covariant	Section 10.5†
`Examples`	small proved and evaluated instances	
`DeciNSSE/*`	trees, order, representations, characterization, readers and holes, hierarchy, bounded-depth compression, hole search	Section 2 and Appendix A

At positive arity the full signature follows the steps of Section 11.2. The procedure for arbitrary trees, `Core.Mixed.decideExistsEntails`, decides satisfiability of both systems, (I0), and (U), and then searches every extracted instance for a hole whose length is bounded by `Core.Words`.

`Self.selfHoleBound`. The theorem `Core.Mixed.existsEntailsT_iff_boundedSelf` proves this search exact; it combines Theorems 4.2 and 9.17 and Corollary 7.8. The procedures for regular and finite trees apply the transfers of Theorems 10.11 and 10.12. The covariant core is described in Appendix H.4. At arity zero the procedures of `Core/Main` dispatch to an enumeration of root labels (`Core.Nullary.existsEntailsT_iff_labels`), which the kernel evaluates (`Examples.nullary_yes`, `Examples.nullary_no`); the general procedures are correct at arity zero as well (`Core.Mixed.decideExistsEntails_nullary_agrees`).

H.3 Correspondence

Table 9. Selected declarations supporting the indicated results of the paper. Where a declaration covers only part of a result, the left cell names that part. A dagger marks a difference in the form of the statement or of the proof, explained in Appendix H.4.

Paper result	Formal counterpart
Systems, Definition 2.4†	`ELit`, `ExistsEntails`, `ExistsEntailsTopFree`, and their versions; `Core.Full.ExistsEntailsT`, `Core.Full.ExistsEntailsTReg`, `Core.Full.ExistsEntailsTFin`
Normalization, first equivalence of Lemma 2.2(2)	`DeciNSSE.treeLe_iff_normalize`
Order over $\{\bot, f\}$, Lemma 2.3	`Core.Full.le_of_constructor_subset`, `Core.Full.constructor_subset_of_le`
Signed correspondence, Lemma 2.5, except the preservation of tree classes	`Core.Full.signed`, `Core.Mixed.FinitePinning.sat_normalize_iff`, `Core.Mixed.FinitePinning.normalize_fixed`, `Core.Mixed.FinitePinning.normalize_decode`, `Core.Mixed.FinitePinning.sat_decode`
Sandwich elimination, Lemma 2.7†	`Core.Full.Flatten.sandwich_iff`, `Core.Full.Flatten.forward`, `Core.Full.Flatten.backward`, `Core.Full.Flatten.finite_iff`, `Core.Full.satisfiable_regular`
Satisfiability and regular-model existence, Theorem 2.9(1)†	`Core.Mixed.decideSatMixed`, `Core.Mixed.satisfiable_regular`, `Core.Mixed.decideSatFin`
Plain entailment, Theorem 2.9(2)†	`nsse_iff_ensse`, `decideEntails`, `decideEntailsReg`, `decideEntailsFin`
Tail, return gap, depth, Theorems 2.15 and 2.19 and Lemma 2.18	`DeciNSSE.RejectedTail.boundedRejectedTail_of_rejectedPath`, `DeciNSSE.RejectedTail.return_gap`, `DeciNSSE.RejectedTail.rejected_depth`
Hole search, Theorem 2.20 and Corollary 2.21	`DeciNSSE.ElementaryBound.exists_short_hole`, `DeciNSSE.RejectedTail.hole_iff_bounded_length`, `DeciNSSE.RejectedTail.holeLengthBound_le`, `DeciNSSE.RejectedTail.decideHole`
Alignment, desubstitution, Lemmas 2.17 and 2.24	`DeciNSSE.LetteredHierarchy.alignment`, `DeciNSSE.LetteredHierarchy.isHole_der_iff`, `DeciNSSE.LetteredHierarchy.isHole_hierOf_iff`, `DeciNSSE.BoundedDepth.card_cores`

Paper result	Formal counterpart
Leaf representation, Theorem 3.2(1), (2), and (3) except the duality of cones	`Core.Mixed.leaf`, `Core.Mixed.leaf_disjoint`, `Core.Mixed.leaf_extension`, `Core.Mixed.decode`, `Core.Mixed.leaf_decode`, `Core.Mixed.leaf_injective`, `Core.Mixed.treeLe_iff_leaf`, `Core.Mixed.leaf_bot`, `Core.Mixed.leaf_top`, `Core.Mixed.leaf_node_nil`, `Core.Mixed.leaf_node_cons`, `Core.Mixed.leaf_root_plus`, `Core.Mixed.leaf_root_minus`
Clause form, Lemma 3.5	`Core.Mixed.sat_iff_sol`, `Core.Mixed.sol_iff_assignment`
Least closures, Lemma 3.8(1) and (4); (2) for $A = \emptyset$ and for single seeds; (5) for the atom sets S_ρ of assignments	`Core.Mixed.Hull`, `Core.Mixed.hull_closed`, `Core.Mixed.hull_facts`, `Core.Mixed.subset_hull`, `Core.Mixed.hull_least`, `Core.Mixed.mu_subset`, `Core.Mixed.cl_subset`, `Core.Mixed.reach_extend`, `Core.Mixed.cl_extend_subset`, `Core.Mixed.live_extend`, `Core.Mixed.sat_iff_mu`, `Core.Mixed.live_iff`, `Core.Mixed.extends_iff_hull`, `Core.Mixed.least_extension`
Duality, extensions first, collision return, Lemmas 3.10 to 3.12	`Core.Mixed.move_dual`, `Core.Mixed.moves_dual`, `Core.Mixed.reach_iff_moves`, `Core.Mixed.collision_return`
Deadness, Corollary 3.13	`Core.Mixed.dead_iff`
Characterization, Theorem 4.2	`Core.Mixed.existsEntailsT_of_unsat`, `Core.Mixed.not_existsEntailsT_of_unsat`, `Core.Mixed.three_conditions_mixed`
Pure forms, Lemma 4.3	`Core.Mixed.condU_iff`, `Core.Mixed.deadU_iff_reach_neg`, `Core.Mixed.condII_iff_moves`
Summaries, lowest point, Lemmas 5.2 and 5.3	`Core.Mixed.signedSys`, `Core.Mixed.moves_iff_signed`, `Core.Automata.summaries_D`, `Core.Automata.summaries_P`, `Core.Automata.summaries_Q`, `Core.Mixed.mem_P_iff_moves`, `Core.Mixed.lowest_point_leaf`
Duality and sign coherence, Lemma 5.4	`Core.Mixed.D_dual`, `Core.Mixed.Q_flip`, `Core.Mixed.sign_coherent`
Least-closure membership, Lemma 5.6†	`Core.Mixed.PF`, `Core.Mixed.mem_PF`, `Core.Mixed.mu_membership`
Roots and the pop bound, Lemma 5.9	`Core.Mixed.popRoot`, `Core.Mixed.P_proper_prefix_neg`, `Core.Mixed.deadU_iff_state`, `Core.Mixed.deadU_prefix`
Deadness from states, Lemma 5.11(1–4)†	`Core.Mixed.deadU_iff_state`, `Core.Mixed.mu_flip_membership`, `Core.Mixed.deadT_iff_state`, `Core.Mixed.deadT_prefix`
Deadness, Theorem 5.12	`Core.Mixed.dead_iff_unit_or_self`, `Core.Mixed.selfEv_return`, `Core.Mixed.return_selfEv_of_not_deadT`
Closure membership, head prefixes, Lemmas 5.15 and 5.16	`Core.Mixed.cl_membership_live`, `Core.Mixed.head_prefix_deadU`
Satisfiability and (I0), Proposition 5.17†	`Core.Mixed.avoids_mu_iff`, `Core.Mixed.consistent_mu_iff`, `Core.Mixed.decideSatMixed`, `Core.Mixed.not_condI0_iff`, `Core.Mixed.condI0_bounded`, `Core.Mixed.decideCondI0`
Source automaton, Lemma 6.3	`Core.Mixed.targetDFA`, `Core.Mixed.targetDFA_run`, `Core.Mixed.targetDFA_deadT`, `Core.Mixed.targetDFA_deadU`, `Core.Mixed.deadT_prefix`, `Core.Mixed.deadU_prefix`, `Core.Mixed.unit_selfEv_iff`

Paper result	Formal counterpart
Unit readers, Lemmas 6.1, 6.5 and 6.6	`Core.Mixed.condU_iff`, `Core.Mixed.unitReader`, `Core.Mixed.unit_absHole_iff`, `Core.Mixed.condU_iff_no_unit_holes`, `Core.Mixed.unit_between`, `Core.Mixed.unit_cone`
Condition (U), Theorem 6.11†	`Core.Mixed.decideCondU`, `DeciNSSE.RejectedTail.decideHole`
Two-word problem, Definitions 7.1 and 7.3	`Core.Words.Self.SelfTwoWordInstance`, `Core.Words.Self.SelfTwoWordInstance.IsHole`, `Core.Words.TwoWordInstance.ForcedHeads`
Heads, Lemmas 7.4 and 7.6	`Core.Words.TwoWordInstance.head_cut_le_J`, `Core.Mixed.extractSelf`, `Core.Mixed.extractSelf_forcedHeads`
Holes and (II), Theorem 7.7 except (2), Corollary 7.8	`Core.Mixed.extractSelf_isHoleS_iff`, `Core.Mixed.condII_iff_no_holes`, `Core.Mixed.mixed_reduction`
Continuation, decomposition, Lemmas 8.1 and 8.3	`Core.Words.LDFA.atCut_append`, `Core.Words.TwoWordInstance.J_append`, `Core.Words.TwoWordInstance.cross_append_iff`, `Core.Words.TwoWordInstance.event_append_iff`, `Core.Words.Self.SelfTwoWordInstance.selfEvent_append_iff`, `Core.Words.periodic_prefix_iff`, `Core.Words.TwoWordInstance.aligned_keeps`, `Core.Words.Self.SelfTwoWordInstance.preHole_complete`, `Core.Words.Self.SelfTwoWordInstance.hole_decomposition`
Kinds of state, Definition 8.5 and Lemma 8.8	`Core.Words.TwoWordInstance.mortal_iff`, `Core.Words.TwoWordInstance.state_kinds`, `Core.Words.TwoWordInstance.mortal_safe_word`, `Core.Words.TwoWordInstance.thin_cyclic_letter`, `Core.Words.TwoWordInstance.thin_lasso_iff`, `Core.Words.TwoWordInstance.thin_mortal_endpoint_short`
Breaker, survival, lassos, Lemmas 8.6, 8.10 and 8.13 and Corollary 8.11	`Core.Words.Self.SelfTwoWordInstance.breakerM`, `Core.Words.Self.SelfTwoWordInstance.preHole_breaker`, `Core.Words.TwoWordInstance.survives_iff_periodic`, `Core.Words.TwoWordInstance.aligned_survives`, `Core.Words.TwoWordInstance.survives_iff_prepend_eq`, `Core.Words.TwoWordInstance.lasso_breaking_prefix`, `Core.Words.Self.SelfTwoWordInstance.lasso_completes`, `Core.Words.TwoWordInstance.aperiodic_survivorFree`, `Core.Words.Self.SelfTwoWordInstance.aperiodic_selfSurvivorFree`
Short pre-holes, Proposition 8.12	`Core.Words.TwoWordInstance.preHoleGraph_correct`, `Core.Words.TwoTapeGraph.short_accepts`, `Core.Words.TwoWordInstance.short_preHole`
Pump, Theorem 8.14†	`Core.Words.Self.SelfTwoWordInstance.pumpGood_short`, `Core.Words.Self.purelyPeriodicShortPreHole_capacity`, `Core.Words.pumpCapacity`
Three cases, Theorem 8.17 and Corollary 8.18†	`Core.Words.Self.three_cases_self`, `Core.Words.Self.nonMortalShortHoleBoundSelf`, `Core.Words.pumpHoleBound`

Paper result	Formal counterpart
Separator word, Lemma 9.4	`Core.Words.Mortal.sepWord`, `Core.Words.Mortal.no_pre_comparison`, `Core.Words.Mortal.cross_comparison_iff`, `Core.Words.Self.target_comparison_iff`, `Core.Words.Mortal.anchored_iff`
Π-reader and depth, Lemmas 9.5 and 9.6†	`Core.Words.Mortal.reader`, `Core.Words.Mortal.card_state`, `Core.Words.Mortal.first_rejected_cut`, `Core.Words.Self.piReader_iff`, `Core.Words.Self.mortal_depth`
Channels, Definition 9.7 and Lemma 9.8	`Anchored.PosChan`, `Anchored.PosHole`, `Anchored.baseChan`, `Anchored.pos_base_iff_comparison`, `Anchored.pos_base_iff_anchored`
Desubstitution, towers, Lemma 9.10 and Corollary 9.11	`Anchored.posDer`, `Anchored.pos_der_iff`, `Anchored.posHole_der_iff`, `Anchored.pos_tower_iff`, `Anchored.pos_hierOf_iff`
Anchored exchange types and non-length-increasing isolated exchange, Lemmas E.3 and E.4†	`Anchored.anchoredType`, `Anchored.card_posRows`, `Anchored.anchoredTypeBound`, `Anchored.posIsolated_merge_spec`
Normal form, dictionaries, Lemma E.5	`Anchored.exists_posNormal_form`, `Anchored.posNormalAt_alphCard_le`, `Anchored.boundary_count_le_of_allowance`, `Anchored.alphCard_le_of_allowance`, `DeciNSSE.ElementaryBound.alphCard_depth_le_one`, `Anchored.exists_bounded_dictionary_posHole`, `Anchored.anchoredDictionaryBound`
Positive unary words, Lemma E.6, with $m' \le m$ in addition	`Anchored.unary_posHole_shorten`
Agreement, Lemma E.7	`Anchored.PosAgree`, `Anchored.PosAgree.posHole_iff`, `Anchored.PosAgree.derive`
Positive summaries, columns, Lemmas E.8 and E.10†	`Anchored.pcolDFA`, `Anchored.pcolDFA_eval`, `Anchored.pos_type_of_psumm`, `Anchored.card_posColumnState`, `Anchored.exists_short_pos_tuple`, `Anchored.exists_short_pos_letters`
Supported compression, Lemma E.11†	`Anchored.shorten_pos_level_supported`, `Anchored.exists_short_posHole_of_dictionary`
Anchored compression, Theorem 9.14 and Lemma E.12†	`Anchored.exists_short_posHole_elementary`, `Anchored.anchoredLengthBound`, `Anchored.anchoredExponent`, `Anchored.anchoredDictionaryBound_le`, `Anchored.anchoredLengthBound_le`
Short holes, Theorems 9.15 and 9.17†	`Core.Words.Self.exists_short_mortalHole`, `Core.Words.Mortal.mortalHoleBound`, `Core.Words.Self.mortalShortHoleBoundSelf`, `Core.Words.Self.shortHoleBoundSelf`
Extensions, two seeds, Lemmas 10.1 and 10.3†	`Core.Mixed.extends_iff_hull`, `Core.Mixed.two_observations_mixed`
Observation pins, Lemma 10.4, without the bound on the number of positions	`Core.Full.observation_bound`, `Core.Mixed.observation_bound_mixed`

Paper result	Formal counterpart
Intersection, Proposition 10.5(1) and the second claim of (3)	`Core.Mixed.sol_sInter`, `Core.Mixed.projected_sInter`, `Core.Mixed.sat_infoInter`, `Core.Mixed.infoInter_bot_top`, `Core.Mixed.allF_not_finite`
Finite-bound pinning, Theorem 10.6	`Core.Mixed.finite_bound_realization` (every arity), from `Core.Mixed.exists_finite_sat_groundBounds` (positive arity); `Core.Mixed.FinitePinning.symmetrize`, `Core.Mixed.FinitePinning.symmetrize_fixed`, `Core.Mixed.FinitePinning.symmetrize_sat`, `Core.Mixed.FinitePinning.symmetrize_pins`
Signed bounds, Lemma 10.7(1), (2), and the first claim of (3)	`Core.Mixed.FinitePinning.sat_normalize_iff`, `Core.Mixed.FinitePinning.pins_normalize_iff`, `Core.Mixed.FinitePinning.signDual_involutive`, `Core.Mixed.FinitePinning.sat_signDual`, `Core.Mixed.FinitePinning.pins_signDual`, `Core.Mixed.FinitePinning.normalize_fixed`, `Core.Mixed.FinitePinning.normalize_decode`, `Core.Mixed.FinitePinning.sat_decode`, `Core.Mixed.FinitePinning.pins_decode`
Median, Lemma 10.8, with symmetry in the first two arguments only	`Support.Median.treeMedian`, `Support.Median.treeMedian_swap`, `Support.Median.treeMedian_self`, `Support.Median.treeMedian_mono`, `Support.Median.treeMedian_node`, `Support.Median.d_treeMedian`, `Support.Median.finiteMedian`, `Support.Median.finiteMedian_toTree`, `Core.Mixed.FinitePinning.sat_median`, `Core.Mixed.FinitePinning.bound_median`
Covariant pinning, Lemma 10.9†	`Core.Full.finiteBoundPinning`, `Core.Full.Pinning.groundPins_iff`, `Core.Full.Pinning.realize_pins`, `Support.RankedExtension.extension_satFin_iff`
Regular trees, Theorem 10.11†	`Core.Mixed.satisfiable_regular`, `Core.Mixed.regular_bound_realization`, `Core.Mixed.regular_class_extension`, `Core.Full.TopGraphs.regularize`, `Core.Mixed.existsEntailsTReg_iff_mixed`
Finite trees, Theorem 10.12 and Corollary 10.13	`Core.Mixed.finite_bound_realization`, `Core.Mixed.finite_class_extension`, `Core.Mixed.existsEntailsTFin_iff_mixed`, `Core.Mixed.decideSatFin`
The signature $\{\bot, f\}$, Theorem 10.16 and Corollary 10.17†	replaced by `Core.three_conditions`, `Core.ss2_reduction`, `Core.existsEntails_iff_bounded`, `Core.existsEntailsReg_iff`, `Core.existsEntailsFin_iff`, `Core.Full.existsEntailsIn_bridge`
Main theorem, Theorem 11.2(1,2)	the six procedures of Appendix H.1 and their correctness theorems
Example 1.1, φ_1 and φ_2 over $\{\bot, f, \top\}$; Remark 10.14, first system at $n = 1$	`Examples.subtree_entails`, `Examples.subtree_not_entails`, `Examples.unaryCycle_classes`

The examples state the claims of Example 1.1 for φ_1 and φ_2 over $\{\bot, f, \top\}$ with f covariant, where the witness and the countermodel are $\top$-free, and the claims of Remark 10.14 for its first system at $n = 1$, for either variance. Not mechanized are Lemma 2.6; item (4) and the duality of cones in item (3) of Theorem 3.2; the signed equivalence of Lemma A.2; the polynomial-time claims of Lemmas 5.6 and 5.11, whose summary sets the formalization computes by finite searches; Lemmas 5.7, 6.9 and 6.10; the gadget of Lemma 10.15 and Theorem 10.16; the bounds in terms of s (Theorem 11.2(3,4), Proposition 11.5, Lemma 9.18, and Corollary 9.19, and the closed form in Theorem 6.11); the counterexample words and countermodels of Definition 11.1 and Lemma 11.3, including the non-extensible least model of Theorem 7.7(2); the results of Section 12 and Appendix G; and the complexity results of Section 13.

H.4 Paper and Formalization

- *Carriers and syntax.* The paper uses sets X and Z of variables. The public statement uses `Fin k` for the visible variables and `Fin (k + h)` for all variables, visible first; the core uses finite types `X` and `X ⊕ Z`. Literals are flat, as in Section 2.3, with `eqTop` for $\top \le u$. Plain entailment uses the flat literals of Zhao [2026], which have no copy literals; `liftConstraint` embeds them. The public deciders use root-label enumeration at arity zero; the class-transfer proofs use the fact that every nullary tree has a finite and regular presentation.
- *Sandwich elimination.* The translation `Core.Full.Flatten.flatten` requires positive arity. For $u \le v$ it uses one fresh witness per ordered pair (u, v) of variables in place of the fresh y, and the variable u itself as the filler in the arguments $1, \dots, n-1$ in place of the fresh $o = \bot$; `Core.Full.Flatten.sandwich_iff` shows that this sandwich still expresses $u \le_c v$. The transfer of solutions is stated over arbitrary trees (`Core.Full.Flatten.forward`, `Core.Full.Flatten.backward`), and the preservation of the tree classes for satisfiability in each class (`Core.Full.Flatten.finite_iff`, `Core.Full.satisfiable_regular`).
- *Satisfiability and* (I0). Satisfiability over arbitrary trees is decided by the criterion of Proposition 5.17(1). Regular solutions and finite satisfiability are obtained through sandwich elimination and the results of Zhao [2026] for systems without copy literals (`Core.Full.satisfiable_regular`, `Core.Full.decideSatFin`); the formalization does not state the signed equivalence of Lemma A.2. Proposition 5.17 decides satisfiability and (I0) in polynomial space; the formal tests are finite searches without a space bound. Forbidden roots are tested on the lower summaries, and the consistency of μ_χ and condition (I0) by words shorter than the number of states of a product of two automata for least closures (`Core.Mixed.consistent_mu_iff`, `Core.Mixed.condI0_bounded`).
- *Plain entailment.* Theorem 2.9(2) is a result of Zhao [2026]. Here the procedures for plain entailment are obtained from the existential ones (`nsse_iff_ensse`), and the equations of Theorem 2.9(2) for regular and finite trees follow from Theorems 10.11 and 10.12.
- *Summary sets.* The set $\mathsf{Fact}^{\uparrow}_{\chi}$ of Lemma 5.6 is `Core.Mixed.PF`, defined as the set of signed variables in the upper summaries of the facts along words of bounded length, and `Core.Mixed.mem_PF` extends this description to words of every length; the least-set description of $\mathsf{Fact}^{\uparrow}_{\chi}$ is not stated. The root-deadness tests of Lemma 5.11 are likewise finite searches over words of bounded length. The polynomial-time claims of Lemmas 5.6 and 5.11 are not mechanized.
- *Condition* (U). The procedure `Core.Mixed.decideCondU` applies the hole search `DeciNSSE.RejectedTail.decideHole` of Zhao [2026] to the unit readers `Core.Mixed.unitReader`. They run the source automaton of Lemma 6.3 with all its N_{U} states, so the search bound is $\mathsf{Len}(N_{\mathrm{U}}, d_{N_{\mathrm{U}}})$ in the notation of Corollary 2.21. The reduced reader and the rejection clock of Lemmas 6.9 and 6.10, which give the sharper bound of Theorem 6.11, are not needed for decidability.

- *Words.* Alphabets are arbitrary finite types α. The final short-hole theorem includes the empty alphabet by handling it separately; the non-mortal intermediate results use an inhabited alphabet. The spine is the first and the target the second word of a hole, as in the paper; the statement of `Core.Mixed.extractSelf_isHoleS_iff` names them t and u. A separator word is a word over `Option` α, with none as $\sharp$. The short-hole statements give $|u| + |t| \le B$ where Theorems 8.17 and 9.17 give $|u| + |t| < B$.
- *The pump.* The formal pump (`Core.Words.Self.SelfTwoWordInstance.pumpGood_short`) counts its deletion signatures in six product types of states, residues modulo the period, and flags. With a period of at most a this gives the capacity `Core.Words.pumpCapacity`, which at (a, b, a', b') is $2a^2ba'b' + a^2a' + abb' + 2aba'b' + aa' + bb' \ge \mathsf{P}(a, b, a', b')$. The non-mortal bound `Core.Words.pumpHoleBound` is $4\max(12aba'b', K) + 6a$ for this capacity K, as L_{nm} is for P, and is again bounded by a polynomial of degree five.
- *The Π-reader.* The formal reader keeps all states of its product types, reachable or not, so it has exactly $1 + N_A N_{A'} + N_B N_{B'} + N_A N_{B'} \ge \bar{N}_I$ states (`Core.Words.Mortal.card_state`). Its depth bound (`Core.Words.Self.mortal_depth`) and the mortal bound `Core.Words.Mortal.mortalHoleBound` use this count in place of $\bar{N}_I$.
- *Isolated exchange.* The formal isolated-exchange theorem (`Anchored.posIsolated_merge_spec`) assumes that the replacement letter has no greater expanded length. This is the specialization used to obtain the normal form.
- *Anchored constants.* The constants $\mathsf{C}^\sharp$, $\mathsf{A}^\sharp$, $\mathsf{G}^\sharp$, and $\Xi^\sharp$ of Eq. (18) are, in this order, `Anchored.anchoredTypeBound`, `Anchored.anchoredAllowance`, `Anchored.anchoredDictionaryBound`, and `Anchored.anchoredExponent`. The positive column state stores the column summary of the monitor and a second complete column summary of the monitor that carries F, R, and S as its target and admissions, where the paper adds only the fields of Lemma E.8; it records the channel test of the first letter of each block, not the letter itself. The stated formal shortening result (`Anchored.exists_short_pos_letters`) supplies subsequences and the agreement of the derived monitors and channels; it does not expose the common deletion of columns and the preservation of first letters stated in Lemma E.10. Its number of states, `Anchored.pcolBound`, is $P(M, k) = \mathrm{col}(M, k)^2\, 2^{2M(M+1)k}\, 3^k \ge \mathrm{col}^\sharp(M, k)$ for a monitor with M states and k tracks, with col from Eq. (14). Supported compression (`Anchored.shorten_pos_level_supported`) expands each letter to at most $P(N + d + 1, g)^j$ letters, in place of $\kappa_\sharp^j$ in Lemma E.11. Hence `Anchored.anchoredLengthBound` is $3(N + d + 1)\, P(N + d + 1, \mathsf{G}^\sharp(N, d))^d \ge \mathsf{Len}^\sharp(N, d)$, and `Anchored.anchoredLengthBound_le` proves the bound $2^{2^{5\Xi^\sharp(N,d)+6}}$ instead of $2^{2^{5\Xi^\sharp(N,d)}}$.
- *The signature $\{\bot, f\}$.* The formal proof does not reduce covariant $\{\bot, f\}$ to $\{\bot, f, \top\}$. Its core represents a tree by the prefix-closed set of its constructor positions (`Core.BFTree`), so that the order is inclusion (Lemma 2.3), and proves the analog of Theorem 4.2 with the conditions `Core.CondI0`, `Core.CondLive`, and `Core.CondII` (`Core.three_conditions`). Here a seed is live iff its closure avoids the forbidden roots, which takes the place of (U), and no self events occur. `Core.ss2_reduction` reduces `Core.CondII` to two-word holes of instances built from four powerset automata (`Core.extract`), and `Core.Words.shortHoleBound`, the case of `Core.Words.Self.shortHoleBoundSelf` with an empty self guard, bounds them. Regular and finite trees follow from regular Horn hulls and finite least solutions (`Core.existsEntailsReg_iff`, `Core.existsEntailsFin_iff`), and `Core.Full.existsEntailsIn_bridge` identifies constructor sets with $\top$-free trees in each tree class.

- *Two seeds.* The formal two-observation lemma (`Core.Mixed.two_observations_mixed`) assumes that the consequent is satisfiable and returns two observations, possibly equal. Unsatisfiable consequents are handled separately in the transfer proofs; the paper includes that case by allowing an empty observation set.
- *Regular transfer.* The paper proves (2) and (3) of Theorem 10.11 with regular least closures (Lemma 5.7). The formal proof follows the finite transfer instead, through the common `Core.Mixed.existsEntailsTIn_iff_of_transfer`. The two observations of `Core.Mixed.two_observations_mixed` become ground bounds (`Core.Mixed.observation_bound_mixed`), and `Core.Full.TopGraphs.regularize` keeps finitely many bounds by regular trees while passing to a regular solution, by adding the graph equations of their presentations, whose solution is unique (`Core.Full.TopGraphs.model_unique`), and applying (1). The same argument with both bounds $\rho(x) \le x \le \rho(x)$ gives (2).
- *Covariant pinning.* The paper encodes a ground bound by one fresh variable per position. The formal proof splits each bound into path pins (`Core.Full.Pinning.groundPins_iff`) and encodes each pin by a ranked chain of fresh variables (`Core.Full.Pinning.PathPin.extension`), whose cycles reflect into the original system (`Support.RankedExtension.extension_cycleClash_iff`).